

CITY UNIVERSITY OF HONG KONG
香港城市大學

**Statistical Learning Theory for Provably
Accurate and Efficient Scientific Machine
Learning Algorithms**

科學機器學習演算法之統計學習理論：可證明的準
確性與效率

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À ma grand-mère, Catherine TONLEU

Abstract

Across numerous scientific and industrial domains, the problem that researchers try to tackle boils down to numerically finding an approximate solution of a partial differential equation (PDE) which reliably satisfies a prescribed degree of accuracy with respect to the true solution. The classical field of Scientific Computing has, over the past decades, been providing efficient and principled ways to design such PDE-solving algorithms. These methods, however, typically scale poorly to high-dimensional problems and suffer from a lack of flexibility to irregular geometries due to their reliance on appropriate meshes. The recently emerging field of Scientific Machine Learning aims to overcome these shortcomings by applying the supervised machine learning methodology to the resolution of PDEs, leveraging the adaptability of (deep) neural networks and other related architectures to highly irregular, high-dimensional problems. Despite promising results and empirically demonstrated advantages over classical solvers, the theoretical underpinnings of most Scientific Machine Learning approaches remain far from fully understood.

In this thesis, we contribute towards a greater theoretical understanding of these algorithms in multiple ways. First, we provide new bounds on VC dimensions, pseudodimensions, and approximation power of hypothesis spaces of higher-order derivatives of fully connected neural networks with hyperbolic tangent and Rectified Quadratic Unit activation. Second, we combine these bounds with learning-theoretic tools to provide non-asymptotic bounds on the error of Physics-Informed Neural Networks (PINNs) trained by empirical risk minimization (ERM) to minimize the residual of a second-order elliptic PDE. Lastly, moving from the theoretical, but not practically computable ERM procedure, we propose a modification of the PINN loss that provably improves the loss landscape of PINNs, leading to guaranteed fast rates of convergence for PINNs trained with finitely many steps of gradient descent. We illustrate all of our theoretical results with numerical experiments on a variety of PDEs, including Poisson, exit-time, and committor problems.

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*One night I dreamed I was walking along the beach with the Lord.
Many scenes from my life flashed across the sky.
In each scene I noticed footprints in the sand.
Sometimes there were two sets of footprints,
other times there were one set of footprints.*

*This bothered me because I noticed
that during the low periods of my life,
when I was suffering from
anguish, sorrow or defeat,
I could see only one set of footprints.*

*So I said to the Lord,
'You promised me Lord,
that if I followed you,
you would walk with me always.
But I have noticed that during the most trying periods of my life
there have only been one set of footprints in the sand.
Why, when I needed you most, you have not been there for me?'*

*The Lord replied,
'The times when you have seen only one set of footprints in the sand,
is when I carried you.'*

— Mary Stevenson, *Footprints in the Sand*

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Chapter 1

Introduction

Over the past few decades, the impressive scientific, technological, and industrial successes of machine learning have taken the world by storm. From content recommendation (Linden et al., 2003; Koren et al., 2009), image classification (Krizhevsky et al., 2012; Tepakbong et al., 2026c), mastering complex strategy games (Silver et al., 2016), predicting protein structure (Jumper et al., 2021), solving olympiad mathematics problems (Trinh et al., 2024), and more recently generating high quality images (Ho et al., 2020) and text (Vaswani et al., 2017), it may appear as if combining a very large amount of data with a comparably large amount of compute allows a computer to learn to solve almost any task. At a high level, most of these systems rely on the same principle. One chooses a family of computable, parameterized functions or algorithms, which acts as an inductive prior, and formulates the task at hand as the problem of finding parameters that optimize a suitable training criterion, typically specified in terms of the available data. The hope is that the underlying data-generating patterns can be accurately captured as the parameters of the model are adjusted, through *training*, to minimize this criterion (Bishop, 2006).

Despite the clear and overwhelming successes of this approach on many data-driven tasks, the theoretical understanding of modern machine learning methods, and in particular those based on artificial neural networks, remains limited. Notable progress has been made on aspects such as expressive power (Zhou, 2003; Yarotsky, 2017; Zhou, 2020), training dynamics (Du et al., 2019; Allen-Zhu et al., 2019), and generalization ability (Koltchinskii, 2011; Schmidt-Hieber, 2020), yet there is still a wide gap between what mathematical theory can guarantee and how these models are used in large scale, high stakes applications (Zhang et al., 2017). The field of *statistical*

learning theory (Vapnik, 2013) aims to narrow this gap by developing a unified framework to understand how learning algorithms interact with data, how well they can be expected to perform on unseen inputs, and how their complexity should scale with the difficulty of the task. Its ultimate goal is not only to explain why existing methods work, but also to guide the design of algorithms that are efficient, reliable, and interpretable (Bartlett and Anthony, 1999; Cucker and Zhou, 2007; Bishop, 2006; Bach, 2024). In the remainder of this chapter we shall see how these questions become particularly pressing in the context of numerical methods for PDEs and their recently proposed scientific machine learning counterparts.

1 From numerical PDEs to scientific machine learning

In many scientific and industrial applications, such as civil and mechanical engineering, network design, climate modelling, or biomedical imaging, reliability, stability, and rigorous uncertainty quantification of the numerical algorithms are non-negotiable. In such settings, off-the-shelf machine learning methods, although extremely powerful as general-purpose prediction tools, rarely come with the type of guarantees that practitioners are accustomed to. Instead, the standard toolbox is provided by the mathematically mature field of numerical analysis (which we will also loosely refer to as *scientific computing*) (Iserles, 1996; Gander and Hrebicek, 2011). A large part of this field is concerned with the numerical solution of partial differential equations (PDEs) that model the evolution of physical, biological, or socio-economic systems.

Even a highly non-exhaustive list of model PDEs already covers a remarkably broad range of applications. For a domain $\Omega \subset \mathbb{R}^d$ and time horizon $T > 0$, typical examples include:

- **Elliptic diffusion and Poisson equations.** Steady-state diffusion of heat or chemical species, electrostatics, and linear elasticity can often be phrased as finding $u: \Omega \rightarrow \mathbb{R}$ such that

$$-\nabla \cdot (a(x)\nabla u(x)) = f(x) \text{ in } \Omega,$$

supplemented with appropriate boundary conditions on $\partial\Omega$. Here a denotes a positive definite diffusion or conductivity tensor and f a source term (Evans, 2022).

- **Parabolic diffusion and heat equations.** Transient heat conduction, diffusion of chemical species, and the evolution of probability densities in some stochastic models are described by parabolic PDEs of the form

$$\partial_t u(t, x) - \nabla \cdot (a(x) \nabla u(t, x)) = f(t, x) \text{ in } (0, T) \times \Omega,$$

with initial data $u(0, x) = u_0(x)$ and suitable boundary conditions (Evans, 2022).

- **Fokker–Planck and Kolmogorov equations.** The evolution and long-time behaviour of stochastic dynamical systems, for instance in molecular dynamics or climate models, are often characterized through linear second-order operators of the form

$$\mathcal{L}u(x) = b(x) \cdot \nabla u(x) + \frac{1}{2} \text{Tr}(\sigma(x)\sigma(x)^\top \nabla^2 u(x))$$

acting on functions on Ω . The associated backward Kolmogorov equation $\mathcal{L}u = -1$ in Ω with Dirichlet boundary conditions on $\partial\Omega$ describes mean exit times, while the adjoint equation for the probability density gives the Fokker–Planck equation (Pavliotis, 2014). Problems of this type will play an important role later in the thesis.

- **Nonlinear conservation laws and fluid dynamics.** Many flow and transport phenomena are governed by nonlinear PDEs, ranging from scalar conservation laws to the incompressible Navier–Stokes equations

$$\partial_t u(t, x) + (u(t, x) \cdot \nabla) u(t, x) = -\nabla p(t, x) + \nu \Delta u(t, x) + f(t, x) \text{ in } (0, T) \times \Omega,$$

together with the incompressibility constraint $\nabla \cdot u(t, x) = 0$. These equations underpin models in aerodynamics, weather and climate prediction, and biomechanics, and they illustrate the strongly nonlinear, multiscale character of many PDEs of interest (Temam, 2024).

The clear mathematical structure of such PDE models has given rise to a broad arsenal of discretization techniques, including finite difference, finite volume, finite element, and spectral methods (Iserles, 1996; Brenner and Scott, 2007). These methods approximate the infinite-dimensional problem by a finite-dimensional linear or nonlinear system and come with a mature theory that quantifies stability and convergence rates in terms of mesh size and regularity of the coefficients and data. This theoretical grounding underpins many industrial software packages and has enabled decades of reliable simulation-based engineering design.

However, classical solvers also face structural limitations. Mesh generation and adaptivity can become complex and costly for highly irregular geometries or moving interfaces. High-fidelity models in uncertainty quantification, optimal control, or Bayesian inverse problems require solving the same PDE for a large number of parameter realizations, which quickly becomes prohibitive when each solve costs as much as a full finite element computation. Moreover, the curse of dimensionality appears in problems where the spatial domain is high dimensional, for example kinetic equations or PDEs posed on configuration spaces, where mesh-based methods scale poorly with dimension (Bungartz and Griebel, 2004; Cottrell et al., 2009).

In parallel with these developments, the recent success of machine learning has prompted a growing interest in combining data-driven models with PDE-based scientific computing. Under the broad label of *scientific machine learning* (SciML) (E and Yu, 2018; Karniadakis et al., 2021), a variety of approaches have been proposed. Physics-informed neural networks minimize a loss built from the PDE residual and boundary conditions, treating the solution u as the output of a neural network (Raissi et al., 2019; Berg and Nyström, 2018). Neural-operator architectures, such as DeepONets and Fourier neural operators, aim directly at learning the solution operator that maps coefficients or forcing terms to solutions (Lu et al., 2021a; Li et al., 2021; Kovachki et al., 2023). Other lines of work use learned surrogates, emulators, or preconditioners to accelerate existing solvers rather than replace them (Benner et al., 2015; Greenfeld et al., 2019). In all of these cases the hope is to exploit the flexibility and expressivity of modern learning architectures to handle complex geometries, parameter dependence, and possibly higher effective dimensions than classical grid-based methods can reach.

Empirical studies for a wide range of benchmark problems are encouraging, yet the reliability and efficiency of SciML methods are still far from being understood at the same level as classical numerical schemes. In particular, rigorous error estimates, stability guarantees, and clear guidance on the choice of architectures and training procedures are often missing or only available in restricted settings (Mishra and Molinaro, 2022; Kutyniok et al., 2022). Providing a learning-theoretic perspective on these questions is the main theme of this thesis.

2 Statistical learning for theoretically principled SciML

In light of the discussion above, current SciML methods occupy an uncomfortable middle ground. Classical PDE solvers come with a mature theory: well-posedness and stability results, a priori and a posteriori error estimates, and asymptotic rates that make the dependence on mesh size, regularity of coefficients, and domain geometry largely transparent. In contrast, most SciML algorithms still behave as sophisticated heuristics. While they can handle complex geometries, parametric dependence, or high-dimensional inputs more flexibly than mesh-based schemes, we typically lack nonasymptotic error guarantees of comparable robustness. It is often unclear how the final error depends on the network architecture, the sampling of training points, or the regularity of the underlying PDE solution, how sensitive the method is to hyperparameter choices, and how far one can trust the learned model away from the specific training configuration.

These difficulties are exacerbated by the way SciML models are trained. Residual-based approaches such as PINNs combine interior residual terms, boundary discrepancies, and sometimes additional regularization into a single loss, leading to highly nonconvex and often severely ill-conditioned optimization problems. Gradient-based training can stagnate, converge extremely slowly, or settle on functions that satisfy parts of the loss while badly violating the PDE or boundary conditions elsewhere (Krishnapriyan et al., 2021; Wang et al., 2021). Similar sensitivities and failure modes arise for neural operators and reduced models. Altogether, there is a clear need for a framework that (i) makes precise what is being learned, (ii) decomposes the different sources of error, and (iii) explains how design choices—architectures, losses, sampling strategies—affect reliability. Statistical learning theory provides such a framework.

Statistical learning theory (Vapnik, 2013; Koltchinskii, 2011) studies how well one can learn from finite data. Very schematically, one chooses a hypothesis class $\mathcal{H}$ of candidate functions (or operators) and a loss functional L , and seeks $f \in \mathcal{H}$ that minimizes the *risk*

$$R(f) = \mathbb{E}[L(f(Z), Z)],$$

where Z denotes the random input (and possibly output) data. Since the distribution of Z is unknown, one observes independent samples $Z_1, \dots, Z_n$

and minimizes instead the empirical risk

$$R_n(f) = \frac{1}{n} \sum_{i=1}^n L(f(Z_i), Z_i).$$

Learning theory then aims to control the *excess risk* $R(f_n) - \inf_{f \in \mathcal{H}} R(f)$ for the output f_n of a learning algorithm, and to understand how this quantity decomposes into approximation error (how well $\mathcal{H}$ can represent the target), statistical error (how much information the sample contains about the best element of $\mathcal{H}$), and optimization error (how far the algorithm is from minimizing R_n).

In the SciML setting, this point of view can be adapted to different algorithmic paradigms. For PINNs and related residual-minimization methods (E et al., 2017; Han et al., 2018; Raissi et al., 2019), the random variable Z can be thought of as a space–time point together with information about which component of the loss (interior, boundary, or auxiliary term) is active; the loss L encodes the squared PDE residual, boundary mismatch, or other constraints at that point. For operator-learning architectures such as DeepONets or neural operators, Z is typically an input–output pair (a, u) , where a is a random coefficient or forcing function and u the corresponding PDE solution computed by a high-fidelity solver (Lu et al., 2021a). In both cases the hypothesis class $\mathcal{H}$ consists of neural networks or neural operators, and the central questions become: how rich $\mathcal{H}$ is for the PDEs of interest, how many samples are needed to achieve a desired accuracy, and how close realistic gradient-based procedures can get to ideal empirical risk minimizers. The rest of this thesis will make these questions precise for concrete problems and architectures.

For classical supervised learning tasks in finite-dimensional spaces, this programme is by now well developed. There exist sharp nonasymptotic generalization bounds that relate the sample size, the regularity or complexity of the target function, and the size or norm of the predictor, often achieving minimax-optimal rates up to logarithmic factors (Bartlett and Anthony, 1999; Koltchinskii, 2011; Bach, 2024). These results are typically phrased in terms of excess risk bounds and oracle inequalities for empirical risk minimization procedures, and they identify how approximation and statistical errors trade off as one varies the hypothesis class or complexity control.

By contrast, the analogous theory for SciML methods remains fragmentary and often far removed from the regimes in which these methods are applied. For PINNs and other residual-minimization schemes, several recent works establish consistency and convergence rates for linear elliptic and parabolic

PDEs under strong smoothness and regularity assumptions (Lu et al., 2022; Jiao et al., 2022; Shin et al., 2023). However, these results typically suffer from one or more of the following limitations:

- they assume, either explicitly or implicitly, that Dirichlet boundary conditions are enforced exactly by the hypothesis space, for instance through a built-in ansatz, or that penalty parameters in soft-constraint formulations have been tuned so that constraints are effectively satisfied (Lu et al., 2022);
- they measure error primarily in relatively weak norms, such as L^2 or energy norms, whereas applications often care about stronger norms which capture more accurately the solution behaviour (Jiao et al., 2022),
- they are formulated at the level of an ideal empirical risk minimizer over the chosen hypothesis class and do not take into account that practical training consists of finite-time gradient-based optimization on a highly nonconvex loss (Kutyniok et al., 2022; Shin et al., 2023).

In particular, existing analyses rarely quantify how the enforcement of boundary conditions through hard constraints or distance-to-boundary ansatzes (Chen et al., 2020; Sukumar and Srivastava, 2022; Berrone et al., 2023) affects approximation and generalization, and they generally do not track the dependence of the constants and rates on these design choices. As a result, current bounds can be difficult to interpret for realistic network sizes, collocation budgets, and training procedures, and they offer limited guidance for choosing architectures, sampling strategies, or boundary-handling mechanisms for a given PDE (Wu et al., 2023b; Jiang et al., 2024; Bonito et al., 2025).

On the approximation side, it has been shown deep neural networks with ReLU or higher-order ReLU^k activations can approximate Sobolev-regular functions with rates that can partially mitigate the curse of dimensionality in certain regimes (Yarotsky, 2017; Siegel and Xu, 2020; Gühring and Raslan, 2021). However, most of these results crucially rely on large network depths and highly structured network constructions. In SciML practice, especially with higher-order activations such as ReLU^k , practitioners often resort to comparatively shallow architectures because deeper networks become unstable or prohibitively expensive to train.

At the same time, SciML losses involve first- and second-order derivatives of the network output, yet our understanding of the capacity of such *derivative* hypothesis classes is still rudimentary: only a few results control complexity

measures for gradients or Hessians of neural networks (Yang et al., 2023; Lu et al., 2022; Lei et al., 2025), and quite few address higher-order derivatives for many architectures used in practice. This lack of derivative-sensitive capacity bounds makes it difficult to turn approximation-theoretic bounds into nonasymptotic error guarantees for PDE-focused empirical risk minimization procedures. One of the aims of this thesis is thus to help fill this gap by developing VC- and pseudo-dimension bounds for derivatives of networks using some of the most popular activations used in SciML.

In parallel, there is a growing literature that focuses explicitly on the optimization pathologies of PINN training. Empirical and theoretical studies document issues such as loss imbalance between interior and boundary terms, extremely ill-conditioned Jacobians of the residual map, and convergence to spurious local minima or flat regions of the loss landscape (Krishnapriyan et al., 2021; Wang et al., 2021). More recently, several works have analysed these phenomena from an operator-conditioning perspective, relating gradient descent dynamics on PINN objectives to the spectrum of PDE-induced operators or to Hessian-like quantities associated with the loss (De Ryck et al., 2024; Rathore et al., 2024; Wang et al., 2022; Chen et al., 2024; Wang et al., 2025). This operator-centric view suggests that successful training is not merely a matter of expressive networks or dense collocation sampling, but also of constructing losses and optimization schemes whose associated operators are well conditioned, for instance via preconditioning or geometry-aware training algorithms (Müller and Zeinhofer, 2023; Schwencke and Furtlehner, 2025; Chen et al., 2025). However, these analyses typically treat training as a numerical optimization problem and only rarely connect improved conditioning to learning-theoretic guarantees or to nonasymptotic error bounds.

Taken together, these observations point to a substantial gap between the level of rigor that numerical analysts expect for PDE solvers and the theory currently available for SciML methods. A central aim of this thesis is to help close this gap for selected classes of problems by (i) developing complexity and approximation bounds that explicitly involve higher-order derivatives of neural networks with activations used in SciML, (ii) deriving nonasymptotic error bounds for PINN-type empirical risk minimization procedures that carefully track the dependence on dimension, regularity, network size, and boundary handling, and (iii) analyzing how augmenting residual-based losses used in SciML with a “noisy data” term can provably improve the conditioning of training and lead to more reliable algorithms.

3 Contributions of this thesis

This thesis is a contribution to the learning-theoretic framework used to analysing SciML algorithms. Specifically, focusing on Physics-informed neural network architectures used to approximate solutions of linear elliptic and parabolic partial differential equations (Karniadakis et al., 2021), we provide new pathways towards sharp and practically relevant theoretical guarantees for these algorithms in numerous ways:

- **VC dimension and pseudo-dimension bounds for higher-order derivatives of ReLU² and tanh networks.** We derive, for the first time, bounds on the VC dimensions and pseudo-dimension of higher-order derivatives of fully-connected neural networks using either ReLU² or tanh activation. These bounds hold for derivatives of order up to 2 in the ReLU² case (after which, the networks are not differentiable anymore), and for derivatives of arbitrarily high order in the tanh case. The precise bounds with full derivation are presented in Chapter 3.
- **$W^{2,\infty}(\Omega)$ approximation error bounds of $C^r(\Omega)$ -smooth functions by shallow ReLU² networks.** Using the recently proposed variation space framework, consisting in studying the approximation abilities of shallow neural networks in the infinite width limit (Bach, 2017; Siegel and Xu, 2020, 2022), we extend a recent result of Yang and Zhou (2025) regarding the $L^\infty(\Omega)$ approximation abilities of shallow ReLU^k networks to $W^{2,\infty}(\Omega)$ approximation bounds for shallow ReLU² networks. The precise statement with complete proof is presented in Chapter 3.
- **A practical criterion for provably $H^2(\Omega)$ -consistent enforcement of Dirichlet boundary conditions.** Although PINNs are generally trained by minimizing a loss corresponding to the sum of a PDE-residual term and a boundary-residual term, we highlight that this methodology is in fact *inconsistent* with error control in $H^2(\Omega)$. We propose instead to multiply the PINN output by a “smooth distance-to-boundary approximation” ρ (Sukumar and Srivastava, 2022), and give sufficient and necessary conditions on ρ to ensure $H^2(\Omega)$ consistency. The precise statement is give in Chapter 3, while the proof together with an algorithm to compute ρ is given in Appendix D.
- **$H^2(\Omega)$ a priori error bounds for “boundary-adapted PINNs” solving elliptic PDEs with Dirichlet boundary conditions.** Combining the three above items with an oracle inequality due to (Lei et al., 2025), we get sharp a priori error bounds in $H^2(\Omega)$ norm for PINN solu-

tions of linear elliptic Dirichlet boundary value problems. The precise statement and its proof is given in Chapter 3, while a detailed discussion on how these results can be extended to parabolic equations is given in Appendix C.

- **A regularization scheme to provably alleviate the “failure modes” of PINNs.** Moving closer to practical concerns, we propose a simple and tractable new computational method to enhance and stabilize the training of PINNs by gradient descent, which is well-known to be extremely challenging (Wang et al., 2021; Krishnapriyan et al., 2021). Our method, which we call “noisy data supervision” is explained in detail, and its loss landscape properties are mathematically proven in Chapter 4.
- **An end-to-end error analysis of “regularized PINNs” trained with finitely many steps of gradient descent.** Combining our learning-theoretic bounds with the guarantees regarding training stability, we derive an end-to-end $L^2(\Omega)$ error analysis for shallow PINNs trained with finitely many steps of (full-batch) gradient descent to solve elliptic and parabolic PDEs. The error bounds are explicit in terms of the number of training steps, neural network architecture, or number of collocation points. The precise statement and proof are given in Chapter 5.

All of the theoretical results are substantiated by numerical experiments, which we present in Chapter 6. Put together, this thesis’ contributions greatly enhance our understanding of the theoretical underpinnings of SciML methods, and pave the way towards truly reliable machine-learning inspired solutions to scientific problems.

4 Organization of the thesis

The remainder of the thesis is structured as follows.

In Chapter 2, we will lay down the learning-theoretic foundations on which the following chapters will be based. We will introduce all the fundamental concepts such as excess risk bounds, oracle inequalities, capacity notions such as VC- and pseudo-dimension, and basic empirical process tools in a form adapted to residual-type and derivative-based losses. This chapter will mostly serve as theoretical background and is intended to make the thesis more self-contained, as well as the arguments that will appear in the following chapters easier to follow.

In Chapter 3, we will apply this learning-theoretic machinery to obtain $H^2(\Omega)$ error bounds for *boundary-adapted* PINNs that solve second-order elliptic Dirichlet problems, with *mean exit time* problems (Pavliotis, 2014) as our main motivation.

In Chapter 4, we will address the optimization pathologies of PINNs by introducing FK-PINNs, where a small number of Feynman–Kac Monte Carlo estimates are added as a data-fidelity term to the usual residual and boundary losses, which we will show, building on recent operator-conditioning analyses of PINN objectives (Krishnapriyan et al., 2021; De Ryck et al., 2024), enjoy greatly enhanced loss landscape and convergence guarantees.

In Chapter 5, we combine all the results from the previous chapters to derive an end-to-end error analysis for FK-PINNs trained with finitely many steps of gradient descent.

In Chapter 6, we present a number of numerical experiment, which illustrate the accuracy of our theoretical results. In particular, we show how the choice of distance-to-boundary function ρ can make or break the training of PINNs. We also show how a simple data supervision term can totally break through severe failure modes of PINNs, confirming that this approach is a promising way towards reliable scientific machine learning.

Lastly, in the Appendices A, B, C, D, and E, we provide additional discussion, results and proofs which complement the main text.

Chapter 2

The Empirical Risk Minimization Framework

Many procedures across statistics and machine learning can be phrased as minimizing an average of data-dependent losses. Least squares and maximum likelihood are classical examples, while M -estimation formalizes the same idea for general objectives (Van Der Vaart and Wellner, 1996). In learning theory this viewpoint appears as *empirical risk minimization* (ERM): its purpose is to turn the optimization of a sample average into guarantees for the corresponding population objective under the unknown data-generating distribution (Vapnik, 2013). The difficulty is that the empirical risk R_n is only a random approximation of the population risk R . Minimizing R_n can therefore overfit unless R_n approximates R uniformly over the candidate class.

In this chapter, we therefore introduce the “standard toolkit” upon which we will build to obtain bounds on the uniform deviations, and further translate them into excess risk bounds. Here, we will not introduce any new result, but rather we will define all the necessary concepts, terminology, and state the theorems we will be relying on for the error analyses in the following chapters. For the sake of exposition and being self-contained, we will provide, whenever possible, proofs of some of the non-trivial results. Some references one can consult to learn more on the topics introduced in this chapter include (Mohri et al., 2018; Sen, 2018; Wainwright, 2019; Bach, 2024) and references within. Lastly, we note that we deliberately choose to keep the presentation mostly problem-agnostic in this chapter, in order to show the general idea. We will highlight how this machinery can be applied for PDE-solving tasks in the last section, as well as the next chapters.

The rest of this chapter is organized as follows: In Section 1 we introduce the learning setup and define the basic concepts such as hypothesis classes, estimators, losses, risks, and excess risk. In Section 2, we formally introduce ERM and prove a basic inequality that expresses the excess risk as a uniform deviation term, and will be the basic ingredient to establish excess risk bounds later down the line. In Sections 3 to 5, we will show how to bound the uniform deviation term under different notions of complexity, from finite classes, to VC type dimensions and then to Rademacher complexity with a contraction principle for Lipschitz compositions. Finally, in Section 6, we discuss in greater detail how the approximation error enters this analysis, and finally explain how PDE-solving tasks can be made to fit into this framework.

Throughout this chapter, $Z, Z_1, \dots, Z_n$ will be i.i.d. random variables with values in a measurable space $(\mathcal{Z}, \mathcal{A})$ and common law P . For a measurable $g : \mathcal{Z} \rightarrow \mathbb{R}$, write $Pg := \mathbb{E}[g(Z)]$ and $P_n g := n^{-1} \sum_{i=1}^n g(Z_i)$. The centred empirical process is $(P_n - P)g := P_n g - Pg$. This notation is standard in empirical process theory (Van Der Vaart and Wellner, 1996; Koltchinskii, 2011).

1 Risk, empirical risk, and excess risk

A unifying viewpoint in modern statistics and machine learning is that an estimator is constructed by optimizing a data-dependent objective. This covers classical least squares and maximum likelihood, as well as regularized and constrained variants (Van Der Vaart and Wellner, 1996; Bach, 2024). The purpose of this section is to define the basic objects in this optimization viewpoint and to isolate a performance criterion that will be controlled in the rest of the chapter.

Definition 2.1. Let $(\mathcal{Z}, \mathcal{A})$ be a measurable space and let P be an unknown probability measure on $\mathcal{Z}$. A sample of size n is an i.i.d. sequence $Z_1, \dots, Z_n$ with law P .

The interpretation of Z depends on the application. In supervised learning one has $Z = (X, Y)$ with input X and label Y . In likelihood-based inference one often takes Z to be a single observation and uses the negative log-likelihood as the loss. Other examples include, for instance, density estimation, inverse problems, and model selection (Mohri et al., 2018; Wainwright, 2019).

Definition 2.2 (Hypothesis class). A hypothesis class is a set $\mathcal{H}$ of measurable functions. Elements of $\mathcal{H}$ are called hypotheses, predictors, or models.

The class $\mathcal{H}$ represents the set of admissible candidates among which a statistician will pick an element to serve as her predictor. Typical examples include linear predictors and generalized linear models, sieve estimators based on finite-dimensional bases, RKHS balls, and neural network classes defined by architectural constraints and norm bounds. Different choices of $\mathcal{H}$ lead to different approximation properties and different generalization behaviour (Mohri et al., 2018).

Definition 2.3. *An estimator with values in $\mathcal{H}$ is defined as a measurable map $\hat{f}_n : \mathcal{Z}^n \rightarrow \mathcal{H}$. With a slight abuse of notation, $\hat{f}_n$ also denotes the random hypothesis $\hat{f}_n(Z_1, \dots, Z_n)$.*

Having now defined the concept of hypothesis class and estimator, we can now introduce that of a *loss function*, which effectively provides the learner with a criterion on which the choice of hypothesis can be made. In supervised learning, losses often have the form $L(f, (x, y)) = \ell(f(x), y)$ for a pointwise loss ℓ . The choice $\ell(\hat{y}, y) = (\hat{y} - y)^2$ yields least squares regression, while $\ell(\hat{y}, y) = \mathbf{1}_{\{\hat{y} \neq y\}}$ yields misclassification risk. In likelihood-based estimation, ℓ is typically the negative log-likelihood. Convex surrogate losses are widely used when the 0–1 loss is computationally inconvenient or statistically unstable (Bach, 2024).

Definition 2.4 (Loss, risk, and empirical risk). *Let $\mathcal{H}$ be a hypothesis class in the sense of Definition 2.2. A loss is a measurable map $L : \mathcal{H} \times \mathcal{Z} \rightarrow \mathbb{R}$. For $f \in \mathcal{H}$ write $L_f(z) := L(f, z)$. The risk and empirical risk are respectively defined by*

$$R(f) := PL_f = \mathbb{E}[L(f, Z)], \text{ and } R_n(f) := P_n L_f = \frac{1}{n} \sum_{i=1}^n L(f, Z_i).$$

In several places, especially when discussing approximation error, it is convenient to compare hypotheses in $\mathcal{H}$ to reference predictors that may not belong to $\mathcal{H}$. To avoid repeatedly restating this point, we will assume when needed that the loss L extends to a larger reference family $\bar{\mathcal{H}} \supset \mathcal{H}$, and that $R(f) := PL_f$ is defined for $f \in \bar{\mathcal{H}}$ with the same meaning.

Although the risk $R(f)$ is the target objective, it depends on the distribution P from Definition 2.1, which is typically unknown and therefore makes $R(f)$ intractable as an optimization criterion. The empirical risk $R_n(f)$ is defined by replacing the expectation by a sample average, therefore making it tractable as an optimization criterion. We sometimes refer to the discrepancy $R(f) - R_n(f) = (P - P_n)L_f$ as the *generalization gap* of f . While for a fixed f , this gap can be reasonably well controlled by standard tools

such as concentration inequalities (Boucheron et al., 2013), whenever f is chosen as a function of the sample itself, those elementary arguments are not enough anymore. Indeed, for data-dependent choices of f , such as the output of empirical risk minimization, we now need a uniform control over $f \in \mathcal{H}$ to obtain meaningful error bounds. To this end, the theory of *empirical processes* has been developed, which bounds suprema of stochastic processes indexed by arbitrary sets such as function classes (Van Der Vaart and Wellner, 1996; Koltchinskii, 2011).

The class of loss functions induced by $(\mathcal{H}, L)$ will appear repeatedly. It is denoted by $\mathcal{L} := \{L_f : f \in \mathcal{H}\}$.

Definition 2.5 (Excess risk). *For an estimator $\hat{f}_n \in \mathcal{H}$, the excess risk relative to $\mathcal{H}$ is*

$$\mathcal{E}(\hat{f}_n; \mathcal{H}) := R(\hat{f}_n) - \inf_{f \in \mathcal{H}} R(f).$$

Excess risk measures the “price paid” for minimizing a data-based approximation of the true risk, by comparing to the best element in $\mathcal{H}$. It is the natural quantity controlled by uniform deviation bounds because it isolates the stochastic error due to the replacement of P by P_n . In the next section, we introduce empirical risk minimization and show how excess risk can be bounded by a single uniform deviation term.

2 Empirical risk minimization and one basic inequality

Given a loss L and a class $\mathcal{H}$, a direct approach is to choose an estimator by minimizing the empirical risk R_n from Definition 2.4. As we discussed, this principle is called *empirical risk minimization* (ERM). It is the learning-theoretic analogue of the plug-in principle for M -estimators in statistics (Van Der Vaart and Wellner, 1996).

Definition 2.6 (Empirical risk minimizer). *An empirical risk minimizer over $\mathcal{H}$ is any estimator $\hat{f}_n$ in the sense of Definition 2.3 such that $R_n(\hat{f}_n) = \inf_{f \in \mathcal{H}} R_n(f)$, whenever the infimum is attained.*

Note that minimizers may not exist, or sometimes fail to be measurable. We will put these technical issues to the side for the remainder of this thesis, as they do not affect the general conclusions provided by our framework. We nonetheless refer the interested (and rigorous!) reader to (Van Der Vaart and Wellner, 1996; Koltchinskii, 2011) and references within for a comprehensive

treatment of these delicate issues. From a much more practice-oriented perspective however, it happens that in most cases, we are not able to *exactly compute* a global minimizer $\hat{f}_n$, but are rather limited to use algorithms such as gradient descent to compute approximate minimizers. This motivates the following definition of an *approximate* risk minimizer.

Definition 2.7 (Approximate ERM). *Let $\varepsilon_{\text{opt}} \geq 0$. An estimator $\hat{f}_n \in \mathcal{H}$ is an ε_{opt} -approximate ERM if $R_n(\hat{f}_n) \leq \inf_{f \in \mathcal{H}} R_n(f) + \varepsilon_{\text{opt}}$.*

Typically, the approximate ERM is the actual model that practitioners work with after having performed their “training” on the empirical risk, it is therefore the estimator for which we would like to have clear guarantees. To this end, we introduce the following lemma, which gives us a way to achieve just that by reducing excess risk to a uniform deviation term plus the optimization error ε_{opt} .

Lemma 2.8 (Basic inequality). *Let $\hat{f}_n$ be an ε_{opt} -approximate ERM in the sense of Definition 2.7. Then*

$$\mathcal{E}(\hat{f}_n; \mathcal{H}) \leq 2 \sup_{f \in \mathcal{H}} |(P_n - P)L_f| + \varepsilon_{\text{opt}}.$$

Proof. Fix $f \in \mathcal{H}$. Using Definition 2.4,

$$R(\hat{f}_n) - R(f) = (P - P_n)L_{\hat{f}_n} + (R_n(\hat{f}_n) - R_n(f)) + (P_n - P)L_f.$$

Definition 2.7 gives $R_n(\hat{f}_n) - R_n(f) \leq \varepsilon_{\text{opt}}$. The remaining two terms are each bounded above by $\sup_{h \in \mathcal{H}} |(P_n - P)L_h|$. Taking the infimum over $f \in \mathcal{H}$ yields the claim. $\square$

Lemma 2.8 shows that the analysis of ERM reduces to controlling the random quantity $\sup_{f \in \mathcal{H}} |(P_n - P)L_f|$. This is the uniform deviation of the loss class $\mathcal{L} = \{L_f : f \in \mathcal{H}\}$ introduced after Definition 2.4. In the following sections, we will present some of the standard machinery available to bound this deviation term, and apply to actual SciML problems in the following chapters.

Corollary 2.9 (Plug-in excess risk bound). *Suppose there exists a deterministic function $\varepsilon_{\text{gen}}(n, \delta)$ such that*

$$\mathbb{P}\left(\sup_{f \in \mathcal{H}} |(P_n - P)L_f| \leq \varepsilon_{\text{gen}}(n, \delta)\right) \geq 1 - \delta.$$

Then with probability at least $1 - \delta$, every ε_{opt} -approximate ERM $\hat{f}_n$ satisfies

$$\mathcal{E}(\hat{f}_n; \mathcal{H}) \leq 2\varepsilon_{\text{gen}}(n, \delta) + \varepsilon_{\text{opt}}.$$

Corollary 2.9 controls excess risk relative to $\mathcal{H}$. In several applications we will compare the learned predictor to a reference function f^* that need not belong to $\mathcal{H}$, for instance a ground-truth solution to a PDE. The next lemma quantifies the corresponding decomposition, which separates estimation and approximation effects.

Lemma 2.10 (Risk decomposition relative to a reference predictor). *Assume that L extends to a reference family $\bar{\mathcal{H}} \supset \mathcal{H}$, and fix any $f^* \in \bar{\mathcal{H}}$. Let $\hat{f}_n$ be an ε_{opt} -approximate ERM over $\mathcal{H}$ in the sense of Definition 2.7. Then*

$$R(\hat{f}_n) - R(f^*) \leq 2 \sup_{f \in \bar{\mathcal{H}}} |(P_n - P)L_f| + \varepsilon_{\text{opt}} + \inf_{f \in \bar{\mathcal{H}}} (R(f) - R(f^*)).$$

Proof. Decompose

$$R(\hat{f}_n) - R(f^*) = (R(\hat{f}_n) - \inf_{f \in \mathcal{H}} R(f)) + (\inf_{f \in \mathcal{H}} R(f) - R(f^*)).$$

The first term is the excess risk $\mathcal{E}(\hat{f}_n; \mathcal{H})$, which is bounded by Lemma 2.8. The second term equals $\inf_{f \in \mathcal{H}} (R(f) - R(f^*))$. $\square$

Note that, as we defined it, the excess risk controls the estimation error only relative to the class $\mathcal{H}$. In particular, it is possible, if the hypothesis class $\mathcal{H}$ is poorly chosen, or not sufficiently rich, for the excess risk to be small while the learned model performs poorly. To capture the performance of an ERM in a more robust way, we can introduce the quantity $R^* := \inf_f R(f)$, which denotes the optimum over a larger reference family (or perhaps even the set of all admissible functions). The error term $R(\hat{f}_n) - R^*$, which now more robustly reflects how $\hat{f}_n$ performs compared to the “best possible” model, decomposes into the aforementioned *estimation error* term $\mathcal{E}(\hat{f}_n; \mathcal{H})$, and an *approximation error* term $\inf_{f \in \mathcal{H}} R(f) - R^*$. We will discuss this classical decomposition in more detail in Section 6.1, at the end of this chapter.

3 Getting started: finite hypothesis classes

Uniform deviation bounds are simplest when $\mathcal{H}$ is finite, since the supremum over $\mathcal{H}$ reduces to a maximum. Although finiteness is not a realistic assumption for most model classes, the argument in this section is worth recording because it illustrates the basic mechanism behind uniform convergence. Indeed, in this toy scenario, elementary concentration inequalities control the generalization gap for a fixed hypothesis, and a union bound upgrades this control to all hypotheses at once. The resulting complexity penalty $\log|\mathcal{H}|$ is

a quantitative form of Occam's principle (also known as *Occam's razor*) and is closely related to structural risk minimization (Vapnik, 2013; Mohri et al., 2018).

Lemma 2.11 (Hoeffding bound for a fixed hypothesis). *Assume $0 \leq L_f(Z) \leq 1$ almost surely. Then for every $\delta \in (0, 1)$,*

$$\mathbb{P}\left(|(P_n - P)L_f| \geq \sqrt{\frac{\log(2/\delta)}{2n}}\right) \leq \delta.$$

Proof. This is Hoeffding's inequality applied to the bounded random variables $L_f(Z_1), \dots, L_f(Z_n)$ (Boucheron et al., 2013). $\square$

Theorem 2.12 (Uniform deviation for finite classes). *Assume $0 \leq L_f(Z) \leq 1$ almost surely for every $f \in \mathcal{H}$, and assume $|\mathcal{H}| < \infty$. Then for every $\delta \in (0, 1)$,*

$$\mathbb{P}\left(\sup_{f \in \mathcal{H}} |(P_n - P)L_f| \leq \sqrt{\frac{\log(2|\mathcal{H}|/\delta)}{2n}}\right) \geq 1 - \delta.$$

Proof. Apply Lemma 2.11 to each $f \in \mathcal{H}$ with $\delta/|\mathcal{H}|$ and take a union bound. $\square$

Corollary 2.13 (Finite-class excess risk bound). *Under the assumptions of Theorem 2.12, every ε_{opt} -approximate ERM $\hat{f}_n$ satisfies, with probability at least $1 - \delta$,*

$$\mathcal{E}(\hat{f}_n; \mathcal{H}) \leq 2\sqrt{\frac{\log(2|\mathcal{H}|/\delta)}{2n}} + \varepsilon_{\text{opt}}.$$

Proof. Combine Theorem 2.12 with Corollary 2.9. $\square$

A common use of finite-class bounds is to analyse discretizations and model selection. For example, if a continuous family $\{f_\theta : \theta \in \Theta\}$ is approximated by a grid $\{\theta_1, \dots, \theta_M\}$, then the penalty $\log|\mathcal{H}|$ becomes $\log M$. This viewpoint underlies many arguments based on covering numbers and nets, where one reduces an infinite class to a finite subset at the price of an approximation error (Mohri et al., 2018). While this approach is attractively simple and highlights some of the main ideas behind mathematical guarantees learning algorithms, it remains too weak for the much more realistic scenario of genuinely infinite classes $\mathcal{H}$. To bypass this limitation, we introduce in the next section other measures of complexity due for the most part to Vapnik and

Chervonenkis (Vapnik, 2013; Van Der Vaart and Wellner, 1996), which quantify the effective richness of a class through the number of distinct behaviours it can realize on finite samples.

4 Pseudo-dimension and VC dimension

The finite-class bound of Section 3 depends on $\log|\mathcal{H}|$, which is not meaningful for the infinite classes that arise in most applications. One way to extend the argument is to replace cardinality by an effective notion of size that reflects how many distinct behaviours a class can exhibit on a finite sample. VC theory, developed by Vapnik and Chervonenkis, does exactly this for indicator-valued classes and yields a uniform law of large numbers controlled by a single integer parameter, called the so-called *VC dimension* (Vapnik, 2013). As we will see, the closely related concept of *pseudo-dimension* extends the same idea to real-valued function classes by replacing set membership with threshold comparisons (Bartlett and Anthony, 1999), yielding similar guarantees for yet broader hypothesis spaces.

4.1 VC dimension

Definition 2.14 (VC dimension). *Let $\mathcal{A}$ be a collection of subsets of $\mathcal{Z}$. A finite set $\{z_1, \dots, z_m\} \subset \mathcal{Z}$ is said to be shattered by $\mathcal{A}$ if for every subset $S \subset \{1, \dots, m\}$ there exists $A \in \mathcal{A}$ such that $z_i \in A$ if and only if $i \in S$. The Vapnik-Chervonenkis dimension (or VC dimension for short) $\text{vc}(\mathcal{A})$ is the largest integer m for which some m -point subset is shattered, with $\text{vc}(\mathcal{A}) = \infty$ if no such m exists (Vapnik, 2013).*

As mentioned earlier, the definition of VC dimension measures the complexity of a hypothesis class not by its cardinal, but by its ability to fit arbitrary patterns. As the definition may appear abstract at first, we give in what follows some examples of VC dimension computations as an illustration of this concept.

Lemma 2.15 (VC dimension of intervals). *Let $\mathcal{A}_{\text{int}} := \{[a, b] \subset \mathbb{R} : a, b \in \mathbb{R}, a \leq b\}$ denote the set of all non-empty closed intervals on $\mathbb{R}$. Then $\text{vc}(\mathcal{A}_{\text{int}}) = 2$.*

Proof. Simply note that any two points $x_1 < x_2$ are always shattered by $\mathcal{A}_{\text{int}}$ since $\{x_1\}$, $\{x_2\}$, and $\{x_1, x_2\}$ are realized by suitable intervals. On the other hand, no set of three points $x_1 < x_2 < x_3$ can be shattered by this class since no interval contains x_1 and x_3 while excluding x_2 . $\square$

Lemma 2.16 (VC dimension of halfspaces). *Let $\mathcal{A}_{\text{hs}} := \{\{x \in \mathbb{R}^d : \langle w, x \rangle + b \geq 0\} : w \in \mathbb{R}^d \text{ and } b \in \mathbb{R}\}$ denote the set of halfspaces. Then $\text{vc}(\mathcal{A}_{\text{hs}}) = d + 1$.*

Proof. For the lower bound, consider the $d+1$ points $0, e_1, \dots, e_d$, where e_j are the standard basis vectors. Given any labeling $y_0, y_1, \dots, y_d \in \{-1, 1\}$, define $b := y_0$ and $w_j := y_j - y_0$. Then $\langle w, 0 \rangle + b = y_0$ and $\langle w, e_j \rangle + b = w_j + b = y_j$. Hence the halfspace $\{x : \langle w, x \rangle + b \geq 0\}$ contains exactly the points with label 1.

For the upper bound, let $x_1, \dots, x_{d+2} \in \mathbb{R}^d$. The vectors $(x_i, 1) \in \mathbb{R}^{d+1}$ are linearly dependent, so there exist coefficients $\alpha_1, \dots, \alpha_{d+2}$, not all zero, such that $\sum_{i=1}^{d+2} \alpha_i x_i = 0$ and $\sum_{i=1}^{d+2} \alpha_i = 0$. Let $I_+ := \{i : \alpha_i > 0\}$ and $I_- := \{i : \alpha_i < 0\}$, which are both nonempty. Consider the labeling that assigns label 1 to I_+ and label 0 to I_- . If it were realized by some halfspace $\{x : \langle w, x \rangle + b \geq 0\}$, then $\langle w, x_i \rangle + b \geq 0$ for $i \in I_+$ and $\langle w, x_i \rangle + b < 0$ for $i \in I_-$. Multiplying by α_i and summing yields $\sum_{i=1}^{d+2} \alpha_i (\langle w, x_i \rangle + b) > 0$. On the other hand, $\sum_{i=1}^{d+2} \alpha_i (\langle w, x_i \rangle + b) = \langle w, \sum_{i=1}^{d+2} \alpha_i x_i \rangle + b \sum_{i=1}^{d+2} \alpha_i = 0$, which is a contradiction. $\square$

Another class for which it is easy to compute the VC-dimension is the set of axis-aligned rectangles in $\mathbb{R}^d$, whose VC dimension equals $2d$ (Vapnik, 2013). As we can see, this concept measures the richness of a hypothesis class in a much more refined way than cardinality alone, as it in some sense quantifies the amount of patterns the class can represent.

4.2 Pseudo-dimension

While VC dimension is well-suited for binary-valued classes of functions, the definition intrinsically can not apply to real-valued classes, making it limiting. For that case, one needs an analogue that captures in a similar way the “thresholding ability” of the class, which we call *pseudo-dimension* (Bartlett and Anthony, 1999).

Definition 2.17 (Pseudo-shattering and pseudo-dimension). *Let $\mathcal{G}$ be a class of real-valued functions on $\mathcal{Z}$. A set $\{z_1, \dots, z_m\} \subset \mathcal{Z}$ is said to be pseudo-shattered by $\mathcal{G}$ if there exist thresholds $t_1, \dots, t_m \in \mathbb{R}$ such that for every subset $S \subset \{1, \dots, m\}$ there exists $g \in \mathcal{G}$ satisfying $g(z_i) \geq t_i$ for $i \in S$ and $g(z_i) < t_i$ for $i \notin S$. The pseudo-dimension $\text{pdim}(\mathcal{G})$ is the largest m for which some m -point subset is pseudo-shattered, with $\text{pdim}(\mathcal{G}) = \infty$ if no such m exists.*

Note that pseudo-dimension is a direct generalization of VC-dimension. Indeed, if $\mathcal{G}$ is indicator-valued, choosing $t_i = 1/2$ shows that pseudo-shattering reduces to shattering, so $\text{pdim}(\mathcal{G}) = \text{vc}(\mathcal{G})$ in that case. As said before, this concept measures the capacity a function class has to fit arbitrary patterns. We now illustrate with a few examples.

Lemma 2.18 (Pseudodimension of affine functions). *Let $\mathcal{G}_{\text{aff}} := \{x \mapsto \langle w, x \rangle + b : w \in \mathbb{R}^d \text{ and } b \in \mathbb{R}\}$ be the set of affine maps on $\mathbb{R}^d$. Then $\text{pdim}(\mathcal{G}_{\text{aff}}) = d + 1$.*

Proof. For the lower bound, consider the points $0, e_1, \dots, e_d$ and thresholds $t_i = 0$. Given any labeling $y_0, y_1, \dots, y_d \in \{-1, 1\}$, define $b := y_0$ and $w_j := y_j - y_0$. Then $g(0) = y_0$ and $g(e_j) = y_j$, so $g(z_i) \geq 0$ holds exactly on the indices with label 1. Hence these $d + 1$ points are pseudo-shattered.

For the upper bound, fix any $d+2$ points $x_1, \dots, x_{d+2} \in \mathbb{R}^d$ and any thresholds $t_1, \dots, t_{d+2} \in \mathbb{R}$. As in the proof of Lemma 2.16, there exist coefficients $\alpha_1, \dots, \alpha_{d+2}$, not all zero, such that $\sum_{i=1}^{d+2} \alpha_i x_i = 0$ and $\sum_{i=1}^{d+2} \alpha_i = 0$. Let $I_+ := \{i : \alpha_i > 0\}$ and $I_- := \{i : \alpha_i < 0\}$. Consider the subset $S = I_+$. If there were $g(x) = \langle w, x \rangle + b$ with $g(x_i) \geq t_i$ for $i \in I_+$ and $g(x_i) < t_i$ for $i \in I_-$, then $\sum_{i=1}^{d+2} \alpha_i (g(x_i) - t_i) > 0$. On the other hand, $\sum_{i=1}^{d+2} \alpha_i g(x_i) = \langle w, \sum_{i=1}^{d+2} \alpha_i x_i \rangle + b \sum_{i=1}^{d+2} \alpha_i = 0$, so $\sum_{i=1}^{d+2} \alpha_i (g(x_i) - t_i) = -\sum_{i=1}^{d+2} \alpha_i t_i$. If $\sum_{i=1}^{d+2} \alpha_i t_i \geq 0$, this is a contradiction, so the labeling $S = I_+$ cannot be realized. If instead $\sum_{i=1}^{d+2} \alpha_i t_i \leq 0$, the same argument applied to $S = I_-$ yields a contradiction. Therefore no $d + 2$ points are pseudo-shattered. $\square$

Lemma 2.19 (Pseudodimension of univariate polynomials). *Let $k \in \mathbb{N}$ and let $\mathcal{G}_k := \{x \mapsto \sum_{j=0}^k a_j x^j : a_0, \dots, a_k \in \mathbb{R}\}$ denote the set of degree k real-polynomials on $\mathbb{R}$. Then $\text{pdim}(\mathcal{G}_k) = k + 1$.*

Proof. For the lower bound, choose $k + 1$ distinct points $x_1, \dots, x_{k+1}$ and thresholds $t_i = 0$. Given any subset $S \subset \{1, \dots, k + 1\}$, define target values $y_i := 1$ if $i \in S$ and $y_i := -1$ otherwise. By Lagrange interpolation there exists a polynomial $p \in \mathcal{G}_k$ with $p(x_i) = y_i$ for all i . This realizes the required comparisons to the thresholds.

For the upper bound, fix any $k + 2$ distinct points $x_1, \dots, x_{k+2}$ and thresholds $t_1, \dots, t_{k+2}$. Consider the linear map $T : \mathbb{R}^{k+1} \rightarrow \mathbb{R}^{k+2}$ that sends the coefficient vector $(a_0, \dots, a_k)$ to the value vector $(p(x_1), \dots, p(x_{k+2}))$. Since $\dim(\mathbb{R}^{k+1}) < \dim(\mathbb{R}^{k+2})$, the image of T is a proper subspace, so there exists a nonzero vector $\alpha \in \mathbb{R}^{k+2}$ orthogonal to the image. Equivalently, $\sum_{i=1}^{k+2} \alpha_i p(x_i) = 0$ for every $p \in \mathcal{G}_k$. Let $I_+ := \{i : \alpha_i > 0\}$ and

$I_- := \{i : \alpha_i < 0\}$. Arguing as in Lemma 2.18, one of the two labelings $S = I_+$ or $S = I_-$ is incompatible with the constraint $\sum_{i=1}^{k+2} \alpha_i p(x_i) = 0$. Hence $\mathcal{G}_k$ cannot pseudo-shatter $k + 2$ points. $\square$

4.3 A uniform deviation theorem

Having introduced our key measures of complexity, we are now ready to state and prove a uniform deviation bound of the type required by Corollary 2.9. As we will see, the argument relies on two parts: the well-known Sauer–Shelah lemma bounds the growth function of a VC class, and the obtained bound is then combined with symmetrization and concentration to control $\sup |(P_n - P)g|$ for bounded classes. Before stating the Sauer–Shelah lemma, we introduce a necessary object:

Definition 2.20 (Growth function). *Let $\mathcal{A}$ be a collection of subsets of $\mathcal{Z}$. For $m \in \mathbb{N}$, the growth function is defined as*

$$\Pi_{\mathcal{A}}(m) := \sup_{z_1, \dots, z_m \in \mathcal{Z}} |\{(A \cap \{z_1, \dots, z_m\}) : A \in \mathcal{A}\}|.$$

We refer the reader to a standard reference, such as (Mohri et al., 2018) for more detail and intuition on the growth function, which is an important combinatorial object in its own right. Having defined it, we can now state and prove our lemma of interest.

Lemma 2.21 (Sauer–Shelah lemma). *Let $\mathcal{A}$ be a collection of subsets of $\mathcal{Z}$ with $\text{vc}(\mathcal{A}) \leq d$, where $d \in \mathbb{N}$. Then for every $m \in \mathbb{N}$,*

$$\Pi_{\mathcal{A}}(m) \leq \sum_{j=0}^d \binom{m}{j}.$$

In particular, if $m \geq d$, then $\Pi_{\mathcal{A}}(m) \leq (em/d)^d$.

Proof. Let $\Pi(d, m)$ denote the supremum of $|\{A \cap S : A \in \mathcal{A}\}|$ over all $S \subset \mathcal{Z}$ with $|S| = m$ under the constraint $\text{vc}(\mathcal{A}) \leq d$. We prove the recursion $\Pi(d, m) \leq \Pi(d, m-1) + \Pi(d-1, m-1)$.

Fix $S \subset \mathcal{Z}$ with $|S| = m$ and pick $z \in S$. Let $\mathcal{A}|_S := \{A \cap S : A \in \mathcal{A}\}$. Split $\mathcal{A}|_S$ into those sets that do not contain z and those that do. The family of intersections that do not contain z is contained in $\mathcal{A}|_{S \setminus \{z\}}$, so it has cardinality at most $\Pi(d, m-1)$.

For the sets that contain z , consider the collection

$$\mathcal{B} := \{B \subset S \setminus \{z\} : B \in \mathcal{A}|_{S \setminus \{z\}} \text{ and } B \cup \{z\} \in \mathcal{A}|_S\}.$$

Then $|\{A \cap S : A \in \mathcal{A}, z \in A\}| = |\mathcal{B}|$. If $\mathcal{B}$ shattered some subset $T \subset S \setminus \{z\}$ with $|T| = d$, then $\mathcal{A}|_S$ would shatter $T \cup \{z\}$ by choosing intersections that include z or exclude z according to the desired label on z . This would contradict $\text{vc}(\mathcal{A}) \leq d$. Hence $\text{vc}(\mathcal{B}) \leq d - 1$, so $|\mathcal{B}| \leq \Pi(d - 1, m - 1)$. This proves the recursion.

With the recursion and the initial values $\Pi(0, m) = 1$ and $\Pi(d, m) = 2^m$ for $d \geq m$, an induction on m yields $\Pi(d, m) \leq \sum_{j=0}^d \binom{m}{j}$. The inequality $\sum_{j=0}^d \binom{m}{j} \leq (em/d)^d$ for $m \geq d$ is standard. $\square$

With this result in hand, we are now ready to state and prove one of the most important theorems of this chapter, which give uniform bounds on the excess risk in terms of the complexity measures we've introduced. For the first one, we show how to bound the error in terms of VC dimension.

Theorem 2.22 (Uniform deviation via VC dimension). *Let $\mathcal{A}$ be a collection of measurable subsets of $\mathcal{Z}$ with $\text{vc}(\mathcal{A}) \leq d$, and let P be the data-generating distribution. Then there exists a universal constant $C > 0$ such that for every $\delta \in (0, 1)$, with probability at least $1 - \delta$,*

$$\sup_{A \in \mathcal{A}} |P_n(\mathbf{1}_A) - P(\mathbf{1}_A)| \leq C \sqrt{\frac{d \log(en) + \log(1/\delta)}{n}}.$$

Proof. Define $\Phi(Z_{1:n}) := \sup_{A \in \mathcal{A}} |(P_n - P)\mathbf{1}_A|$. Replacing one observation Z_i changes $P_n(\mathbf{1}_A)$ by at most $1/n$ for every A , hence changes Φ by at most $1/n$. McDiarmid's inequality therefore yields $\Phi \leq \mathbb{E}[\Phi] + \sqrt{\log(1/\delta)/(2n)}$ with probability at least $1 - \delta$ (Boucheron et al., 2013).

It remains to bound $\mathbb{E}[\Phi]$. Let $Z'_1, \dots, Z'_n$ be an i.i.d. ghost sample, independent of $Z_1, \dots, Z_n$, and let P'_n be its empirical measure. Since $P(\mathbf{1}_A) = \mathbb{E}[P'_n(\mathbf{1}_A) | Z_{1:n}]$, Jensen's inequality gives

$$\mathbb{E}[\Phi] \leq \mathbb{E} \left[\sup_{A \in \mathcal{A}} |P_n(\mathbf{1}_A) - P'_n(\mathbf{1}_A)| \right].$$

Condition on $Z_{1:n}, Z'_{1:n}$ and let $\sigma_1, \dots, \sigma_n$ be i.i.d. Rademacher variables, independent of all data. By exchangeability of the pairs (Z_i, Z'_i) ,

$$\mathbb{E}[\Phi] \leq \mathbb{E} \left[\mathbb{E}_\sigma \left[\sup_{A \in \mathcal{A}} \left| \frac{1}{n} \sum_{i=1}^n \sigma_i (\mathbf{1}_A(Z_i) - \mathbf{1}_A(Z'_i)) \right| \right] \right] \leq 2 \mathbb{E} \left[\mathbb{E}_\sigma \left[\sup_{A \in \mathcal{A}} \left| \frac{1}{n} \sum_{i=1}^n \sigma_i \mathbf{1}_A(Z_i) \right| \right] \right].$$

Condition on $Z_{1:n}$. The set of vectors $\{(\mathbf{1}_A(Z_1), \dots, \mathbf{1}_A(Z_n)) : A \in \mathcal{A}\}$ has cardinality at most $\Pi_{\mathcal{A}}(n)$ by Definition 2.20. For each fixed binary vector $a \in \{0, 1\}^n$, the random variable $\sum_{i=1}^n \sigma_i a_i$ is subgaussian with variance proxy $\sum_{i=1}^n a_i^2 \leq n$, so Massart's lemma (see Lemma A.4 in the Appendix)

$$\mathbb{E}_{\sigma} \left[\max_a \left| \frac{1}{n} \sum_{i=1}^n \sigma_i a_i \right| \right] \leq \sqrt{\frac{2 \log(2\Pi_{\mathcal{A}}(n))}{n}}.$$

Taking expectation over the sample and using Lemma 2.21 gives $\log \Pi_{\mathcal{A}}(n) \leq d \log(en)$ up to a change of constants. Combining the bounds yields the claim. $\square$

We would now like to get similar guarantees when VC dimension is replaced by pseudodimension, thus allowing to control the error of more general function classes. For this, it suffices to make the following simple observation:

Lemma 2.23 (Fixed-threshold classes inherit VC dimension). *Let $\mathcal{G}$ be a class of real-valued functions on $\mathcal{Z}$, and fix $t \in \mathbb{R}$. Define $\mathcal{A}_t := \{z \in \mathcal{Z} : g(z) \geq t\} : g \in \mathcal{G}\}$. Then $\text{vc}(\mathcal{A}_t) \leq \text{pdim}(\mathcal{G})$.*

Proof. If $\mathcal{A}_t$ shattered $\{z_1, \dots, z_m\}$, then $\mathcal{G}$ would pseudo-shatter $\{z_1, \dots, z_m\}$ with thresholds $t_1 = \dots = t_m = t$. $\square$

We are now ready to extend Theorem 2.22 from VC classes to real-valued classes via pseudo-dimension.

Theorem 2.24 (Uniform deviation via pseudo-dimension). *Assume $0 \leq L_f(Z) \leq 1$ almost surely for every $f \in \mathcal{H}$, and let $\mathcal{L} := \{L_f : f \in \mathcal{H}\}$. If $\text{pdim}(\mathcal{L}) \leq d$, then there exists a universal constant $C > 0$ such that for every $\delta \in (0, 1)$,*

$$\mathbb{P} \left(\sup_{f \in \mathcal{H}} |(P_n - P)L_f| \leq C \sqrt{\frac{d \log(en) + \log(1/\delta)}{n}} \right) \geq 1 - \delta.$$

Proof. Fix an integer $m \geq 2$. For $u \in [0, 1]$, the identity $\lfloor mu \rfloor = \sum_{k=1}^{m-1} \mathbf{1}_{\{u \geq k/m\}}$ implies

$$\frac{\lfloor mL_f(z) \rfloor}{m} = \frac{1}{m} \sum_{k=1}^{m-1} \mathbf{1}_{\{L_f(z) \geq k/m\}} \text{ for every } z \in \mathcal{Z}.$$

Moreover, $\lfloor mL_f(z) \rfloor / m \leq L_f(z) < \lfloor mL_f(z) \rfloor / m + 1/m$, so

$$\sup_{f \in \mathcal{H}} |(P_n - P)L_f| \leq \sup_{f \in \mathcal{H}} \left| (P_n - P) \frac{\lfloor mL_f \rfloor}{m} \right| + \frac{2}{m}.$$

Using the expansion above and the inequality $\frac{1}{m} \sum_{k=1}^{m-1} a_k \leq \max_{1 \leq k < m} a_k$ for nonnegative a_k , we obtain

$$\sup_{f \in \mathcal{H}} \left| (P_n - P) \frac{\lfloor mL_f \rfloor}{m} \right| \leq \max_{1 \leq k < m} \sup_{f \in \mathcal{H}} |(P_n - P) \mathbf{1}_{\{L_f \geq k/m\}}|.$$

For each $k \in \{1, \dots, m-1\}$, define the set system $\mathcal{A}_{k,m} := \{\{z : L_f(z) \geq k/m\} : f \in \mathcal{H}\}$. By Lemma 2.23, $\text{vc}(\mathcal{A}_{k,m}) \leq \text{pdim}(\mathcal{L}) \leq d$. Applying Theorem 2.22 to each $\mathcal{A}_{k,m}$ with confidence $\delta/(m-1)$ and taking a union bound over k yields that, with probability at least $1 - \delta$,

$$\max_{1 \leq k < m} \sup_{f \in \mathcal{H}} |(P_n - P) \mathbf{1}_{\{L_f \geq k/m\}}| \leq C_1 \sqrt{\frac{d \log(en) + \log(m/\delta)}{n}}$$

for a universal constant C_1 . Choosing $m = n$ and absorbing $\log n$ into $d \log(en)$ up to constants yields the stated bound. $\square$

Corollary 2.25 (Excess risk bound via pseudo-dimension). *Under the assumptions of Theorem 2.24, every ε_{opt} -approximate ERM $\hat{f}_n$ satisfies, with probability at least $1 - \delta$,*

$$\mathcal{E}(\hat{f}_n; \mathcal{H}) \leq 2C \sqrt{\frac{d \log(en) + \log(1/\delta)}{n}} + \varepsilon_{\text{opt}}.$$

Proof. Combine Theorem 2.24 with Corollary 2.9. $\square$

Pseudo-dimension is useful when explicit shattering arguments are available. For composite real-valued losses, however, Rademacher complexity is often more stable under sums and Lipschitz compositions. In the next section, we introduce this alternative notion of complexity and show how it can be used to establish similar learning guarantees.

5 Rademacher complexity, contraction, and chaining

VC type bounds quantify complexity through the number of threshold patterns a class can realize. For real-valued loss classes, where this fairly geometric and combinatorial approach is not tractable, an alternative is to measure complexity directly through the empirical process $(P_n - P)g$ (Van Der Vaart and Wellner, 1996; Koltchinskii and Panchenko, 2000). This idea naturally

leads one to the concept of *Rademacher complexity*, which is in many cases a much more powerful and effective measure of complexity, as it is well suited for symmetrization arguments algebraic operations such as sums, products with bounded factors, and Lipschitz compositions (Koltchinskii, 2011). We start by recalling its definition:

Definition 2.26 (Empirical and expected Rademacher complexity). *Let $\mathcal{G}$ be a class of measurable functions $g : \mathcal{Z} \rightarrow \mathbb{R}$. For a fixed sample $Z_1, \dots, Z_n$, we call Rademacher complexity the quantity*

$$\hat{\mathfrak{R}}_n(\mathcal{G}) := \mathbb{E}_\sigma \left[\sup_{g \in \mathcal{G}} \left| \frac{1}{n} \sum_{i=1}^n \sigma_i g(Z_i) \right| \right],$$

where $\sigma_1, \dots, \sigma_n$ are i.i.d. Rademacher variables taking values in $\{-1, 1\}$. The expected Rademacher complexity is then $\mathfrak{R}_n(\mathcal{G}) := \mathbb{E}[\hat{\mathfrak{R}}_n(\mathcal{G})]$.

In some refined bounds, it is convenient to work with a one-sided variant that drops the absolute value inside the supremum.

Definition 2.27. *For a fixed sample $Z_1, \dots, Z_n$, define the one-sided empirical Rademacher complexity*

$$\hat{\mathfrak{R}}_n^+(\mathcal{G}) := \mathbb{E}_\sigma \left[\sup_{g \in \mathcal{G}} \frac{1}{n} \sum_{i=1}^n \sigma_i g(Z_i) \right],$$

and its expectation $\mathfrak{R}_n^+(\mathcal{G}) := \mathbb{E}[\hat{\mathfrak{R}}_n^+(\mathcal{G})]$.

Note that $\hat{\mathfrak{R}}_n^+(\mathcal{G}) \leq \hat{\mathfrak{R}}_n(\mathcal{G})$ by definition.

The next lemma is the essential bridge between uniform deviations and Rademacher means. It is a statement purely about the empirical process itself, and does not rely on the optimality of any estimator.

Lemma 2.28. *Let $\mathcal{G}$ be a class of measurable functions $g : \mathcal{Z} \rightarrow \mathbb{R}$ such that $\mathbb{E}[|g(Z)|] < \infty$ for every $g \in \mathcal{G}$. Then*

$$\mathbb{E} \left[\sup_{g \in \mathcal{G}} |(P_n - P)g| \right] \leq 2\mathfrak{R}_n(\mathcal{G}).$$

Proof. Let $Z'_1, \dots, Z'_n$ be an i.i.d. ghost sample, independent of $Z_1, \dots, Z_n$, and write P'_n for its empirical measure. Since $Pg = \mathbb{E}[P'_n g \mid Z_{1:n}]$, Jensen's inequality gives

$$\mathbb{E} \left[\sup_{g \in \mathcal{G}} |(P_n - P)g| \right] \leq \mathbb{E} \left[\sup_{g \in \mathcal{G}} |(P_n - P'_n)g| \right].$$

Condition on $Z_{1:n}, Z'_{1:n}$. Let $\sigma_1, \dots, \sigma_n$ be i.i.d. Rademacher variables, independent of all data. By exchangeability of the pairs (Z_i, Z'_i) ,

$$\mathbb{E} \left[\sup_{g \in \mathcal{G}} |(P_n - P'_n)g| \right] = \mathbb{E} \left[\mathbb{E}_\sigma \left[\sup_{g \in \mathcal{G}} \left| \frac{1}{n} \sum_{i=1}^n \sigma_i (g(Z_i) - g(Z'_i)) \right| \right] \right].$$

By the triangle inequality,

$$\left| \frac{1}{n} \sum_{i=1}^n \sigma_i (g(Z_i) - g(Z'_i)) \right| \leq \left| \frac{1}{n} \sum_{i=1}^n \sigma_i g(Z_i) \right| + \left| \frac{1}{n} \sum_{i=1}^n \sigma_i g(Z'_i) \right|.$$

Taking the supremum over $g \in \mathcal{G}$, then $\mathbb{E}_\sigma$, and finally expectation over $(Z_{1:n}, Z'_{1:n})$ yields

$$\mathbb{E} \left[\sup_{g \in \mathcal{G}} |(P_n - P'_n)g| \right] \leq 2\mathfrak{R}_n(\mathcal{G}),$$

which completes the proof. $\square$

The symmetrization lemma provides an expectation bound. A high-probability bound follows by combining it with a bounded differences inequality:

Theorem 2.29 (Uniform deviation via Rademacher complexity). *Let $\mathcal{G}$ be a class of measurable functions $g : \mathcal{Z} \rightarrow \mathbb{R}$ such that $|g| \leq B$ pointwise for every $g \in \mathcal{G}$. Then for every $\delta \in (0, 1)$, with probability at least $1 - \delta$,*

$$\sup_{g \in \mathcal{G}} |(P_n - P)g| \leq 2\mathfrak{R}_n(\mathcal{G}) + 2B \sqrt{\frac{\log(1/\delta)}{2n}}.$$

Proof. Define $\Phi(Z_{1:n}) := \sup_{g \in \mathcal{G}} |(P_n - P)g|$. Replacing one observation Z_i by an independent copy changes $P_n g$ by at most $2B/n$ for each fixed g , hence changes Φ by at most $2B/n$. McDiarmid's inequality yields

$$\Phi \leq \mathbb{E}[\Phi] + 2B \sqrt{\frac{\log(1/\delta)}{2n}}$$

with probability at least $1 - \delta$. Lemma 2.28 bounds $\mathbb{E}[\Phi]$ by $2\mathfrak{R}_n(\mathcal{G})$, which gives the claim. $\square$

For localized analyses, it is often useful to keep track of both an envelope bound and a variance bound. The next lemma gives a one-sided uniform deviation inequality in terms of $\hat{\mathfrak{R}}_n^+(\mathcal{G})$, together with explicit Bernstein-type remainder terms.

Lemma 2.30. *Let $\mathcal{G}$ be a class of measurable functions $g : \mathcal{Z} \rightarrow \mathbb{R}$ such that $\|g\|_\infty \leq G$ and $\text{Var}(g(Z)) \leq V$ for every $g \in \mathcal{G}$. Then for every $t > 0$, with probability at least $1 - e^{-t}$,*

$$\sup_{g \in \mathcal{G}} (P_n g - P g) \leq 3 \hat{\mathfrak{R}}_n^+(\mathcal{G}) + \sqrt{\frac{2Vt}{n}} + \frac{4Gt}{3n}.$$

Moreover, with probability at least $1 - 2e^{-t}$,

$$\sup_{g \in \mathcal{G}} (P g - P_n g) \leq 6 \hat{\mathfrak{R}}_n^+(\mathcal{G}) + \sqrt{\frac{2Vt}{n}} + \frac{23Gt}{3n}.$$

Proof. This result is proved in (Bartlett et al., 2005, Theorem 3). $\square$

Theorem 2.29 gives yet another bound for the excess risk of ERM solutions. For it to be actually useful however, one needs bounds on $\mathfrak{R}_n(\mathcal{G})$. A key tool to achieve that is the contraction principle, which shows that Rademacher complexity does not increase much under Lipschitz nonlinearities.

Lemma 2.31 (Contraction for Lipschitz compositions). *Let $\psi : \mathbb{R} \rightarrow \mathbb{R}$ be L -Lipschitz and satisfy $\psi(0) = 0$. Then for every sample $Z_1, \dots, Z_n$,*

$$\hat{\mathfrak{R}}_n(\psi \circ \mathcal{G}) \leq L \hat{\mathfrak{R}}_n(\mathcal{G}),$$

where $\psi \circ \mathcal{G} := \{\psi \circ g : g \in \mathcal{G}\}$.

Proof. A complete proof is given in Appendix A, Section 1.1. $\square$

In later chapters, the loss will often depend on several features of a hypothesis at once, for instance a finite collection of derivatives. The next lemma is a convenient vector-valued extension that turns an ℓ_1 -Lipschitz dependence into a sum of Rademacher complexities of the induced coordinate classes.

Lemma 2.32. *Fix a sample $Z_1, \dots, Z_n$. Let $m \in \mathbb{N}$, and for each $j \in \{1, \dots, m\}$ let $\mathcal{G}_j$ be a class of real-valued functions on $\mathcal{Z}$. Let $\mathcal{F}$ be an index set, and assume that for each $f \in \mathcal{F}$ we are given a vector-valued map $z \mapsto (g_{f,1}(z), \dots, g_{f,m}(z))$ with $g_{f,j} \in \mathcal{G}_j$. For each $i \in \{1, \dots, n\}$, let $\psi_i : \mathbb{R}^m \rightarrow \mathbb{R}$ be L -Lipschitz with respect to $\|\cdot\|_1$, and satisfy $\psi_i(0) = 0$. Then*

$$\mathbb{E}_\sigma \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \sigma_i \psi_i(g_{f,1}(Z_i), \dots, g_{f,m}(Z_i)) \right] \leq L \sum_{j=1}^m \hat{\mathfrak{R}}_n(\mathcal{G}_j).$$

Proof. The proof is given in Appendix A, Section 1.2. See also (Ledoux and Talagrand, 2013) where the arguments are largely taken from. $\square$

5.1 Covering numbers and Dudley's bound

Theorem 2.29 reduces uniform deviations to bounds on $\mathfrak{R}_n(\mathcal{G})$. A standard route is via covering numbers and chaining. Although this machinery may not always be the sharpest tool for the structured hypothesis classes used later in this thesis, it still provides a general baseline for converting geometric size information into Rademacher bounds (Koltchinskii and Panchenko, 2000; Koltchinskii, 2006).

Covering numbers make the passage from infinite classes to finite approximations explicit. Fixing the sample $Z_1, \dots, Z_n$, an ε -cover provides a finite set of representative functions that approximates $\mathcal{G}$ in the empirical norm. This allows one to reduce the Rademacher supremum over $\mathcal{G}$ to a supremum over a finite class, plus an approximation term of order ε (Zhou, 2002). Dudley's chaining argument refines this idea by combining covers across multiple scales, which leads to an entropy integral bound (Sen, 2018).

Definition 2.33 (Empirical L_2 seminorm and covering number). *Fix a sample $Z_1, \dots, Z_n$. For measurable $g : \mathcal{Z} \rightarrow \mathbb{R}$, define $\|g\|_{L_2(P_n)} := (P_n g^2)^{1/2}$. For $\varepsilon > 0$, the covering number $N(\varepsilon, \mathcal{G}, \|\cdot\|_{L_2(P_n)})$ is the smallest $M \in \mathbb{N}$ such that there exist $g_1, \dots, g_M \in \mathcal{G}$ with the property that for every $g \in \mathcal{G}$ there exists $j \in \{1, \dots, M\}$ satisfying $\|g - g_j\|_{L_2(P_n)} \leq \varepsilon$.*

Theorem 2.34 (A one-scale covering bound). *Assume $\sup_{g \in \mathcal{G}} \|g\|_{L_2(P_n)} \leq c$. Then for every $\varepsilon > 0$,*

$$\hat{\mathfrak{R}}_n(\mathcal{G}) \leq \varepsilon + c \sqrt{\frac{2 \log(2N(\varepsilon, \mathcal{G}, \|\cdot\|_{L_2(P_n)}))}{n}}.$$

Consequently,

$$\hat{\mathfrak{R}}_n(\mathcal{G}) \leq \inf_{\varepsilon > 0} \left(\varepsilon + c \sqrt{\frac{2 \log(2N(\varepsilon, \mathcal{G}, \|\cdot\|_{L_2(P_n)}))}{n}} \right).$$

Proof. The proof is given in Appendix A, Section 2. □

Theorem 2.35 (Dudley entropy integral bound). *Assume $|g| \leq 1$ pointwise for every $g \in \mathcal{G}$. Then for every sample $Z_1, \dots, Z_n$,*

$$\hat{\mathfrak{R}}_n(\mathcal{G}) \leq \inf_{\alpha \in (0,1]} \left(4\alpha + \frac{12}{\sqrt{n}} \int_{\alpha}^1 \sqrt{\log N(\varepsilon, \mathcal{G}, \|\cdot\|_{L_2(P_n)})} d\varepsilon \right).$$

Proof. A complete proof is given in Appendix A, Section 2. □

In later chapters we will also use a localized variant, which bounds the Rademacher complexity of the subset of $\mathcal{G}$ with small empirical $L_2(P_n)$ norm.

Lemma 2.36. *Fix a sample $Z_1, \dots, Z_n$ and let $\gamma > 0$. Then*

$$\hat{\mathfrak{R}}_n(\{g \in \mathcal{G} : \|g\|_{L_2(P_n)} \leq \gamma\}) \leq \inf_{0 < \beta < \gamma} \left(4\beta + \frac{12}{\sqrt{n}} \int_{\beta}^{\gamma} \sqrt{\log N(\varepsilon, \mathcal{G}, \|\cdot\|_{L_2(P_n)})} d\varepsilon \right).$$

Proof. We prove the result in Appendix A, Section 2.3. $\square$

In addition to empirical L_2 covers, it is sometimes convenient to control complexity through covers in the ℓ_{∞} metric on evaluation vectors. This is particularly useful when pseudo-dimension bounds are available.

Definition 2.37 (ℓ_{∞} covering numbers on samples). *Fix $n \in \mathbb{N}$ and let $Z_{1:n} = (Z_1, \dots, Z_n)$. For a class $\mathcal{G}$ of functions $g : \mathcal{Z} \rightarrow \mathbb{R}$, define the evaluation set*

$$\mathcal{G}|_{Z_{1:n}} := \{(g(Z_1), \dots, g(Z_n)) : g \in \mathcal{G}\} \subset \mathbb{R}^n.$$

On $\mathbb{R}^n$ consider the metric $d_{\infty}(a, a') := \max_{1 \leq i \leq n} |a_i - a'_i|$. For $\varepsilon > 0$, let $\mathcal{C}(\varepsilon, A, d_{\infty})$ denote the smallest cardinality of an ε -cover of $A \subset \mathbb{R}^n$ with respect to d_{∞} . We then define

$$\mathcal{C}_{\infty}(\varepsilon, \mathcal{G}, n) := \sup_{Z_{1:n}} \mathcal{C}(\varepsilon, \mathcal{G}|_{Z_{1:n}}, d_{\infty}).$$

Proposition 2.38 (Covering numbers from pseudo-dimension). *Let $\mathcal{G}$ be a class of functions from $\mathcal{Z}$ to $[0, B]$ with $\text{pdim}(\mathcal{G}) \leq d$. Then for every $n \geq d$ and every $\varepsilon \in (0, B)$,*

$$\mathcal{C}_{\infty}(\varepsilon, \mathcal{G}, n) \leq \left(\frac{eBn}{\varepsilon d} \right)^d.$$

Proof. The proof is given in Appendix A, Section 2.4. $\square$

Corollary 2.39 (Dudley bound via ℓ_{∞} covers). *Let $\mathcal{G}$ be a class of functions $g : \mathcal{Z} \rightarrow \mathbb{R}$ with $|g| \leq B$ pointwise. Then for every $n \in \mathbb{N}$,*

$$\mathfrak{R}_n(\mathcal{G}) \leq \inf_{0 < \alpha < B} \left(4\alpha + \frac{12}{\sqrt{n}} \int_{\alpha}^B \sqrt{\log(2\mathcal{C}_{\infty}(\varepsilon, \mathcal{G}, n))} d\varepsilon \right).$$

Proof. Fix $Z_{1:n}$ and define $\mathcal{G}_B := \{g/B : g \in \mathcal{G}\}$. Then $|h| \leq 1$ pointwise for every $h \in \mathcal{G}_B$, and $\hat{\mathfrak{R}}_n(\mathcal{G}) = B \hat{\mathfrak{R}}_n(\mathcal{G}_B)$. Applying Theorem 2.35 to $\mathcal{G}_B$ and changing variables yields

$$\hat{\mathfrak{R}}_n(\mathcal{G}) \leq \inf_{\alpha \in (0, B]} \left(4\alpha + \frac{12}{\sqrt{n}} \int_{\alpha}^B \sqrt{\log N(\varepsilon, \mathcal{G}, \|\cdot\|_{L_2(P_n)})} d\varepsilon \right).$$

Any d_{∞} -cover of $\mathcal{G}|_{Z_{1:n}}$ is also an $L_2(P_n)$ -cover since $\|g-h\|_{L_2(P_n)} \leq d_{\infty}(g|_{Z_{1:n}}, h|_{Z_{1:n}})$. Hence $N(\varepsilon, \mathcal{G}, \|\cdot\|_{L_2(P_n)}) \leq \mathcal{C}_{\infty}(\varepsilon, \mathcal{G}, n)$. Taking expectations over $Z_{1:n}$ yields the claim. $\square$

The bounds introduced in this chapter so far control estimation error with respect a fixed class $\mathcal{H}$ only. To turn them into meaningful problem-level guarantees one also needs approximation information, and one must relate the chosen risk to the quantity of interest. This is the topic of the next section.

6 Approximation error and problem structure

Sections 2 to 5 provide generalization bounds that control estimation error for empirical minimizers over a fixed class $\mathcal{H}$. To translate such bounds into a statement about the quality of the learned predictor in a concrete problem, we also need to control the approximation power of our hypothesis class. Indeed, even with infinite data, restricting to $\mathcal{H}$ introduces a bias if the target cannot be represented well within $\mathcal{H}$ (Cucker and Zhou, 2007).

Beyond approximation guarantees, we also typically require information on the underlying problem's structure, meaning a link between the chosen risk and the quantity of interest. In supervised learning this link can come from strong convexity or calibration inequalities. In inverse problems and PDE settings it is typically provided by stability estimates for the forward model. We will now illustrate these two points in a bit more detail in what follows.

6.1 Approximation and estimation

Let $R^* := \inf_f R(f)$ denote the optimal risk over a reference family that contains $\mathcal{H}$, such as all measurable predictors in supervised learning. For any estimator $\hat{f}_n \in \mathcal{H}$,

$$R(\hat{f}_n) - R^* = \mathcal{E}(\hat{f}_n; \mathcal{H}) + \left(\inf_{f \in \mathcal{H}} R(f) - R^* \right),$$

where $\mathcal{E}(\hat{f}_n; \mathcal{H})$ is the excess risk from Definition 2.5. The first term is an estimation error controlled by the generalization bounds introduced in this chapter. The second term is an approximation error controlled by approximation theory for the chosen hypothesis class and by modelling choices (Bach, 2024).

Control of the approximation error is strongly dependent on the choice of hypothesis space and information on the set of minimizers f^* achieving the optimal risk. In the context of PDE-learning, it is often assumed, or known from PDE regularity theory, that the optimizer lies in a certain smoothness space, such as Hölder or Sobolev spaces (Evans, 2022). For hypothesis classes based on neural networks and kernels, there is an extremely rich literature on approximation power over such smoothness spaces (Cucker and Zhou, 2007; Steinwart and Christmann, 2008; Yarotsky, 2017; Gühring and Raslan, 2021). We will study and prove such bounds specifically for neural networks hypothesis spaces in the next chapters. We will however not be focusing on kernel methods in this thesis. Since they constitute an extremely significant class of learning algorithms, we include in Appendix A some essential results and exposition showing how the learning-theoretic framework applies specifically to kernel-based methods, and how those would apply to scientific problems such as PDE-learning.

6.2 Residual objectives as risks

We now conclude this chapter by making explicit how the ERM framework can be applied to obtain theoretical guarantees on residual minimization for PDE problems. The technical details and results for specific settings will be provided in the next chapters.

Consider an equation $\mathcal{L}u = f$ on Ω with boundary condition $\mathcal{B}u = g$ on $\partial\Omega$. For a candidate v , define the interior residual $r_v(x) := (\mathcal{L}v)(x) - f(x)$ for $x \in \Omega$ and the boundary mismatch $b_v(z) := (\mathcal{B}v)(z) - g(z)$ for $z \in \partial\Omega$. Let μ_{int} be a distribution on Ω and μ_{bd} a distribution on $\partial\Omega$, and fix weights $\lambda_{\text{int}}, \lambda_{\text{bd}} > 0$. We define the population residual risk

$$R(v) := \lambda_{\text{int}} \mathbb{E}_{X \sim \mu_{\text{int}}} [|r_v(X)|^2] + \lambda_{\text{bd}} \mathbb{E}_{Z \sim \mu_{\text{bd}}} [|b_v(Z)|^2].$$

Given independent samples $X_1, \dots, X_{n_\Omega} \sim \mu_{\text{int}}$ and $Z_1, \dots, Z_{n_{\partial\Omega}} \sim \mu_{\text{bd}}$, the corresponding empirical objective is

$$R_{n_\Omega, n_{\partial\Omega}}(v) := \lambda_{\text{int}} \frac{1}{n_\Omega} \sum_{i=1}^{n_\Omega} |r_v(X_i)|^2 + \lambda_{\text{bd}} \frac{1}{n_{\partial\Omega}} \sum_{j=1}^{n_{\partial\Omega}} |b_v(Z_j)|^2.$$

Minimizing $R_{n_\Omega, n_{\partial\Omega}}$ over a hypothesis class is therefore an ERM problem. This viewpoint underlies physics-informed neural networks and related residual minimization methods (Raissi et al., 2019; Karniadakis et al., 2021). From classical PDE theory, it is known that, up to regularity assumptions on the PDE at hand, there exists a unique solution to the PDE with known regularity, typically expressed in terms of Sobolev or Hölder classes (Evans, 2022). In particular, this means that the approximation error bounds for smooth functions can be directly used to bound the approximation error of this PDE-problem.

Regarding the estimation, or statistical term, the last bit that is missing from our learning-theoretic formalism is what we often call a *structural assumption*: that is, a quantitative link between the residual risk and a physically meaningful error in the solution. For well-posed problems, this link is provided by stability of the forward model, typically in the form of an estimate

$$\|v - u^*\|_{\mathcal{U}} \leq C_{\text{stab}} \left(\|\mathcal{L}v - f\|_{\mathcal{Y}} + \|\mathcal{B}v - g\|_{\mathcal{Z}} \right),$$

where u^* is the true solution and $\mathcal{U}, \mathcal{Y}, \mathcal{Z}$ are problem-dependent normed spaces (Gilbarg and Trudinger, 1977). In such cases, controlling the empirical-to-population gap for $R_{n_\Omega, n_{\partial\Omega}}$, together with approximation information for the hypothesis class, yields a bound on the residual terms, and stability then turns that bound into a bound on $\|v - u^*\|_{\mathcal{U}}$ (Lu et al., 2022). We will make this precise in the next chapters.

We point out that the distributions μ_{int} and μ_{bd} are part of the method design, and can impact the error analysis. since they specify how residuals are weighted and sampled. For instance, with i.i.d. sampling, $(P_n - P)$ is a sampling fluctuation and the tools from this chapter apply once one controls the induced loss class (Koltchinskii and Panchenko, 2000). On the other hand, with deterministic collocation sets, the discrepancy between $R_{n_\Omega, n_{\partial\Omega}}$ and R is a quadrature error and other bounds are needed (Mishra and Molinaro, 2022; Wang et al., 2023). With other sampling strategies, such as adaptive sampling, the points are dependent, and the analysis yet again requires different techniques than the i.i.d. theory developed above (Wu et al., 2023a).

We lastly mention that, although the techniques introduced in this chapter are powerful, they do not control the *optimization error*, due to computing the ERM approximately only, through algorithms like gradient descent (Polyak, 1963). Unfortunately, getting optimization to reliably converge to non-spurious local minima is one of the biggest challenges in current SciML (Wang et al., 2021). We will discuss practical strategies to provably overcome these limitations in the coming chapters.

Chapter 3

Boundary-adapted PINNs for elliptic Dirichlet problems

In this chapter, we will show how error guarantees for Physics-Informed Neural Networks (PINNs) (Raissi et al., 2019) solving linear elliptic equations with Dirichlet boundary conditions can be established. The main example that will be the guiding motivation for this work will be the problem of computing *mean-exit times* (MET), which quantify how long a system takes to reach or escape a certain state of interest (Pavliotis, 2014). For such problems, especially in “small-noise regime”, the exact enforcement of Dirichlet boundary conditions is often necessary to ensure numerical stability and satisfactory solution accuracy (Sukumar and Srivastava, 2022). Addressing a gap in the literature, we rigorously show for the first time how exact boundary conditions enforcement in PINNs can be achieved while providing sharp convergence guarantees. We also uncover sufficient conditions on the boundary-enforcing functions that ensure consistency in the process. As we will see, the learning theory toolkit from Chapter 2 will play a crucial role in our analysis. This chapter is largely based on (Tepakbong et al., 2026a). We discuss in Appendix C how the analysis performed in this chapter can be extended to parabolic PDEs.

1 Introduction

1.1 Mean escape time

The Mean Escape Time problem (also known as Mean First Passage Time problem) is a specific instance in the broader class of *First Passage Problems*,

for which the task is to quantify the time it takes for a stochastic process $(X_t)_{t \geq 0}$ evolving in a domain $\Omega \subseteq \mathbb{R}^d$ to cross a certain barrier or reach a certain set, given information about its dynamics and its initial position or distribution. First Passage Problems have been the subject of extensive both theoretical and applied research over the years, due to their importance in a plethora of scientific fields, such as, e.g., chemistry, where they're used to model the inverse reaction rate, in engineering, where they serve to estimate the reliability of certain system configurations, in biology, where they model the firing of a synapse or in ecology, where they model population dynamics of species in a given environment. We refer the reader to (van Kampen, 2007; Holcman and Schuss, 2015; Schuss et al., 2007; Lancaster, 1972; Redner, 2001) and references within for a more complete overview of the First Passage Problems literature and its broad range of applications.

Due to the intrinsically stochastic nature of First Passage Problems, one can in practice only compute either the distribution of the first passage time, or its moments. The Mean Escape Time (MET) problem precisely consists in computing the first moment of this first passage time, where the set of interest is given by the boundary $\partial\Omega$ of a bounded domain $\Omega \subseteq \mathbb{R}^d$. More formally, for all $x \in \Omega$, we define $\tau(x)$ as the expected value of the first time $t > 0$ for which the process $(X_t)_{t \geq 0}$ hits the boundary $\partial\Omega$, where $(X_t)_{t \geq 0}$ solves the Stochastic Differential Equation (SDE):

$$dX_t = b(X_t, t)dt + \varsigma(X_t, t)dW_t, \quad X_0 = x,$$

where the coefficients ς and b are assumed to be known, and W is a standard Brownian motion in $\mathbb{R}^d$.

Outside of the simplest dynamics ς, b , and geometries Ω , closed-form expressions of τ are not tractable and one has to resort to numerical methods. This proves to be a very challenging and computationally intensive task in general, as the main options are either to rely on Monte Carlo sampling, which requires a large number of long time trajectories for each initial state, or to rely on expert-guided solution ansatz (e.g. WKB approximation in small noise limit), which require a lot of a priori knowledge while lacking generality and arbitrary accuracy. In addition, it is important to note that for many SDEs of interest, such as the ones which describe the transitions from different metastable states in molecular dynamics, the crossing of the boundary $\partial\Omega$ typically constitutes a *rare event*, which informally means that $\tau(x) \gg 0$ for all x away from the boundary (Freidlin and Wentzell, 2012; Schütte et al., 2023; Budhiraja and Dupuis, 2019; Dupuis et al., 2015). This makes both classic Monte-Carlo, and ansatz-based estimation of the MET extremely difficult, even in small dimensions.

A potential solution to work around these issues is to use classical results from probability theory and recast the MET problem as an elliptic PDE. It is indeed known—provided that the coefficients b and ς are sufficiently regular—that the time-dependent density of X_t , solves the Fokker-Planck equation. From there, one can algebraically derive that τ is the solution of a second-order elliptic PDE with homogenous Dirichlet boundary conditions (Pavliotis, 2014; Schuss et al., 2007). This alternative formulation of the MET as a PDE solution is attractive, as it allows to bypass the aforementioned challenges by use of numerical PDE solvers. However the sharp boundary layer of τ near $\partial\Omega$ typically makes those numerical solutions hard to compute, and to scale to higher dimensions. Physics-Informed Neural Networks (PINNs), with their added flexibility, and scalability thus naturally appear as a natural candidate to bypass these issues. We will thus investigate in this work, mostly from a theoretical perspective, whether PINNs can indeed constitute viable solutions to solve Mean Escape Time problems, by providing a complete a priori error analysis for PINN solutions of general second-order elliptic problems.

1.2 Physics-informed neural networks

Physics-Informed Neural Networks (PINNs) have emerged as a powerful mesh-free paradigm for approximating PDE solutions by embedding the equations directly into the neural network training process (Raissi et al., 2019). In this approach, a neural network is trained to minimize the PDE residual evaluated at discrete collocation points within the domain, supplemented by terms enforcing boundary and initial conditions. This residual-based formulation enables PINNs to efficiently solve a myriad of scientific problems, particularly in high-dimensional or irregular domains where traditional methods like finite elements struggle due to meshing complexities (Cuomo et al., 2022). Closely related are variational methods, such as the Deep Ritz method, which instead minimize a discretized energy functional derived from the PDE’s weak form (E and Yu, 2018).

A critical aspect of PINNs is the enforcement of boundary conditions, which can significantly influence solution accuracy and physical meaningfulness (Chen et al., 2020). Common strategies include “soft” enforcement, where boundary discrepancies are penalized in the loss function, and “hard” enforcement, which constructs the neural network output to satisfy conditions exactly through multiplicative or additive ansatzes (Sukumar and Srivastava, 2022). While computationally simpler, soft methods may suffer from loss imbalance, leading to poor boundary adherence unless penalty weights have

gone through an expensive hyperparameter tuning process (Krishnapriyan et al., 2021). While hard enforcement mitigates these issues by ensuring exact satisfaction, it may also lead to reduced expressivity of the neural network and instabilities during training if the multiplicative ansatz has been poorly chosen.

1.3 Our contributions

Motivated by the Mean Escape Time problem, which features sharp boundary layers that are typically difficult to learn using soft constraints, we study “boundary-enforced PINNs” for solving second-order elliptic PDEs. Specifically, we consider functions of the form $x \mapsto \rho(x) \cdot u_\theta(x)$, where ρ vanishes on $\partial\Omega$ and u_θ is a fully connected neural network (FCNN) parameterized by θ . The main contributions of this chapter can be summarized as follows:

- By a classical estimate of (Agmon et al., 1959), we observe that adding an $L^2(\partial\Omega)$ penalty to the PDE residual is inconsistent with obtaining $H^2(\Omega)$ a priori error estimates. Instead, we show in Proposition 3.9 that the only way to guarantee such $H^2(\Omega)$ a priori estimates for general PINNs is to multiply the neural network ansatz by a *smooth distance approximation normalized to first order* (See Definition 3.6). We will refer to PINNs constructed this way as *boundary-adapted PINNs*.
- We prove in Theorem 3.18 new bounds on the VC dimension of up to second-order derivatives of ReLU^2 and $\tanh$ fully connected neural networks. These bounds enable us to apply an oracle inequality due to (Lei et al., 2025), which in turn allows control of the statistical error without imposing restrictive bounds on the magnitude of the network weights. This makes our theoretical results more relevant for practical PINN implementations.
- In Theorem 3.24, we derive a complete a priori error analysis for boundary-adapted PINN solutions of second-order elliptic PDEs using either ReLU^2 or $\tanh$ activations. Our analysis combines the aforementioned statistical error bounds with $W^{2,\infty}(\Omega)$ approximation error bounds for shallow ReLU^2 and $\tanh$ networks. In the ReLU^2 case, the bounds are also new and are derived by generalizing variation space-based arguments to higher derivatives (Siegel and Xu, 2022; Yang and Zhou, 2025). To the best of our knowledge, these are the first rigorous error bound for PINNs that explicitly accounts for the enforcement of boundary conditions.

In Chapter 6, we will numerically illustrate the theoretical findings in this

chapter, and highlight the critical role that the choice of ρ plays when solving Mean Escape Time problems. As we will see, our results confirm that poorly chosen distance-approximation ansatzes and soft penalty approaches yield significantly worse solutions than boundary-adapted PINNs, in agreement with empirical observations in (Berrone et al., 2023).

1.4 Related works

Numerical solutions of escape time problems

Classical approaches to learn First Passage Times moments and distributions have traditionally involved either Monte Carlo methods with importance sampling schemes to speed up convergence (Heidelberger et al., 1996; Cérou and Guyader, 2007; Drugowitsch, 2016; Xu et al., 2017; Budhiraja and Dupuis, 2019; Nakayama and Tuffin, 2019). Another popular approach is the use of so-called *asymptotic perturbative methods*, to approximate the solution in small noise limit (typically the matched asymptotic expansion with WKB ansatz with a “singular” term near the boundary layer, and a “regular” term away from it), which often leads to series expansion solutions with recursive formulas for the coefficients solvable in closed-form (Siegert, 1951; Assaf and Meerson, 2010; Walter et al., 2021; Simpson et al., 2021; Carr et al., 2022).

Alternatively, one may exploit the PDE formulation of the first passage problem and derive numerical solutions either by spectral methods or by directly solving the PDE, using a parametrization which allows for faster convergence (Voss and Voss, 2008; Boehm et al., 2022) when available. We note however, that despite this PDE formulation, the use of Physics-Informed Neural Networks to solve for first passage times has not received much attention in the literature, at the exception of a few earlier works (Nagel et al., 2022; Qian et al., 2024), which successfully compute first passage time distributions for reliability engineering problems.

Theoretical guarantees for physics-informed machine learning

While a number of works, such as (Shin et al., 2020; Lu et al., 2021a; Jiao et al., 2022; Shin et al., 2023), have provided some theoretical guarantees for PINN solutions of elliptic problems, our understanding of these algorithms remains far from complete. Indeed, besides the challenging issue of training, for which only few results are known (Rathore et al., 2024; Luo and Yang, 2024; Jiao et al., 2025), it appears that the influence of “hard constraints” on PINN solutions is not yet covered by existing analyses.

Indeed, in (Shin et al., 2020; Lu et al., 2022; Shin et al., 2023; Luo and Yang, 2024), the authors assume either explicitly or implicitly that the FCNNs in their hypothesis spaces satisfy exactly the Dirichlet boundary conditions, which is of course not valid in most cases. To carry out the error analysis without having to rely on this assumption, other authors have considered either replacing the $L^2(\Omega)$ penalty on the loss function with a Sobolev penalty of order sufficiently large to ensure consistency (Wu et al., 2023b; Jiang et al., 2024; Bonito et al., 2025), or simply getting a priori error estimates in a weaker norm, such as $H^{1/2}(\Omega)$ (Jiao et al., 2022). We are thus the first to address, in this work, the issue of computing $H^2(\Omega)$ error bounds when the FCNNs in the hypothesis space has been multiplied with a “hard-constraint” function, enforcing the Dirichlet boundary condition exactly.

1.5 Notations

We now introduce the main notational conventions that we will be using throughout this chapter.

Multi-index notation

We denote by $\mathbb{N} := \{0, 1, 2, \dots\}$ the set of natural numbers. For a positive $d \in \mathbb{N}$, we call any element of $\alpha \in \mathbb{N}^d$ a multi-index. We define $\|\alpha\|_1 := \sum_{i=1}^d \alpha_i$, $\alpha! = \prod_{i=1}^d \alpha_i!$ and for all $x \in \mathbb{R}^d$, $x^\alpha := \prod_{i=1}^d x_i^{\alpha_i}$. For $\alpha, \beta \in \mathbb{N}^d$, we denote the partial order $\alpha \leq \beta \iff \alpha_i \leq \beta_i$ for all $1 \leq i \leq d$. We further denote the binomial coefficient:

$$\binom{\beta}{\alpha} := \prod_{i=1}^d \binom{\beta_i}{\alpha_i}.$$

And likewise, for $k \in \mathbb{N}$, we denote the multinomial coefficient:

$$\binom{k}{\alpha} := \frac{k!}{\alpha!}.$$

Lastly, for a domain $\Omega \subseteq \mathbb{R}^d$, and a function $f : \Omega \rightarrow \mathbb{R}$, we denote by

$$\partial^\alpha f := \frac{\partial^{\|\alpha\|_1} f}{\partial x_1^{\alpha_1} \dots \partial x_d^{\alpha_d}}$$

its derivative of order α , understood in either the classical, weak, or distributional sense.

Sobolev and Lebesgue spaces

For positive $d \in \mathbb{N}$, $1 \leq p \leq \infty$ and $\Omega \subseteq \mathbb{R}^d$, we denote by $L^p(\Omega)$ the usual Lebesgue space of equivalence classes of functions equipped with the usual norm:

$$\|f\|_{L^p(\Omega)} := \begin{cases} \left(\int_{\Omega} |f(x)|^p dx \right)^{1/p}, & 1 \leq p < \infty \\ \text{ess sup}_{x \in \Omega} |f(x)|, & p = \infty. \end{cases}$$

Likewise, for $r \in \mathbb{N}$, we define the Sobolev space $W^{r,p}(\Omega)$ in the standard way:

$$W^{r,p}(\Omega) := \{f \in L^p(\Omega) : \partial^\alpha f \in L^p(\Omega) \text{ for all } \alpha \in \mathbb{N}^d : \|\alpha\|_1 \leq r\}.$$

Norms

For a vector $x \in \mathbb{R}^d$ and $1 \leq p \leq \infty$, we define its ℓ_p norm in the standard way:

$$\|x\|_p := \begin{cases} \left(\sum_{i=1}^d |x_i|^p \right)^{1/p}, & 1 \leq p < \infty \\ \max_{1 \leq i \leq d} |x_i|, & p = \infty. \end{cases}$$

For $r \in \mathbb{N}$, and $\Omega \subseteq \mathbb{R}^d$ open, we choose the following definition of the $W^{r,p}(\Omega)$ norm:

$$\|f\|_{W^{r,p}(\Omega)} := \begin{cases} \left(\sum_{\alpha \in \mathbb{N}^d : \|\alpha\|_1 \leq r} \|\partial^\alpha f\|_{L^p(\Omega)}^p \right)^{1/p}, & 1 \leq p < \infty \\ \max_{\alpha \in \mathbb{N}^d : \|\alpha\|_1 \leq r} \|\partial^\alpha f\|_{L^\infty(\Omega)}, & p = \infty. \end{cases}$$

It is well known that $W^{r,p}(\Omega)$ equipped with this norm is a complete normed space. In the case where $p = 2$, $W^{r,2}(\Omega)$ is also a Hilbert space and we will denote it $H^r(\Omega)$ instead.

2 Problem setup

We start by formally introducing the Mean Escape Time (MET) problem, as it will serve as a concrete example to illustrate our theory. We stress however that the results we will show apply to all second-order elliptic PDEs satisfying appropriate regularity conditions.

Let $d \geq 2$ be an integer, and for all $x \in \mathbb{R}^d$, denote by X_t^x the unique — up to standard regularity conditions — solution of the stochastic differential

equation (Pavliotis, 2014)

$$dX_t^x = b(X_t^x)dt + \varsigma(X_t^x)dW_t, \quad X_0 = x \quad (3.1)$$

where (W_t) is a standard Brownian motion in $\mathbb{R}^d$, while $b : \mathbb{R}^d \rightarrow \mathbb{R}^d$ and $\varsigma : \mathbb{R}^d \rightarrow \mathbb{R}^{d \times d}$ are respectively called the *drift* and *diffusion* coefficients of the stochastic process X_t^x .

Given a bounded, open domain $\Omega \subset \mathbb{R}^d$ with smooth boundary $\partial\Omega$, we define for all $x \in \Omega$ the *first passage time*, or *first exit time* as the time when X_t^x first exits Ω :

$$\tau_\Omega^x := \inf\{t \geq 0 : X_t^x \notin \Omega\}.$$

We then define the *mean escape time* (also known as mean exit time or mean first passage time) as the expected value of τ_Ω^x :

$$\tau(x) := \mathbb{E}\tau_\Omega^x = \mathbb{E}[\inf\{t \geq 0 : X_t^x \notin \Omega\}]. \quad (3.2)$$

2.1 An elliptic boundary value problem for the mean escape time

It is well-known that τ solves the following boundary value problem:

Theorem 3.1 (Theorem 7.1 in (Pavliotis, 2014)). *The mean exit time is given by the solution of the boundary value problem*

$$\begin{cases} \mathcal{L}\tau(x) &= -1, & x \in \Omega \\ \tau(x) &= 0, & x \in \partial\Omega \end{cases} \quad (3.3)$$

where $\mathcal{L}$ is the generator of the stochastic process (3.1), whose expression is given by

$$\mathcal{L} = \sum_{i=1}^d b_i(x) \frac{\partial}{\partial x_i} + \frac{1}{2} \sum_{i,j=1}^d \Sigma_{ij}(x) \frac{\partial^2}{\partial x_i \partial x_j}, \quad (3.4)$$

where $\Sigma(x) := \varsigma(x)\varsigma(x)^T$ is the diffusion matrix.

Let $r \geq 3$ be a natural number and $0 \leq \kappa \leq 1$, which we will respectively refer to as the *smoothness order* (or *exponent*) and the *Hölder exponent*. We will assume that the operator $\mathcal{L}$ and the domain Ω satisfy the following assumptions:

A.1 Ω is a $C^{r+2,\kappa}$ domain, in the sense that $\partial\Omega$ is $C^{r+2,\kappa}$.

A.2 The diffusion matrix Σ is *strictly* (or *uniformly*) *elliptic*: there exists $\lambda > 0$ such that

$$\xi^T \Sigma(x) \xi \geq \lambda \|\xi\|_2^2, \quad \text{for all } x \in \Omega \text{ and } \xi \in \mathbb{R}^d$$

A.3 The coefficients $(b_j)_{1 \leq j \leq d}, (\Sigma_{ij})_{1 \leq i, j \leq d}$ of $\mathcal{L}$ all belong to $C^{r, \kappa}(\bar{\Omega})$.

We refer the reader to standard references such as (Gilbarg and Trudinger, 1977) for the definitions of $C^{r+2, \kappa}$ domains and $C^{r, \kappa}(\bar{\Omega})$ functions. Under Assumptions (A.1) — (A.3), we have from standard elliptic PDE theory that the MET τ is well-defined as the unique classical solution of (3.3).

Theorem 3.2. *If Assumptions (A.1) — (A.3) hold, then the boundary value problem (3.3) has a unique solution τ lying in $C^{r+2, \kappa}(\bar{\Omega})$.*

Proof. See Theorems 6.14 and 6.19 in (Gilbarg and Trudinger, 1977). $\square$

Since Ω is bounded, we will assume for convenience in all of the following that Ω is contained in either the unit ball $\mathbb{B}^d$ or the unit cube $[0, 1]^d$, which can be done without loss of generality by scaling and translation of the coordinate system.

Remark 3.3. Once again, we point out that our results still hold if (3.3) and (3.4) are respectively replaced by

$$\mathcal{L}\tau(x) = f(x), \quad (x \in \Omega), \quad \tau(x) = g(x) \quad (x \in \partial\Omega),$$

and

$$\mathcal{L} = c(x) + \sum_{i=1}^d b_i(x) \frac{\partial}{\partial x_i} + \sum_{i,j=1}^d \Sigma_{ij}(x) \frac{\partial^2}{\partial x_i \partial x_j},$$

where $f : \Omega \rightarrow \mathbb{R}, g : \bar{\Omega} \rightarrow \mathbb{R}$, and $c : \Omega \rightarrow \mathbb{R}$ are functions with sufficient regularity.

To fix ideas, let's consider a simple, but representative example: an Ornstein-Uhlenbeck process in the unit ball of $\mathbb{R}^d$ with constant drift and noise coefficients. In this case, X_t solves the SDE

$$dX_t = -\theta X_t dt + \sqrt{2\varepsilon} dW_t. \quad (3.5)$$

In the small ε regime, the theory of singular perturbations gives a limit-behaviour for the MET, namely $\tau_\varepsilon(x) \sim_{\varepsilon \rightarrow 0^+} C \exp\left(\theta \frac{1 - \|x\|_2^2}{\varepsilon^2}\right)$ for all x at least $\sqrt{\varepsilon}$ away from $\partial\Omega$ (Verhulst, 2005; Freidlin and Wentzell, 2012). This

small-noise regime, which can be interpreted as the dynamics of a particle trapped in a potential well with very high energy barrier, is of high interest, e.g., in computational chemistry. However, this sharp growth of the solution over such a thin boundary layer implies that PINNs are most likely to struggle in satisfying the Dirichlet conditions, if they are not explicitly enforced.

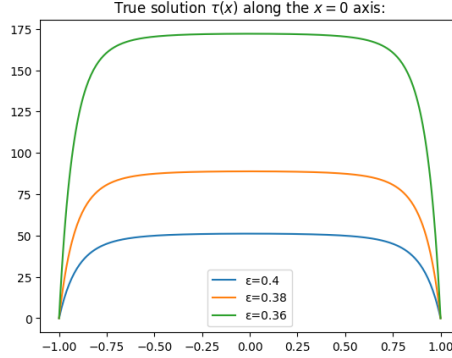


Figure 3.1: Projection on the $\{(x, y) : x = 0\}$ axis of the MET for the SDE (3.5) in the 2D unit ball, with $\theta = 1$ and varying $\varepsilon > 0$. The closed-form expression for this MET has been derived in (Kersting et al., 2023).

2.2 Physics-informed neural networks

An approach to solve boundary value problems (BVPs) such as (3.3) consists in using Physics-Informed Neural Networks (PINNs). Popularized in (Raissi et al., 2019), the idea is to use neural networks as solution ansatz to the BVP, and minimize the PDE residual by using gradient-based algorithms. More specifically, for a PDE such as (3.3), the residual is given by

$$\mathcal{R}(u) = \frac{1}{|\Omega|} \int_{\Omega} |\mathcal{L}u(x) + 1|^2 dx + \frac{1}{\sigma(\partial\Omega)} \int_{\partial\Omega} u(x)^2 dx,$$

where $|\cdot|$ and $\sigma(\cdot)$ respectively denote the Lebesgue measure in $\mathbb{R}^d$ and the $d-1$ dimensional Hausdorff measure. The PINN approach in this setting consists in sampling n independent realizations of the uniform distribution respectively on Ω and $\partial\Omega$, that is, $x_1, \dots, x_n \in \Omega$, and $x_{n+1}, \dots, x_{2n} \in \partial\Omega$ and minimizing the empirical residual:

$$\hat{\theta} \in \operatorname{argmin}_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^n (\mathcal{L}u_{\theta}(x_i) + 1)^2 + \frac{\lambda}{n} \sum_{i=1}^n u_{\theta}(x_{n+i})^2,$$

where u_{θ} is a neural network parametrized by $\theta \in \Theta$, and $\lambda > 0$ is a hyper-parameter which regulates the strength of the boundary constraint.

Standard PINN loss is inconsistent

As we saw in Chapter 2, learning theory gives us tools to control the excess residual $\mathcal{R}(\hat{u}) - \mathcal{R}(\tau)$, where $\hat{u} \equiv u_{\hat{\theta}}$ is a minimizer of the empirical residual, in terms of the sample size n by decomposing the residual into an approximation error term and a statistical error term. It is however natural to ask whether such residual error bounds can provide any guarantees on the *actual* error $\|\hat{u} - \tau\|$, where $\|\cdot\|$ is a Sobolev norm appropriate for this problem. Since we are considering the strong formulation of the BVP (3.3), we will use the $H^2(\Omega)$ norm as our metric. Note however that for weak or energy-based formulations, weaker Sobolev norms might be used instead. Likewise, it is important to understand whether the PDE error $\|\hat{u} - \tau\|$ can control the excess risk $\mathcal{R}(\hat{u}) - \mathcal{R}(\tau)$ in any way. These questions boil down to asking whether the residual $\mathcal{R}$ satisfies a so-called *strong-convexity estimate* of the form

$$\|\hat{u} - \tau\|^2 \lesssim \mathcal{R}(\hat{u}) - \mathcal{R}(\tau) \lesssim \|\hat{u} - \tau\|^2, \quad (3.6)$$

where $\lesssim$ denotes an inequality which holds up to a universal multiplicative constant. Strong convexity estimates such as (3.6) are closely related to a priori estimates for elliptic PDEs, and are highly non-trivial in general. Since τ solves the BVP (3.3), we get for any $u \in H^2(\Omega)$:

$$\begin{aligned} \mathcal{R}(u) - \mathcal{R}(\tau) &= \frac{1}{|\Omega|} \int_{\Omega} |\mathcal{L}u(x) + 1|^2 dx + \frac{1}{\sigma(\partial\Omega)} \int_{\partial\Omega} u(x)^2 dx \\ &= \frac{1}{|\Omega|} \int_{\Omega} |\mathcal{L}u(x) - \mathcal{L}\tau(x)|^2 dx + \frac{1}{\sigma(\partial\Omega)} \int_{\partial\Omega} (u(x) - \tau(x))^2 dx \\ &= \frac{1}{|\Omega|} \|\mathcal{L}(u - \tau)\|_{L^2(\Omega)}^2 + \frac{1}{\sigma(\partial\Omega)} \|u - \tau\|_{L^2(\partial\Omega)}^2 \\ &\lesssim \|u - \tau\|_{H^2(\Omega)}^2 + \|u - \tau\|_{L^2(\partial\Omega)}^2, \end{aligned}$$

where the last inequality holds under standard regularity assumptions on the coefficients of $\mathcal{L}$. Obtaining a comparable lower bound, however, is a much more delicate matter, which is addressed by the following classical theorem:

Theorem 3.4 (Adapted from Theorem 15.2 in (Agmon et al., 1959)). *Let $\mathcal{L}$ be a uniformly elliptic operator of order 2 whose coefficients are in $C(\bar{\Omega})$, and suppose that $\partial\Omega$ is of class C^2 . If u is a solution of the boundary value problem*

$$\begin{cases} \mathcal{L}u &= f & \text{in } \Omega \\ u &= g & \text{in } \partial\Omega \end{cases} \quad (3.7)$$

such that $\|u\|_{H^2(\Omega)}$, $\|f\|_{L^2(\Omega)}$, and $\|g\|_{H^{3/2}(\partial\Omega)}$ are all finite, then we have the estimate

$$\|u\|_{H^2(\Omega)} \lesssim \|f\|_{L^2(\Omega)} + \|g\|_{H^{3/2}(\partial\Omega)} + \|u\|_{L^2(\Omega)}.$$

Furthermore, if the BVP (3.7) has a unique solution in $H^2(\Omega)$, the additional $\|u\|_{L^2(\Omega)}$ term can be omitted, leading to the estimate:

$$\|u\|_{H^2(\Omega)} \lesssim \|f\|_{L^2(\Omega)} + \|g\|_{H^{3/2}(\partial\Omega)}.$$

From Theorem 3.4, provided that conditions such as our Assumptions (A.1)—(A.3) hold, we get the estimate

$$\|\hat{u} - \tau\|_{H^2(\Omega)} \lesssim \|\mathcal{L}(\hat{u} - \tau)\|_{L^2(\Omega)} + \|\hat{u} - \tau\|_{H^{3/2}(\partial\Omega)}. \quad (3.8)$$

Hence we see that the residual loss for PINN is inconsistent with the strong convexity estimate (3.6) required to translate bounds on $\mathcal{R}(\hat{u}) - \mathcal{R}(\tau)$ into bounds on the error $\|\hat{u} - \tau\|_{H^2(\Omega)}$. To circumvent this issue, one could replace the inconsistent $\|\cdot\|_{L^2(\partial\Omega)}$ penalty term by a $\|\cdot\|_{H^s(\partial\Omega)}$ term with $s \geq 3/2$, which would fit the required lower bound. However, this approach is often impractical, as the accurate computation of higher-order (including fractional) Sobolev norms on arbitrary manifolds tends to be challenging. Moreover, incorporating such norms directly into Neural Network training can significantly increase computational complexity and implementation difficulty, especially when relying on automatic differentiation. We note that it is still possible to get error estimates using $L^2(\partial\Omega)$, or appropriate $W^{s,p}(\partial\Omega)$ penalty. This however typically results in estimates in weaker norms, which may fail to capture the full complexity of the PDE solution (Jiao et al., 2022; Bonito et al., 2025).

A more practical alternative is to enforce the homogenous Dirichlet condition directly, which is typically achieved by multiplying the Neural Network outputs by a smooth approximation of the distance to boundary function. This approach, known as the *conforming method*, has been first proposed in (Lagaris et al., 1998, 2000), and recently further developed in (Berg and Nyström, 2018; Sukumar and Srivastava, 2022) with promising numerical results. This is the approach we will consider in this study.

2.3 Boundary-adapted PINN architectures

In the physics-informed machine learning literature, it is standard to take the PDE-solving ansatz as a fully-connected neural network (FCNN) with a sufficiently smooth activation function (E and Yu, 2018; Raissi et al., 2019; Lu et al., 2021b; De Ryck et al., 2021). This leads to hypothesis spaces of the following form:

Definition 3.5 (Fully Connected Neural Networks). *Let $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ be an activation function, $L \in \mathbb{N}^+$, $\mathbf{m} := \{m_i\}_{1 \leq i \leq L} \subset \mathbb{N}^+$, $m_0 = d$, and $B, M > 0$. We denote by $\Sigma_{\mathbf{m}_{1:L}}^\sigma(B)$ the class of FCNN defined on $\mathbb{R}^d$*

$$\Sigma_{\mathbf{m}_{1:L}}^\sigma(B) = \{ \mathbb{R}^d \ni x \mapsto c \cdot h_L(x) : h_0(x) = x, h_{i+1}(x) = \sigma(A_i h_i(x) + b_i), A_i \in \mathbb{R}^{m_i \times m_{i-1}}, b_i \in \mathbb{R}^{m_i}, c \in \mathbb{R}^{m_L}, \|A_i\|_{\infty, \infty} \leq B, \|b_i\|_\infty \leq B, i = 1, \dots, L \} \quad (3.9)$$

where σ is applied entry-wise, and $\|b\|_\infty$, $\|A\|_{\infty, \infty}$ both denote the entry-wise maximum absolute value of respectively $b \in \mathbb{R}_i^{m_i}$ and $A \in \mathbb{R}^{m_i \times m_{i-1}}$. If we further require that $\|c\|_\infty \leq M$, we will denote the class as $\Sigma_{\mathbf{m}_{1:L}, M}^\sigma(B)$, following notation from (He et al., 2023; Siegel and Xu, 2022, 2024). Lastly, we will denote $\Sigma_{\mathbf{m}_{1:L}}^\sigma(\infty)$ the space of FCNNs with no norm restrictions on the parameters.

Although the hypothesis space $\Sigma_{\mathbf{m}_{1:L}, M}^\sigma(B)$ generally enjoys a lot of desirable properties regarding expressiveness and complexity, the functions in this space do not, in general, satisfy the homogeneous Dirichlet boundary condition (3.3). As we discussed in the previous subsection, this implies that a control of the $H^2(\Omega)$ a priori error with this hypothesis space is not possible. To circumvent this issue in the error analysis, authors have either worked under the assumption of exact satisfaction of boundary conditions (Lu et al., 2022), which is not realistic, or considered PDEs on boundary-less submanifolds of $\mathbb{R}^d$, such as the sphere (Lei et al., 2025), which is restrictive.

Sufficient condition on the distance function for consistency

To make up for the aforementioned limitations, we propose to exactly enforce the boundary conditions by multiplying the outputs of the FCNN hypothesis space by *smooth distance approximations*, as proposed in (Sukumar and Srivastava, 2022):

Definition 3.6 (Smooth distance approximation). *Given $\rho : \bar{\Omega} \rightarrow \mathbb{R}$ and $r \in \mathbb{N}^+$, we say that ρ is an r -smooth approximate distance function normalized to the first order if:*

- $\rho \in C^r(\bar{\Omega})$,
- $\rho(x) = 0$ for all $x \in \partial\Omega$,
- $\rho(x) > 0$ for all $x \in \Omega$,
- $\frac{\partial \rho}{\partial \nu} = 1$,

where $\nu \equiv \boldsymbol{\nu}$ is the unit inward vector on $\partial\Omega$, and $\partial\rho/\partial\nu := \nabla\rho \cdot \boldsymbol{\nu}$ denotes the unit normal derivative with respect to ν on $\partial\Omega$. We will also call such a function ρ an r -smooth distance approximation.

(Sukumar and Srivastava, 2022) and references within give a rich framework under which one can construct smooth distance approximations for a wide variety of domain geometries. We will give examples of explicit constructions in Chapter 6 and in the Appendix D. Given such a distance approximation, we can then define the hypothesis space of *Boundary-enforced* Neural Networks:

Definition 3.7 (Boundary-enforced FCNNs). *Let $\rho : \bar{\Omega} \rightarrow \mathbb{R}$ be a smooth distance approximation, $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ be an activation function, $L \in \mathbb{N}^+$, $\mathbf{m} := \{m_i\}_{1 \leq i \leq L} \subset \mathbb{N}^+$, and $Q, B, M > 0$. We denote by $\mathcal{F}(\rho, L, \mathbf{m}, \sigma, Q, B, M)$ the hypothesis space of functions defined on Ω by*

$$\mathcal{F}(\rho, L, \mathbf{m}, \sigma, Q, B, M) = \left\{ \Omega \ni x \mapsto \rho(x)f(x) : f \in \Sigma_{\mathbf{m}_{1:L}, M}^\sigma(B) \text{ and } \max_{\|\alpha\|_1 \leq 2} \|\partial^\alpha f\|_{L^\infty} \leq Q \right\}. \quad (3.10)$$

As in Definition 3.5, we may let $B = \infty$ or $M = \infty$ to signify that the weights are not bounded.

As we will see later, we in fact do not need to impose restrictions on the FCNN weights to carry our error analysis, but we still provide sufficient weight bounds for the sake of comparison with similar results in the literature.

Remark 3.8. It is straightforward to modify Definition 3.7 to handle boundary conditions of the form $\tau(x) = g(x)$ for $x \in \partial\Omega$ and g defined on $\partial\Omega$: we can just add $g \cdot \mathbb{1}_{\partial\Omega}$ to the functions in (3.10). All of our results thus extend straightforwardly to that case.

Assuming the activation function σ is sufficiently regular, it follows immediately that all “boundary-enforced” PINNs satisfy the homogeneous Dirichlet boundary condition. Consequently, the strong convexity estimate (3.6) holds, which is necessary for our error analysis to carry through.

However, multiplying by ρ introduces additional challenges from a learning theory perspective. Specifically, to control the approximation error of the hypothesis spaces defined in (3.10), we must control the Sobolev norm of the function $f - \tau/\rho$, where f lies in the FCNN space $\Sigma_{\mathbf{m}_{1:L}}^\sigma(B)$. Since Definition 3.6 alone does not govern the interaction between ρ and τ , it is unclear whether the quotient $\tau(x)/\rho(x)$ is even well-defined as a function for all $x \in \partial\Omega$, let alone whether τ/ρ possesses any Sobolev regularity.

The following Proposition, perhaps surprisingly, guarantees ρ being an $(r+2)$ -smooth distance approximation is both a sufficient and necessary condition for the quotient τ/ρ to be well-defined and C^r -smooth up to the boundary:

Proposition 3.9. *If Assumptions (A.1), (A.2), and (A.3) hold such that τ is well-defined as the unique solution of (3.3), and if $\rho : \bar{\Omega} \rightarrow \mathbb{R}$ is an $(r+2)$ -smooth distance approximation as per Definition 3.6, then $\tau/\rho \in C^r(\bar{\Omega})$.*

In light of Proposition 3.9, whose proof is given in Appendix D, we will refer to functions of the form (3.10), where ρ is an $(r+2)$ -smooth distance approximation, as *boundary-adapted FCNNs*.

The $C^r(\bar{\Omega})$ regularity of τ/ρ implies that it can be extended to a $W^{r,\infty}(\mathbb{R}^d)$ function, which will be helpful for theoretical analysis:

Lemma 3.10 (Extension lemma). *Under the assumptions of Proposition 3.9, there exist a function $u \in W^{r,\infty}(\mathbb{R}^d)$ and a constant $C_{ext} > 0$ which depends solely on r, d and Ω , such that*

- $u = \tau/\rho$ in $\bar{\Omega}$,
- u is supported on a compact subset of $\mathbb{R}^d$,
-

$$\|u\|_{W^{r,\infty}(\mathbb{R}^d)} \leq C_{ext} \|\tau/\rho\|_{W^{r,\infty}(\bar{\Omega})}. \quad (3.11)$$

Proof. By Proposition 3.9, we have $\tau/\rho \in C^r(\bar{\Omega})$. The result thus follows from (Gilbarg and Trudinger, 1977, Theorem 7.25). $\square$

2.4 Excess risk decomposition

We now formally establish the learning-theoretic framework of this work, in accordance with the groundwork laid out in Chapter 2. Throughout this chapter, we will consider FCNNs using either the Rectified Quadratic Unit (ReQU), or the hyperbolic tangent (tanh) as activation functions. These choices ensure that the FCNN hypothesis spaces defined in (3.10) are contained in the Sobolev space $W^{2,\infty}(\Omega) \subset H^2(\Omega)$, and represent two of the most widely used activation functions in scientific machine learning. Accordingly, we consider the activation functions defined as

$$\sigma : x \mapsto \begin{cases} x^2, & x > 0 \\ 0, & x \leq 0 \end{cases}, \quad \text{or } \sigma : x \mapsto \tanh(x) := \frac{e^x - e^{-x}}{e^x + e^{-x}}, \quad (3.12)$$

and will provide an error analysis for each of choice.

With these definitions established, we define the residual of a function $u \in H^2(\Omega)$ as

$$\mathcal{R}(u) = \frac{1}{|\Omega|} \int_{\Omega} |\mathcal{L}u(x) + 1|^2 dx. \quad (3.13)$$

And likewise, its empirical counterpart as

$$\mathcal{R}_n(u) = \frac{1}{n} \sum_{i=1}^n |\mathcal{L}u(x_i) + 1|^2, \quad (3.14)$$

where the $x_1, \dots, x_n$ are independent samples drawn uniformly on Ω .

Let $\mathcal{F} \subset H^2(\Omega)$ be a hypothesis space of functions and denote by

$$u_n \in \arg \min_{u \in \mathcal{F}} \mathcal{R}_n(u), \quad u_{\mathcal{F}} \in \arg \min_{u \in \mathcal{F}} \mathcal{R}(u),$$

the minimizers of the empirical and population risk respectively. Recall that our goal is to control the error $\|u_n - \tau\|_{H^2(\Omega)}$, which, as justified in the previous sections, can be controlled by $\mathcal{R}(u_n) - \mathcal{R}(\tau)$ up to a constant factor independent of n . To that end, we proceed by decomposing the error as follows:

$$\begin{aligned} \mathcal{R}(u_n) - \mathcal{R}(\tau) &= [\mathcal{R}(u_n) - \mathcal{R}_n(u_n)] + [\mathcal{R}_n(u_n) - \mathcal{R}_n(u_{\mathcal{F}})] + [\mathcal{R}_n(u_{\mathcal{F}}) - \mathcal{R}(u_{\mathcal{F}})] \\ &\quad + [\mathcal{R}(u_{\mathcal{F}}) - \mathcal{R}(\tau)] \\ &\leq [\mathcal{R}(u_n) - \mathcal{R}_n(u_n)] + [\mathcal{R}_n(u_n) - \mathcal{R}_n(u_{\mathcal{F}})] + [\mathcal{R}_n(u_{\mathcal{F}}) - \mathcal{R}(u_{\mathcal{F}})] \\ &\quad + [\mathcal{R}(u_{\mathcal{F}}) - \mathcal{R}(\tau)] + [\mathcal{R}_n(u_{\mathcal{F}}) - \mathcal{R}_n(u_n)]. \end{aligned}$$

Which, after rearranging, yields the inequality:

$$\begin{aligned} \mathcal{R}(u_n) - \mathcal{R}(\tau) &\leq [\mathcal{R}(u_{\mathcal{F}}) - \mathcal{R}(\tau)] \\ &\quad + [\mathcal{R}_n(u_{\mathcal{F}}) - \mathcal{R}_n(\tau) - \mathcal{R}(u_{\mathcal{F}}) + \mathcal{R}(\tau)] \\ &\quad + [\mathcal{R}(u_n) - \mathcal{R}(\tau) - \mathcal{R}_n(u_n) + \mathcal{R}_n(\tau)]. \end{aligned} \quad (3.15)$$

The residual is thus controlled by the three summands on the right-hand side of (3.15). The first one being the *approximation error*, and the last one being the *statistical error*. While the second summand can be bounded by means of classical concentration inequalities, the two other terms require a detailed analysis, which we provide in the next section.

3 Main results

3.1 Approximation error bounds

We first give a lemma which bounds the Sobolev distance between ρf and g in terms of the Sobolev distance between f and g/ρ . The proof is given in the appendix.

Lemma 3.11. *Let $f, g \in W^{r,\infty}(\Omega)$, $\rho \in C^r(\Omega)$ be such that $\rho f \in W^{r,\infty}(\Omega)$ and $g/\rho \in W^{r,\infty}(\Omega)$. The following estimate holds*

$$\|\rho f - g\|_{W^{r,\infty}(\Omega)} \leq C_{\text{mult}} \left\| f - \frac{g}{\rho} \right\|_{W^{r,\infty}(\Omega)}, \quad (3.16)$$

where

$$C_{\text{mult}} = \max_{\substack{\alpha \in \mathbb{N}^d \\ \|\alpha\|_1 \leq r, \beta \leq \alpha}} \binom{\alpha}{\beta} \cdot \sum_{k=0}^r \binom{k+d-1}{k} (k+1)^{d^2} \cdot \|\rho\|_{W^{r,\infty}(\Omega)}.$$

We now introduce an upper bound for the norm of the operator $\mathcal{L} : H^2(\Omega) \rightarrow L^2(\Omega)$ in terms of its coefficients, which will be necessary to control the approximation error term in (3.15):

Lemma 3.12. *For any $u \in H^2(\Omega)$, we have the estimate*

$$\|\mathcal{L}u\|_{L^2(\Omega)}^2 \leq C_{\mathcal{L}} \|u\|_{H^2(\Omega)}^2,$$

where

$$C_{\mathcal{L}} = \max \left\{ 2d \cdot \max_{1 \leq i \leq d} \|b_i\|_{L^\infty(\Omega)}^2, \frac{d^2}{2} \cdot \max_{1 \leq i, j \leq d} \|\Sigma_{ij}\|_{L^\infty(\Omega)}^2 \right\}$$

Proof. By direct computation:

$$\begin{aligned} \|\mathcal{L}u\|_{L^2(\Omega)}^2 &= \left\| \sum_{i=1}^d b_i \frac{\partial u}{\partial x_i} + \frac{1}{2} \sum_{i,j=1}^d \Sigma_{ij} \frac{\partial^2 u}{\partial x_i \partial x_j} \right\|_{L^2(\Omega)}^2 \\ &\leq 2d \sum_{i=1}^d \left\| b_i \frac{\partial u}{\partial x_i} \right\|_{L^2(\Omega)}^2 + \frac{d^2}{2} \sum_{i,j=1}^d \left\| \Sigma_{ij} \frac{\partial^2 u}{\partial x_i \partial x_j} \right\|_{L^2(\Omega)}^2 \end{aligned}$$

and we conclude by applying Hölder's inequality in each summand. $\square$

We can now state the approximation error bounds of smooth functions using respectively ReQU and tanh activation. We begin by the former, for which we give a proof in the Section 4:

Theorem 3.13. *Let $r, N \geq 2$ be two integers, and let $u \in W^{r,\infty}(\mathbb{R}^d)$. Take the hypothesis space of ReQU FCNNs as $\mathcal{F}_{NN} := \Sigma_{\mathbf{m}_{1:L},M}^\sigma(B)$ with*

- $\sigma : x \mapsto \max\{0, x\}^2$,
- $L = 1$,
- $\|\mathbf{m}\|_\infty = N^d$,
- $B = B_0$,
- $M = M_N(u)$, where $M_N(u) \leq C_{d,r} (1 + M_*(N)) \|u\|_{W^{r,\infty}(\mathbb{R}^d)}$ with

$$M_*(N) = \begin{cases} N^{\frac{d+5-2r}{2}}, & r < (d+5)/2, \\ \sqrt{\log N}, & r = (d+5)/2 \text{ and } r \notin 2\mathbb{N} \\ 1, & r = (d+5)/2 \text{ and } r \in 2\mathbb{N} \\ 1, & r > (d+5)/2. \end{cases}$$

where $B_0 > 0$ and $C_{d,r} > 0$ depend only on d and r . Then there exists a constant $M_{\text{ReQU}} > 0$, also dependent on r and d only, such that

$$\inf_{f_{NN} \in \mathcal{F}_{NN}} \|f_{NN} - u\|_{W^{2,\infty}(\mathbb{B}^d)} \leq M_{\text{ReQU}} \frac{\log(N)^{1/2}}{N^{\gamma(r)}} \|u\|_{W^{r,\infty}(\mathbb{R}^d)}, \quad (3.17)$$

where

$$\gamma(r) = \min \left\{ r - 2, \frac{d+1}{2} \right\}.$$

Theorem 3.13 is a generalization of the recently established sharp rates of approximation for shallow ReLU^k networks (Yang and Zhou, 2025) to higher order Sobolev norms. We also have an analogous approximation error bound for FCNNs with tanh activation function, due to (De Ryck et al., 2021).

Theorem 3.14 (Adapted from (De Ryck et al., 2021), Theorem 5.1). *Let $r \geq 2$, $N \geq 3d/2$ be two integers, and let $u \in C^r([0,1]^d)$. Take the hypothesis space of tanh FCNNs as $\mathcal{F}_{NN} = \Sigma_{\mathbf{m}_{1:L},M}^\sigma(B)$, with*

- $\sigma : x \mapsto \tanh(x)$
- $L = 2$,
- $\|\mathbf{m}\|_\infty \leq \max\{3(ed)^{r-1} \lceil r/2 \rceil, 15(5N)^d \lceil (d+2)/2 \rceil\}$,
- $B = M \lesssim C_\rho^{-r/2} N^{d(d+r^2+4)/2} (r(r+2))^{3r(r+2)}$,
- $C_\rho = 2^7 \cdot 3^6 \cdot \max_{0 \leq \ell \leq 2} \{(3d/2)^{r-\ell} / (r-\ell)!\}$.

We have the approximation error bound:

$$\inf_{f_{NN} \in \mathcal{F}_{NN}} \|f_{NN} - u\|_{W^{2,\infty}([0,1]^d)} \leq \ln^2(\beta N^{r+d+2}) \frac{M_{\tanh}}{N^{(r-2)}} \|u\|_{W^{r,\infty}(\mathbb{R}^d)}, \quad (3.18)$$

where

$$\beta := \frac{2^{d+3} \sqrt{d} \max \left\{ 1, \|u\|_{W^{r,\infty}([0,1]^d)}^{1/2} \right\}}{\min \{1, \sqrt{C_\rho}\}}, \quad M_{\tanh} := 3^d C_\rho.$$

From Theorems 3.13 and 3.14, together with the lemmas preceding them, we deduce as a corollary the following approximation error bound for our boundary-adapted Neural Networks hypothesis space.

Corollary 3.15. *Let $r, N \geq 2$ be two integers. Assume that Assumptions (A.1), (A.2), (A.3) hold, and that $\rho: \bar{\Omega} \rightarrow \mathbb{R}$ is an $(r+2)$ -smooth distance approximation. Denote by $u \in W^{r,\infty}(\mathbb{R}^d)$ an extension of $\tau/\rho \in C^r(\bar{\Omega})$ to $\mathbb{R}^d$, and denote the hypothesis space of boundary-adapted FCNNs as $\mathcal{F} = \mathcal{F}(\rho, L, \mathbf{m}, \sigma, Q, B, M)$.*

- *If σ is the ReQU activation, the parameters $L, \mathbf{m}, M = M(u)$ and B are chosen as in Theorem 3.13, and*

$$Q \geq \left(1 + \sqrt{M_{\text{ReQU}}}\right) \|\tau/\rho\|_{W^{r,\infty}(\mathbb{R}^d)}, \quad (3.19)$$

then the approximation error of the hypothesis space $\mathcal{F}$ satisfies the inequality:

$$\mathcal{R}(h_{\mathcal{F}}) - \mathcal{R}(\tau) \leq C_{\text{ReQU}} \|\tau/\rho\|_{W^{r,\infty}(\Omega)}^2 \frac{1}{N^{2\gamma(r)}}, \quad (3.20)$$

where $\gamma(r)$ is defined in Theorem 3.13, $h_{\mathcal{F}} \in \inf_{h \in \mathcal{F}} \mathcal{R}(h)$, and

$$C_{\text{ReQU}} := C_{\mathcal{L}} C_{\text{mult}}^2 (d+2)^2 M_{\text{ReQU}}^2.$$

- *If σ is the tanh activation, the parameters $L, \mathbf{m}, M$ and B are chosen as in Theorem 3.14, and*

$$Q \geq \left(1 + \sqrt{M_{\tanh}}\right) \|\tau/\rho\|_{W^{r,\infty}(\mathbb{R}^d)}, \quad (3.21)$$

then the approximation error of the hypothesis space $\mathcal{F}$ satisfies the inequality:

$$\mathcal{R}(h_{\mathcal{F}}) - \mathcal{R}(\tau) \leq C_{\tanh} \|\tau/\rho\|_{W^{r,\infty}(\Omega)}^2 \frac{\ln^4(\beta N^{r+d+2})}{N^{2(r-2)}}, \quad (3.22)$$

where $h_{\mathcal{F}} \in \inf_{h \in \mathcal{F}} \mathcal{R}(h)$, β is as defined in Theorem 3.14, and

$$C_{\tanh} := C_{\mathcal{L}} C_{\text{mult}}^2 (d+2)^2 M_{\tanh}^2.$$

Proof of Corollary 3.15. We only give the proof for the ReQU activation, as the one for tanh is identical. Let $h \in \mathcal{F}$. By definition, there exists $\hat{u} \in \Sigma_{\mathbf{m}_1:L,M}^\sigma$ such that $h = \rho \cdot \hat{u}$. We have

$$\begin{aligned} \mathcal{R}(\rho \cdot \hat{u}) - \mathcal{R}(\tau) &= \frac{1}{|\Omega|} \|\mathcal{L}(\rho \cdot \hat{u} - \tau)\|_{L^2(\Omega)}^2 \leq \frac{C_{\mathcal{L}}}{|\Omega|} \|\rho \cdot \hat{u} - \tau\|_{H^2(\Omega)}^2 \\ &\leq C_{\mathcal{L}} C_{\text{mult}}^2 (d+2)^2 \|\hat{u} - \tau/\rho\|_{W^{2,\infty}(\Omega)}^2 \\ &\leq C_{\mathcal{L}} C_{\text{mult}}^2 (d+2)^2 \|\hat{u} - u\|_{W^{2,\infty}(\Omega)}^2, \end{aligned}$$

where $u \in W^{r,\infty}(\mathbb{R}^d)$ is the extension of τ/ρ , which is guaranteed to exist by Lemma 3.10, and we used Lemmas 3.12, 3.11, together with the fact that $\|f\|_{H^2(\Omega)}^2 \leq (d+2)^2 \|f\|_{W^{2,\infty}(\Omega)}^2$.

By applying Theorem 3.13, it follows that by taking the infimum over $\mathcal{F}$, we have

$$\mathcal{R}(h_{\mathcal{F}}) - \mathcal{R}(\tau) \leq C_{\text{ReQU}} \|u\|_{W^{r,\infty}(\mathbb{R}^d)}^2 N^{-2\gamma(r)},$$

where

$$C_{\text{ReQU}} := C_{\mathcal{L}} C_{\text{mult}}^2 (d+2)^2 M_{\text{ReQU}}^2.$$

All that remains is to check that the $W^{2,\infty}$ norm of $h_{\mathcal{F}}$ is bounded by $(1 + \sqrt{M_{\text{ReQU}}}) \|\tau/\rho\|_{W^{r,\infty}(\mathbb{R}^d)}$. Simply observe that by the triangle inequality

$$\begin{aligned} \|h_{\mathcal{F}}\|_{W^{2,\infty}(\Omega)} &\leq \|\tau/\rho\|_{W^{2,\infty}(\Omega)} + \|h_{\mathcal{F}} - \tau/\rho\|_{W^{2,\infty}(\Omega)} \\ &\leq \|\tau/\rho\|_{W^{r,\infty}(\mathbb{R}^d)} (1 + \sqrt{M_{\text{ReQU}}}) \end{aligned}$$

This shows the desired inequality and, since the case $\sigma \equiv \tanh$ can be proved the exact same way, completes the proof. $\square$

The approximation error bounds from Theorem 3.13 and 3.14 are valid for shallow networks, which makes them more relevant for scientific computing tasks, where they are reported to be noticeably more stable to train. However, there is an incompressible gap in approximation speed between shallow ReQU and tanh networks, which it is possible to recover at the cost of a moderate increase in depth. This is the content of the following Theorem:

Theorem 3.16. *Let $r, N \geq 2$ be two integers and $B > 0$ be a real number, and let $u \in C^r(\mathbb{R}^d)$. Take the hypothesis space of ReQU FCNNs as $\mathcal{F}_{NN} := \Sigma_{\mathbf{m}_1:L,M}^\sigma(\tilde{B})$ with*

- $\sigma : x \mapsto \max\{0, x\}^2$
- $L = 3 + 2\lceil \log_2(2d) + \log_2(r-1) \rceil$,

- $\|\mathbf{m}\|_\infty = \max\{2(rN)^d, dN^d\},$
- $M = M(u) = \sup_{x_0 \in [0,1]^d} B^{-2r} \left(\frac{r+1}{2}\right)^{2d+r+1} \left(\sum_{\alpha \in \mathbb{N}^d: \|\alpha\|_1 \leq r-1} \left| \frac{\partial^\alpha u(x_0)}{\alpha!} \right| \right),$
- $\tilde{B} = \max\{N, B\}.$

We have the approximation error bound

$$\inf_{f_{NN} \in \mathcal{F}_{NN}} \|f_{NN} - u\|_{W^{2,\infty}([0,1]^d)} \leq \frac{M_{\text{ReQU}}}{N^{(r-2)}} \|u\|_{W^{r,\infty}(\mathbb{R}^d)}, \quad (3.23)$$

where

$$M_{\text{ReQU}} := 3 \cdot 2^{d+3} \binom{2+d}{2} \max_{0 \leq k \leq 2} \frac{(d/2)^{r+k-2}}{(r+k-2)!}.$$

The remarkable fact about Theorem 3.16 is that the depth increase required to match the tanh rate is only logarithmic in smoothness and dimension, and most importantly is independent on N . This is achieved by combining classical partition of unity arguments (Yarotsky, 2017; Gühring and Raslan, 2021) with recent exact representation results for ReLU^k networks (He et al., 2023). A proof of Theorem 3.16 will be given in Appendix B.

3.2 Statistical error bounds

We now focus on the second and third summands in inequality (3.15), which are error terms incurred by approximating the PINN risk with a random sample. We compute new bounds on VC dimensions of hypothesis spaces of higher order derivatives of FCNNs, which together with tools from empirical process theory literature allow us to bound these two terms.

VC dimension and pseudodimension bounds for higher derivatives of FCNNs

As discussed in Chapter 2, an important element we need to obtain non-asymptotic excess risk bounds by means of oracle inequalities is an estimation of the Vapnik-Chervonenkis dimension (VC dimension) of the hypothesis space of interest. We start by recalling its definition, together with the closely related concept of *pseudo-dimension*. We refer the reader to Section 4 for more extended discussion and illustration of these concepts.

Definition 3.17 (VC-dimension and pseudo-dimension). *Given a function class $\mathcal{G} : \mathcal{X} \rightarrow \mathbb{R}$ and a set $X = \{x_i\}_{i=1}^m$ of m points in the input space $\mathcal{X}$, let $\text{sign}(\mathcal{G}) = \{\text{sign}(g) : g \in \mathcal{G}\}$ be the set of binary functions $\mathcal{X} \rightarrow \{0, 1\}$ induced by $\mathcal{G}$, where we denote $\text{sign}(t) := \mathbf{1}_{\{t > 0\}}$. If $\text{sign}(\mathcal{G})$ can compute all*

dichotomies of X , that is, $|\{(\text{sign}(g(x)))_{x \in X} : g \in \mathcal{G}\}| = 2^m$, we say that $\mathcal{G}$ shatters X .

The Vapnik-Chervonenkis dimension (or VC-dimension) of $\mathcal{G}$ is defined as the size of the largest shattered subset of $\mathcal{X}$, and denoted by $\text{VCDim}(\mathcal{G})$.

Moreover, we say that X is pseudo-shattered by $\mathcal{G}$, if there are real numbers $r_1, r_2, \dots, r_m$, such that for each $v \in \{0, 1\}^m$, there exists a function $g_v \in \mathcal{G}$ with $\text{sign}(g_v(x_i) - r_i) = v_i$ for $1 \leq i \leq m$. The pseudo-dimension of $\mathcal{G}$, denoted by $\text{PDim}(\mathcal{G})$, is defined as the maximum cardinality of a subset X of $\mathcal{X}$ that is pseudo-shattered by $\mathcal{G}$.

For a hypothesis space $\mathcal{F}$ of functions defined on $\mathbb{R}^d$ and for a multi-index $\alpha \in \mathbb{N}^d$, we denote by $\partial^\alpha \mathcal{F} := \{\partial^\alpha f : f \in \mathcal{F}\}$ the space of α -th order partial derivatives, understood in the weak sense if necessary. We have the following VC dimension bounds.

Theorem 3.18. *Let $r \geq 2$, $\rho : \bar{\Omega} \rightarrow \mathbb{R}$ be a $C^r(\bar{\Omega})$ function, and denote by $\mathcal{F} \equiv \mathcal{F}(\rho, L, \mathbf{m}, \sigma, Q, \infty, \infty)$ the FCNN hypothesis space as defined in (3.10) with activation σ and arbitrary parameter Q . Denote by $\mathcal{S} \equiv \mathcal{S}(\mathcal{F}) = \sum_{\ell=1}^L m_\ell(m_{\ell-1} + 1)$ the number of parameters in $\mathcal{F}$.*

- If σ is the ReQU activation, then we have

$$\sup_{\substack{\alpha \in \mathbb{N}^d \\ \|\alpha\|_1 \leq 2}} \text{VCDim}(\partial^\alpha \mathcal{F}) = \mathcal{O}(L^2 \mathcal{S} \log_2 [L \|\mathbf{m}\|_\infty \log_2 \|\mathbf{m}\|_\infty]). \quad (3.24)$$

- If σ is the tanh activation, then we have

$$\sup_{\substack{\alpha \in \mathbb{N}^d \\ \|\alpha\|_1 \leq r}} \text{VCDim}(\partial^\alpha \mathcal{F}) = \mathcal{O}(L^2 \|\mathbf{m}\|_\infty^2 \mathcal{S}^2). \quad (3.25)$$

Note, in particular, that these bounds do not depend on the upper bound Q .

The VC dimension bounds provided by Theorem 3.18 are, to the best of our knowledge, new in the literature. Bounds for hypothesis spaces of (higher-order) derivatives of ReLU or ReLU-ReLU^k networks have appeared in previous works (Yang et al., 2023; Lei et al., 2025), based on a seminal result of Bartlett et al. (2019). These analyses however do not extend to our setup of ReQU FCNNs multiplied by a smooth function. Furthermore, the tanh activation not being a piecewise polynomial, controlling VC dimensions of higher derivatives of tanh networks requires a fundamentally different approach. We give a complete proof of Theorem 3.18 in Section 6.

Remark 3.19. Note that in the VC dimension bounds above, ρ is not required to be a distance approximation, but merely a $C^r(\bar{\Omega})$ function. In particular, applying Theorem 3.18 with $\rho \equiv 1$ yields VC dimension bounds for up to second order derivatives of FCNNs with ReQU activation, and for *any order* derivatives of FCNNs with tanh activation function.

We now recall the following bound on the pseudodimension in relation to the VC dimension:

Lemma 3.20 ((Bartlett and Anthony, 1999), Theorem 14.1). *For a class of neural networks $\mathcal{G}$ with a fixed architecture and fixed activation functions, it holds,*

$$\text{Pdim}(\mathcal{G}) \leq \text{VCdim}(\mathcal{G}_0), \quad (3.26)$$

where

$$\mathcal{G}_0 := \left\{ [g_0 : \mathbb{R}^d \times \mathbb{R} \rightarrow \mathbb{R}; (x, y) \mapsto g(x) - y] : g \in \mathcal{G} \right\}$$

is the set of functions defined on $\mathbb{R}^{d+1}$ by “thresholding” the outputs of the original neural network hypothesis space.

Remark 3.21. The proof of Lemma 3.20 uses only the definitions of VC-dimension and pseudo-dimension and does not rely on any specific properties of neural networks. In fact, for any real-valued function class $\mathcal{H}$ one has

$$\text{Pdim}(\mathcal{H}) = \text{VCdim}(\mathcal{H}_0), \text{ where } \mathcal{H}_0 := \{(x, y) \mapsto h(x) - y : h \in \mathcal{H}\}.$$

Hence inequality (3.26) holds for arbitrary classes $\mathcal{H}$ of real-valued functions as well, and not only for neural networks.

By applying Remark 3.21 to $\mathcal{H} := \partial^\alpha \mathcal{F}$, we obtain the following lemma:

Lemma 3.22 (Pseudo-dimension bounds from VC-dimension bounds). *Let $\text{sign}(t) := \mathbf{1}_{\{t>0\}}$ and σ be either the tanh or ReQU activation. Then for every multi-index α with $\|\alpha\|_1 \leq 2$,*

$$\text{Pdim}(\partial^\alpha \mathcal{F}) = \text{VCdim}((\partial^\alpha \mathcal{F})_0) = \text{VCdim}(\sigma \circ (\partial^\alpha \mathcal{F})_0),$$

where $(\partial^\alpha \mathcal{F})_0 := \{(x, y) \mapsto \partial^\alpha f(x) - y : f \in \mathcal{F}\}$. Moreover, for any such α ,

$$\text{VCdim}(\sigma \circ (\partial^\alpha \mathcal{F})_0) \lesssim \sup_{\|\beta\|_1 \leq 2} \text{VCdim}(\partial^\beta \mathcal{F}),$$

with an absolute implicit constant which may depend on σ only. In particular, Theorem 3.18 implies the following pseudo-dimension bounds:

$$\sup_{\substack{\alpha \in \mathbb{N}^d \\ \|\alpha\|_1 \leq 2}} \text{Pdim}(\partial^\alpha \mathcal{F}) = \begin{cases} \mathcal{O}(L^2 \mathcal{S} \log_2[L \|\mathbf{m}\|_\infty \log_2 \|\mathbf{m}\|_\infty]), & \sigma = \text{ReQU}, \\ \mathcal{O}(L^2 \|\mathbf{m}\|_\infty^2 \mathcal{S}^2), & \sigma = \text{tanh}. \end{cases} \quad (3.27)$$

Proof. The first equality is Remark 3.21 applied to $\mathcal{H} = \partial^\alpha \mathcal{F}$. For both $\sigma \equiv \tanh$ and $\sigma \equiv \text{ReLU}$, we have $\sigma(t) > 0$ if and only if $t > 0$, hence $\text{sign}(\partial^\alpha f(x) - y) = \text{sign}(\sigma(\partial^\alpha f(x) - y))$ holds for all pairs (x, y) , which gives the second equality. The final comparison follows by inspecting the proof of Theorem 3.18: adjoining one extra input variable y and composing once with the fixed map $(z, y) \mapsto \sigma(z - y)$ changes the underlying computation description by at most an absolute constant, so the same VC-dimension upper bounds apply up to a factor which may depend on σ only. $\square$

An oracle inequality for the statistical error

The importance of these complexity measure bounds is highlighted in the following oracle inequality, which directly translates bounds on VC and pseudo-dimension into high-probability error bounds on the statistical error.

Theorem 3.23. *Let $r \geq 2$, and assume that Assumptions (A.1), (A.2), (A.3) hold, so that τ is well-defined as the unique solution of (3.3). Let $\rho \in C^2(\bar{\Omega})$ satisfy $\rho = 0$ on $\partial\Omega$, and denote by $\mathcal{F} \equiv \mathcal{F}(\rho, L, \mathbf{m}, \sigma, Q, B, M)$ the FCNN hypothesis space as defined in (3.10). Define*

$$Q_0 := \max \left(4Q \|\rho\|_{W^{2,\infty}}, \|\tau\|_{W^{2,\infty}} \right). \quad (3.28)$$

For all $t \geq 0$, $n \geq \max_{\|\alpha\|_1 \leq 2} \text{Pdim}(\partial^\alpha \mathcal{F}) \vee \frac{8e^2 Q_0^2}{C_{11}+3}$, and $u_n \in \text{argmin}_{u \in \mathcal{F}} \mathcal{R}_n(u)$, we have with probability at least $1 - \exp(-t)$

$$\mathcal{R}(u_n) - \mathcal{R}(\tau) \leq \mathcal{R}(u_{\mathcal{F}}) - \mathcal{R}(\tau) + \frac{\text{VC}_{\mathcal{F}}}{n} \log n + \frac{t}{n} + \frac{\exp(-t)}{\log(n/\log n)}. \quad (3.29)$$

Here, $u_{\mathcal{F}} \in \text{argmin}_{u \in \mathcal{F}} \mathcal{R}(u)$, $C_{11} > 0$ is a constant satisfying the estimate (3.8):

$$\|u - \tau\|_{H^2(\Omega)}^2 \leq C_{11} \|\mathcal{L}(u - \tau)\|_{L^2(\Omega)}^2,$$

and we used the notation $\text{VC}_{\mathcal{F}} := \max_{\|\alpha\|_1 \leq 2} \text{VCdim}(\partial^\alpha \mathcal{F})$.

Theorem 3.23 follows from following the localized Rademacher-complexity analysis and peeling argument introduced in (Lei et al., 2025) to prove their Theorem 4. We will nonetheless, for the sake of completeness, give a complete proof in Section 6. Theorems 3.23, 3.13 and 3.14 combined together give us an explicit upper bound for the excess risk, from which we can establish our a priori error bounds.

3.3 Main result: a priori error estimates for PINN estimators of the mean escape time

We are now ready to state and prove the main result of this chapter, which is a non-asymptotic a priori error estimate for numerical solution of the Mean Escape Time learned with “Boundary-Adapted” PINNs.

Theorem 3.24. *Let $r \geq 3$, and assume that Assumptions (A.1), (A.2), (A.3) hold, so that τ is well-defined as the unique solution of (3.3). Let $\rho : \bar{\Omega} \rightarrow \mathbb{R}$ be an $(r+2)$ -smooth distance approximation, and denote by $\mathcal{F}_n \equiv \mathcal{F}(\rho, L, \mathbf{m}_n, \sigma, Q, \infty, \infty)$ the FCNN hypothesis space as defined in (3.10) with activation σ and no restrictions on the weights and parameters magnitudes. Let Q_0 be defined as in Equation (3.28). For all $n \in \mathbb{N}$, let $u_n \in \operatorname{argmin}_{u \in \mathcal{F}_n} \mathcal{R}_n(u)$ denote a minimizer of the empirical residual (3.14). The following holds:*

- *If σ is the ReQU activation, and the parameters $L, \mathbf{m}_n, Q$ are chosen as*

$$L = 1, \quad \|\mathbf{m}_n\|_\infty = r^d n^{\frac{d}{2(d+\gamma(r))}}, \quad Q = (1 + \sqrt{M_{\text{ReQU}}}) \|\tau/\rho\|_{W^{r,\infty}(\Omega)},$$

where

$$\gamma(r) = \min \left\{ r - 2, \frac{d+1}{2} \right\},$$

then for all $n \geq N_{\text{ReQU}} \vee \frac{8e^2 Q_0^2}{C_{11+3}}$, we have with probability greater than $1 - \exp(-n^{d/(d+\gamma(r))})$ that

$$\begin{cases} \mathcal{R}(u_n) - \mathcal{R}(\tau) & \leq c_1 n^{-\frac{\gamma(r)}{d+\gamma(r)}} \log^2(n \log n), \text{ and} \\ \|u_n - \tau\|_{H^2(\Omega)} & \leq c_2 n^{-\frac{\gamma(r)}{2(d+\gamma(r))}} \log(n \log n), \end{cases}$$

where $N_{\text{ReQU}}, c_1, c_2 > 0$ are constants which depend only on $\mathcal{L}, \tau, \rho, r, d$ and Ω .

- *If σ is the ReQU activation, and the parameters $L, \mathbf{m}_n, Q$ are chosen as*

$$L = 3 + 2 \lceil \log_2(2d) + \log_2(r-1) \rceil, \quad \|\mathbf{m}_n\|_\infty = r^d n^{\frac{d}{2(d+r-2)}}, \quad Q = (1 + \sqrt{M_{\text{ReQU}}}) \|\tau/\rho\|_{W^{r,\infty}(\Omega)},$$

then for all $n \geq N_{\text{ReQU}} \vee \frac{8e^2 Q_0^2}{C_{11+3}}$, we have with probability greater than $1 - \exp(-n^{d/(d+r-2)})$ that

$$\begin{cases} \mathcal{R}(u_n) - \mathcal{R}(\tau) & \leq c_1 n^{-\frac{r-2}{d+r-2}} \log^2(n \log n), \text{ and} \\ \|u_n - \tau\|_{H^2(\Omega)} & \leq c_2 n^{-\frac{r-2}{2(d+r-2)}} \log(n \log n), \end{cases}$$

where $N_{\text{ReQU}}, c_1, c_2 > 0$ are constants which depend only on $\mathcal{L}, \tau, \rho, r, d$ and Ω .

- If σ is the $\tanh$ activation, and the parameters $L, \mathbf{m}_n, Q$ are chosen as

$$L = 3, \quad \|\mathbf{m}_n\|_\infty = 3d5^{d+1}n^{\frac{d}{2(3d+r-2)}}, \quad Q = (1 + \sqrt{M_{\tanh}})\|\tau/\rho\|_{W^{r,\infty}(\Omega)},$$

then for all $n \geq N_{\tanh} \vee \frac{8e^2Q_0^2}{C_{11}+3}$, we have with probability greater than $1 - \exp(-n^{d/(3d+r-2)})$ that

$$\begin{cases} \mathcal{R}(u_n) - \mathcal{R}(\tau) & \leq \kappa_1 n^{-\frac{r-2}{3d+r-2}} \log^4(n \log n), \text{ and} \\ \|u_n - \tau\|_{H^2(\Omega)} & \leq \kappa_2 n^{-\frac{r-2}{2(3d+r-2)}} \log^2(n \log n), \end{cases}$$

where $N_{\tanh}, \kappa_1, \kappa_2 > 0$ are constants which depend only on $\mathcal{L}, \tau, \rho, r, d$ and Ω .

Proof. We give a detailed proof for the ReQU case, and we sketch at the end how the argument needs to be modified to handle the $\tanh$ activation. Let $a > 0$ be an exponent to be chosen later. By applying the ReQU version of Corollary 3.15 with $N \equiv n^a$, we find that for $\mathcal{F} \equiv \mathcal{F}(\rho, L, \mathbf{m}, \sigma, Q, \infty, \infty)$ with $\sigma \equiv \text{ReQU}$ and

$$L = 1, \quad \|\mathbf{m}\|_\infty = r^d n^{ad}, \quad Q = (1 + \sqrt{M_{\text{ReQU}}})\|\tau/\rho\|_{W^{r,\infty}(\Omega)},$$

we have for $u_{\mathcal{F}} \in \arg\min_{u \in \mathcal{F}} \mathcal{R}(u)$ the approximation error bound

$$\mathcal{R}(u_{\mathcal{F}}) - \mathcal{R}(\tau) \leq C_{\text{ReQU}} \|\tau/\rho\|_{W^{r,\infty}(\Omega)}^2 \frac{1}{n^{2a\gamma(r)}} =: \frac{C_1}{n^{2a\gamma(r)}}.$$

On the other hand, the number of parameters of $\mathcal{F}$ satisfies

$$\mathcal{S} := \sum_{\ell=1}^L m_\ell(m_{\ell-1} + 1) \leq (\|\mathbf{m}\|_\infty + \|\mathbf{m}\|_\infty^2)L = \mathcal{O}(\|\mathbf{m}\|_\infty^2 L).$$

Hence, by Theorem 3.18, the VC dimension of $\mathcal{F}$ and its higher order derivatives satisfies:

$$\begin{aligned} \text{VC}_{\mathcal{F}} &:= \max_{\|\alpha\|_1 \leq 2} \text{VCdim}(\partial^\alpha \mathcal{F}) \leq \mathcal{O}(L^3 \|\mathbf{m}\|_\infty^2 \log_2[L \|\mathbf{m}\|_\infty \log_2 \|\mathbf{m}\|_\infty]) \\ &\leq C_2 n^{2ad} \log(n \log n). \end{aligned}$$

For our choice of L and $\|\mathbf{m}\|_\infty$, we have by equation (3.27) that $\max_{\|\alpha\|_1 \leq 2} \text{Pdim}(\partial^\alpha \mathcal{F}) \leq C n^{2ad} \log(n \log n)$. Furthermore, for $a < 1/(2d)$, we also have $C n^{2ad} \log(n \log n) \leq$

n for all $n \geq N_{\text{ReQU}}$. Therefore the error decomposition (3.15) together with the oracle inequality from Theorem 3.23 give that, for all $n \geq N_{\text{ReQU}} \vee \frac{8e^2 Q_0^2}{C_{11}+3}$ and for all $t > 0$, we have with probability at least $1 - \exp(-t)$:

$$\mathcal{R}(u_n) - \mathcal{R}(\tau) \leq \frac{3C_1}{n^{2a\gamma(r)}} \log(n) + \frac{t}{n} + C_2 n^{2ad-1} \log^2(n \log n) + \frac{\exp(-t)}{\log(n/\log n)}.$$

We thus pick $a = (2(\gamma(r) + d))^{-1}$ and $t = n^b$ with $b = d/(d + \gamma(r))$ to obtain the inequality

$$\mathcal{R}(u_n) - \mathcal{R}(\tau) \leq (3C_1 + C_2 + 2) n^{-\frac{\gamma(r)}{d+\gamma(r)}} \log^2(n \log n).$$

Finally, we recover the desired H^2 a priori error estimate by applying inequality (3.8), which, letting $C_3 := 3C_1 + C_2 + 2$ yields:

$$\|u_n - \tau\|_{H^2(\Omega)} \leq \sqrt{C_{11}C_3} n^{-\frac{\gamma(r)}{2(d+\gamma(r))}} \log(n \log n).$$

This concludes the proof for the $\sigma \equiv \text{ReQU}$ case. The analogous bound for the $\sigma \equiv \tanh$ case is obtained the same way, taking into account that in this case the approximation error bound (3.22) involves an additional logarithmic term, and likewise, the VC dimension bound (3.25) scales differently. $\square$

Theorem 3.24 both in the deep ReQU and shallow tanh setting thus gives fast polynomial rates of convergence $\mathcal{O}(n^{-\frac{r-2}{2(d+r-2)}})$ for the a priori $H^2(\Omega)$ error of PINN solutions for the Mean Escape Time, which matches the fast rates obtained in (Lei et al., 2025) for PINN solvers of PDEs on spheres, and is very close to the minimax optimal $\mathcal{O}(n^{-\frac{2r-4}{d+4r-8}})$ obtained in (Lu et al., 2022). This result shows that, if the regularity r is much larger than the dimension d —which is often the case for Mean Escape Time problems—PINN estimators can approximate high-dimensional Mean Escape Time solutions with H^2 accuracy proportional to the inverse square root of the number of collocation points, thus bypassing the curse of dimensionality. For shallow ReQU networks however, we get a saturation of the convergence rate when the smoothness exponent is large, suggesting that shallow ReQU networks may not be suitable for PDEs where smoothness is much larger than dimension. We highlight again that, as our proof reveals, Theorem 3.24 can be extended to any elliptic PDE with Dirichlet boundary conditions on a regular domain with $C^{r+2}(\bar{\Omega})$ solution.

4 Approximation error analysis

We will now prove Theorem 3.13 for approximation with shallow ReQU networks. The argument is based off the integral representation of infinitely

wide neural networks (Barron, 2002), which was refined to ReLU activation by Bach (2017) and later extended to broader activations (Siegel and Xu, 2020, 2022). In particular, our Theorem 3.13 extends Yang and Zhou (2025)'s results to the case of higher order Sobolev smoothness. Before proceeding with the error analysis, we give the proof of Lemma 3.11.

Proof of Lemma 3.11. Let $u \in C^r(\Omega)$ and $v \in W^{r,\infty}(\Omega)$. By the generalized Leibniz rule for differentiation of products, we have

$$\begin{aligned}
\|uv\|_{W^{r,\infty}(\Omega)} &:= \max_{\|\alpha\|_1 \leq r} \|\partial^\alpha(uv)\|_{L^\infty(\Omega)} = \max_{\|\alpha\|_1 \leq r} \left\| \sum_{\beta: \beta \leq \alpha} \binom{\alpha}{\beta} (\partial^{\alpha-\beta} u) (\partial^\beta v) \right\|_{L^\infty(\Omega)} \\
&\leq \max_{\|\alpha\|_1 \leq r} \sum_{\beta: \beta \leq \alpha} \binom{\alpha}{\beta} \|\partial^{\alpha-\beta} u\|_{L^\infty(\Omega)} \|\partial^\beta v\|_{L^\infty(\Omega)} \\
&\leq C_1 \max_{\|\alpha\|_1 \leq r} \sum_{\beta: \beta \leq \alpha} \|\partial^\beta v\|_{L^\infty(\Omega)} \\
&\leq \|v\|_{W^{r,\infty}(\Omega)} \cdot C_1 \sum_{k=0}^r \sum_{\|\alpha\|_1=k} |\{\beta : \|\beta\|_1 \leq \alpha\}| \\
&\leq \|v\|_{W^{r,\infty}(\Omega)} \cdot C_1 \sum_{k=0}^r \binom{k+d-1}{k} (k+1)^{d^2}, \tag{3.30}
\end{aligned}$$

where the constant C_1 is defined as

$$C_1 := \max_{\substack{\alpha \in \mathbb{N}^d \\ \|\alpha\|_1 \leq r, \beta \leq \alpha}} \binom{\alpha}{\beta} \|u\|_{W^{r,\infty}(\Omega)},$$

and we used the fact that for any $k \in \mathbb{N}$,

$$|\{\gamma \in \mathbb{N}^d : \|\gamma\|_1 \leq k\}| = \binom{k+d}{k},$$

together with the inequality

$$|\{\gamma \in \mathbb{N}^d : \gamma \leq \alpha\}| \leq \prod_{i=1}^d (\alpha_i + 1)^d \leq (\|\alpha\|_1 + 1)^{d^2}.$$

The claimed result thus follows by applying the estimate (3.30) to $u \equiv \rho \in C^r(\Omega)$ and $v \equiv f - g/\rho \in W^{r,\infty}(\Omega)$. $\square$

4.1 Preliminaries

In this subsection we recall the main notions from harmonic analysis on the sphere that will be used in the proof of the approximation rates and we introduce the variation spaces associated with shallow ReQU networks. Throughout we fix an integer $d \geq 2$. We write $\mathbb{S}^d \subset \mathbb{R}^{d+1}$ for the unit sphere endowed with its standard surface measure τ_d normalized by $\tau_d(\mathbb{S}^d) = 1$ and we write $\mathbb{B}^d \subset \mathbb{R}^d$ for the closed unit ball.

Geometry and spectral theory on the sphere

We start by recalling the spherical harmonic decomposition of $L^2(\mathbb{S}^d)$ and the associated spectral data of the Laplace–Beltrami operator. We refer the reader to (Dai and Xu, 2013) and references within for a thorough overview on this topic.

Definition 3.25 (Spherical harmonics and spectral decomposition). *For $n \in \mathbb{N}$ let $\mathcal{H}_n$ denote the space of real-valued homogeneous harmonic polynomials of degree n on $\mathbb{R}^{d+1}$ and let Y_n denote the set of their restrictions to the unit sphere. The elements of Y_n are called spherical harmonics of degree n . The Laplace–Beltrami operator Δ on $\mathbb{S}^d$ satisfies*

$$\Delta Y = -\lambda_n Y, \text{ where } \lambda_n := n(n + d - 1) \text{ and } Y \in Y_n,$$

so Y_n is the eigenspace associated with $-\lambda_n$.

The spaces $(Y_n)_{n \geq 0}$ are pairwise orthogonal in $L^2(\mathbb{S}^d, \tau_d)$ and they span $L^2(\mathbb{S}^d)$. We will denote by $\mathcal{P}_n : L^2(\mathbb{S}^d) \rightarrow Y_n$ the corresponding orthogonal projector. The dimension of Y_n is equal to 1 if $n = 0$ and equal to

$$N(d, n) = \frac{2n + d - 1}{n} \binom{n + d - 2}{d - 1},$$

whenever $n \geq 1$.

The next definition fixes the coordinate based C^m -norm on $\mathbb{S}^d$ that we use in combination with covariant derivatives.

Definition 3.26 (Definition of the C^m -norm on $\mathbb{S}^d$). *For any sufficiently regular $u : \mathbb{S}^d \rightarrow \mathbb{R}$ we denote by $\|u\|_{C^m(\mathbb{S}^d)}$ the standard C^m -norm defined via a finite smooth atlas. More concretely, fix a finite smooth atlas $(U_\alpha, \varphi_\alpha)_{\alpha=1}^L$ on $\mathbb{S}^d$ and a smooth partition of unity $(\chi_\alpha)_{\alpha=1}^L$ subordinate to this atlas. For $u \in C^m(\mathbb{S}^d)$ we set*

$$\|u\|_{C^m(\mathbb{S}^d)} := \max_{1 \leq \alpha \leq L} \max_{|\beta| \leq m} \|\partial^\beta ((\chi_\alpha u) \circ \varphi_\alpha^{-1})\|_{L^\infty(\varphi_\alpha(U_\alpha))}.$$

Different choices of atlas and partition of unity lead to equivalent norms. For the above choice there exists a constant $c_m \geq 1$, depending only on d and m , such that

$$c_m^{-1} \sum_{k=0}^m \|\nabla^k u\|_{L^\infty(\mathbb{S}^d)} \leq \|u\|_{C^m(\mathbb{S}^d)} \leq c_m \sum_{k=0}^m \|\nabla^k u\|_{L^\infty(\mathbb{S}^d)},$$

where ∇ is the Levi–Civita connection associated with the round metric (Taylor, 2023).

Convolution and spectral multipliers on $\mathbb{S}^d$

We next recall the addition formula, the definition of spherical convolution and the Funk–Hecke formula.

Lemma 3.27 (Addition formula and Gegenbauer polynomials). *For each $n \in \mathbb{N}_0$ fix an orthonormal basis $\{Y_{n,\ell}\}_{\ell=1}^{N(d,n)}$ of Y_n in $L^2(\mathbb{S}^d, \tau_d)$. There exists a univariate polynomial $P_n : [-1, 1] \rightarrow \mathbb{R}$ (the Gegenbauer polynomial of degree n) with $P_n(1) = 1$ such that*

$$\sum_{\ell=1}^{N(d,n)} Y_{n,\ell}(u) Y_{n,\ell}(v) = N(d, n) P_n(u \cdot v), \text{ for all } u, v \in \mathbb{S}^d.$$

In particular $|P_n(t)| \leq 1$ for all $t \in [-1, 1]$ and $n \in \mathbb{N}_0$ (Dai and Xu, 2013, Theorem 1.2.6).

The addition formula shows that the projector $\mathcal{P}_n$ can be written as a convolution with a zonal kernel. We now fix our notion of convolution on the sphere.

Definition 3.28 (Convolution on the sphere). *Let $c_d > 0$ be the normalizing constant*

$$c_d^{-1} = \int_{-1}^1 (1 - t^2)^{(d-2)/2} dt$$

and define the probability measure ϱ on $[-1, 1]$ with density $c_d(1 - t^2)^{(d-2)/2}$. For $f \in L^1(\mathbb{S}^d, \tau_d)$ and a measurable function $g : [-1, 1] \rightarrow \mathbb{R}$ with $g \in L^1([-1, 1], \varrho)$ we define the convolution

$$(f * g)(u) := \int_{\mathbb{S}^d} f(v) g(u \cdot v) d\tau_d(v), \text{ for } u \in \mathbb{S}^d.$$

Thus g is a zonal kernel, in the sense that it depends only on the inner product $u \cdot v$.

Convolution by a zonal kernel acts diagonally on the spherical harmonic decomposition.

Lemma 3.29 (Funk–Hecke formula). *Let $g \in L^1([-1, 1], \varrho)$ and define coefficients*

$$\hat{g}(n) := \frac{\omega_{d-1}}{\omega_d} \int_{-1}^1 g(t) P_n(t) (1-t^2)^{(d-2)/2} dt, \text{ for all } n \in \mathbb{N}_0,$$

where P_n is the Gegenbauer polynomial from Lemma 3.27 and ω_d denotes the surface area of $\mathbb{S}^d$. Then for every $f \in L^2(\mathbb{S}^d, \tau_d)$ and every $n \in \mathbb{N}_0$,

$$\mathcal{P}_n(f * g) = \hat{g}(n) \mathcal{P}_n f.$$

In other words convolution with a zonal kernel is a Fourier multiplier on the spherical harmonic side (Dai and Xu, 2013, Theorem 2.1.3).

Moduli of smoothness on $\mathbb{S}^d$

We now recall the notion of spherical translations and moduli of smoothness, which will allow us to quantify smoothness of functions on $\mathbb{S}^d$ in a way that interacts well with convolution.

Definition 3.30 (Spherical translations and moduli of smoothness). *For $u \in \mathbb{S}^d$ let*

$$\mathbb{S}_u^\perp := \{v \in \mathbb{S}^d : u \cdot v = 0\} \cong \mathbb{S}^{d-1}.$$

For $\theta \in [0, \pi]$ and $f \in L^1(\mathbb{S}^d, \tau_d)$ the spherical translation (or spherical mean) operator is defined by

$$T_\theta f(u) := \int_{\mathbb{S}_u^\perp} f(u \cos \theta + v \sin \theta) d\tau_{d-1}(v), \text{ for } u \in \mathbb{S}^d,$$

where τ_{d-1} is the normalized surface measure on $\mathbb{S}_u^\perp$. One has

$$\mathcal{P}_n(T_\theta f) = P_n(\cos \theta) \mathcal{P}_n f, \text{ for } n \in \mathbb{N}_0.$$

Let $\alpha > 0$ and $0 < \theta < \pi$. We define the fractional difference operator Δ_θ^α by spectral calculus through

$$\mathcal{P}_n(\Delta_\theta^\alpha f) := (1 - P_n(\cos \theta))^{\alpha/2} \mathcal{P}_n f, \text{ for } n \in \mathbb{N}_0.$$

For $1 \leq p < \infty$ and $f \in L^p(\mathbb{S}^d)$, or for $p = \infty$ and $f \in C(\mathbb{S}^d)$, the α -th order modulus of smoothness of f is

$$\omega_\alpha(f, t)_p := \sup_{0 < \theta \leq t} \|\Delta_\theta^\alpha f\|_{L^p(\mathbb{S}^d)}, \text{ where } 0 < t < \pi.$$

For even integer orders the modulus of smoothness can equivalently be written in terms of finite linear combinations of the translations $T_{j\theta}$.

Lemma 3.31. *Let $s \in \mathbb{N}$ and $1 \leq p \leq \infty$. There exists a constant $C_{s,p} > 0$ such that for every $f \in L^p(\mathbb{S}^d)$ and every $0 < t < \pi$,*

$$\begin{aligned} C_{s,p}^{-1} \sup_{0 < \theta \leq t} \left\| \sum_{j=0}^{2s} (-1)^j \binom{2s}{j} T_{j\theta} f \right\|_{L^p(\mathbb{S}^d)} &\leq \omega_{2s}(f, t)_p \\ &\leq C_{s,p} \sup_{0 < \theta \leq t} \left\| \sum_{j=0}^{2s} (-1)^j \binom{2s}{j} T_{j\theta} f \right\|_{L^p(\mathbb{S}^d)}. \end{aligned}$$

Proof. See (Ditzian, 2006) or (Dai and Xu, 2013, Section 10.1). $\square$

Lemma 3.32 (Marchaud inequality on $\mathbb{S}^d$). *Let $0 < s < r$ and let $1 \leq p \leq \infty$. There exists a constant $C > 0$, depending only on d, r, s , and p , such that for every f in the domain of $\omega_r(\cdot, \cdot)_p$ and every $0 < t < \pi$,*

$$\omega_s(f, t)_p \leq C t^s \int_t^\pi \frac{\omega_r(f, u)_p}{u^{s+1}} du.$$

Proof. See (Dai and Xu, 2013, Chapter 10) $\square$

Corollary 3.33. *Let $0 < s < r$ and let $1 \leq p \leq \infty$. Suppose there exists $A > 0$ such that $\omega_r(f, u)_p \leq A u^\alpha$ for all $0 < u < \pi$ and some $\alpha \in (0, r]$. Then there exists $C > 0$, depending only on d, r, s , and p , such that for all $0 < t < \pi$,*

$$\omega_s(f, t)_p \leq C A \begin{cases} t^\alpha, & \alpha < s, \\ t^s \log(e/t), & \alpha = s, \\ t^s, & \alpha > s. \end{cases}$$

Proof. Combine Lemma 3.32 with the bound $\omega_r(f, u)_p \leq A u^\alpha$ and evaluate the resulting one-dimensional integral. $\square$

On the Euclidean side we will work with Hölder spaces on the unit ball.

Definition 3.34 (Hölder spaces on $\mathbb{R}^d$ and on $\mathbb{B}^d$). *Let $m \in \mathbb{N}_0$ and $\kappa \in (0, 1]$. The Hölder space $C^{m,\kappa}(\mathbb{R}^d)$ consists of all functions $f \in C^m(\mathbb{R}^d)$ such that the seminorm*

$$|f|_{C^{m,\kappa}(\mathbb{R}^d)} := \max_{|\gamma|=m} \sup_{x \neq y} \frac{|\partial^\gamma f(x) - \partial^\gamma f(y)|}{\|x - y\|_2^\kappa}$$

is finite. The norm on $C^{m,\kappa}(\mathbb{R}^d)$ is

$$\|f\|_{C^{m,\kappa}(\mathbb{R}^d)} := \max \left\{ \max_{|\gamma| \leq m} \|\partial^\gamma f\|_{L^\infty(\mathbb{R}^d)}, |f|_{C^{m,\kappa}(\mathbb{R}^d)} \right\}.$$

We define $C^{m,\kappa}(\mathbb{B}^d)$ as the space of restrictions to $\mathbb{B}^d$ of functions in $C^{m,\kappa}(\mathbb{R}^d)$, equipped with the norm

$$\|f\|_{C^{m,\kappa}(\mathbb{B}^d)} := \inf \{ \|g\|_{C^{m,\kappa}(\mathbb{R}^d)} : g \in C^{m,\kappa}(\mathbb{R}^d), g = f \text{ on } \mathbb{B}^d \}.$$

For $\alpha > 0$ we set $a := \lceil \alpha \rceil - 1$ and $\kappa := \alpha - a \in (0, 1]$. We denote the unit ball of $C^{a,\kappa}(\mathbb{B}^d)$ by

$$\mathcal{H}^\alpha := \{ f \in C^{a,\kappa}(\mathbb{B}^d) : \|f\|_{C^{a,\kappa}(\mathbb{B}^d)} \leq 1 \}.$$

Variation spaces for shallow ReQU networks

We now define the infinite-width ReQU variation spaces on $\mathbb{B}^d$ and on $\mathbb{S}^d$.

Definition 3.35 (ReQU activation and variation space on $\mathbb{B}^d$). *We denote for the remaining of this section the ReQU activation as $\sigma : t \mapsto \max\{0, t\}^2$. Let μ be a finite signed Radon measure on $\mathbb{S}^d$. For $x \in \mathbb{B}^d$ we consider the integral representation*

$$f(x) = \int_{\mathbb{S}^d} \sigma((x^\top, 1) \cdot v) d\mu(v).$$

We define the variation norm of f by

$$\gamma(f) := \inf \{ \|\mu\|_{\text{TV}} : f(x) = \int_{\mathbb{S}^d} \sigma((x^\top, 1) \cdot v) d\mu(v) \text{ for all } x \in \mathbb{B}^d \},$$

where the infimum runs over all finite signed Radon measures μ on $\mathbb{S}^d$ and $\|\mu\|_{\text{TV}}$ denotes the total variation of μ . For $M > 0$ we define the ReQU variation space on $\mathbb{B}^d$ by

$$\mathcal{V}_\sigma(M) := \{ f : \mathbb{B}^d \rightarrow \mathbb{R} : \gamma(f) \leq M \}.$$

This space can be viewed as the infinite-width limit of one-hidden-layer ReQU networks with a constraint on the ℓ_1 -norm of the first-layer weights ([Bach, 2017](#)).

Definition 3.36 (Spherical ReQU variation space). *Let μ be a finite signed Radon measure on $\mathbb{S}^d$. For $u \in \mathbb{S}^d$ we set*

$$g(u) = \int_{\mathbb{S}^d} \sigma(u \cdot v) d\mu(v).$$

The variation norm of g is

$$\gamma(g) := \inf\{\|\mu\|_{\text{TV}} : g(u) = \int_{\mathbb{S}^d} \sigma(u \cdot v) d\mu(v) \text{ for all } u \in \mathbb{S}^d\}.$$

For $M > 0$ we define

$$\mathcal{G}_\sigma(M) := \{g : \mathbb{S}^d \rightarrow \mathbb{R} : \gamma(g) \leq M\}.$$

If μ is absolutely continuous with respect to τ_d with density $\phi \in L^1(\mathbb{S}^d, \tau_d)$ then

$$g(u) = \int_{\mathbb{S}^d} \sigma(u \cdot v) \phi(v) d\tau_d(v) = (\phi * \sigma)(u),$$

where the convolution is that of Definition 3.28.

Several alternative but equivalent characterizations of variation spaces for shallow networks can be found in (Savarese et al., 2019; Ongie et al., 2020; Siegel and Xu, 2020). In the present work we will only use the definition above, which is tailored to the harmonic analysis on $\mathbb{S}^d$ recalled in this subsection.

4.2 Approximation error of ReQU variation space

Having defined the necessary objects, we are now ready to carry out our error analysis for shallow ReQU networks. In this section, we begin by looking at the approximation power of the variation space corresponding to the infinite-width limit of ReQU networks. The first result we will need is

Proposition 3.37 (Elliptic control of covariant derivatives by powers of the Laplace–Beltrami operator on $\mathbb{S}^d$). *Let $d \in \mathbb{N}$ and let $\mathbb{S}^d \subset \mathbb{R}^{d+1}$ denote the unit sphere endowed with the standard metric g and associated Laplace–Beltrami operator Δ (Taylor, 2023). For $m \in \mathbb{N}$, define*

$$\|u\|_{W^{m,\infty}(\mathbb{S}^d)} := \max_{0 \leq k \leq m} \|\nabla^k u\|_{L^\infty(\mathbb{S}^d)},$$

where $\nabla^k u$ denotes the k -th covariant derivative of u and the pointwise norm of tensors is induced by g . Moreover, define the Laplacian-based norm

$$\|u\|_{\Delta,m,\infty} := \sum_{j=0}^m \|(-\Delta)^{j/2} u\|_{L^\infty(\mathbb{S}^d)},$$

where $(-\Delta)^{j/2}$ is defined by spectral functional calculus for the nonnegative self-adjoint operator $-\Delta$ on $L^2(\mathbb{S}^d)$. Then for each $m \in \mathbb{N}$ there exists a constant $C_m > 0$ such that

$$\|u\|_{W^{m,\infty}(\mathbb{S}^d)} \leq C_m \|u\|_{\Delta,m,\infty} \text{ for all } u \in C^\infty(\mathbb{S}^d).$$

Before proving Proposition 3.37, we recall the classical Schauder estimates on the sphere $\mathbb{S}^d$, which is a special case of a compact manifold without boundary:

Lemma 3.38 (Schauder estimates on the sphere). *For every integer $k \geq 0$ and every $\alpha \in [0, 1]$ there exists $C_{k,\alpha} > 0$ such that*

$$\|u\|_{C^{k+2,\alpha}(\mathbb{S}^d)} \leq C_{k,\alpha} \left(\|\Delta u\|_{C^{k,\alpha}(\mathbb{S}^d)} + \|u\|_{C^0(\mathbb{S}^d)} \right) \text{ for all } u \in C^\infty(\mathbb{S}^d). \quad (3.31)$$

In particular, for $\alpha = 0$ and each integer $k \geq 0$ there exists $C_k > 0$ such that

$$\|u\|_{C^{k+2}(\mathbb{S}^d)} \leq C_k \left(\|\Delta u\|_{C^k(\mathbb{S}^d)} + \|u\|_{C^0(\mathbb{S}^d)} \right) \text{ for all } u \in C^\infty(\mathbb{S}^d). \quad (3.32)$$

Proof. See, e.g. (Taylor, 2023, Chapter 5, Theorem 1.3). $\square$

We are now ready to carry out our proof.

Proof of Proposition 3.37. We first derive bounds in the coordinate based C^m -norms. By Lemma 3.38 with $\alpha = 0$, for each integer $k \geq 0$ there exists $C_k > 0$ such that

$$\|u\|_{C^{k+2}(\mathbb{S}^d)} \leq C_k \left(\|\Delta u\|_{C^k(\mathbb{S}^d)} + \|u\|_{C^0(\mathbb{S}^d)} \right) \text{ for all } u \in C^\infty(\mathbb{S}^d).$$

We prove by induction on $r \in \mathbb{N}$ that there exist constants $A_r > 0$ such that

$$\|u\|_{C^{2r}(\mathbb{S}^d)} \leq A_r \sum_{j=0}^r \|\Delta^j u\|_{C^0(\mathbb{S}^d)} \text{ for all } u \in C^\infty(\mathbb{S}^d). \quad (3.33)$$

For $r = 1$, inequality (3.33) is precisely (3.32) with $k = 0$ and $A_1 := C_0$. Assume that (3.33) holds for some $r \geq 1$. Setting $k = 2r$ in (3.32) gives

$$\|u\|_{C^{2r+2}(\mathbb{S}^d)} \leq C_{2r} \left(\|\Delta u\|_{C^{2r}(\mathbb{S}^d)} + \|u\|_{C^0(\mathbb{S}^d)} \right).$$

Applying the induction hypothesis (3.33) to $v := \Delta u$ yields

$$\|\Delta u\|_{C^{2r}(\mathbb{S}^d)} = \|v\|_{C^{2r}(\mathbb{S}^d)} \leq A_r \sum_{j=0}^r \|\Delta^j v\|_{C^0(\mathbb{S}^d)} = A_r \sum_{j=0}^r \|\Delta^{j+1} u\|_{C^0(\mathbb{S}^d)}.$$

Hence

$$\|u\|_{C^{2r+2}(\mathbb{S}^d)} \leq C_{2r} A_r \sum_{j=1}^{r+1} \|\Delta^j u\|_{C^0(\mathbb{S}^d)} + C_{2r} \|u\|_{C^0(\mathbb{S}^d)}.$$

Choosing $A_{r+1} \geq C_{2r} \max\{1, A_r\}$ we obtain

$$\|u\|_{C^{2r+2}(\mathbb{S}^d)} \leq A_{r+1} \sum_{j=0}^{r+1} \|\Delta^j u\|_{C^0(\mathbb{S}^d)},$$

which is (3.33) with r replaced by $r+1$. Thus (3.33) holds for all $r \in \mathbb{N}$.

Let now $m \in \mathbb{N}$ be arbitrary and write either $m = 2r$ or $m = 2r+1$. If $m = 2r$ then (3.33) gives

$$\|u\|_{C^m(\mathbb{S}^d)} = \|u\|_{C^{2r}(\mathbb{S}^d)} \leq A_r \sum_{j=0}^r \|\Delta^j u\|_{C^0(\mathbb{S}^d)}.$$

If $m = 2r+1$ then $\|u\|_{C^m(\mathbb{S}^d)} \leq \|u\|_{C^{2r+2}(\mathbb{S}^d)}$, hence by (3.33) with r replaced by $r+1$,

$$\|u\|_{C^m(\mathbb{S}^d)} \leq \|u\|_{C^{2r+2}(\mathbb{S}^d)} \leq A_{r+1} \sum_{j=0}^{r+1} \|\Delta^j u\|_{C^0(\mathbb{S}^d)}.$$

In both cases there exists $B_m > 0$ such that

$$\|u\|_{C^m(\mathbb{S}^d)} \leq B_m \sum_{j=0}^{\lceil m/2 \rceil} \|\Delta^j u\|_{C^0(\mathbb{S}^d)} \text{ for all } u \in C^\infty(\mathbb{S}^d). \quad (3.34)$$

Since $\mathbb{S}^d$ is compact, the C^0 - and L^∞ -norms coincide, so $\|\cdot\|_{C^0(\mathbb{S}^d)} = \|\cdot\|_{L^\infty(\mathbb{S}^d)}$.

Next, we express the powers Δ^j in terms of powers of $-\Delta$. The operator $-\Delta$ is nonnegative and self-adjoint on $L^2(\mathbb{S}^d)$, with a discrete spectrum. For each integer $j \geq 0$ we have

$$\Delta^j = (-1)^j (-\Delta)^j = (-1)^j (-\Delta)^{2j/2},$$

hence

$$\|\Delta^j u\|_{L^\infty(\mathbb{S}^d)} = \|(-\Delta)^{2j/2} u\|_{L^\infty(\mathbb{S}^d)}.$$

The indices $2j$ that appear here satisfy $0 \leq 2j \leq 2\lceil m/2 \rceil \leq 2m$, so in particular they all belong to $\{0, \dots, m\}$ when $m \geq 1$ after possibly enlarging the constant. Therefore

$$\sum_{j=0}^{\lceil m/2 \rceil} \|\Delta^j u\|_{L^\infty(\mathbb{S}^d)} = \sum_{j=0}^{\lceil m/2 \rceil} \|(-\Delta)^{2j/2} u\|_{L^\infty(\mathbb{S}^d)} \leq \sum_{\ell=0}^m \|(-\Delta)^{\ell/2} u\|_{L^\infty(\mathbb{S}^d)} = \|u\|_{\Delta, m, \infty}.$$

Combining this with (3.34) gives

$$\|u\|_{C^m(\mathbb{S}^d)} \leq B_m \|u\|_{\Delta, m, \infty} \text{ for all } u \in C^\infty(\mathbb{S}^d). \quad (3.35)$$

To conclude, we relate the coordinate based C^m -norm to the covariant derivative norm $\|u\|_{W^{m,\infty}(\mathbb{S}^d)}$. By Definition 3.26 there exists $c_m \geq 1$ such that

$$\|u\|_{C^m(\mathbb{S}^d)} \leq c_m \sum_{k=0}^m \|\nabla^k u\|_{L^\infty(\mathbb{S}^d)} \text{ for all } u \in C^\infty(\mathbb{S}^d).$$

Since

$$\|u\|_{W^{m,\infty}(\mathbb{S}^d)} = \max_{0 \leq k \leq m} \|\nabla^k u\|_{L^\infty(\mathbb{S}^d)} \leq \sum_{k=0}^m \|\nabla^k u\|_{L^\infty(\mathbb{S}^d)},$$

we obtain

$$\|u\|_{W^{m,\infty}(\mathbb{S}^d)} \leq c_m \|u\|_{C^m(\mathbb{S}^d)}.$$

Combining this with (3.35) and absorbing the constants into a new constant $C_m > 0$ yields

$$\|u\|_{W^{m,\infty}(\mathbb{S}^d)} \leq C_m \|u\|_{\Delta,m,\infty} \text{ for all } u \in C^\infty(\mathbb{S}^d),$$

which completes the proof. $\square$

Transfer between the ball and the sphere

We now introduce a geometric transfer operator that allows us to move between functions on the unit sphere and functions on the unit ball. Throughout this subsection we keep the dimension $d \geq 1$ fixed and we work with the sphere $\mathbb{S}^d \subset \mathbb{R}^{d+1}$ and the ball $\mathbb{B}^d \subset \mathbb{R}^d$.

Definition 3.39 (Transfer operator from the sphere to the ball). *Let*

$$\Omega := \{u = (u_1, \dots, u_{d+1}) \in \mathbb{S}^d : u_{d+1} \geq 2^{-1/2}\} \subset \mathbb{S}^d$$

be the upper spherical cap. For $x \in \mathbb{B}^d$ we define

$$\Phi(x) := \frac{1}{\sqrt{1 + \|x\|_2^2}} \begin{pmatrix} x \\ 1 \end{pmatrix} \in \Omega, \text{ and } \psi(x) := 1 + \|x\|_2^2.$$

For a bounded function $g : \mathbb{S}^d \rightarrow \mathbb{R}$ we define the transfer operator $\mathcal{T}$ by

$$(\mathcal{T}g)(x) := \psi(x) g(\Phi(x)), \text{ for all } x \in \mathbb{B}^d.$$

The map Φ is a smooth embedding of $\mathbb{B}^d$ into Ω and the factor ψ compensates for the homogeneity of the ReQU activation.

The next lemma shows that $\mathcal{T}$ does not increase Sobolev norms by more than a controlled constant once we measure smoothness of g in terms of powers of the Laplace–Beltrami operator.

Lemma 3.40 (Transfer of Sobolev norms via $\mathcal{T}$). *Let $r \in \mathbb{N}_0$. There exists a constant $C_{r,d} > 0$ such that for every smooth function $g : \mathbb{S}^d \rightarrow \mathbb{R}$,*

$$\|\mathcal{T}g\|_{W^{r,\infty}(\mathbb{B}^d)} \leq C_{r,d} \sum_{j=0}^r \|(-\Delta)^{j/2} g\|_{L^\infty(\mathbb{S}^d)}.$$

Proof. Let $r \in \mathbb{N}_0$ and let $\beta \in \mathbb{N}_0^d$ with $|\beta| \leq r$. By Leibniz' rule and the multivariate chain rule we can write

$$\partial_x^\beta(\mathcal{T}g)(x) = \sum_{|\gamma| \leq |\beta|} a_{\beta,\gamma}(x) \partial_u^\gamma g(\Phi(x)), \text{ for all } x \in \mathbb{B}^d,$$

where the sum is finite, the multi-indices γ range over $\mathbb{N}_0^{d+1}$ and the coefficient functions $a_{\beta,\gamma}$ are smooth on $\mathbb{B}^d$ and bounded together with all their derivatives. Since $\mathbb{B}^d$ is compact there exists a constant $A_{r,d} > 0$ such that

$$\max_{|\beta| \leq r} \|\partial_x^\beta(\mathcal{T}g)\|_{L^\infty(\mathbb{B}^d)} \leq A_{r,d} \max_{|\gamma| \leq r} \|\partial_u^\gamma g\|_{L^\infty(\mathbb{S}^d)}.$$

The right-hand side is controlled by the covariant derivatives of g via Definition 3.26 and then by the Laplacian-based norm through Proposition 3.37 with $m = r$. This yields

$$\max_{|\gamma| \leq r} \|\partial_u^\gamma g\|_{L^\infty(\mathbb{S}^d)} \leq C'_{r,d} \sum_{j=0}^r \|(-\Delta)^{j/2} g\|_{L^\infty(\mathbb{S}^d)},$$

for a constant $C'_{r,d}$ depending only on r and d . Combining the last two displays and recalling the definition of the $W^{r,\infty}$ -norm on $\mathbb{B}^d$ gives the desired estimate with $C_{r,d} := A_{r,d} C'_{r,d}$. $\square$

The following proposition connects the spherical and Euclidean variation spaces through the transfer operator.

Proposition 3.41 (Transfer of ReQU variation space between sphere and ball). *Let σ be the ReQU activation defined in Definition 3.35. Then the following hold.*

(i) *For every $M > 0$ and every $g \in \mathcal{G}_\sigma(M)$ the function $\mathcal{T}g$ belongs to $\mathcal{V}_\sigma(M)$ and*

$$\gamma(\mathcal{T}g) \leq \gamma(g) \leq M.$$

(ii) *Let $\alpha > 0$ and set $a := \lfloor \alpha \rfloor$ and $\kappa := \alpha - a \in (0, 1]$. There exists a constant $C_\alpha > 0$ such that for every $h \in \mathcal{H}^\alpha$ there is a function $\tilde{h} \in C^{a,\kappa}(\mathbb{S}^d)$ such that $\mathcal{T}\tilde{h} = h$ on $\mathbb{B}^d$,*

$$\tilde{h}(-u) = -\tilde{h}(u), \text{ for all } u \in \mathbb{S}^d,$$

and

$$\|\tilde{h}\|_{L^\infty(\mathbb{S}^d)} + \|\tilde{h}\|_{C^{a,\kappa}(\mathbb{S}^d)} \leq C_\alpha.$$

Proof. (i) Let $g \in \mathcal{G}_\sigma(M)$. By Definition 3.36 there exists a finite signed Radon measure μ on $\mathbb{S}^d$ with $\|\mu\|_{\text{TV}} \leq M$ such that

$$g(u) = \int_{\mathbb{S}^d} \sigma(u \cdot v) d\mu(v), \text{ for all } u \in \mathbb{S}^d.$$

Let $x \in \mathbb{B}^d$. Since $\Phi(x) \in \mathbb{S}^d$ we have

$$(\mathcal{T}g)(x) = \psi(x) g(\Phi(x)) = \psi(x) \int_{\mathbb{S}^d} \sigma(\Phi(x) \cdot v) d\mu(v).$$

By the definition of Φ ,

$$\Phi(x) \cdot v = \frac{1}{\sqrt{1 + \|x\|_2^2}} (x^\top, 1) \cdot v,$$

and the homogeneity of σ gives

$$\sigma(\Phi(x) \cdot v) = \frac{1}{(1 + \|x\|_2^2)} \sigma((x^\top, 1) \cdot v).$$

Since $\psi(x) = 1 + \|x\|_2^2$ we obtain

$$(\mathcal{T}g)(x) = \int_{\mathbb{S}^d} \sigma((x^\top, 1) \cdot v) d\mu(v), \text{ for all } x \in \mathbb{B}^d.$$

Thus $\mathcal{T}g$ admits a ReQU variation representation on $\mathbb{B}^d$ with the same measure μ . By Definition 3.35,

$$\gamma(\mathcal{T}g) \leq \|\mu\|_{\text{TV}} \leq M,$$

which proves the claim in (i).

(ii) Let $h \in \mathcal{H}^\alpha$ and set $a := \lfloor \alpha \rfloor$ and $\kappa := \alpha - a$. By Definition 3.34 there exists $\hat{h} \in C^{a,\kappa}(\mathbb{R}^d)$ with $\hat{h} = h$ on $\mathbb{B}^d$ and $\|\hat{h}\|_{C^{a,\kappa}(\mathbb{R}^d)} \leq 2$. We first define a function $\tilde{h}_+$ on the spherical cap Ω by

$$\tilde{h}_+(u) := u_{d+1}^2 \hat{h}\left(\frac{u'}{u_{d+1}}\right), \text{ where } u = (u', u_{d+1}) \in \Omega,$$

where $u' = (u_1, \dots, u_d)^\top$. For $x \in \mathbb{B}^d$ we have

$$\Phi(x) = \frac{1}{\sqrt{1 + \|x\|_2^2}} \begin{pmatrix} x \\ 1 \end{pmatrix},$$

hence

$$\Phi(x)' = \frac{x}{\sqrt{1 + \|x\|_2^2}}, \quad \Phi(x)_{d+1} = \frac{1}{\sqrt{1 + \|x\|_2^2}}, \quad \text{and} \quad \frac{\Phi(x)'}{\Phi(x)_{d+1}} = x.$$

It follows that

$$\tilde{h}_+(\Phi(x)) = \Phi(x)_{d+1}^2 \hat{h}(x) = \frac{1}{1 + \|x\|_2^2} h(x),$$

and therefore

$$(\mathcal{T}\tilde{h}_+)(x) = \psi(x) \tilde{h}_+(\Phi(x)) = (1 + \|x\|_2^2) \frac{1}{1 + \|x\|_2^2} h(x) = h(x), \quad \text{for all } x \in \mathbb{B}^d.$$

The map $u \mapsto u'/u_{d+1}$ is smooth on Ω and all its derivatives are bounded in terms of d . Since $\hat{h} \in C^{a,\kappa}(\mathbb{R}^d)$ with bounded norm, repeated application of the chain rule shows that $\tilde{h}_+ \in C^{a,\kappa}(\Omega)$ and

$$\|\tilde{h}_+\|_{C^{a,\kappa}(\Omega)} \leq C'_\alpha \|\hat{h}\|_{C^{a,\kappa}(\mathbb{R}^d)} \leq 2C'_\alpha.$$

for a constant C'_α depending only on α and d . Let $-\Omega := \{-u : u \in \Omega\}$ and define $\tilde{h}_-$ on $-\Omega$ by

$$\tilde{h}_-(u) := -\tilde{h}_+(-u), \quad \text{for } u \in -\Omega.$$

Set $U := \Omega \cup (-\Omega)$ and define $\tilde{h}_0$ on U by $\tilde{h}_0 = \tilde{h}_+$ on Ω and $\tilde{h}_0 = \tilde{h}_-$ on $-\Omega$. Then $\tilde{h}_0(-u) = -\tilde{h}_0(u)$ for all $u \in U$, and $\tilde{h}_0 \in C^{a,\kappa}(U)$ with

$$\|\tilde{h}_0\|_{C^{a,\kappa}(U)} \leq C'_\alpha \|\hat{h}\|_{C^{a,\kappa}(\mathbb{R}^d)}.$$

We extend $\tilde{h}_0$ from U to a function $\bar{h} \in C^{a,\kappa}(\mathbb{S}^d)$ by a linear extension operator for Hölder spaces on smooth domains (Fefferman, 2006; Taylor, 2023, Section 4.4). Finally, define

$$\tilde{h}(u) := \frac{1}{2}(\bar{h}(u) - \bar{h}(-u)), \quad \text{for } u \in \mathbb{S}^d.$$

Then $\tilde{h} \in C^{a,\kappa}(\mathbb{S}^d)$, $\tilde{h}(-u) = -\tilde{h}(u)$ for all $u \in \mathbb{S}^d$, and $\tilde{h} = \tilde{h}_0$ on U . In particular $\mathcal{T}\tilde{h} = h$ on $\mathbb{B}^d$ since $\Phi(\mathbb{B}^d) \subset \Omega$. Moreover,

$$\|\tilde{h}\|_{L^\infty(\mathbb{S}^d)} + \|\tilde{h}\|_{C^{a,\kappa}(\mathbb{S}^d)} \leq C_\alpha$$

for a constant C_α independent of h . The identity $\mathcal{T}\tilde{h} = h$ on $\mathbb{B}^d$ continues to hold since the extension coincides with the original $\tilde{h}_+$ on Ω , which contains the image of $\Phi(\mathbb{B}^d)$. $\square$

Moduli of smoothness for lifted functions and their Laplacian powers

In the remainder of this subsection we fix $\alpha > 0$ and set $a := \lfloor \alpha \rfloor$ and $\theta := \alpha - a \in (0, 1]$. We consider functions $h \in \mathcal{H}^\alpha$ and associated lifts $\tilde{h}$ given by Proposition 3.41(ii). The next lemma quantifies the spherical smoothness of $\tilde{h}$ in terms of the moduli of smoothness from Definition 3.30.

Lemma 3.42 (Modulus of smoothness of the lifted function). *Let $\alpha > 0$ and let $s \in \mathbb{N}$ satisfy $\alpha \leq 2s$. There exists a constant $C_{\alpha,s} > 0$ such that for every $h \in \mathcal{H}^\alpha$ and every lift $\tilde{h}$ given by Proposition 3.41 we have*

$$\omega_{2s}(\tilde{h}, t)_\infty \leq C_{\alpha,s} t^\alpha, \text{ for all } 0 < t < \pi.$$

Proof. Set $a := \lfloor \alpha \rfloor$ and $\kappa := \alpha - a \in (0, 1]$. By construction $\tilde{h} \in C^{a,\kappa}(\mathbb{S}^d)$ with a norm bounded by a constant depending only on α and d . By Lemma 3.31 there exists $C > 0$ such that for every $0 < t < \pi$,

$$\omega_{2s}(\tilde{h}, t)_\infty \leq C \sup_{0 < \theta \leq t} \sup_{u \in \mathbb{S}^d} \left| \sum_{j=0}^{2s} (-1)^j \binom{2s}{j} T_{j\theta} \tilde{h}(u) \right|.$$

Fixing $u \in \mathbb{S}^d$ and $v \in \mathbb{S}_u^\perp$ we introduce the one-dimensional function

$$f_{u,v}(\varphi) := \tilde{h}(u \cos \varphi + v \sin \varphi), \text{ for } \varphi \in [0, \pi].$$

Using the definition of $T_{j\theta}$ we obtain

$$\left| \sum_{j=0}^{2s} (-1)^j \binom{2s}{j} T_{j\theta} \tilde{h}(u) \right| \leq \sup_{v \in \mathbb{S}_u^\perp} \left| \sum_{j=0}^{2s} (-1)^j \binom{2s}{j} f_{u,v}(j\theta) \right|.$$

The map $\varphi \mapsto u \cos \varphi + v \sin \varphi$ is smooth on $[0, \pi]$ with bounded derivatives of all orders, hence $f_{u,v} \in C^{a,\kappa}([0, \pi])$ and

$$\|f_{u,v}\|_{C^{a,\kappa}([0,\pi])} \leq C'_\alpha \|\tilde{h}\|_{C^{a,\kappa}(\mathbb{S}^d)} \leq C''_\alpha,$$

for constants C'_α, C''_α independent of h, u and v . The finite difference operator of order $2s$ applied at 0 can be written as

$$\sum_{j=0}^{2s} (-1)^j \binom{2s}{j} f_{u,v}(j\theta) = \Delta_\theta^{2s} f_{u,v}(0),$$

where the right-hand side is the standard one-dimensional finite difference. Let $r := 2s$ and recall $\alpha = a + \kappa$. Since $f_{u,v} \in C^{a,\kappa}([0, \pi])$ and $r \geq a + 1$,

set $g := f_{u,v}^{(a)}$. Then g is κ -Hölder, so $\|\Delta_\theta g\|_{L^\infty} \leq [g]_{C^{0,\kappa}} \theta^\kappa$. Using $\Delta_\theta \phi(x) = \int_0^\theta \phi'(x+t) dt$ iterated a times gives

$$\|\Delta_\theta^{a+1} f_{u,v}\|_{L^\infty} \leq C \theta^{a+\kappa} \|f_{u,v}\|_{C^{a,\kappa}([0,\pi])}.$$

Finally, $\Delta_\theta^r = \Delta_\theta^{r-a-1} \Delta_\theta^{a+1}$ and $\|\Delta_\theta^m\|_{L^\infty \rightarrow L^\infty} \leq 2^m$, hence

$$\sup_{0 < \theta \leq t} |\Delta_\theta^{2s} f_{u,v}(0)| \leq C t^\alpha \|f_{u,v}\|_{C^{a,\kappa}([0,\pi])} \leq C t^\alpha.$$

Taking the supremum over u and v and combining the previous bounds yields

$$\omega_{2s}(\tilde{h}, t)_\infty \leq C_\alpha t^\alpha$$

for all $0 < t < \pi$. □

We next extend this estimate to Laplacian powers of the lift.

Lemma 3.43 (Moduli of smoothness for Laplacian derivatives of the lift). *Let $\alpha > 0$, set $a := \lfloor \alpha \rfloor$, and let $s \in \mathbb{N}$ satisfy $\alpha \leq 2s$. For every integer ℓ with $0 \leq \ell \leq a$ there exists a constant $C_{\alpha,\ell,s} > 0$ such that for every $h \in \mathcal{H}^\alpha$ and every associated lift $\tilde{h}$,*

$$\omega_{2s}((-\Delta)^{\ell/2} \tilde{h}, t)_\infty \leq C_{\alpha,\ell,s} t^{\alpha-\ell}, \text{ for all } 0 < t < \pi.$$

Proof. Let $\ell \in \{0, \dots, a\}$. The operator $(-\Delta)^{\ell/2}$ is a classical elliptic pseudodifferential operator of order ℓ on the compact manifold $\mathbb{S}^d$ (Taylor, 2023). It maps $C^\alpha(\mathbb{S}^d)$ continuously into $C^{\alpha-\ell}(\mathbb{S}^d)$. Since $\tilde{h}$ has uniformly bounded $C^\alpha(\mathbb{S}^d)$ norm by Proposition 3.41, we have a uniform bound on $\|(-\Delta)^{\ell/2} \tilde{h}\|_{C^{\alpha-\ell}(\mathbb{S}^d)}$. Applying Lemma 3.42 to $(-\Delta)^{\ell/2} \tilde{h}$ with exponent $\alpha - \ell$ yields the claim. □

Rates of approximation in Laplacian-based norms

We now introduce the zonal kernels used to approximate functions on the sphere and recall the approximation result that links their action to the moduli of smoothness.

Definition 3.44 (Spectral cutoff kernels and spherical polynomials). *Let $\eta \in C^\infty([0, \infty))$ satisfy $\eta(t) = 1$ for $0 \leq t \leq 1$ and $\eta(t) = 0$ for $t \geq 2$. For each integer $m \geq 1$ we define the zonal kernel $L_m : [-1, 1] \rightarrow \mathbb{R}$ by*

$$L_m(t) := \sum_{n=0}^{\infty} \eta\left(\frac{n}{m}\right) N(d, n) P_n(t), \text{ for all } t \in [-1, 1],$$

where $N(d, n)$ and P_n are as in Definition 3.25 and Lemma 3.27. Since $\eta(n/m) = 0$ for $n \geq 2m$ the sum is finite. For a continuous function f on $\mathbb{S}^d$ we set

$$g_m := f * L_m.$$

Then g_m is a spherical polynomial of degree at most $2m - 1$ and depends linearly on f .

The following lemma extracts from (Dai and Xu, 2013) the error bounds that we need.

Lemma 3.45 (Approximation of lifted functions in Laplacian-based norms). *Let $\alpha > 0$ and set $a := \lfloor \alpha \rfloor$. Let $h \in \mathcal{H}^\alpha$ and let $\tilde{h}$ be the associated lift from Proposition 3.41. For each integer $m \geq 1$ define $g_m := \tilde{h} * L_m$ as in Definition 3.44. Then for every integer j with $0 \leq j \leq a$ there exists a constant $C_{\alpha,j} > 0$, independent of h and m , such that*

$$\|(-\Delta)^{j/2}(\tilde{h} - g_m)\|_{L^\infty(\mathbb{S}^d)} \leq C_{\alpha,j} m^{-(\alpha-j)}.$$

Proof. Let $0 \leq j \leq a$ and set $\varphi_j := (-\Delta)^{j/2}\tilde{h}$. Let $s_* \in \mathbb{N}$ be the smallest integer such that $\alpha \leq 2s_*$. By Lemma 3.43 with $s = s_*$ there exists $C''_{\alpha,j} > 0$ such that

$$\omega_{2s_*}(\varphi_j, t)_\infty \leq C''_{\alpha,j} t^{\alpha-j}, \text{ for } 0 < t < \pi,$$

where s_* is the smallest integer with $\alpha \leq 2s_*$. By (Dai and Xu, 2013, Theorem 10.3.2) together with the equivalence between K -functionals and moduli of smoothness in (Dai and Xu, 2013, Theorem 10.4.1), applied with $r = 2s_*$, and $p = \infty$, there exists a constant $C_{2s_*} > 0$ such that, for every $f \in C(\mathbb{S}^d)$ and every integer $m \geq 1$, one has

$$\|f - f * L_m\|_{L^\infty(\mathbb{S}^d)} \leq C_{2s_*} \omega_{2s_*}(f, m^{-1})_\infty.$$

Taking $f = \varphi_j$ and writing $C''_{\alpha,j} := C_{2s_*}$, we obtain for all $m \geq 1$,

$$\|\varphi_j - \varphi_j * L_m\|_{L^\infty(\mathbb{S}^d)} \leq C''_{\alpha,j} \omega_{2s_*}(\varphi_j, m^{-1})_\infty \leq C''_{\alpha,j} C'_{\alpha,j} m^{-(\alpha-j)}.$$

Since convolution commutes with $(-\Delta)^{j/2}$ we have

$$\varphi_j * L_m = (-\Delta)^{j/2}(\tilde{h} * L_m) = (-\Delta)^{j/2}g_m$$

and therefore

$$\|(-\Delta)^{j/2}(\tilde{h} - g_m)\|_{L^\infty(\mathbb{S}^d)} = \|\varphi_j - \varphi_j * L_m\|_{L^\infty(\mathbb{S}^d)} \leq C_{\alpha,j} m^{-(\alpha-j)}$$

with $C_{\alpha,j} := C''_{\alpha,j} C'_{\alpha,j}$. This proves the lemma. $\square$

The next estimate lets us convert control of the L^2 modulus of smoothness of f into an L^2 bound on the fractional Laplacian of its filtered approximation $f * L_m$. We will use it to bound Sobolev-type norms of $f * L_m$ in terms of approximation quantities.

Lemma 3.46. *Let $s > 0$. There exists a constant $C_{s,d} > 0$ such that for every $f \in L^2(\mathbb{S}^d)$ and every integer $m \geq 1$,*

$$\|(-\Delta)^{s/2}(f * L_m)\|_{L^2(\mathbb{S}^d)} \leq C_{s,d} m^s \omega_s(f, m^{-1})_2.$$

Proof. By (Dai and Xu, 2013, Theorem 10.4.1) one has $K_s(f, t)_2 \leq C \omega_s(f, t)_2$, and by (Dai and Xu, 2013, Theorem 10.3.2) (with $\kappa = 0$, $p = 2$) one has $n^{-s} \|(-\Delta_{\mathbb{S}^d})^{s/2}(L_n f)\|_2 \leq C K_s(f, t)_2$ whenever $n \asymp t^{-1}$. Take $t = m^{-1}$, $n \asymp m$, and identify $f * L_m$ with the corresponding $L_n f$. $\square$

Having computed the rates of approximation, all that is left now is to bound the variation norm of the spherical approximants g_m in terms of m . This can be done by using the bounds on the Fourier coefficients of the ReQU function, as was done in (Yang and Zhou, 2025).

Lemma 3.47. *Let $\alpha > 0$, set $\alpha_c := \frac{d+5}{2}$, and let $h \in \mathcal{H}^\alpha$ with lift $\tilde{h}$. For $m \geq 1$ define $g_m := \tilde{h} * L_m$. There exist constants $C_1, C_2 > 0$ (depending only on d and α) such that:*

(i) *If $\alpha > \alpha_c$ or if $\alpha = \alpha_c$ is an even integer, then*

$$\sup_{m \geq 1} \gamma(g_m) \leq C_1.$$

(ii) *If $\alpha = \alpha_c$ is not an even integer, then for all $m \geq 2$,*

$$\gamma(g_m) \leq C_2 \sqrt{\log m}.$$

(iii) *If $\alpha < \alpha_c$, then for all $m \geq 1$,*

$$\gamma(g_m) \leq C_2 m^{\alpha_c - \alpha}.$$

Proof. First note that by Proposition 3.41(ii) we may assume that $\tilde{h}$ is odd, meaning $\tilde{h}(-u) = -\tilde{h}(u)$ for all $u \in \mathbb{S}^d$. Since g_m is obtained from $\tilde{h}$ by a spectral multiplier, g_m is odd as well, hence $\mathcal{P}_n g_m = 0$ for all even n . Furthermore, by Definition 3.44 and Lemma 3.29 the spherical harmonic expansion of g_m is

$$g_m = \sum_{n=0}^{\infty} \eta\left(\frac{n}{m}\right) \mathcal{P}_n \tilde{h}.$$

On the other hand, any $g \in \mathcal{G}_\sigma(M)$ with representing measure μ satisfies $\mathcal{P}_n g = \hat{\sigma}(n) \mathcal{P}_n \phi$, where ϕ is an L^2 -density of μ when it exists and where $\hat{\sigma}(n)$ are the Gegenbauer-Fourier coefficients of σ . These coefficients are explicitly known and satisfy $\hat{\sigma}(n) = 0$ for all even n with $n \geq 4$ and $|\hat{\sigma}(n)| \asymp n^{-(d+5)/2}$ for the non-vanishing indices n , as shown in (Bach, 2017, Appendix D.2). This implies that for a continuous function g with spherical harmonic coefficients $(\mathcal{P}_n g)_{n \geq 0}$ we have the estimate

$$\gamma(g)^2 \lesssim \|\phi\|_{L^2(\mathbb{S}^d)}^2 \lesssim \sum_{\hat{\sigma}(n) \neq 0} |\hat{\sigma}(n)|^{-2} \|\mathcal{P}_n g\|_{L^2(\mathbb{S}^d)}^2.$$

Applying this bound to $g \equiv g_m$ yields

$$\gamma(g_m)^2 \lesssim \sum_{n \geq 0} |\hat{\sigma}(n)|^{-2} \|\mathcal{P}_n g_m\|_{L^2(\mathbb{S}^d)}^2.$$

Since $|\hat{\sigma}(n)|^{-2} \lesssim (1+n)^{d+5}$ on the non-vanishing indices n , we have

$$\gamma(g_m)^2 \lesssim \sum_{n \geq 0} (1+n)^{d+5} \|\mathcal{P}_n g_m\|_2^2.$$

Moreover $g_m = \tilde{h} * L_m$ is a spherical polynomial of degree $\leq 2m-1$, so $\mathcal{P}_n g_m = 0$ for $n \geq 2m$ and therefore

$$\gamma(g_m)^2 \lesssim \sum_{n=0}^{2m-1} (1+n)^{d+5} \|\mathcal{P}_n g_m\|_2^2.$$

Finally, on $\mathbb{S}^d$ the eigenvalue of $-\Delta$ on degree- n harmonics is $\lambda_n = n(n+d-1) \asymp (1+n)^2$, hence for $s = (d+5)/2$,

$$\|(-\Delta)^{s/2} g_m\|_2^2 = \sum_{n=0}^{2m-1} \lambda_n^s \|\mathcal{P}_n g_m\|_2^2 \asymp \sum_{n=0}^{2m-1} (1+n)^{d+5} \|\mathcal{P}_n g_m\|_2^2.$$

Combining these estimates gives

$$\gamma(g_m)^2 \lesssim 1 + \|(-\Delta)^{s/2} g_m\|_{L^2(\mathbb{S}^d)}^2, \text{ with } s = \frac{d+5}{2}.$$

We now bound $\|(-\Delta)^{\alpha_c/2} g_m\|_{L^2(\mathbb{S}^d)}$ in terms of m . If $\alpha = \alpha_c$ and α_c is an even integer, choose $s = \alpha_c/2$ in Lemma 3.42 and use $\|\cdot\|_{L^2} \leq \|\cdot\|_{L^\infty}$ to obtain $\omega_{\alpha_c}(\tilde{h}, t)_2 \lesssim t^{\alpha_c}$. Otherwise choose an integer $s \in \mathbb{N}$ such that $\alpha \leq 2s$ and $\alpha_c < 2s$, and apply Lemma 3.42 with this s to obtain $\omega_{2s}(\tilde{h}, t)_2 \lesssim t^\alpha$.

In the case $\alpha \neq \alpha_c$ or $\alpha = \alpha_c$ not an even integer, Corollary 3.33 with $(r, s) = (2s, \alpha_c)$ yields

$$\omega_{\alpha_c}(\tilde{h}, m^{-1})_2 \lesssim \begin{cases} m^{-\alpha}, & \alpha < \alpha_c, \\ m^{-\alpha_c} \log m, & \alpha = \alpha_c \text{ and } \alpha_c \notin 2\mathbb{N}, \\ m^{-\alpha_c}, & \alpha > \alpha_c. \end{cases}$$

Applying Lemma 3.46 with $f = \tilde{h}$ and $s = \alpha_c$ gives

$$\|(-\Delta)^{\alpha_c/2} g_m\|_{L^2(\mathbb{S}^d)} \lesssim m^{\alpha_c} \omega_{\alpha_c}(\tilde{h}, m^{-1})_2,$$

which is bounded uniformly in m if $\alpha > \alpha_c$, grows like $\log m$ if $\alpha = \alpha_c$ with $\alpha_c \notin 2\mathbb{N}$, and grows like $m^{\alpha_c - \alpha}$ if $\alpha < \alpha_c$. Combining this with $\gamma(g_m)^2 \lesssim 1 + \|(-\Delta)^{\alpha_c/2} g_m\|_{L^2(\mathbb{S}^d)}^2$ yields (i), (ii), and (iii). $\square$

Approximation rates for the ReQU variation space in $W^{q,\infty}$

We are now ready to state the main approximation result for the infinite-width ReQU variation space in Sobolev norms on the ball.

Theorem 3.48. *Fix $d \in \mathbb{N}$ and $\sigma(t) = \max\{0, t\}^2$. Let $\alpha > 0$ and set $a := \lfloor \alpha \rfloor$. Let q be an integer with $0 \leq q \leq a$, set $s := \alpha - q$, and define the critical index $\alpha_c := (d+5)/2$. Then there exist constants $c, C > 0$ (depending only on d, α, q) such that for all sufficiently large M :*

(supercritical) *If $\alpha > \alpha_c$ or if $\alpha = \alpha_c$ is an even integer, then there exists $M_0 < \infty$ such that every $h \in \mathcal{H}^\alpha$ belongs to $\mathcal{V}_\sigma(M_0)$ and*

$$\inf_{f \in \mathcal{V}_\sigma(M_0)} \|h - f\|_{W^{q,\infty}(\mathbb{B}^d)} = 0.$$

(critical) *If $\alpha = \alpha_c$ is not an even integer, then*

$$\sup_{h \in \mathcal{H}^\alpha} \inf_{f \in \mathcal{V}_\sigma(M)} \|h - f\|_{W^{q,\infty}(\mathbb{B}^d)} \leq C \exp(-c s M^2).$$

(subcritical) *If $\alpha < \alpha_c$, then*

$$\sup_{h \in \mathcal{H}^\alpha} \inf_{f \in \mathcal{V}_\sigma(M)} \|h - f\|_{W^{q,\infty}(\mathbb{B}^d)} \leq C M^{-\frac{s}{\alpha_c - \alpha}}.$$

Proof. Let $h \in \mathcal{H}^\alpha$ and let $\tilde{h}$ be the associated lift from Proposition 3.41. For each integer $m \geq 1$ let $g_m := \tilde{h} * L_m$ be the spherical polynomial from

Definition 3.44 and set $f_m := \mathcal{T}g_m$ on $\mathbb{B}^d$. By Lemma 3.45 with $j \leq q$ and Lemma 3.40 there exists $C_0 > 0$ such that

$$\|h - f_m\|_{W^{q,\infty}(\mathbb{B}^d)} = \|\mathcal{T}(\tilde{h} - g_m)\|_{W^{q,\infty}(\mathbb{B}^d)} \leq C_0 \sum_{j=0}^q \|(-\Delta)^{j/2}(\tilde{h} - g_m)\|_{L^\infty(\mathbb{S}^d)} \leq C'_0 m^{-(\alpha-q)}$$

with a constant C'_0 independent of h and m . Since $s = \alpha - q$, this gives

$$\|h - f_m\|_{W^{q,\infty}(\mathbb{B}^d)} \leq C'_0 m^{-s}.$$

On the other hand, Proposition 3.41(i) and Lemma 3.47 show that the variation norm of f_m satisfies the same bounds as that of g_m up to a universal multiplicative constant. In the supercritical regime $\alpha > \alpha_c$ or $\alpha = \alpha_c$ even, Lemma 3.47(i) implies that $\gamma(g_m)$ remains uniformly bounded in m . Choose representing measures μ_m for g_m such that $\|\mu_m\|_{\text{TV}} \leq C$ uniformly in m . By Banach–Alaoglu there exists a subsequence, not relabeled, and a finite signed Radon measure μ on $\mathbb{S}^d$ such that μ_m converges to μ in the weak-* topology and $\|\mu\|_{\text{TV}} \leq C$. By Lemma 3.45 with $j = 0$ we have $g_m \rightarrow \tilde{h}$ uniformly on $\mathbb{S}^d$. For each fixed $u \in \mathbb{S}^d$ the function $v \mapsto \sigma(u \cdot v)$ is continuous and bounded, hence

$$\tilde{h}(u) = \lim_{m \rightarrow \infty} g_m(u) = \lim_{m \rightarrow \infty} \int_{\mathbb{S}^d} \sigma(u \cdot v) d\mu_m(v) = \int_{\mathbb{S}^d} \sigma(u \cdot v) d\mu(v).$$

Thus $\tilde{h} \in \mathcal{G}_\sigma(M_0)$ for $M_0 := C$, and therefore $h = \mathcal{T}\tilde{h} \in \mathcal{V}_\sigma(M_0)$. This shows the supercritical case.

In the critical regime $\alpha = \alpha_c$ with α not an even integer, Lemma 3.47(ii) gives $\gamma(g_m) \leq C_1 \sqrt{\log m}$ for $m \geq 2$. Choosing m such that $M \asymp \sqrt{\log m}$ we obtain $m \asymp \exp(cM^2)$ for some $c > 0$ and hence

$$\|h - f_m\|_{W^{q,\infty}(\mathbb{B}^d)} \leq C'_0 m^{-s} \leq C \exp(-csM^2),$$

for all sufficiently large M . This proves the critical case.

Finally, in the subcritical regime $\alpha < \alpha_c$, Lemma 3.47(iii) gives $\gamma(g_m) \leq C_2 m^{\alpha_c - \alpha}$ for $m \geq 1$. Choosing m as a function of M so that $M \asymp m^{\alpha_c - \alpha}$ yields $m \asymp M^{1/(\alpha_c - \alpha)}$ and therefore

$$\|h - f_m\|_{W^{q,\infty}(\mathbb{B}^d)} \leq C'_0 m^{-s} \leq C M^{-s/(\alpha_c - \alpha)},$$

where we absorbed all constants into C . This proves the subcritical case. Taking the infimum over all $f \in \mathcal{V}_\sigma(M)$ and the supremum over $h \in \mathcal{H}^\alpha$ yields the stated bounds. $\square$

4.3 From variation space to finite-width networks

In this subsection we pass from the infinite-width approximation results for the ReQU variation space established in Theorem 3.48 to approximation guarantees for actual finite-width shallow networks. The key point is that the same discrete measure that approximates the representing measure of a target function in the variation space automatically approximates all derivatives up to order two, so that the sampling argument can be formulated in the Banach space $W^{2,\infty}(\mathbb{B}^d)$ instead of only $L^\infty(\mathbb{B}^d)$.

Definition 3.49 (Finite-width ReQU networks). *Fix $d \in \mathbb{N}$ and let $\sigma : t \mapsto \max\{0, t\}^2$ be the ReQU activation. Let $N \in \mathbb{N}$ and $M > 0$. We consider the subclass of fully connected networks from Definition 3.5 given by $L = 1$, $\|\mathbf{m}\|_\infty \leq N$ and $\|c\|_\infty \leq M$, with a fixed inner-layer bound $B_0 > 0$ that depends only on d . In other words, we take $\Sigma_{\mathbf{m}_1:L,M}^\sigma(B_0)$ with $L = 1$ and $m_1 \leq N$ as the class of shallow ReQU networks with width at most N and outer coefficients bounded by M . Each such network can be written in the form*

$$f_N(x) = \sum_{j=1}^{m_1} a_j \sigma(w_j^\top x + b_j),$$

where $m_1 \leq N$, the inner parameters satisfy $\|w_j\|_\infty \leq B_0$ and $|b_j| \leq B_0$, and the outer coefficients satisfy $|a_j| \leq M$ for $1 \leq j \leq m_1$. The constant B_0 is chosen so that, for every $v \in \mathbb{S}^d$, the feature map used in Definition 3.36 is realized by some (w, b) with $\|w\|_\infty \leq B_0$ and $|b| \leq B_0$.

By Definitions 3.35 and 3.36, every function $f \in \mathcal{V}_\sigma(M)$ admits a representation

$$f(x) = \int_{\mathbb{S}^d} \sigma((x^\top, 1) \cdot v) d\mu(v), \text{ for all } x \in \mathbb{B}^d,$$

for some signed Radon measure μ on $\mathbb{S}^d$ with $\|\mu\|_{\text{TV}} \leq M$. Replacing μ by a discrete measure $\mu_N = \sum_{j=1}^{m_1} a_j \delta_{v_j}$ with at most $m_1 \leq N$ atoms produces a one-hidden-layer network of the form in Definition 3.49. The total variation of μ_N is $\|\mu_N\|_{\text{TV}} = \sum_{j=1}^{m_1} |a_j|$, so that $\|\mu_N\|_{\text{TV}} \leq M$ implies $|a_j| \leq M$ for every j and therefore $f_N \in \Sigma_{\mathbf{m}_1:L,M}^\sigma(B_0)$.

We next state the result that allows us to approximate a given function in the ReQU variation space by a finite-width one-hidden-layer network with control in the $W^{2,\infty}$ -norm.

Lemma 3.50 (Sampling a ReQU variation function in $W^{2,\infty}$). *Fix $d \in \mathbb{N}$ and the ReQU activation $\sigma : t \mapsto \max\{0, t\}^2$. There exists a constant $C_d > 0$ such that the following holds. For every $M > 0$ and every $f \in \mathcal{V}_\sigma(M)$ in*

the sense of Definition 3.35, there exists a shallow ReQU network f_N of the form described in Definition 3.49, with width at most N and outer coefficient bound at most M , such that

$$\|f - f_N\|_{W^{2,\infty}(\mathbb{B}^d)} \leq C_d M N^{-\theta_d}, \text{ where } \theta_d = \frac{1}{2} + \frac{1}{2d}.$$

Proof. This is an immediate consequence of Theorem 3 in (Siegel, 2025). Their dictionary $\mathbb{P}_2^d$ is the collection of ReQU ridge functions of the form $x \mapsto \sigma(\omega \cdot x + b)$ on $\mathbb{B}^d$ with ω on the unit sphere and b in a bounded interval. Its variation ball $\mathcal{K}_1(\mathbb{P}_2^d)$ coincides, up to an absolute constant that is absorbed into C_d , with the ReQU variation ball $\mathcal{V}_\sigma(M)$ from Definition 3.35. Specializing Theorem 3 in (Siegel, 2025) to $k = 2$, $m = 2$ and $\Omega = \mathbb{B}^d$ yields the stated rate in the $W^{2,\infty}$ -norm, with the width parameter N equal to the number of dictionary atoms. The realization of these atoms by networks in Definition 3.49 is ensured by the choice of B_0 . $\square$

Having established approximation error bounds of smooth functions by the variation space in Theorem 3.48, and approximation bounds of variation space elements by finite-width networks in Lemma 3.50, all that is left is combining these results to prove our final bound. We simply recall a well-known fact to relate Sobolev and Hölder spaces before proceeding:

Lemma 3.51. *Let $r \in \mathbb{N}$ with $r \geq 1$ and let $u \in W^{r,\infty}(\mathbb{R}^d)$. Then $u|_{\mathbb{B}^d}$ has a representative in $C^{r-1,1}(\mathbb{B}^d)$ and there exists a constant $C_{d,r}$.*

$$\|u\|_{C^{r-1,1}(\mathbb{B}^d)} \leq C_{d,r} \|u\|_{W^{r,\infty}(\mathbb{R}^d)}.$$

Proof. See any standard analysis text, such as (Evans, 2022). $\square$

Proof of Theorem 3.13. Let $u \in W^{r,\infty}(\mathbb{R}^d)$. By Lemma 3.51, the restriction $h := u|_{\mathbb{B}^d}$ belongs to $C^{r-1,1}(\mathbb{B}^d)$ and satisfies $\|h\|_{C^{r-1,1}(\mathbb{B}^d)} \leq C_{d,r} \|u\|_{W^{r,\infty}(\mathbb{R}^d)}$. In particular, $h \in \mathcal{H}^r(\mathbb{B}^d)$. Now set $\alpha := r$ (so $h \in \mathcal{H}^\alpha(\mathbb{B}^d) = C^{r-1,1}(\mathbb{B}^d)$), fix $N \geq 2$, and let $M > 0$ to be specified later. Theorem 3.48 (with $q = 2$) provides, for each such M , a function $f \in \mathcal{V}_\sigma(M)$ that approximates h in $W^{2,\infty}(\mathbb{B}^d)$ with a rate depending on the regime determined by r .

In the subcritical regime $\alpha < \alpha_c$, Theorem 3.48 yields

$$\|h - f\|_{W^{2,\infty}(\mathbb{B}^d)} \leq C_{d,\alpha} M^{-\frac{s}{\alpha_c - \alpha}} \|u\|_{W^{r,\infty}(\mathbb{R}^d)},$$

for all M large enough with $s = \alpha - 2$, $\alpha_c = (d+5)/2$. Applying Lemma 3.50 to f with width parameter N^d gives a one-hidden-layer network f_N with

width at most N^d and outer coefficient bound at most M such that

$$\|f - f_N\|_{W^{2,\infty}(\mathbb{B}^d)} \leq C_d M N^{-d\theta_d}, \text{ where } \theta_d = \frac{1}{2} + \frac{1}{2d}.$$

The triangle inequality gives

$$\|h - f_N\|_{W^{2,\infty}(\mathbb{B}^d)} \leq C_{d,\alpha} M^{-\frac{s}{\alpha_c - \alpha}} \|u\|_{W^{r,\infty}(\mathbb{R}^d)} + C_d M N^{-d\theta_d}.$$

Optimizing the right-hand side over M gives M of order $N^{d\theta_d s / (\alpha_c - \alpha)}$, which leads to the rate $N^{-(\alpha-2)}$ and the corresponding choice $M_N(h)$ stated in the theorem. This proves (3.17) with $\gamma(r) = r - 2$ in the subcritical regime.

In the critical regime $\alpha = \alpha_c$ that is not covered by the supercritical inclusion, Theorem 3.48 gives an infinite-width error bounded by $C_{d,\alpha} \exp(-c_\alpha M^2) \|u\|_{W^{r,\infty}(\mathbb{R}^d)}$. Combining this with Lemma 3.50 and choosing M growing logarithmically with N yields

$$\|h - f_N\|_{W^{2,\infty}(\mathbb{B}^d)} \leq C'_{d,\alpha} N^{-d\theta_d} \|u\|_{W^{r,\infty}(\mathbb{R}^d)}$$

up to logarithmic factors. This corresponds to $\gamma(r) = (d+1)/2$ in the critical regime.

In the supercritical regime of Theorem 3.48, namely when $\alpha > \alpha_c$ or when $\alpha = \alpha_c$ is an even integer, every $h \in \mathcal{H}^\alpha$ belongs to a fixed variation ball $\mathcal{V}_\sigma(M_0)$, with M_0 independent of h , and the infinite-width approximation error in $W^{2,\infty}(\mathbb{B}^d)$ is zero once $M \geq M_0$. Applying Lemma 3.50 with $M = M_0$ yields

$$\|h - f_N\|_{W^{2,\infty}(\mathbb{B}^d)} \leq C_d M_0 N^{-d\theta_d} \|u\|_{W^{r,\infty}(\mathbb{R}^d)},$$

which again gives (3.17) with $\gamma(r) = (d+1)/2$.

Putting the three cases together, and recalling that $\theta_d = 1/2 + 1/(2d)$, we obtain the stated expression of $\gamma(r)$ and the corresponding bounds on $M_N(h)$. The networks constructed in Lemma 3.50 belong to the class in Definition 3.49, so that the hypothesis space $\mathcal{F}_{NN}$ in the theorem is a particular choice of such networks with width at most N^d and outer coefficient bound at most $M_N(h)$. This completes the proof. $\square$

5 Proof of VC dimension bounds

In this section, we prove the VC dimension bounds stated in Theorem 3.18. A technicality we need to overcome is the presence of the multiplicative factor ρ , which induces extra terms in the derivatives expressions, and therefore makes VC dimension analysis of the Neural Network derivatives alone insufficient. We begin by addressing this technicality in Lemma 3.52 below.

5.1 Invariance under multiplication by parameter-independent factor

The boundary-enforced hypothesis spaces of Definition 3.7 are obtained by multiplying a network by a fixed function ρ which does not depend on the parameter vector $\boldsymbol{\theta}$. The next lemma makes precise why this operation does not affect any of the VC dimension bounds we derive in this section. The definition of Pfaffian chains and functions will be given in Section 5.3.

Lemma 3.52. *Let $r \in \mathbb{N}^+$, let $\Omega \subseteq \mathbb{R}^d$ be open, and let $\rho \in C^r(\bar{\Omega})$ be independent of $\boldsymbol{\theta} \in \mathbb{R}^p$. Let $\mathcal{F}$ be any class of C^r functions on Ω parametrized by $\boldsymbol{\theta}$, and define*

$$\rho\mathcal{F} := \{x \mapsto \rho(x)\phi(x, \boldsymbol{\theta}) : \phi(\cdot, \boldsymbol{\theta}) \in \mathcal{F}\}.$$

Then for every multi-index $\alpha \in \mathbb{N}^d$ with $\|\alpha\|_1 \leq r$ and every $\phi \in \mathcal{F}$,

$$\partial^\alpha(\rho\phi)(x, \boldsymbol{\theta}) = \sum_{\beta: \beta \leq \alpha} \binom{\alpha}{\beta} (\partial^{\alpha-\beta}\rho)(x) \partial^\beta\phi(x, \boldsymbol{\theta}). \quad (3.36)$$

In particular, for every fixed $x_0 \in \bar{\Omega}$, the map $\boldsymbol{\theta} \mapsto \partial^\alpha(\rho\phi)(x_0, \boldsymbol{\theta})$ is a finite linear combination (with coefficients depending only on ρ and x_0) of the maps $\boldsymbol{\theta} \mapsto \partial^\beta\phi(x_0, \boldsymbol{\theta})$ with $\beta \leq \alpha$.

Moreover, the following closure properties hold:

- *If each $\boldsymbol{\theta} \mapsto \partial^\beta\phi(x_0, \boldsymbol{\theta})$ is a polynomial in $\boldsymbol{\theta}$ of degree at most D_β , then $\boldsymbol{\theta} \mapsto \partial^\alpha(\rho\phi)(x_0, \boldsymbol{\theta})$ is a polynomial of degree at most $\max_{\beta \leq \alpha} D_\beta$.*
- *If each $\boldsymbol{\theta} \mapsto \partial^\beta\phi(x_0, \boldsymbol{\theta})$ is Pfaffian with respect to a common Pfaffian chain of length ℓ and degrees (γ, κ_β) , then $\boldsymbol{\theta} \mapsto \partial^\alpha(\rho\phi)(x_0, \boldsymbol{\theta})$ is Pfaffian with respect to the same chain, of degrees*

$$(\gamma, \max_{\beta \leq \alpha} \kappa_\beta).$$

Proof. Equation (3.36) is the Leibniz rule. Fixing x_0 makes each factor $(\partial^{\alpha-\beta}\rho)(x_0)$ a constant independent of $\boldsymbol{\theta}$, hence the stated linear-combination property. The polynomial and Pfaffian statements follow immediately from the fact that both classes are closed under addition and multiplication by constants, and that these operations do not increase the relevant degree parameters (polynomial degree in $\boldsymbol{\theta}$, or Pfaffian β -degree). $\square$

Because both the polynomial degree and respectively the Pfaffian chain degree control the VC dimension of FCNNs with ReQU and respectively tanh

activation, Lemma 3.52 allows us to directly translate VC dimension bounds on FCNN derivatives hypothesis spaces to VC-dimension bounds on derivatives of boundary-enforced FCNNs hypothesis spaces, as we will now see.

5.2 ReQU FCNNs

We aim to derive bounds on the VC dimension of the hypothesis spaces defined by α -th order partial derivatives of fully connected neural networks (FCNNs) with ReQU activations, where $\alpha \in \mathbb{N}^d$ and $\|\alpha\| \leq 2$. This problem is challenging because the compositional structure of FCNNs leads to highly intricate derivative expressions, making it difficult to apply standard VC dimension bounding techniques. Although (Yang et al., 2023) obtained optimal VC dimension and pseudodimension bounds for FCNNs with ReLU activations, and (Lei et al., 2025) established VC dimension bounds for higher-order derivatives of deep convolutional ReLU networks composed with a shallow ReLU^k network, these results do not directly extend to the ReQU setting. We therefore develop in this section novel bounds tailored specifically for FCNNs with ReQU activations and their derivatives.

To estimate the VC dimension of FCNNs with piecewise polynomial activation, we first need these essential Lemmas:

Lemma 3.53 ((Bartlett and Anthony, 1999), Theorem 8.3). *Let $p_1, \dots, p_M$ be polynomials in $W \leq M$ variables with degree at most D . Define the sign function as $\text{sign}(x) := \mathbb{1}_{x>0}$, and let*

$$K := \left| \left\{ (\text{sign}(p_1(\theta)), \dots, \text{sign}(p_M(\theta))) \mid \theta \in \mathbb{R}^W \right\} \right|$$

be the number of sign vectors that can be realized by $p_1, \dots, p_M$. We have $K \leq 2(2eMD/W)^W$.

Lemma 3.54 ((Bartlett et al., 2019), Lemma 18). *Suppose that $2^m \leq 2^t(mq/w)^w$ for some $q \geq 16$ and $m \geq w \geq t \geq 0$. Then, $m \leq t + w \log_2(2q \log_2 q)$.*

Before starting our analysis, we begin by recalling the bound on the VC dimension of standard ReQU FCNNs spaces, which follows directly from (Bartlett et al., 2019, Theorem 7):

Proposition 3.55. *Let $\sigma : x \mapsto \max\{0, x\}^2$ be the ReQU activation function, and denote by $\mathcal{F} = \Sigma_{\mathbf{m}_{1:L}}^\sigma(\infty)$ the hypothesis space of ReQU FCNNs with no restriction on their weights, as per Definition 3.5. We have the estimate*

$$\text{VCdim}(\mathcal{F}) = \mathcal{O}(\mathcal{S}L[L + \log_2(L\|\mathbf{m}\|_\infty)]),$$

where $\mathcal{S} = \sum_{i=1}^L m_i(m_{i-1} + 1)$ is the number of parameters in the architecture.

First order derivatives

For convenience, denote $\sigma_2 : x \mapsto \max\{0, x\}^2$ the ReQU activation, and $\sigma_1 : x \mapsto 2\max\{0, x\}$ its derivative. Recall that an element $\phi \in \Sigma_{\mathbf{m}_{1:L}}^{\sigma_2}(\infty)$ can be written as

$$\phi : \mathbf{x} \mapsto \mathbf{W}_{L+1}\sigma_2(\mathbf{W}_L\sigma_2(\cdots\sigma_2(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1)\cdots) + \mathbf{b}_L) + b_{L+1}.$$

For any $1 \leq i \leq d$, the partial derivative with respect to the i -th coordinate can thus be written as

$$\begin{aligned} \partial_i \phi : \mathbf{x} \mapsto & \mathbf{W}_{L+1}\sigma_1(\mathbf{W}_L\sigma_2(\cdots\sigma_2(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1)\cdots) + \mathbf{b}_L) \\ & \cdot \mathbf{W}_L\sigma_1(\cdots\sigma_2(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1)\cdots) \\ & \cdots \mathbf{W}_2\sigma_1(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1)(\mathbf{W}_1)_i, \end{aligned} \quad (3.37)$$

where $(\mathbf{W})_i$ denotes the i -th column of $\mathbf{W}$. Note that for fixed $\mathbf{x} \in \mathbb{R}^d$, $\partial_i \phi$ can be seen as a function of the parameters (weights and biases), in which case we can denote it $\partial_i \phi(\mathbf{x}, \cdot)$.

Following an approach similar to (Bartlett et al., 2019; Yang et al., 2023), our goal is to find a partition $\mathcal{P}$ of the parameter space $\mathbb{R}^S$ such that, for fixed $x_1, \dots, x_M$, and any $P \in \mathcal{P}$, the functions $\partial_i \phi(x_1, \cdot), \dots, \partial_i \phi(x_M, \cdot)$ are polynomials on P . Indeed if we denote

$$K = \left| \left\{ (\text{sign}(\partial_i \phi(x_1, \theta)), \dots, \text{sign}(\partial_i \phi(x_M, \theta))) \mid \theta \in \mathbb{R}^S \right\} \right|,$$

we have

$$K \leq \sum_{P \in \mathcal{P}} \left| \left\{ (\text{sign}(\partial_i \phi(x_1, \theta)), \dots, \text{sign}(\partial_i \phi(x_M, \theta))) \mid \theta \in P \right\} \right|.$$

Therefore, by applying Lemmas 3.53 and 3.54, we can deduce from the cardinality of the partition $\mathcal{P}$ a bound on the VC dimension. We begin by defining a family of functions with respect to the parameter $\theta \in \mathbb{R}^S$:

$$\begin{aligned} \mathbb{F}_0 &:= \{(\mathbf{W}_1)_i, \mathbf{W}_1\mathbf{x} + \mathbf{b}_1\} \\ \mathbb{F}_1 &:= \{(\mathbf{W}_1)_i, \mathbf{W}_2\sigma_1(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1), \mathbf{W}_2\sigma_2(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2\} \\ \mathbb{F}_2 &:= \{(\mathbf{W}_1)_i, \mathbf{W}_2\sigma_1(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1), \mathbf{W}_3\sigma_1(\mathbf{W}_2\sigma_2(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2), \mathbf{W}_3\sigma_2(\mathbf{W}_2\sigma_2(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2) \\ &\quad \vdots \\ \mathbb{F}_L &:= \{(\mathbf{W}_1)_i, \mathbf{W}_2\sigma_1(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1), \dots, \mathbf{W}_{L+1}\sigma_1(\mathbf{W}_L\sigma_2(\cdots\sigma_2(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1)\cdots) + \mathbf{b}_L)\}. \end{aligned} \quad (3.38)$$

We will build the partition recursively, through a sequence $\mathcal{P}_0, \mathcal{P}_1, \dots, \mathcal{P}_L$ which is successively refined over each layer. These successive refinements will satisfy the following properties:

1. $|\mathcal{P}_0| = 1$, and for all $1 \leq \ell \leq L$,

$$\frac{|\mathcal{P}_\ell|}{|\mathcal{P}_{\ell-1}|} \leq 2 \left(\frac{2eMm_\ell(2^{\ell+1} - 1)}{\sum_{i=1}^\ell m_i(m_{i-1} + 1)} \right)^{\sum_{i=1}^\ell m_i(m_{i-1} + 1)}. \quad (3.39)$$

2. For all $0 \leq \ell \leq L - 1$, and all $P \in \mathcal{P}_\ell$, each element $f \in \mathbb{F}_\ell$, when restricted to $\boldsymbol{\theta} \in P$, is a polynomial function in $\sum_{i=1}^\ell m_i(m_{i-1} + 1)$ variables of degree at most $2^{\ell+1} - 1$.
3. If we denote by $\{f_\ell^{(L)}\}_{0 \leq \ell \leq L+1}$ the elements of $\mathbb{F}_L$, then for all $P \in \mathcal{P}_L$, the restriction to $\boldsymbol{\theta} \in P$ of $f_\ell^{(L)}$ is a polynomial in $\sum_{i=1}^\ell m_i(m_{i-1} + 1)$ variables of degree at most $2^{\ell+1} - 1$.

We begin by defining $\mathcal{P}_0 := \{\mathbb{R}^S\}$, which clearly satisfies items 1 and 2 above, as both $\boldsymbol{\theta} \mapsto (\mathbf{W}_1)_i$ and $\boldsymbol{\theta} \mapsto \mathbf{W}_1 \mathbf{x} + \mathbf{b}_1$ are affine in $\mathbf{W}_1, \mathbf{b}_1$.

For the induction step, assume that $\mathcal{P}_0, \dots, \mathcal{P}_{\ell-1}$ satisfying items 1 and 2 have already been defined, so that our task is now to construct $\mathcal{P}_\ell$. For integers $1 \leq h \leq m_\ell$, $1 \leq j \leq M$, and $P \in \mathcal{P}_{\ell-1}$, denote by $p_{h, \mathbf{x}_j, P}(\boldsymbol{\theta})$ the restriction to $\boldsymbol{\theta} \in P$ of the function describing the input of the h -th unit in the ℓ -th layer, in response to $\mathbf{x}_j$. By induction hypothesis, this is a polynomial in $\sum_{i=1}^{\ell-1} m_i(m_{i-1} + 1)$ variables of degree at most $2^{\ell+1} - 1$. By Lemma 3.53, the collection of polynomials

$$\{p_{h, \mathbf{x}_j, P}(\boldsymbol{\theta}) \mid 1 \leq h \leq m_\ell, 1 \leq j \leq M\}$$

can realize at most

$$\Pi := 2 \left(\frac{2eMm_\ell(2^{\ell+1} - 1)}{\sum_{i=1}^\ell m_i(m_{i-1} + 1)} \right)^{\sum_{i=1}^\ell m_i(m_{i-1} + 1)}$$

distinct sign patterns as $\boldsymbol{\theta}$ varies in $\mathbb{R}^S$. We can thus partition the parameter space $\mathbb{R}^S$ into Π many regions, such that all of these polynomials have constant sign within each region. We can intersect all the elements of this partition with P to obtain a partition of P into Π subregions over which each polynomial has constant sign. By performing this operation for all $P \in \mathcal{P}_{\ell-1}$, we thus obtain our desired partition $\mathcal{P}_\ell$, which clearly satisfies inequality (3.39).

We now check that $\mathcal{P}_\ell$ satisfies item 2. To that end, first observe that if we label $f_0^{(\ell)}, f_1^{(\ell)}, \dots, f_{\ell+1}^{(\ell)}$ the elements of $\mathbb{F}_\ell$, we have that $f_i^{(\ell)} \in \mathbb{F}_{\ell-1}$ for all $0 \leq i \leq \ell - 1$. Hence, because the partition $\mathcal{P}_\ell$ is finer than $\mathcal{P}_{\ell-1}$, it follows

simply by induction hypothesis that for all $P \in \mathcal{P}_\ell$, and $i \leq \ell - 1$, $f_i(\boldsymbol{\theta})$ is a polynomial in $\sum_{i=1}^{\ell-1} m_i(m_{i-1} + 1)$ variables and of degree at most $2^\ell - 1$ when restricted to $\boldsymbol{\theta} \in P$.

Secondly, note that $f_\ell^{(\ell)}$ and $f_{\ell+1}^{(\ell)}$ are obtained by composing a function of the form $\boldsymbol{\theta} \mapsto \mathbf{W}_{\ell+1}\sigma_{12}\mathbf{x} + \mathbf{b}_{\ell+1}$ with $f_\ell^{(\ell-1)}$. Since we've already established that $f_\ell^{(\ell-1)}$ is a polynomial in $\sum_{i=1}^{\ell-1} m_i(m_{i-1} + 1)$ variables, of degree at most $2^\ell - 1$, and of fixed sign as $\boldsymbol{\theta}$ varies in $P \in \mathcal{P}_\ell$, it follows that $f_\ell^{(\ell)}$ and $f_{\ell+1}^{(\ell)}$ are also polynomials in $\sum_{i=1}^\ell m_i(m_{i-1} + 1)$ variables and of degree at most $2^{\ell+1} - 1$, which implies that item 2 is satisfied.

Likewise, observe that in the last step, all the L first elements $f_0^{(L)}, \dots, f_{L-1}^{(L)}$ of $\mathbb{F}_L$ respectively correspond to last element of $\mathbb{F}_0, \dots, \mathbb{F}_{L-1}$, which as we've established for all $0 \leq \ell \leq L - 1$ are polynomials in $\sum_{i=1}^\ell m_i(m_{i-1} + 1)$ variables of degree at most $2^{\ell+1} - 1$ when restricted to $P \in \mathcal{P}_L$. The last two elements of $\mathbb{F}_L$ being obtained by composing functions of the form $\boldsymbol{\theta} \mapsto \mathbf{W}_{\ell+1}\sigma_{12}\mathbf{x} + \mathbf{b}_{\ell+1}$ with $f_L^{(L-1)}$, we conclude that item 3 is also satisfied, hence the partitions $\mathcal{P}_0, \dots, \mathcal{P}_L$ satisfy all the desired properties.

We are now ready to bound the VC dimension of ReQU network derivatives. First, note that by equation (3.37), $\partial_i \phi$ is exactly equal to the product of elements of all the elements of $\mathbb{F}_L$. This implies that for all $P \in \mathcal{P}_L$, $\partial_i \phi(\mathbf{x}, \boldsymbol{\theta})$ is a polynomial in $\sum_{i=1}^L m_i(m_{i-1} + 1)$ variables and of degree at most $\sum_{\ell=0}^L (2^\ell - 1) \leq 2^{L+2} - 1$ when restricted to $\boldsymbol{\theta} \in P$. Hence for all $P \in \mathcal{P}_L$, we have

$$K_L := \left| \left\{ (\text{sign}(\partial_i \phi(\mathbf{x}_1, \boldsymbol{\theta})), \dots, \text{sign}(\partial_i \phi(\mathbf{x}_M, \boldsymbol{\theta}))) \mid \boldsymbol{\theta} \in P \right\} \right| \leq 2 \left(\frac{2eMm_\ell(2^{L+2} - 1)}{\sum_{i=1}^L m_i(m_{i-1} + 1)} \right)^{\sum_{i=1}^L m_i(m_{i-1} + 1)}$$

On the other hand, we can recursively apply inequality (3.39) to find:

$$|\mathcal{P}_L| \leq \prod_{\ell=1}^L 2 \left(\frac{2eMm_\ell(2^{\ell+1} - 1)}{\sum_{i=1}^\ell m_i(m_{i-1} + 1)} \right)^{\sum_{i=1}^\ell m_i(m_{i-1} + 1)}.$$

It follows that if we denote

$$\Pi_{\partial_i \mathcal{F}}(M) := \sum_{P \in \mathcal{P}_L} \left| \left\{ (\text{sign}(\partial_i \phi(\mathbf{x}_1, \boldsymbol{\theta})), \dots, \text{sign}(\partial_i \phi(\mathbf{x}_M, \boldsymbol{\theta}))) \mid \boldsymbol{\theta} \in P \right\} \right|,$$

we get

$$\begin{aligned}
\Pi_{\partial_i \mathcal{F}}(M) &\leq K_L \prod_{\ell=1}^L 2 \left(\frac{2eMm_\ell(2^{\ell+1} - 1)}{\sum_{i=1}^\ell m_i(m_{i-1} + 1)} \right)^{\sum_{i=1}^\ell m_i(m_{i-1} + 1)} \\
&\leq \prod_{\ell=1}^{L+1} 2 \left(\frac{2eMm_\ell(2^{\ell+1} - 1)}{\sum_{i=1}^\ell m_i(m_{i-1} + 1)} \right)^{\sum_{i=1}^\ell m_i(m_{i-1} + 1)} \\
&\leq 2^{L+1} \left(\frac{2eM \sum_{\ell=1}^{L+1} m_\ell(2^{\ell+1} - 1)}{\sum_{\ell=1}^{L+1} \sum_{i=1}^\ell m_i(m_{i-1} + 1)} \right)^{\sum_{\ell=1}^{L+1} \sum_{i=1}^\ell m_i(m_{i-1} + 1)} \\
&\leq 2^{L+1} \left(\frac{4eM \|\mathbf{m}\|_\infty 2^{L+2}}{(L+1)\mathcal{S}} \right)^{(L+1)\mathcal{S}},
\end{aligned}$$

where we used the weighted AM-GM inequality in the second line. From the definition of the VC dimension, it follows that

$$2^{\text{VCdim}(\partial_i \mathcal{F})} = \Pi_{\partial_i \mathcal{F}}(\text{VCdim}(\partial_i \mathcal{F})) \leq 2^{L+1} \left(\frac{4e \text{VCdim}(\partial_i \mathcal{F}) \|\mathbf{m}\|_\infty 2^{L+2}}{(L+1)\mathcal{S}} \right)^{(L+1)\mathcal{S}}.$$

Hence, by Lemma 3.54, we conclude that

$$\begin{aligned}
\text{VCdim}(\partial_i \mathcal{F}) &\leq L + 1 + (L+1)\mathcal{S} \log_2 (2^{L+5} e \|\mathbf{m}\|_\infty \log_2 [2^{L+4} e \|\mathbf{m}\|_\infty]) \\
&= \mathcal{O}(L^2 \mathcal{S} \log_2 [L \|\mathbf{m}\|_\infty \log_2 \|\mathbf{m}\|_\infty]).
\end{aligned}$$

Second order derivatives

Again for convenience, denote $\sigma_2 : x \mapsto \max\{0, x\}^2$ the ReQU activation, $\sigma_1 : x \mapsto 2 \max\{0, x\}$ its derivative, and $\sigma_0 : x \mapsto 2 \text{sign}(x) = 2 \mathbb{1}_{x>0}$ its second derivative. Recall from Equation (3.37) that an element of $\partial_i \mathcal{F}$ is of the form

$$\partial_i \phi : \mathbf{x} \mapsto \prod_{\ell=0}^{L+1} f_\ell^{(L)}(\mathbf{x}),$$

where $f_\ell^{(L)}$ is the ℓ -th element of $\mathbb{F}_L$ as defined in (3.38). Therefore, a repeated application of the product rule yields that for all $1 \leq i, j \leq d$, an element of $\partial_{ij} \mathcal{F}$ is of the form

$$\partial_{ij} \phi : \mathbf{x} \mapsto \sum_{\ell=0}^{L+1} (\partial_j f_\ell^{(L)}(\mathbf{x})) \cdot \prod_{\substack{0 \leq k \leq L+1 \\ k \neq \ell}} f_k^{(L)}(\mathbf{x}), \quad (3.40)$$

where

$$\partial_j f_\ell^{(L)}(\mathbf{x}) = \begin{cases} 0, & \ell = 0, \\ \left[\prod_{k=1}^L \mathbf{W}_{k+1} \sigma_0(f_k^{k-1}(\mathbf{x})) \right] \cdot (\mathbf{W}_1)_j & 1 \leq \ell \leq L+1, \end{cases}$$

and again, $f_m^{(n)}$ is the m -th element of $\mathbb{F}_n$ as defined in (3.38). We will proceed similarly as before. Denote for all $1 \leq \ell \leq L+1$ $g_\ell(\mathbf{x}) := \partial_j f_\ell^{(L)}(\mathbf{x})$, which we notice is a polynomial of degree 1 in the parameters, due to σ_0 being (piecewise) constant, and define the associated function families

$$\mathbb{F}_k^{(\ell)} := \begin{cases} \mathbb{F}_k, & 0 \leq k \leq L-1, \\ \{g_\ell(\mathbf{x})\} \cup \mathbb{F}_k, & k = L. \end{cases} \quad (3.41)$$

Note that $\partial_{ij}\phi$ is equal to the sum for $1 \leq \ell \leq L+1$ of the products of the elements of $\mathbb{F}_\ell^{(\ell)}$. In particular, by choosing the sequence of partitions $\mathcal{P}_0, \mathcal{P}_1, \dots, \mathcal{P}_L$ which satisfy conditions 1, 2 and 3 exactly identical as in the previous subsection, we have that for all $P \in \mathcal{P}_L$, the restriction to $\boldsymbol{\theta} \in P$ of $\partial_{ij}\phi(\mathbf{x}, \boldsymbol{\theta})$ is a polynomial in $\sum_{i=1}^L m_i(m_{i-1}+1)$ variables and of degree at most $\sum_{\ell=0}^L 2^{\ell+1} - 1 \leq 2^{L+2} - 1$. We can thus reproduce the same calculations to deduce that

$$\text{VCdim}(\partial_{ij}\mathcal{F}) = \mathcal{O}(L^2 \mathcal{S} \log_2 [L \|\mathbf{m}\|_\infty \log_2 \|\mathbf{m}\|_\infty]).$$

By Lemma 3.52, the same VC-dimension bounds apply to boundary-enforced ReQU FCNNs, hence the proof is complete. $\square$

5.3 tanh FCNNs

We now turn our attention to the case of FCNNs with tanh activation. The hyperbolic tangent not being a piecewise polynomial, there is no way to extend the analysis in the last section to this case. We thus begin by introducing the class of so-called *Pfaffian functions* (Khovanskii, 1991), together with related results which will be crucial to carry our analysis.

VC dimension bounds for Pfaffian functions

Definition 3.56 (Pfaffian Functions). *A Pfaffian chain of order $\ell \geq 1$ and degree $\gamma \geq 1$, in an open domain $U \subseteq \mathbb{R}^d$, is a sequence of real-valued analytic functions $f_1, f_2, \dots, f_\ell$ over U , satisfying the differential equations*

$$\frac{\partial f_j(\mathbf{x})}{\partial x_i} = g_{ij}(\mathbf{x}, f_1(\mathbf{x}), \dots, f_j(\mathbf{x})) \quad 1 \leq j \leq \ell, \quad (3.42)$$

where the functions $g_{ij}(\mathbf{x}, y_1, \dots, y_j)$ are polynomials in $\mathbf{x} \in U$ and $y_1, \dots, y_j \in \mathbb{R}$ of degree at most γ . A function

$$f : \mathbf{x} \mapsto P\left(\mathbf{x}, f_1(\mathbf{x}), \dots, f_\ell(\mathbf{x})\right),$$

where $P(\mathbf{x}, y_1, \dots, y_\ell)$ is a polynomial of degree not exceeding β , is called a Pfaffian function of format (γ, β, ℓ) . We might alternatively say that f is Pfaffian of order ℓ and degree (γ, β) .

Pfaffian functions constitute an extremely broad class which includes most analytic functions one may encounter in practical applications (Khovanskiĭ, 1991). For instance, sigmoidal activation functions such as the arctangent $\arctan$, the logistic sigmoid sig , and the hyperbolic tangent $\tanh$ are Pfaffian functions, with

$$\text{format}(\arctan) = (3, 1, 2), \quad \text{format}(\text{sig}) = (2, 1, 1), \quad \text{format}(\tanh) = (2, 1, 1).$$

Pfaffian functions were first introduced in (Khovanskiĭ, 1991) to generalize Bézout’s theorem to bound the number of solutions to systems of polynomial equations. The theory of Pfaffian function has been applied to bounding the VC dimension of FCNNs using Pfaffian activation functions first in the seminal work (Karpinski and Macintyre, 1997), and more recently for other architectures such as Graph Neural Networks and Recurrent Neural Networks (D’ Inverno et al., 2025; Scarselli et al., 2018).

Following the framework proposed in (Karpinski and Macintyre, 1997) we see a classifier as an object described by a logical formula constructed from Pfaffian equations: concretely, for some integer $\bar{s} \geq 1$, we let $\phi_1, \dots, \phi_{\bar{s}}$ be a collection of C^∞ functions from $\mathbb{R}^{d+p}$ to $\mathbb{R}$, and we suppose that for each $\mathbf{x} \in \mathbb{R}^d, \boldsymbol{\theta} \in \mathbb{R}^p$, there is a logical expression $\Phi(\mathbf{x}, \boldsymbol{\theta})$ built using the operators *and* and *or* from simpler conditions (atomic predicates) of the form $\phi_i(\mathbf{x}, \boldsymbol{\theta}) = 0$ and $\phi_i(\mathbf{x}, \boldsymbol{\theta}) > 0$, which describes our classifier. In the case of a $\tanh$ FCNN, the decision rule is fully specified by the logical formula $\phi_1(\mathbf{x}, \boldsymbol{\theta}) \geq 0$, where we take $\bar{s} = 1$ and ϕ_1 is taken as the network’s output. With this formalism, the prediction is then given by 1 if the formula is true and 0 otherwise.

We then define the hypothesis space $\mathcal{F}_\Phi := \{\mathbf{x} \mapsto \Phi(\mathbf{x}, \boldsymbol{\theta}) \mid \boldsymbol{\theta} \in \mathbb{R}^p\}$ in the usual manner, and likewise its VC dimension $\text{VCdim}(\mathcal{F}_\Phi)$ as the maximum possible cardinality of a set shattered by $\mathcal{F}_\Phi$. As it turns out, $\text{VCdim}(\mathcal{F}_\Phi)$ can be bounded by studying the topological properties of the inverse image—in the parameter space—of the functions ϕ_i (Karpinski and Macintyre, 1997).

More precisely, for integers $1 \leq u, z \leq p$, and $\mathbf{x}_1, \dots, \mathbf{x}_z$ arbitrary points in $\mathbb{R}^d$, define the mapping $\mathbf{T} : \mathbb{R}^p \rightarrow \mathbb{R}^u$, $u \leq p$, by

$$\mathbf{T}(\boldsymbol{\theta}) = [\bar{\phi}_1(\boldsymbol{\theta}), \dots, \bar{\phi}_u(\boldsymbol{\theta})], \quad (3.43)$$

where each $\bar{\phi}_r(\boldsymbol{\theta})$ is obtained by evaluating one of the functions ϕ_i at one of the points $\mathbf{x}_j$. That is, for each $1 \leq r \leq u$, there exist indices i and j such that $\bar{\phi}_r(\boldsymbol{\theta}) = \phi_i(\mathbf{x}_j, \boldsymbol{\theta})$. Then, the following theorem holds:

Theorem 3.57 ((Karpinski and Macintyre, 1997), Theorem 2). *Assume that there exists a positive integer B which uniformly bounds the number of connected components of the inverse image $\mathbf{T}^{-1}([\epsilon_1, \dots, \epsilon_u])$, for any $1 \leq u, z \leq p$, any choice of points $\mathbf{x}_1, \dots, \mathbf{x}_z$, and any $[\epsilon_1, \dots, \epsilon_u]$ which is a regular value of $\mathbf{T}$. Then, the following VC dimension bound on $\mathcal{F}_\Phi$ holds:*

$$\text{VCdim}(\mathcal{F}_\Phi) \leq 2 \log B + p(16 + 2 \log \bar{s}).$$

Theorem 3.57 thus provides a bound on the VC dimension of $\mathcal{F}_\Phi$ in terms of the number p of parameters, the total number $\bar{s}$ of atomic functions ϕ_i , and the constant B bounding the number of connected components of $\mathbf{T}^{-1}([\epsilon_1, \dots, \epsilon_u])$. For this result to be useful however, one crucially needs a concrete estimate on the quantity B . This is where the following bound for systems of Pfaffian equations comes into play:

Theorem 3.58 ((Gabrielov and Vorobjov, 2004), Corollary 3.3). *Consider a system of equations*

$$\bar{q}_1(\boldsymbol{\theta}) = 0, \quad \dots, \quad \bar{q}_k(\boldsymbol{\theta}) = 0,$$

where each $\bar{q}_i$, for $1 \leq i \leq k$, is a Pfaffian function defined on a domain $G \subseteq \mathbb{R}^p$. Assume that these functions share a common Pfaffian chain of length ℓ and have maximum degrees (γ, β) . Then, the number of connected components of the set

$$\{\boldsymbol{\theta} \in G \mid \bar{q}_1(\boldsymbol{\theta}) = 0, \dots, \bar{q}_k(\boldsymbol{\theta}) = 0\}$$

is bounded by

$$2^{\frac{\ell(\ell-1)}{2}+1}(\gamma + 2\beta - 1)^{p-1} \left((2p-1)(\gamma + \beta) - 2p + 2 \right)^\ell.$$

Theorem 3.58 is in fact a Corollary of an earlier result from (Khovanskii, 1991). Combining Theorems 3.57 and 3.58, together with knowledge on the Pfaffian format of the tanh function, we can deduce the following VC dimension bounds for tanh FCNNs:

Proposition 3.59 ((Karpinski and Macintyre, 1997), Theorem 7). *Let $\sigma : x \mapsto \tanh(x)$ be the tanh activation function, and denote by $\mathcal{F} = \Sigma_{\mathbf{m}_{1:L}}^\sigma(\infty)$ the hypothesis space of tanh FCNNs with no restriction on their weights, as per Definition 3.5. We have the estimate*

$$\text{VCdim}(\mathcal{F}) = \mathcal{O}(L^2 \|\mathbf{m}\|_\infty^2 \mathcal{S}^2),$$

where $\mathcal{S} = \sum_{i=1}^L m_i(m_{i-1} + 1)$ is the number of parameters in the architecture.

Because it will be a helpful reference for later, we give a proof of Proposition 3.59 below, which is merely a modern and slightly more fleshed-out presentation of the original argument in (Karpinski and Macintyre, 1997).

Proof. Let $\phi \in \mathcal{F}$. We start by building a Pfaffian chain to which it belongs. To do so, we proceed iteratively, layer by layer: in the first layer, we have m_1 neurons, hence for $1 \leq i \leq m_1$, let

$$g_i^{(1)}(\mathbf{x}, \boldsymbol{\theta}) := (\mathbf{W}_1 \mathbf{x})_i + (\mathbf{b}_1)_i, \quad h_i^{(1)}(\mathbf{x}, \boldsymbol{\theta}) := \sigma(g_i^{(1)}(\mathbf{x}, \boldsymbol{\theta})),$$

where $(\mathbf{W})_i$ is the i -th column of $\mathbf{W}$, and $(\mathbf{b}_1)_i$ the i -th coordinate of $\mathbf{b}$. Now, for the next step, let $2 \leq \ell \leq L$ and assume that we have already defined $g_i^{(k)}$ and $h_i^{(k)}$ for all $1 \leq i \leq m_k$ and $1 \leq k \leq \ell - 1$, and let for all $1 \leq i \leq m_\ell$:

$$g_i^{(\ell)}(\mathbf{x}, \boldsymbol{\theta}) := \sum_{j=1}^{m_{\ell-1}} (\mathbf{W}_\ell)_{ij} h_j^{(\ell-1)}(\mathbf{x}, \boldsymbol{\theta}) + (\mathbf{b}_\ell)_i, \quad h_i^{(\ell)}(\mathbf{x}, \boldsymbol{\theta}) := \sigma(g_i^{(\ell)}(\mathbf{x}, \boldsymbol{\theta})). \quad (3.44)$$

Fix $\phi \in \mathcal{F}$ and consider it as a real-analytic function on $U := \mathbb{R}^{d+p}$ with variables $(\mathbf{x}, \boldsymbol{\theta})$. Define the ordered list of functions

$$\mathcal{C}(\mathbf{x}, \boldsymbol{\theta}) := \left(g_i^{(\ell)}(\mathbf{x}, \boldsymbol{\theta}), h_i^{(\ell)}(\mathbf{x}, \boldsymbol{\theta}) \right)_{1 \leq \ell \leq L, 1 \leq i \leq m_\ell},$$

where $h_i^{(\ell)} = \sigma(g_i^{(\ell)})$ and the $g_i^{(\ell)}$ are given by (3.44). Its length is

$$\ell_0 := 2 \sum_{\ell=1}^L m_\ell \leq 2L \|\mathbf{m}\|_\infty.$$

We claim that $\mathcal{C}$ is a Pfaffian chain on U of degree $\gamma := 3L$. In particular, the network output ϕ is a Pfaffian function of format (γ, β, ℓ_0) with $\beta = 1$. Indeed, let v be any coordinate among $(x_1, \dots, x_d, \theta_1, \dots, \theta_p)$. For $\ell = 1$, $g_i^{(1)}(\mathbf{x}, \boldsymbol{\theta}) = (\mathbf{W}_1 \mathbf{x})_i + (\mathbf{b}_1)_i$ is affine in $(\mathbf{x}, \boldsymbol{\theta})$, hence $\partial_v g_i^{(1)}$ is a polynomial of degree at most 1 in $(\mathbf{x}, \boldsymbol{\theta})$. Moreover,

$$\partial_v h_i^{(1)} = (1 - (h_i^{(1)})^2) \partial_v g_i^{(1)},$$

so $\partial_v h_i^{(1)}$ is a polynomial in $(\mathbf{x}, \boldsymbol{\theta}, h_i^{(1)})$ of degree at most 3. Now, assume inductively that for all layers $1 \leq k \leq \ell - 1$ the derivatives $\partial_v h_j^{(k)}$ are polynomials in $(\mathbf{x}, \boldsymbol{\theta})$ and earlier chain elements of degree at most $3k$. Since

$$g_i^{(\ell)} = \sum_{j=1}^{m_{\ell-1}} (W_{\ell})_{ij} h_j^{(\ell-1)} + (b_{\ell})_i,$$

we have

$$\partial_v g_i^{(\ell)} = \sum_{j=1}^{m_{\ell-1}} (\partial_v (W_{\ell})_{ij}) h_j^{(\ell-1)} + \sum_{j=1}^{m_{\ell-1}} (W_{\ell})_{ij} \partial_v h_j^{(\ell-1)} + \partial_v (b_{\ell})_i.$$

Here $\partial_v (W_{\ell})_{ij}$ and $\partial_v (b_{\ell})_i$ are either 0 or 1. Thus $\partial_v g_i^{(\ell)}$ is a polynomial in $(\mathbf{x}, \boldsymbol{\theta})$ and earlier chain elements of degree at most $1 + 3(\ell - 1) = 3\ell - 2$. Finally,

$$\partial_v h_i^{(\ell)} = (1 - (h_i^{(\ell)})^2) \partial_v g_i^{(\ell)},$$

so $\partial_v h_i^{(\ell)}$ is polynomial in $(\mathbf{x}, \boldsymbol{\theta})$ and chain elements up to $h_i^{(\ell)}$ of degree at most $(3\ell - 2) + 2 = 3\ell$. This shows that the chain is of degree $\gamma = 3L$, as claimed.

We now bound, uniformly over $1 \leq u, z \leq p$ (with $p = \mathcal{S}$), the number of connected components of $\mathbf{T}^{-1}([\epsilon_1, \dots, \epsilon_u])$ for regular values, where $\mathbf{T}$ is defined in (3.43). Fix z points $\mathbf{x}_1, \dots, \mathbf{x}_z$ and define $\bar{\phi}_r(\boldsymbol{\theta}) = \phi(\mathbf{x}_{j_r}, \boldsymbol{\theta})$ for $r = 1, \dots, u$. As we've just established, $(\mathbf{x}, \boldsymbol{\theta}) \mapsto \phi(\mathbf{x}, \boldsymbol{\theta})$ is Pfaffian on $\mathbb{R}^{d+p}$ with format $(\gamma, 1, \ell_0)$. Freezing $\mathbf{x} = \mathbf{x}_j$ preserves the Pfaffian structure, so each $\bar{\phi}_r : \mathbb{R}^p \rightarrow \mathbb{R}$ is Pfaffian of format $(\gamma, 1, \ell_0)$. Concatenating the z chains corresponding to the z evaluation points yields a common Pfaffian chain for $\bar{\phi}_1, \dots, \bar{\phi}_u$ of length

$$\ell := z \ell_0 \leq 2z \sum_{\ell'=1}^L m_{\ell'} \leq 2pL \|\mathbf{m}\|_{\infty}.$$

Since translating by constants does not change the format, we may apply Theorem 3.58 to the system $\bar{\phi}_1(\boldsymbol{\theta}) - \epsilon_1 = 0, \dots, \bar{\phi}_u(\boldsymbol{\theta}) - \epsilon_u = 0$ to obtain a bound

$$B(z) \leq 2^{\frac{\ell(\ell-1)}{2}+1} (\gamma + 1)^{p-1} \left((2p-1)(\gamma + 1) - 2p + 2 \right)^{\ell}.$$

The right-hand side is nondecreasing in ℓ , hence in z , and it does not depend on the particular choice of points nor on the regular value. Therefore, taking $z = p$ yields a single constant $B := B(p)$ that satisfies the uniformity requirement in Theorem 3.57 for all $1 \leq u, z \leq p$. Applying Theorem 3.57 with $\bar{s} = 1$ concludes the proof. $\square$

Higher order derivatives

We now extend the above argument to mixed derivatives of arbitrary order. Let $r \geq 1$ and $\alpha \in \mathbb{N}^d$ be a multi-index with $\|\alpha\|_1 \leq r$. Recall the layerwise quantities $g_i^{(\ell)}(\mathbf{x}, \boldsymbol{\theta})$ and $h_i^{(\ell)}(\mathbf{x}, \boldsymbol{\theta}) = \sigma(g_i^{(\ell)}(\mathbf{x}, \boldsymbol{\theta}))$ defined by (3.44), where $\sigma = \tanh$. We begin by a simple lemma showing that Pfaffian functions are closed under differentiation with explicit control on their derivatives' format.

Lemma 3.60. *Let f be Pfaffian on an open set $U \subseteq \mathbb{R}^n$ with format (γ, β, ℓ) , given by a Pfaffian chain $f_1, \dots, f_\ell$ of degree γ and a polynomial representation $f(\mathbf{x}) = P(\mathbf{x}, f_1(\mathbf{x}), \dots, f_\ell(\mathbf{x}))$ with $\deg P \leq \beta$. Then for each coordinate x_i , the partial derivative $\partial_{x_i} f$ is Pfaffian with respect to the same chain and has format $(\gamma, \gamma + \beta - 1, \ell)$.*

Proof. Follows by differentiating $P(\mathbf{x}, f_1(\mathbf{x}), \dots, f_\ell(\mathbf{x}))$ and noting that each $\partial_{x_i} f_j$ is a polynomial in $(\mathbf{x}, f_1, \dots, f_j)$ of degree at most γ . $\square$

Lemma 3.60 together with the analysis developed in the preceding section yields the following proposition:

Proposition 3.61 (Input-derivatives of tanh FCNNs are Pfaffian). *Fix $r \geq 0$ and a multi-index $\alpha \in \mathbb{N}^d$ with $\|\alpha\|_1 \leq r$. Let $\phi \in \mathcal{F}$ and view $(\mathbf{x}, \boldsymbol{\theta}) \mapsto \phi(\mathbf{x}, \boldsymbol{\theta})$ as an analytic function on $\mathbb{R}^{d+p}$. Then $(\mathbf{x}, \boldsymbol{\theta}) \mapsto \partial_{\mathbf{x}}^\alpha \phi(\mathbf{x}, \boldsymbol{\theta})$ is Pfaffian on $\mathbb{R}^{d+p}$ with respect to the same chain $\mathcal{C}$, of order $\ell_0 \leq 2L\|\mathbf{m}\|_\infty$ and degrees (γ, β_r) , where*

$$\gamma := 3L, \text{ and } \beta_r := 1 + r(\gamma - 1).$$

Proof. By repeating the argument used to prove Proposition 3.59, ϕ is Pfaffian of format $(\gamma, 1, \ell_0)$. Applying Lemma 3.60 iteratively $\|\alpha\|_1$ times with respect to the input variables then yields the claim. $\square$

We are now ready to prove the claimed VC-dimension bound: fix $\alpha \in \mathbb{N}^d$ with $\|\alpha\|_1 \leq r$ and consider $\partial_{\mathbf{x}}^\alpha \mathcal{F} := \{\mathbf{x} \mapsto \partial_{\mathbf{x}}^\alpha \phi(\mathbf{x}, \boldsymbol{\theta}) : \boldsymbol{\theta} \in \mathbb{R}^p\}$. We repeat the proof of Proposition 3.59 with $\bar{\phi}_r(\boldsymbol{\theta}) := (\partial_{\mathbf{x}}^\alpha \phi)(\mathbf{x}_j, \boldsymbol{\theta})$. By Proposition 3.61, each $\bar{\phi}_r$ is Pfaffian with degrees (γ, β_r) , and after concatenation over $z \leq p$ sample points we obtain a common chain of length $\ell \leq 2pL\|\mathbf{m}\|_\infty$. Applying Theorems 3.58 and 3.57 thus yields

$$\text{VCdim}(\partial_{\mathbf{x}}^\alpha \mathcal{F}) = \mathcal{O}(L^2 \|\mathbf{m}\|_\infty^2 \mathcal{S}^2),$$

where the dependence on r only enters through β_r inside logarithmic factors. By Lemma 3.52, the same bound holds for the boundary-enforced tanh networks as well, as desired. $\square$

6 Proof of oracle inequality

As mentioned earlier, this proof follows for the most part the arguments given in (Lei et al., 2025) and is mostly provided for the sake of making this thesis self-contained. We will therefore at times refer to results established in Chapter 2 and omit some straightforward details. Throughout this section, P denotes the uniform distribution on Ω , and $X, X_1, \dots, X_n$ are i.i.d. with law P . Recall that for $(x, u) \in \Omega \times H^2(\Omega)$ we defined

$$L(x, u) := |\mathcal{L}u(x) + 1|^2, \quad (3.45)$$

so that the true and empirical residuals are

$$\mathcal{R}(u) = \mathbb{E}_P[L(X, u)], \text{ and } \mathcal{R}_n(u) = \frac{1}{n} \sum_{i=1}^n L(X_i, u),$$

in agreement with (3.13) and (3.14). We write

$$g_u(x) := L(x, u) - L(x, \tau) = |\mathcal{L}(u - \tau)(x)|^2,$$

so that

$$\mathcal{R}(u) - \mathcal{R}(\tau) = \mathbb{E}_P[g_u(X)].$$

We will also make repeated use of the Agmon-Douglis-Nirenberg estimate (3.8): there exists $C_{11} > 0$ such that for all $v \in H^2(\Omega)$ with $v|_{\partial\Omega} = 0$,

$$\|v\|_{H^2(\Omega)}^2 \leq C_{11} \|\mathcal{L}v\|_{L^2(\Omega)}^2. \quad (3.46)$$

Since τ solves (3.3) with homogeneous Dirichlet condition and every $u \in \mathcal{F}$ is of the form $u = \rho f$ with the same boundary factor ρ that vanishes on $\partial\Omega$, we have $u - \tau \in H^2(\Omega) \cap H_0^1(\Omega)$ for all $u \in \mathcal{F}$, and (3.46) applies to $v = u - \tau$.

6.1 Bernstein's bound

We begin by a lemma which will allow us to control the second summand in the error decomposition (3.15).

Lemma 3.62. *Let $\mathcal{F} \subset W^{2,\infty}(\Omega)$ be a family of functions such that $\|u\|_{W^{2,\infty}(\Omega)} \leq Q$ for some $Q > 0$ and all $u \in \mathcal{F}$, and assume that Assumptions (A.1), (A.2), (A.3) hold, so that τ is well defined as the unique solution of (3.3). Then for all $u \in \mathcal{F} \cup \{\tau\}$, we have*

$$\mathbb{E}_P [(L(X, u) - L(X, \tau))^2] \leq C_0 \mathbb{E}_P [L(X, u) - L(X, \tau)] = C_0 (\mathcal{R}(u) - \mathcal{R}(\tau)), \quad (3.47)$$

where P is the uniform distribution on Ω ,

$$C_0 = \left[1 + Q \left(\sum_{i=1}^d \|b_i\|_{L^\infty(\Omega)} + \sum_{i,j=1}^d \|\Sigma_{ij}\|_{L^\infty(\Omega)}/2 \right) \right]^2,$$

and for all $(x, u) \in \Omega \times H^2(\Omega)$, we denote

$$L(x, u) := |\mathcal{L}u(x) + 1|^2.$$

Furthermore, again reusing notation from (3.15), we have for all $t > 0$ that with probability greater than $1 - e^{-t}$:

$$\mathcal{R}_n(u_{\mathcal{F}}) - \mathcal{R}_n(\tau) - \mathcal{R}(u_{\mathcal{F}}) + \mathcal{R}(\tau) \leq \mathcal{R}(u_{\mathcal{F}}) - \mathcal{R}(\tau) + \frac{7C_0 t}{n}. \quad (3.48)$$

Proof. First, observe that (3.47) holds trivially for $u = \tau$, and that since τ is the solution of (3.3), $L(x, \tau) = 0$ for all $x \in \Omega$. Hence for all $u \in \mathcal{F}$, we have:

$$\begin{aligned} \mathbb{E}_P [(L(X, u) - L(X, \tau))^2] &= \mathbb{E}_P [(L(X, u))^2] = \frac{1}{|\Omega|} \int_{\Omega} |\mathcal{L}u(x) + 1|^4 \\ &\leq \frac{C_0}{|\Omega|} \int_{\Omega} |\mathcal{L}u(x) + 1|^2 \\ &= C_0(\mathcal{R}(u) - \mathcal{R}(\tau)). \end{aligned}$$

This proves inequality (3.47), which together with Bernstein's inequality (Boucheron et al., 2013, Theorem 2.10) yields that for all $t > 0$, with probability at least $1 - e^{-t}$:

$$\begin{aligned} \mathcal{R}_n(u_{\mathcal{F}}) - \mathcal{R}_n(\tau) - \mathcal{R}(u_{\mathcal{F}}) + \mathcal{R}(\tau) &\leq \sqrt{\frac{2C_0 \mathbb{E}_P [L(X, u_{\mathcal{F}}) - L(X, \tau)] t}{n}} + \frac{2C_0 t}{3n} \\ &\leq \mathbb{E}_P [L(X, u_{\mathcal{F}}) - L(X, \tau)] + \frac{7C_0 t}{n} \\ &= \mathcal{R}(u_{\mathcal{F}}) - \mathcal{R}(\tau) + \frac{7C_0 t}{n}, \end{aligned}$$

where we used Young's product inequality. The proof is thus complete. $\square$

6.2 Uniform $W^{2,\infty}$ -bounds and effect of multiplication ρ

Recall from Definition 3.7 that

$$\mathcal{F} \equiv \mathcal{F}(\rho, L, \mathbf{m}, \sigma, Q_{\text{net}}, B, M) = \left\{ \Omega \ni x \mapsto u(x) := \rho(x)f(x) \mid f \in \Sigma_{\mathbf{m}_{1:L}, M}^\sigma(B), \max_{\|\alpha\|_1 \leq 2} \|\partial^\alpha f\|_{L^\infty(\Omega)} \leq Q_{\text{net}} \right\},$$

where $\rho : \bar{\Omega} \rightarrow \mathbb{R}$ is a fixed $C^2(\bar{\Omega})$ function.

The oracle inequality of Theorem 3.23 is formulated in terms of a uniform $W^{2,\infty}$ -bound on the hypothesis class $\mathcal{F}$ and on τ . We now explicit this by quantifying the effect of multiplying by ρ .

Lemma 3.63 (Simple product-rule bound for $u = \rho f$). *Let $\rho \in C^2(\bar{\Omega})$ and define*

$$\|\rho\|_{2,\infty} := \max_{\alpha \in \mathbb{N}^d: \|\alpha\|_1 \leq 2} \|\partial^\alpha \rho\|_{L^\infty(\Omega)}.$$

Then for every $f \in W^{2,\infty}(\Omega)$ and $u = \rho f$ we have

$$\|u\|_{W^{2,\infty}(\Omega)} \leq 4 \|\rho\|_{2,\infty} \|f\|_{W^{2,\infty}(\Omega)}. \quad (3.49)$$

Proof. Straightforward application of Leibniz rule. $\square$

As a consequence, if $\max_{\|\alpha\|_1 \leq 2} \|\partial^\alpha f\|_\infty \leq Q_{\text{net}}$ for all f in the underlying FCNN class, then for all $u = \rho f \in \mathcal{F}$ we have

$$\|u\|_{W^{2,\infty}(\Omega)} \leq Q_{\text{class}}, \text{ and } Q_{\text{class}} := 4 \|\rho\|_{2,\infty} Q_{\text{net}}.$$

Since $\tau \in C^{r+2,\kappa}(\bar{\Omega}) \subset W^{2,\infty}(\Omega)$ by Assumptions (A.1)–(A.3), we set

$$Q := \max \left\{ Q_{\text{class}}, \|\tau\|_{W^{2,\infty}(\Omega)} \right\}. \quad (3.50)$$

In what follows, all bounds stated “for $u \in \mathcal{F} \cup \{\tau\}$ with $\|u\|_{W^{2,\infty}} \leq Q$ ” refer to this Q , which depends only on ρ , Q_{net} , and the PDE coefficients, but not on n or t .

Remark 3.64 (Effect of the boundary factor on complexity). For each multi-index α with $\|\alpha\|_1 \leq 2$, the map $u = \rho f \mapsto \partial^\alpha u$ can be written, by the product rule, as a linear combination of derivatives $\partial^\beta f$ with $\|\beta\|_1 \leq 2$ and fixed coefficient functions built from derivatives of ρ . On any finite sample $\{x_1, \dots, x_n\} \subset \Omega$ this is a linear transformation of the vectors $(\partial^\beta f(x_i))_{i,\beta}$. Therefore the pseudo-dimensions and VC-dimensions of the derivative classes $\partial^\alpha \mathcal{F}$ differ from those of the corresponding derivative classes of the underlying FCNNs by at most absolute multiplicative constants depending on ρ and d . The global quantity

$$\text{VC}_{\mathcal{F}} := \max_{\|\alpha\|_1 \leq 2} \text{VCdim}(\partial^\alpha \mathcal{F})$$

used in Theorem 3.23 is defined for the actual boundary-enforced class $\mathcal{F}$ and therefore already absorbs this ρ -dependence. In particular, Theorem 3.18 applies directly to $\mathcal{F}$ as defined above.

6.3 Lipschitz property of the loss and a contraction lemma

We first control the sensitivity of $L(x, u)$ with respect to the derivatives of u . Recall that, by (3.50), we have $\|u\|_{W^{2,\infty}(\Omega)} \leq Q$ for all $u \in \mathcal{F} \cup \{\tau\}$.

Lemma 3.65 (Lipschitz continuity of the loss). *Assume that $\mathcal{F} \subset W^{2,\infty}(\Omega)$ satisfies $\|u\|_{W^{2,\infty}(\Omega)} \leq Q$ for all $u \in \mathcal{F} \cup \{\tau\}$, with Q as in (3.50). Then there exists a constant $C_{\text{Lip}} > 0$, depending only on Q , d , and the coefficients $(b_i)_{1 \leq i \leq d}$, $(\Sigma_{ij})_{1 \leq i,j \leq d}$ of $\mathcal{L}$, such that for all $x \in \Omega$ and $u_1, u_2 \in \mathcal{F} \cup \{\tau\}$,*

$$|L(x, u_1) - L(x, u_2)| \leq C_{\text{Lip}} \sum_{\alpha \in \mathbb{N}^d: \|\alpha\|_1 \leq 2} |\partial^\alpha u_1(x) - \partial^\alpha u_2(x)|. \quad (3.51)$$

Proof. For $u \in H^2(\Omega)$ we have

$$\mathcal{L}u(x) = \sum_{i=1}^d b_i(x) \partial_i u(x) + \frac{1}{2} \sum_{i,j=1}^d \Sigma_{ij}(x) \partial_{ij}^2 u(x).$$

Write

$$B_1 := \sum_{i=1}^d \|b_i\|_{L^\infty(\Omega)}, \quad B_2 := \frac{1}{2} \sum_{i,j=1}^d \|\Sigma_{ij}\|_{L^\infty(\Omega)}.$$

Then for $u \in \mathcal{F} \cup \{\tau\}$ and $x \in \Omega$,

$$|\mathcal{L}u(x)| \leq B_1 \max_{1 \leq i \leq d} |\partial_i u(x)| + B_2 \max_{1 \leq i,j \leq d} |\partial_{ij}^2 u(x)| \leq (B_1 + B_2) Q,$$

hence

$$|\mathcal{L}u_1(x) + \mathcal{L}u_2(x) + 2| \leq 2(B_1 + B_2)Q + 2 =: C_1.$$

Moreover,

$$\begin{aligned} |\mathcal{L}u_1(x) - \mathcal{L}u_2(x)| &\leq \sum_{i=1}^d \|b_i\|_\infty |\partial_i u_1(x) - \partial_i u_2(x)| + \frac{1}{2} \sum_{i,j=1}^d \|\Sigma_{ij}\|_\infty |\partial_{ij}^2 u_1(x) - \partial_{ij}^2 u_2(x)| \\ &\leq C_2 \sum_{\|\alpha\|_1 \leq 2} |\partial^\alpha u_1(x) - \partial^\alpha u_2(x)| \end{aligned}$$

for a suitable constant $C_2 > 0$ depending only on the coefficient bounds and d . Since

$$L(x, u) = |\mathcal{L}u(x) + 1|^2,$$

we have

$$L(x, u_1) - L(x, u_2) = (\mathcal{L}u_1(x) + \mathcal{L}u_2(x) + 2)(\mathcal{L}u_1(x) - \mathcal{L}u_2(x)),$$

and therefore

$$|L(x, u_1) - L(x, u_2)| \leq C_1 C_2 \sum_{\|\alpha\|_1 \leq 2} |\partial^\alpha u_1(x) - \partial^\alpha u_2(x)|.$$

The claim follows with $C_{\text{Lip}} := C_1 C_2$. $\square$

Now define the centered hypothesis spaces

$$\tilde{\mathcal{F}} := \mathcal{F} - \tau := \{u - \tau : u \in \mathcal{F}\}, \text{ and } \partial^\alpha \tilde{\mathcal{F}} := \{\partial^\alpha(u - \tau) : u \in \mathcal{F}\}.$$

We next use the vector-valued contraction Lemma 2.32 from Chapter 2 to relate the empirical Rademacher complexity of the loss class to that of the derivative classes.

Lemma 3.66 (Contraction for the loss). *Let $\mathbf{X} = (X_1, \dots, X_n)$ be a sample from P . With C_{Lip} as in Lemma 3.65 we have*

$$\text{Rad}(L \circ \mathcal{F}, \mathbf{X}) \leq C_{\text{Lip}} \sum_{\|\alpha\|_1 \leq 2} \text{Rad}(\partial^\alpha \tilde{\mathcal{F}}, \mathbf{X}).$$

Proof. For $u_1, u_2 \in \mathcal{F} \cup \{\tau\}$ we can rewrite (3.51) as

$$|L(x, u_1) - L(x, u_2)| \leq C_{\text{Lip}} \sum_{\|\alpha\|_1 \leq 2} |\partial^\alpha(u_1 - \tau)(x) - \partial^\alpha(u_2 - \tau)(x)|.$$

Lemma 2.32 applied to the map

$$(g_\alpha)_{\|\alpha\|_1 \leq 2} \mapsto \frac{1}{n} \sum_{i=1}^n \varepsilon_i (L(X_i, \tau + g)(X_i) - L(X_i, \tau)),$$

where $g(x) = u(x) - \tau(x)$ and $g_\alpha(x) = \partial^\alpha g(x)$, yields

$$\begin{aligned} \text{Rad}(L \circ \mathcal{F}, \mathbf{X}) &\leq C_{\text{Lip}} \text{Rad}\left(\sum_{\|\alpha\|_1 \leq 2} \partial^\alpha \tilde{\mathcal{F}}, \mathbf{X}\right) \\ &\leq C_{\text{Lip}} \sum_{\|\alpha\|_1 \leq 2} \text{Rad}(\partial^\alpha \tilde{\mathcal{F}}, \mathbf{X}), \end{aligned}$$

as claimed. $\square$

6.4 Localized complexity analysis and peeling

Still following (Lei et al., 2025), for $\gamma > 0$, define the localized classes:

$$\begin{aligned}\mathcal{F}_\gamma &:= \{u \in \mathcal{F} : \mathbb{E}_P g_u(X) = \mathcal{R}(u) - \mathcal{R}(\tau) \leq \gamma\}, \\ \tilde{\mathcal{F}}_\gamma &:= \{u - \tau : u \in \mathcal{F}_\gamma\}, \\ \mathcal{G}_\gamma &:= \{g_u : u \in \mathcal{F}_\gamma\}, \\ \partial^\alpha \tilde{\mathcal{F}}_\gamma &:= \{\partial^\alpha u - \partial^\alpha \tau : u \in \mathcal{F}_\gamma\}, \text{ for all } \|\alpha\|_1 \leq 2.\end{aligned}$$

Given a sample $\mathbf{X} = (X_1, \dots, X_n)$ we define, for a function class $\mathcal{G}$ and $g \in \mathcal{G}$,

$$\|g\|_{L^2(\mathbf{X})}^2 := \frac{1}{n} \sum_{i=1}^n g(X_i)^2, \text{ and } \mathbb{E}_{\mathbf{X}} g := \frac{1}{n} \sum_{i=1}^n g(X_i).$$

The following arguments will rely heavily on results introduced in Chapter 2. We particularly refer the reader to Section 5 where all such results are discussed at length.

We will also use the Bernstein bound for the second moment of the excess loss given in Lemma 3.62. That is, of the fact that for all $u \in \mathcal{F} \cup \{\tau\}$,

$$\mathbb{E}_P[(L(X, u) - L(X, \tau))^2] \leq C_0 \mathbb{E}_P[L(X, u) - L(X, \tau)] = C_0(\mathcal{R}(u) - \mathcal{R}(\tau)), \quad (3.52)$$

with $C_0 > 0$ depending only on Q , on the coefficients of $\mathcal{L}$, and on ρ through Q .

We next show that on a risk level set $\mathcal{F}_\gamma$, the empirical L^2 -norms of the derivatives of $u - \tau$ are controlled by $\sqrt{\gamma}$.

For notational convenience define

$$N_{\mathcal{L}} := \#\left\{\alpha \in \mathbb{N}^d : \|\alpha\|_1 \leq 2\right\} = 1 + d + \frac{d(d+1)}{2}.$$

Lemma 3.67. *Let $t > 0$ and suppose*

$$\gamma \geq \max_{\|\alpha\|_1 \leq 2} \left\{ 12Q \operatorname{Rad}(\partial^\alpha \tilde{\mathcal{F}}_\gamma, \mathbf{X}), \frac{8C_{11}Q^2t}{n}, \frac{16Q^2t}{3n} \right\}.$$

Then with probability at least $1 - N_{\mathcal{L}}e^{-t}$,

$$\max_{\|\alpha\|_1 \leq 2} \sup_{u \in \mathcal{F}_\gamma} \|\partial^\alpha u - \partial^\alpha \tau\|_{L^2(\mathbf{X})} \leq \sqrt{(C_{11} + 3)\gamma}. \quad (3.53)$$

Proof. Fix a multi-index α with $\|\alpha\|_1 \leq 2$ and consider

$$\partial^\alpha \tilde{\mathcal{F}}_\gamma^2 := \{g^2 : g \in \partial^\alpha \tilde{\mathcal{F}}_\gamma\} = \{(\partial^\alpha u - \partial^\alpha \tau)^2 : u \in \mathcal{F}_\gamma\}.$$

By the $W^{2,\infty}$ -bound $\|u\|_{W^{2,\infty}} \leq Q$ for $u \in \mathcal{F}$ and τ , we have $\|\partial^\alpha u - \partial^\alpha \tau\|_\infty \leq 2Q$ and hence

$$\|(\partial^\alpha u - \partial^\alpha \tau)^2\|_\infty \leq (2Q)^2 = 4Q^2.$$

Moreover,

$$\begin{aligned} \sum_{\|\beta\|_1 \leq 2} \mathbb{E}_P[(\partial^\beta u - \partial^\beta \tau)^2] &= \|u - \tau\|_{H^2(\Omega)}^2 \\ &\leq C_{11} \|\mathcal{L}(u - \tau)\|_{L^2(\Omega)}^2 = C_{11} \mathbb{E}_P[g_u(X)] \leq C_{11} \gamma, \end{aligned}$$

so in particular

$$\mathbb{E}_P[(\partial^\alpha u - \partial^\alpha \tau)^2] \leq C_{11} \gamma.$$

Therefore

$$\text{Var}((\partial^\alpha u - \partial^\alpha \tau)^2) \leq \mathbb{E}_P[(\partial^\alpha u - \partial^\alpha \tau)^4] \leq 4Q^2 \mathbb{E}_P[(\partial^\alpha u - \partial^\alpha \tau)^2] \leq 4C_{11} Q^2 \gamma.$$

Apply Lemma 2.30 to the class $\mathcal{G} := \partial^\alpha \tilde{\mathcal{F}}_\gamma^2$. With $G = 4Q^2$ and $V = 4C_{11} Q^2 \gamma$, we obtain, with probability at least $1 - e^{-t}$,

$$\begin{aligned} \sup_{u \in \mathcal{F}_\gamma} (\|\partial^\alpha u - \partial^\alpha \tau\|_{L^2(\mathbf{X})}^2 - \mathbb{E}_P[(\partial^\alpha u - \partial^\alpha \tau)^2]) \\ \leq 3 \text{Rad}(\partial^\alpha \tilde{\mathcal{F}}_\gamma^2, \mathbf{X}) + \sqrt{\frac{8C_{11} Q^2 \gamma t}{n}} + \frac{16Q^2 t}{3n}. \end{aligned}$$

Since $|\partial^\alpha u - \partial^\alpha \tau| \leq 2Q$ and the map $z \mapsto z^2$ is $4Q$ -Lipschitz on $[-2Q, 2Q]$, a contraction argument gives

$$\text{Rad}(\partial^\alpha \tilde{\mathcal{F}}_\gamma^2, \mathbf{X}) \leq 4Q \text{Rad}(\partial^\alpha \tilde{\mathcal{F}}_\gamma, \mathbf{X}).$$

By the assumed lower bound on γ we have

$$3 \text{Rad}(\partial^\alpha \tilde{\mathcal{F}}_\gamma^2, \mathbf{X}) \leq \gamma, \quad \sqrt{\frac{8C_{11} Q^2 \gamma t}{n}} \leq \gamma, \quad \text{and} \quad \frac{16Q^2 t}{3n} \leq \gamma.$$

Hence, with probability at least $1 - e^{-t}$,

$$\begin{aligned} \sup_{u \in \mathcal{F}_\gamma} \|\partial^\alpha u - \partial^\alpha \tau\|_{L^2(\mathbf{X})}^2 &\leq \sup_{u \in \mathcal{F}_\gamma} \mathbb{E}_P[(\partial^\alpha u - \partial^\alpha \tau)^2] + 3\gamma \\ &\leq C_{11} \gamma + 3\gamma = (C_{11} + 3)\gamma. \end{aligned}$$

A union bound over all multi-indices with $\|\alpha\|_1 \leq 2$ yields (3.53) with probability at least $1 - N_{\mathcal{L}} e^{-t}$. $\square$

We now control the localized empirical Rademacher complexity of $\mathcal{G}_\gamma$ in terms of the pseudo-dimensions of the derivative classes $\partial^\alpha \mathcal{F}$.

Lemma 3.68. *Let $\mathbf{X} = (X_1, \dots, X_n)$ be i.i.d. from P , and suppose there exist $C_{\text{loc}} > 0$ and $\gamma > 0$ such that*

$$\begin{aligned} \max_{\|\alpha\|_1 \leq 2} \sup_{u \in \mathcal{F}_\gamma} \|\partial^\alpha u - \partial^\alpha \tau\|_{L^2(\mathbf{X})} &\leq C_{\text{loc}} \sqrt{\gamma}, \\ (1/n)^2 &\leq \gamma, \\ \max_{\|\alpha\|_1 \leq 2} \text{Pdim}(\partial^\alpha \mathcal{F}) \vee (4eQ/C_{\text{loc}})^2 &\leq n. \end{aligned} \quad (3.54)$$

Then

$$\max_{\|\alpha\|_1 \leq 2} \text{Rad}(\partial^\alpha \tilde{\mathcal{F}}_\gamma, \mathbf{X}) \lesssim \sqrt{\frac{\text{VC}_{\mathcal{F}}}{n}} \gamma \log n, \quad (3.55)$$

where $\text{VC}_{\mathcal{F}} := \max_{\|\alpha\|_1 \leq 2} \text{VCdim}(\partial^\alpha \mathcal{F})$. Moreover, for any $t > 0$, with probability at least $1 - 2e^{-t}$,

$$\begin{aligned} \sup_{u \in \mathcal{F}_\gamma} (\mathbb{E}_P g_u - \mathbb{E}_{\mathbf{X}} g_u) &\lesssim C_{\text{Lip}} N_{\mathcal{L}} \sqrt{\frac{\text{VC}_{\mathcal{F}}}{n}} \gamma \log n \\ &\quad + \sqrt{\frac{2C_0 \gamma t}{n}} + \frac{23C_0 t}{3n}, \end{aligned} \quad (3.56)$$

where C_{Lip} is from Lemma 3.65 and C_0 from Lemma 3.62.

Proof. Consider $\mathcal{G}_\gamma = \{g_u : u \in \mathcal{F}_\gamma\}$. For each $u \in \mathcal{F}_\gamma$ we have $g_u = L(\cdot, u) - L(\cdot, \tau)$ and $L(\cdot, \tau) = 0$, so by the proof of Lemma 3.62, together with the bound $\|u\|_{W^{2,\infty}} \leq Q$, there exists a constant $\tilde{C} > 0$ (depending only on Q , on the coefficients of $\mathcal{L}$, and on ρ through Q) such that

$$\|g_u\|_\infty \leq \tilde{C} =: G, \text{ and } \text{Var}(g_u) \leq \mathbb{E}[g_u^2] \leq C_0 \mathbb{E}[g_u] \leq C_0 \gamma =: V.$$

Applying Lemma 2.30 to $\mathcal{G}_\gamma$ yields, with probability at least $1 - 2e^{-t}$,

$$\sup_{u \in \mathcal{F}_\gamma} (\mathbb{E}_P g_u - \mathbb{E}_{\mathbf{X}} g_u) \leq 6 \text{Rad}(\mathcal{G}_\gamma, \mathbf{X}) + \sqrt{\frac{2C_0 \gamma t}{n}} + \frac{23C_0 t}{3n}.$$

To bound $\text{Rad}(\mathcal{G}_\gamma, \mathbf{X})$ we first use Lemma 3.66:

$$\text{Rad}(\mathcal{G}_\gamma, \mathbf{X}) \leq C_{\text{Lip}} \sum_{\|\alpha\|_1 \leq 2} \text{Rad}(\partial^\alpha \tilde{\mathcal{F}}_\gamma, \mathbf{X}).$$

For each fixed α the localization condition (3.54) implies that

$$\partial^\alpha \tilde{\mathcal{F}}_\gamma \subset \left\{ h : \Omega \rightarrow \mathbb{R} : h = \partial^\alpha u - \partial^\alpha \tau, \ u \in \mathcal{F}, \ \|h\|_{L^2(\mathbf{X})} \leq C_{\text{loc}} \sqrt{\gamma} \right\}.$$

Hence Lemma 2.36 gives

$$\begin{aligned} \text{Rad}(\partial^\alpha \tilde{\mathcal{F}}_\gamma, \mathbf{X}) &\leq \text{Rad}\left(\{h : \|h\|_{L^2(\mathbf{X})} \leq C_{\text{loc}} \sqrt{\gamma}\}, \mathbf{X}\right) \\ &\leq \inf_{0 < \beta < C_{\text{loc}} \sqrt{\gamma}} \left\{ 4\beta + \frac{12}{\sqrt{n}} \int_\beta^{C_{\text{loc}} \sqrt{\gamma}} \sqrt{\log \mathcal{N}(\delta, \partial^\alpha \mathcal{F}, \|\cdot\|_{L^2(\mathbf{X})})} d\delta \right\}. \end{aligned}$$

Since $\|h\|_{L^2(\mathbf{X})} \leq \|h\|_\infty$ and $\|\partial^\alpha u\|_\infty \leq Q$ for all $u \in \mathcal{F}$, we may apply Lemma 2.38 with $M = Q$ and obtain, for $n \geq \text{Pdim}(\partial^\alpha \mathcal{F})$ and any $\beta \in (0, C_{\text{loc}} \sqrt{\gamma})$,

$$\begin{aligned} 4\beta + \frac{12}{\sqrt{n}} \int_\beta^{C_{\text{loc}} \sqrt{\gamma}} \sqrt{\log \mathcal{N}(\delta, \partial^\alpha \mathcal{F}|_{X_1, \dots, X_n}, \|\cdot\|_\infty)} d\delta \\ \leq 4\beta + \frac{12}{\sqrt{n}} \int_\beta^{C_{\text{loc}} \sqrt{\gamma}} \sqrt{\text{Pdim}(\partial^\alpha \mathcal{F}) \log \left(\frac{2enQ}{\delta \text{Pdim}(\partial^\alpha \mathcal{F})} \right)} d\delta. \end{aligned}$$

Choosing $\beta = C_{\text{loc}} \sqrt{\text{Pdim}(\partial^\alpha \mathcal{F}) \gamma / n} \leq C_{\text{loc}} \sqrt{\gamma}$ and using $n \geq (4eQ/C_{\text{loc}})^2$ and $\gamma \geq (1/n)^2$, one checks (see the analogous computation in (Lei et al., 2025, Appendix A)) that

$$\text{Rad}(\partial^\alpha \tilde{\mathcal{F}}_\gamma, \mathbf{X}) \lesssim \sqrt{\frac{\text{Pdim}(\partial^\alpha \mathcal{F}) \gamma}{n} \log n}.$$

By the pseudo-dimension/VC-dimension relation of Lemma 3.20 and the definition of $\text{VC}_{\mathcal{F}} = \max_{\|\alpha\|_1 \leq 2} \text{VCdim}(\partial^\alpha \mathcal{F})$, we have $\text{Pdim}(\partial^\alpha \mathcal{F}) \lesssim \text{VC}_{\mathcal{F}}$ uniformly in α , hence this upper bound is in turn

$$\lesssim \sqrt{\frac{\text{VC}_{\mathcal{F}}}{n} \gamma \log n},$$

uniformly over α , which proves (3.55). Thus

$$\text{Rad}(\mathcal{G}_\gamma, \mathbf{X}) \lesssim C_{\text{Lip}} N_{\mathcal{L}} \sqrt{\frac{\text{VC}_{\mathcal{F}}}{n} \gamma \log n}.$$

Plugging this into the deviation bound from Lemma 2.30 concludes the proof of (3.56). $\square$

Following (Lei et al., 2025), we now introduce the *critical radius* at which the localized complexity balances the scale γ .

Definition 3.69 (Critical radius). *For fixed $t > 0$ define*

$$\gamma_* := \inf \left\{ \gamma > 0 : \forall \gamma' \geq \gamma, \gamma' \geq \max_{\|\alpha\|_1 \leq 2} \left\{ 12Q \operatorname{Rad}(\partial^\alpha \tilde{\mathcal{F}}_{\gamma'}), \frac{8C_{11}Q^2t}{n}, \frac{16Q^2t}{3n} \right\} \right\}.$$

Lemma 3.70 (Upper bound on the critical radius). *Assume $t \geq 1$ and*

$$n \geq \max_{\|\alpha\|_1 \leq 2} \operatorname{Pdim}(\partial^\alpha \mathcal{F}) \vee \frac{8e^2Q^2}{C_{11} + 3}.$$

Then

$$\gamma_* \lesssim Q^2 \left(\frac{\operatorname{VC}_{\mathcal{F}}}{n} \log n + \frac{t}{n} + N_{\mathcal{L}} e^{-t} \right). \quad (3.57)$$

Proof. By Definition 3.69, for any $\varepsilon > 0$ there exists $\gamma \in [\gamma_* - \varepsilon, \gamma_*]$ such that

$$\gamma < \max_{\|\alpha\|_1 \leq 2} \left\{ 12Q \operatorname{Rad}(\partial^\alpha \tilde{\mathcal{F}}_\gamma, \mathbf{X}), \frac{8C_{11}Q^2t}{n}, \frac{16Q^2t}{3n} \right\}.$$

Letting $\varepsilon \rightarrow 0$ yields

$$\gamma_* \leq \max_{\|\alpha\|_1 \leq 2} \left\{ 12Q \operatorname{Rad}(\partial^\alpha \tilde{\mathcal{F}}_{\gamma_*}, \mathbf{X}), \frac{8C_{11}Q^2t}{n}, \frac{16Q^2t}{3n} \right\}.$$

Define the event

$$E := \left\{ \max_{\|\alpha\|_1 \leq 2} \sup_{u \in \mathcal{F}_{2\gamma_*}} \|\partial^\alpha u - \partial^\alpha \tau\|_{L^2(\mathbf{X})} \leq \sqrt{2(C_{11} + 3)\gamma_*} \right\}.$$

By Lemma 3.67 (applied with γ replaced by $2\gamma_*$ and parameter t) and the definition of γ_* , we have $\mathbb{P}(E) \geq 1 - N_{\mathcal{L}} e^{-t}$ for all $t \geq 1$.

On E , condition (3.54) holds with γ replaced by $2\gamma_*$ and $C_{\text{loc}} = \sqrt{2(C_{11} + 3)}$. Since $\gamma_* \geq 16Q^2t/(3n) \geq (1/n)^2$ by definition (for n large enough, as ensured by the hypothesis on n), the assumptions of Lemma 3.68 are satisfied and thus

$$\mathbb{E} \left[\mathbf{1}_E \max_{\|\alpha\|_1 \leq 2} \operatorname{Rad}(\partial^\alpha \tilde{\mathcal{F}}_{\gamma_*}, \mathbf{X}) \right] \lesssim \sqrt{\frac{\operatorname{VC}_{\mathcal{F}}}{n}} \gamma_* \log n.$$

Therefore

$$\begin{aligned}
\gamma_* &\leq \max_{\|\alpha\|_1 \leq 2} \left\{ 12Q \operatorname{Rad}(\partial^\alpha \tilde{\mathcal{F}}_{\gamma_*}, \mathbf{X}), \frac{8C_{11}Q^2t}{n}, \frac{16Q^2t}{3n} \right\} \\
&\lesssim 12Q \mathbb{E} \left[\mathbf{1}_E \max_{\|\alpha\|_1 \leq 2} \operatorname{Rad}(\partial^\alpha \tilde{\mathcal{F}}_{\gamma_*}, \mathbf{X}) \right] + 24Q^2 \mathbb{P}(E^c) + \frac{8C_{11}Q^2t}{n} + \frac{16Q^2t}{3n} \\
&\lesssim 12Q \sqrt{\frac{\operatorname{VC}_{\mathcal{F}}}{n}} \gamma_* \log n + 24Q^2 N_{\mathcal{L}} e^{-t} + \frac{8C_{11}Q^2t}{n} + \frac{16Q^2t}{3n}.
\end{aligned}$$

This is a quadratic inequality in $\sqrt{\gamma_*}$, which after solving yields (3.57). $\square$

We now complete the proof of Theorem 3.23 via a peeling argument.

Proof of Theorem 3.23. Fix $t \geq 0$ and set

$$t' := t + \log((N_{\mathcal{L}} + 3) \log_2(C_{\text{Lip}}Qn/\log n)) \geq 1.$$

Let γ_* be the critical radius of Definition 3.69 with parameter t' . Take any

$$\gamma > \gamma_* \vee \frac{\log n}{n},$$

to be specified. Recall the localized class

$$\mathcal{F}_\gamma = \{u \in \mathcal{F} : \mathbb{E}_P g_u \leq \gamma\}.$$

We decompose $\mathcal{F}$ into dyadic shells

$$\mathcal{F} = \mathcal{F}_\gamma \cup (\mathcal{F}_{2\gamma} \setminus \mathcal{F}_\gamma) \cup \cdots \cup (\mathcal{F}_{2^\ell \gamma} \setminus \mathcal{F}_{2^{\ell-1} \gamma}),$$

where ℓ is the largest integer such that $2^\ell \gamma \leq C_{\text{Lip}}Q$. This ensures $\ell \lesssim \log_2(C_{\text{Lip}}Qn/\log n)$.

Suppose that for some $1 \leq j \leq \ell$ we have $u_n \in \mathcal{F}_{2^j \gamma}$. Since $2^j \gamma > \gamma_*$, Lemma 3.67 applied with γ replaced by $2^j \gamma$ and parameter t' guarantees that, with probability at least $1 - N_{\mathcal{L}} e^{-t'}$,

$$\max_{\|\alpha\|_1 \leq 2} \|\partial^\alpha u_n - \partial^\alpha \tau\|_{L^2(\mathbf{X})} \leq \sqrt{2^j(C_{11} + 3)\gamma}.$$

Thus condition (3.54) of Lemma 3.68 holds with $C_{\text{loc}} = \sqrt{2(C_{11} + 3)}$ and γ replaced by $2^j \gamma$, and our assumptions on n ensure that $n \geq \max_{\|\alpha\|_1 \leq 2} \operatorname{Pdim}(\partial^\alpha \mathcal{F}) \vee (4eQ/C_{\text{loc}})^2$. Moreover $2^j \gamma \geq \gamma_* \geq (1/n)^2$.

Lemma 3.68 (with γ replaced by $2^j\gamma$ and parameter t') yields, with probability at least $1 - 2e^{-t'}$,

$$\begin{aligned} \mathbb{E}_P g_{u_n} - \mathbb{E}_{\mathbf{X}} g_{u_n} &\lesssim C_{\text{Lip}} N_{\mathcal{L}} \sqrt{\frac{\text{VC}_{\mathcal{F}}}{n} 2^j \gamma \log n} \\ &\quad + \sqrt{\frac{2C_0 2^j \gamma t'}{n}} + \frac{23C_0 t'}{3n}. \end{aligned}$$

On the other hand, by Lemma 3.62 applied to $u_{\mathcal{F}}$, with probability at least $1 - e^{-t'}$,

$$\mathbb{E}_{\mathbf{X}} g_{u_{\mathcal{F}}} - \mathbb{E}_P g_{u_{\mathcal{F}}} \leq \mathbb{E}_P g_{u_{\mathcal{F}}} + \frac{7C_0 t'}{n}.$$

Since u_n minimizes $\mathcal{R}_n$ over $\mathcal{F}$, we have $\mathbb{E}_{\mathbf{X}} g_{u_n} \leq \mathbb{E}_{\mathbf{X}} g_{u_{\mathcal{F}}}$ and therefore, on the intersection of these events,

$$\begin{aligned} \mathbb{E}_P g_{u_n} &\leq \mathbb{E}_{\mathbf{X}} g_{u_n} + C_{\text{Lip}} N_{\mathcal{L}} \sqrt{\frac{\text{VC}_{\mathcal{F}}}{n} 2^j \gamma \log n} + \sqrt{\frac{2C_0 2^j \gamma t'}{n}} + \frac{23C_0 t'}{3n} \\ &\leq \mathbb{E}_{\mathbf{X}} g_{u_{\mathcal{F}}} + C_{\text{Lip}} N_{\mathcal{L}} \sqrt{\frac{\text{VC}_{\mathcal{F}}}{n} 2^j \gamma \log n} + \sqrt{\frac{2C_0 2^j \gamma t'}{n}} + \frac{23C_0 t'}{3n} \\ &\leq 2 \mathbb{E}_P g_{u_{\mathcal{F}}} + C_{\text{Lip}} N_{\mathcal{L}} \sqrt{\frac{\text{VC}_{\mathcal{F}}}{n} 2^j \gamma \log n} + \sqrt{\frac{2C_0 2^j \gamma t'}{n}} + \frac{23C_0 t'}{3n} + \frac{7C_0 t'}{n}. \end{aligned}$$

Suppose we choose γ so that for all $1 \leq j \leq \ell$,

$$\begin{aligned} 2 \mathbb{E}_P g_{u_{\mathcal{F}}} + C_{\text{Lip}} N_{\mathcal{L}} \sqrt{\frac{\text{VC}_{\mathcal{F}}}{n} 2^j \gamma \log n} + \sqrt{\frac{2C_0 2^j \gamma t'}{n}} \\ + \frac{23C_0 t'}{3n} + \frac{7C_0 t'}{n} \leq 2^{j-1} \gamma. \end{aligned} \tag{3.58}$$

Then, on an event of probability at least $1 - (N_{\mathcal{L}} + 3)e^{-t'}$, we have $\mathbb{E}_P g_{u_n} \leq 2^{j-1}\gamma$, i.e. $u_n \in \mathcal{F}_{2^{j-1}\gamma}$. Iterating this argument shell-by-shell from $2^\ell\gamma$ down to γ and applying a union bound over j shows that with probability at least $1 - \ell(N_{\mathcal{L}} + 3)e^{-t'}$ we have $u_n \in \mathcal{F}_\gamma$, i.e.

$$\mathcal{R}(u_n) - \mathcal{R}(\tau) = \mathbb{E}_P g_{u_n} \leq \gamma.$$

It remains to specify γ . Condition (3.58) holds for all $1 \leq j \leq \ell$ if

$$2 \mathbb{E}_P g_{u_{\mathcal{F}}} + \frac{23C_0 t'}{3n} + \frac{7C_0 t'}{n} \leq 2^{j-2}\gamma, \quad \forall 1 \leq j \leq \ell,$$

and

$$C_{\text{Lip}} N_{\mathcal{L}} \sqrt{\frac{\text{VC}_{\mathcal{F}}}{n} 2^j \gamma \log n} + \sqrt{\frac{2C_0 2^j \gamma t'}{n}} \leq 2^{j-2} \gamma, \quad \forall 1 \leq j \leq \ell.$$

A sufficient choice is

$$\gamma = c_1 \frac{\text{VC}_{\mathcal{F}}}{n} \log n + c_2 \frac{t'}{n} + 4 \mathbb{E}_P g_{u_{\mathcal{F}}} + \frac{\log n}{n} + \gamma_*,$$

for suitable absolute constants $c_1, c_2 > 0$ depending only on C_{Lip}, C_0 and $N_{\mathcal{L}}$. Using the upper bound (3.57) on γ_* from Lemma 3.70, and recalling that $t' = t + \log((N_{\mathcal{L}} + 3) \log_2(C_{\text{Lip}} Q n / \log n))$, we obtain

$$\gamma \lesssim \mathbb{E}_P g_{u_{\mathcal{F}}} + \frac{\text{VC}_{\mathcal{F}}}{n} \log n + \frac{t}{n} + \frac{e^{-t}}{\log(n/\log n)}.$$

Finally, since $\ell \lesssim \log_2(C_{\text{Lip}} Q n / \log n)$ by construction of the peeling, we have $\ell(N_{\mathcal{L}} + 3)e^{-t'} \leq e^{-t}$, so that the above bound on γ holds with probability at least $1 - e^{-t}$. Recalling that $\mathbb{E}_P g_{u_{\mathcal{F}}} = \mathcal{R}(u_{\mathcal{F}}) - \mathcal{R}(\tau)$, we obtain

$$\mathcal{R}(u_n) - \mathcal{R}(\tau) \leq \mathcal{R}(u_{\mathcal{F}}) - \mathcal{R}(\tau) + C \left(\frac{\text{VC}_{\mathcal{F}}}{n} \log n + \frac{t}{n} + \frac{e^{-t}}{\log(n/\log n)} \right)$$

for some universal constant $C > 0$ independent of n and t (but depending on ρ through the generic constants). Absorbing C into the definitions of $\text{VC}_{\mathcal{F}}$ and the numerical constants in front of t/n and $e^{-t}/\log(n/\log n)$ yields precisely the inequality (3.29) stated in Theorem 3.23. $\square$

7 Conclusion and discussion

In this chapter we have applied the learning-theoretic methodology laid out in Chapter 2 to derive close to minimax-optimal $H^2(\Omega)$ a priori error bounds for boundary-adapted PINNs solving linear elliptic PDEs with Dirichlet boundary conditions, using tanh or ReQU activations. A key aspect of the analysis is that it tracks the effect of the smooth distance approximation ρ that enforces the boundary conditions by multiplication. The results rely on ρ being a *smooth distance approximation*, for which constructive algorithms are available (Sukumar and Srivastava, 2022).

Building on these results, several extensions are natural:

- **Parabolic equations.** as we show in Appendix C, extension of these results to linear parabolic PDEs with Dirichlet boundary conditions can be done by following the same overall strategy, provided that “strong-convexity estimate” akin to (3.6), are available.

- **More general boundary conditions.** For variants of the MET problem, such as the *narrow escape* problem, or escape problems with traps which are common in computational biology (Schuss et al., 2007; Holcman and Schuss, 2015), the quantity of interest may solve an elliptic or parabolic PDE with mixed (Robin, Neumann, Dirichlet...) boundary conditions. Although (Sukumar and Srivastava, 2022) have built modified PINN architectures to satisfy these boundary conditions exactly, understanding the approximation capability of these modified PINNs remains open.
- **Adaptation to boundary layer.** For certain problems, such as escape problems arising from Langevin diffusions in the small-noise $\varepsilon \rightarrow 0^+$ regime, the theory of WKB asymptotics shows that solutions have leading-order behaviour of the form $\exp(C/\varepsilon)$ away from the boundary (Freidlin and Wentzell, 2012). Designing PINN architecture which can accurately and reliably capture this boundary-layer behaviour would be an extremely valuable contribution for many applied fields.
- **General non-linear PDEs.** Naturally, one may wonder whether a priori error bounds akin to Theorem 3.24 could be established for PINNs solving any more general class of (non-linear) PDEs. Unfortunately, due to the reliance on existence, regularity and stability results such as (3.6), we can not expect this question to have a positive answer. However, the general proof strategy could in principle still be applied, as long as such estimates have been demonstrated.

Beyond these extensions, more practical considerations regarding the significance of our result remain: the a priori error bounds shown in this chapter solely hold for the empirical risk minimizer, and do not address the error incurred due to minimizing the empirical risk with finitely many steps of (typically, some variant of) gradient descent. Knowing how delicate the issue of training PINNs to convergence is (Wang et al., 2021, 2022; Krishnapriyan et al., 2021), the analysis therefore feels slightly incomplete. To address this important limitation and work towards true end-to-end theoretical guarantees for PINNs, we will introduce in Chapter 4 a modification of the standard PINN loss which regularizes the loss landscape and leads to provable convergence of gradient descent in a finite number of steps.

Chapter 4

From Theory to Practice: Provably Convergent Training Algorithms for Scientific Machine Learning

In Chapter 3, we have derived a priori error bounds for PINNs which were defined as Empirical Risk Minimizers of the PDE residual. Although important, the significance of our results remains rather limited by the fact that in practice, neural networks are almost never guaranteed to minimize the excess risk, and are rather the output of finitely many steps of gradient descent. Furthermore, it is now widely documented in the SciML literature that training physics-informed neural networks to convergence using gradient descent is notoriously difficult, and the training often gets stuck in awfully poor local minima – a phenomenon sometimes referred to as *failure mode* of PINNs (Krishnapriyan et al., 2021). Quantifying how the optimization procedure affects the overall error bounds, and providing a way to mitigate these training pathologies are thus essential bridges to cross in order to make the results of Chapter 3 of relevance to practitioners.

This is precisely the question we will tackle in this chapter: we will introduce *FK-PINNs*, a simple modification of the PINN objective that provably improves the loss landscape conditioning, and guarantees convergence of gradient descent under technical conditions which we will make explicit. More specifically, while Rathore et al. (2024) have shown that the condition number of the PINN objective increases like $\mathcal{O}(N^\beta)$ with the number N of collocation points, we will show that the FK-PINNs we introduce enjoy a uni-

formly bounded condition number when the right conditions are met. This chapter is largely based off the previous work (Tepakbong et al., 2026b).

1 Introduction

Partial differential equations (PDEs) underpin much of computational science and engineering. While classical numerical methods, such as finite elements, are mature and accurate, their cost grows quickly with dimension and geometric complexity, making large-scale or high-resolution simulations prohibitively difficult in numerous practical scenarios. Physics-informed neural networks (PINNs) offer a mesh-free alternative: a neural network u_θ approximates the solution, and training is driven by a loss that penalizes PDE residuals and boundary-condition violations at sampled collocation points (Raissi et al., 2019). PINNs have been investigated in a variety of settings, including forward and inverse problems, parameter estimation, and coupled multi-physics models (Cuomo et al., 2022).

Although the idea of enforcing physical constraints within expressive neural networks is conceptually appealing, many studies report severe training pathologies. For moderately stiff or high-frequency problems, optimization can be extremely slow and may fail to reduce the residual even with expressive architectures (Krishnapriyan et al., 2021; Wang et al., 2021). Moreover, the training dynamics are highly sensitive to hyperparameters, sampling strategies, and loss weights, hence small changes in either of these can lead to vastly different outcomes. Various heuristics, such as adaptive sampling, curriculum training, residual/gradient-based re-weighting or domain decomposition have been proposed to alleviate these issues (Wu et al., 2023a; Krishnapriyan et al., 2021). Although such approaches have been shown to help in specific cases, there is still no generally reliable way to obtain well-conditioned, robust PINN objectives.

1.1 An operator-conditioning perspective on PINN training

Recent work investigates this issue through an *operator-conditioning* and loss landscape lens. In particular, De Ryck et al. (2024) analyze gradient descent on PINN objectives and relate the dynamics to the spectrum of the Hermitian square $\mathcal{L}^*\mathcal{L}$ of the underlying PDE operator $\mathcal{L}$. Closely related analyses likewise tie PINN training difficulty to the conditioning of PDE-induced operators, their associated kernels, or PINN loss Hessians, and ex-

plain through theoretical and numerical results the resulting optimization pathologies (Rathore et al., 2024; Wang et al., 2022; Chen et al., 2024; Wang et al., 2025). In this view, a “Hessian-like” operator governs training: when it is ill-conditioned—with widely spread eigenvalues and nearly null modes—some parameter directions evolve orders of magnitude more slowly than others, leading to stagnation and failure to fit the residual even when the target solution lies well within the hypothesis class.

This operator-centric view suggests that successful training of PINNs is not just a matter of expressive networks or dense collocation sampling, but also of constructing losses and optimization schemes whose associated operators are well conditioned. Several prior works have indeed proposed to modify the traditional gradient-based neural network training algorithms by schemes which take into account the natural function space geometry associated with PINNs (Müller and Zeinhofer, 2023; Schwencke and Furtlehner, 2025; Chen et al., 2025; Wang et al., 2022, 2025). In many cases, these modified training schemes can be interpreted as applying a preconditioning on the operator (De Ryck et al., 2024).

Complementary to these optimizer-based approaches, we instead propose to modify the training objective itself: we design *preconditioned* PINN losses that augment the standard residual and boundary terms with lightweight “mass” or data-anchoring penalties, which effectively precondition the operator governing the training dynamics. In this chapter, we propose to construct these penalties using probabilistic representations of PDE solutions via the Feynman–Kac formula, yielding a simple, mesh-free modification of the PINN loss that is completely agnostic to architecture and optimizer choice. This choice is motivated by the family of linear elliptic and parabolic problems we are interested in, for which such representations are available (Karatzas and Shreve, 2014). We stress however, that the idea could be extended to any such “supervised data” term.

1.2 Feynman–Kac supervision as operator preconditioning

For a broad class of linear second-order elliptic and parabolic PDEs, the Feynman–Kac (FK) formula represents the solution $u^*(x)$ as the expectation of a functional of a stochastic process $(X_t)_{t \geq 0}$ solving a certain SDE (Øksendal and Øksendal, 2003; Karatzas and Shreve, 2014). For an appropriate choice of SDE and stopping time, $u^*(x)$ is given by an expectation over trajectories of X_t started at x , with the functional involving source terms, potentials,

and boundary data along the path. Approximating this expectation by simulating finitely many time-discretized trajectories from x and averaging the corresponding FK functional yields a natural Monte Carlo estimator of $u^*(x)$.

Classical probabilistic numerical methods for PDEs exploit this idea by simulating many trajectories at spatial points of interest to approximate the solution field (Sabelfeld, 1991; Billaud-Friess et al., 2018). In contrast, we use FK only as a *sparse* source of supervision: we incorporate a small number of Monte Carlo FK estimates into the PINN loss as anchors, augmenting the standard residual and boundary terms with a penalty on discrepancies between u_θ and these estimates at a few interior points. We refer to the resulting model as a *Feynman–Kac-Preconditioned Physics-Informed Neural Network* (FK-PINN), and show that the added term acts as an operator preconditioner, contributing a mass-like component that improves conditioning and stabilizes gradient descent even for relatively small FK budgets.

1.3 Contributions

The contributions of this chapter can be summarized as follows:

1. **Feynman–Kac-Preconditioned PINN objective.** We introduce FK-PINNs, which augment the standard PINN loss with a data-fidelity term based on Monte Carlo FK estimates of the solution at a small set of interior domain points. The objective is mesh-free and acts as a simple drop-in replacement compatible with standard PINN architectures and training pipelines. Of course, the idea can be extended verbatim to any other source of data supervision.
2. **Operator-preconditioning analysis.** Building on the operator-conditioning framework of De Ryck et al. (2024) and Rathore et al. (2024), we rigorously show, in the finite-width setting of standard PINNs rather than an infinite-width NTK limit, that the FK data-fidelity term induces a positive semidefinite mass operator that shifts the spectrum of the Hermitian square of the PDE operator. For suitable FK weights and point distributions, this yields a bounded and substantially improved condition number for the resulting Hessian-like operator, even for modest Monte Carlo budgets.
3. **Learning-theoretic error decomposition.** Building on the results from Chapters 2 and 3, we develop a learning-theoretic analysis of FK-PINNs trained by gradient descent. Under standard regularity assumptions, we derive high-probability $L^2(\Omega)$ -error bounds between the true solution u^* and an FK-PINN after T steps of gradient descent. The

bounds are non-asymptotic and depend explicitly on the problem parameters.

4. **Empirical illustrations on canonical PDE problems.** We complement our theory with experiments on Poisson, Schrödinger, mean exit time, and committor problems. FK-PINNs exhibit improved training dynamics, better-conditioned operator spectra, lower solution error, and remain stable on instances where baseline PINNs suffer catastrophic optimization failures, all while using relatively small FK budgets. Detailed numerical results are reported in Chapter 6.

2 Background and problem setup

2.1 PDE setting and function spaces

For the sake of illustration, throughout this Chapter we will assume to be considering linear second-order elliptic and parabolic PDEs on a bounded Lipschitz domain $\Omega \subset \mathbb{R}^d$ with Dirichlet boundary conditions. As the analysis will reveal later, neither the linearity of $\mathcal{L}$ nor the Dirichlet condition are used anywhere in the proofs. With this out of the way, let $\mathcal{L}$ be a linear second-order differential operator with bounded measurable coefficients, and consider

$$\mathcal{L}u(x) = f(x), \quad x \in \Omega, \quad (4.1)$$

with boundary condition

$$u(x) = g(x), \quad x \in \partial\Omega. \quad (4.2)$$

We assume that $\mathcal{L}$ is uniformly elliptic and that g is sufficiently regular. Under these standard assumptions, there exists a unique weak solution $u^* \in H^1(\Omega)$ with $u^*|_{\partial\Omega} = g$. See (Evans, 2022) or Section 3.1 for a precise statement.

In this chapter, we will restrict our attention to operators $\mathcal{L}$ for which the solution u^* admits a Feynman–Kac representation in terms of an associated stochastic differential equation (SDE) (Øksendal and Øksendal, 2003). This class covers many important PDE models used in applications (E et al., 2021). We also note that our basic construction has natural analogues for broader classes of nonlinear PDEs that admit similar stochastic representations (E et al., 2017), as we will show in Section 3 and Appendix E. The ideas discussed do however extend straightforwardly to any PDE for which supervision data is available, as we will see.

2.2 Standard PINN approximation of PDE solutions

Let $\{u_\theta : \theta \in \Theta\}$ be a parametric class of functions implemented by a fully connected neural network with depth L , layer-wise widths $(m_\ell)_{\ell=0}^L$ and activation $\sigma \equiv \tanh$, mapping $x \in \Omega \subset \mathbb{R}^d$ to $u_\theta(x) \in \mathbb{R}$. Physics-informed neural networks (PINNs) approximate $u^\star$ by minimizing a loss that penalizes PDE residuals in the interior and violations of boundary conditions (Raissi et al., 2019).

Let $\{x_i^{\text{int}}\}_{i=1}^{N_{\text{int}}} \subset \Omega$ be interior collocation points and $\{x_j^{\partial\Omega}\}_{j=1}^{N_{\partial\Omega}} \subset \partial\Omega$ boundary collocation points, drawn i.i.d. from the uniform distribution on Ω and $\partial\Omega$. We define the empirical PDE and boundary losses

$$\mathcal{R}_{\text{PDE}}(\theta) := \frac{1}{N_{\text{int}}} \sum_{i=1}^{N_{\text{int}}} |\mathcal{L}u_\theta(x_i^{\text{int}}) - f(x_i^{\text{int}})|^2, \quad (4.3)$$

$$\mathcal{R}_{\partial\Omega}(\theta) := \frac{1}{N_{\partial\Omega}} \sum_{j=1}^{N_{\partial\Omega}} |u_\theta(x_j^{\partial\Omega}) - g(x_j^{\partial\Omega})|^2. \quad (4.4)$$

The standard PINN objective is

$$\mathcal{R}_{\text{PINN}}(\theta) := \mathcal{R}_{\text{PDE}}(\theta) + \lambda_{\partial\Omega} \mathcal{R}_{\partial\Omega}(\theta), \quad (4.5)$$

where $\lambda_{\partial\Omega} > 0$ balances PDE residuals and boundary accuracy. In practice, $\mathcal{R}_{\text{PINN}}$ is minimized using (stochastic) gradient-based optimizers (e.g., Adam), with automatic differentiation to compute $\mathcal{L}u_\theta(x)$.

2.3 Operator-conditioning view of PINN Training

Recent work analyzes PINN training through the conditioning of PDE-induced operators (Rathore et al., 2024; De Ryck et al., 2024). We denote by H a positive semidefinite “Hessian-like” operator associated with $\mathcal{R}_{\text{PINN}}$ at a local minimizer $\theta^\star$, such as the empirical Gauss–Newton matrix. Near $\theta^\star$, gradient descent is well approximated by a linear dynamics governed by H , so directions corresponding to small eigenvalues converge very slowly and give rise to the optimization pathologies observed in practice (Wang et al., 2022; De Ryck et al., 2024; Rathore et al., 2024).

To circumvent this, we propose to augment the standard PINN objective with a Feynman–Kac based data-fidelity term. At the second-order structure-level, this induces an additional positive semidefinite “mass” operator M , so that the Hessian-like operator becomes

$$H_{\text{FK}} := H + \lambda_{\text{FK}} M, \quad (4.6)$$

with $\lambda_{\text{FK}} > 0$ a weight. Building on the operator-conditioning analyses of Rathore et al. (2024); De Ryck et al. (2024), we study in Section 5 how the spectrum and conditioning of H_{FK} compare to those of H .

2.4 Related Work

We now survey some of the most relevant literature around these issues.

Mitigating the failure modes of PINNs. Ever since their introduction by Raissi et al. (2019), a growing body of work documents a number of severe training difficulties when training PINNs, including spectral bias, failure to minimize residuals, and sensitivity to sampling and loss weights (Krishnapriyan et al., 2021; Cuomo et al., 2022). Finding methods to mitigate these issues is an active field of research, and proposed methods include adaptive collocation, residual-based reweighting, domain decomposition, and specialized architectures (Wang et al., 2021; Wu et al., 2023a; Wang et al., 2024).

Operator- and loss-landscape-based analyses. A complementary line of work analyzes PINN training through the operators induced by the PDE and the loss, relating gradient descent dynamics and convergence rates to the spectrum and conditioning of $\mathcal{L}^*\mathcal{L}$ or NTK-like operators (Wang et al., 2022; De Ryck et al., 2024). Recent such studies (Rathore et al., 2024; Chen et al., 2024; Wang et al., 2025) further show how ill-conditioned differential operators translate into ill-conditioned PINN objectives and motivate preconditioned or second-order optimization schemes. Our work adopts this operator-centric viewpoint but improves conditioning by modifying the PINN objective itself.

Monte Carlo methods for PDEs. Classical probabilistic numerical methods exploit Feynman–Kac representations to compute PDE solutions via Monte Carlo simulation of associated SDEs (Sabelfeld, 1991; Billaud-Friess et al., 2018; Yu and Mascagni, 2022). These approaches yield pointwise estimates of $u^*(x)$ with controllable bias and variance, but typically do not use neural function approximators. In contrast, we use Feynman–Kac supervision as an *operator preconditioner* for PINNs and analyze its effect on both the conditioning of the induced Hessian-like operator and the resulting generalization error.

3 Feynman–Kac representation and Monte Carlo supervision

3.1 Feynman–Kac representation for linear PDEs

We briefly recall the celebrated Feynman–Kac (FK) representation at the core of our Monte Carlo supervision: consider the boundary value problem (4.1)–(4.2) for the operator $\mathcal{L}$ introduced in Section 2. Under conditions ensuring that (4.1)–(4.2) admits a Feynman–Kac representation (Øksendal and Øksendal, 2003; Karatzas and Shreve, 2014), there exists a diffusion process $(X_t)_{t \geq 0}$ with generator $\mathcal{L}$ and exit time $\tau := \inf\{t \geq 0 : X_t \notin \Omega\}$, such that for every $x \in \Omega$,

$$u^*(x) = \mathbb{E}_x \left[\int_0^\tau r(X_t) dt + h(X_\tau) \right], \quad (4.7)$$

where r is a running functional and h is a terminal functional determined by the coefficients of $\mathcal{L}$ and the data (f, g) . Explicit expressions for r and h , together with analogous stochastic representation formulas for more general PDE families are detailed in Appendix E.

This probabilistic representation is classical in the theory of diffusion processes and underlies many probabilistic numerical methods for elliptic and parabolic PDEs (Pavliotis, 2014). It naturally covers, for example, mean exit time problems and committor functions describing transition probabilities between metastable sets in stochastic dynamics (E and Vanden-Eijnden, 2006).

3.2 Monte Carlo estimator of the FK functional

To obtain pointwise supervision values for the PINN, we approximate the FK functional in (4.7) by Monte Carlo (MC) simulation of the associated diffusion. Starting from a point $x \in \Omega$, we simulate N_{MC} independent trajectories with a time step $\Delta t > 0$, using a standard Euler–Maruyama scheme (Gobet, 2016), and stop each trajectory when it exits Ω or when a truncation time $T_{\text{max}} > 0$ is reached.

Let $X_n^{(m)}$ denote the n -th point along the m -th simulated trajectory, and let $\hat{\tau}^{(m)}$ be the corresponding (discrete) stopping index. Our Monte Carlo approximation of $u^*(x)$ is

$$\hat{u}^{\text{MC}}(x) = \frac{1}{N_{\text{MC}}} \sum_{m=1}^{N_{\text{MC}}} \left(\sum_{n=0}^{\hat{\tau}^{(m)}-1} r(X_n^{(m)}) \Delta t + h(X_{\hat{\tau}^{(m)}}^{(m)}) \right). \quad (4.8)$$

Algorithm 1 : Monte Carlo Feynman–Kac estimator at a point x

Require: Point $x \in \Omega$; number of trajectories N_{MC} ; time step Δt ; truncation time T_{max} ; drift b and diffusion σ ; running and terminal functionals r, h .

```

1: for  $m = 1, \dots, N_{\text{MC}}$  do
2:   Initialize  $X_0^{(m)} \leftarrow x, S^{(m)} \leftarrow 0, n \leftarrow 0$ .
3:   while  $X_n^{(m)} \in \Omega$  and  $n\Delta t < T_{\text{max}}$  do
4:     Sample  $\Delta W_n^{(m)} \sim \mathcal{N}(0, \Delta t \mathbf{I})$ .
5:     Update  $X_{n+1}^{(m)} \leftarrow X_n^{(m)} + b(X_n^{(m)}) \Delta t + \sigma(X_n^{(m)}) \Delta W_n^{(m)}$ .
6:     Update  $S^{(m)} \leftarrow S^{(m)} + r(X_n^{(m)}) \Delta t$ .
7:     Increment  $n \leftarrow n + 1$ .
8:   end while
9:   Set  $\hat{\tau}^{(m)} \leftarrow n$  and update  $S^{(m)} \leftarrow S^{(m)} + h(X_{\hat{\tau}^{(m)}}^{(m)})$ .
10: end for
11: return  $\hat{u}^{\text{MC}}(x) \leftarrow \frac{1}{N_{\text{MC}}} \sum_{m=1}^{N_{\text{MC}}} S^{(m)}$ .

```

In words, along each trajectory we approximate the FK integral by a simple time Riemann sum of r , add the terminal term h at the (possibly truncated) exit point, and then average across all trajectories. [Algorithm 1](#) summarizes the simulation procedure.

Under mild conditions on the diffusion and on $(\Delta t, T_{\text{max}})$, the estimator $\hat{u}^{\text{MC}}(x)$ has variance $\text{Var}[\hat{u}^{\text{MC}}(x)] = \mathcal{O}(1/N_{\text{MC}})$ and a bias controlled by Δt and T_{max} . Precise bias–variance bounds and tail estimates for $\hat{u}^{\text{MC}}(x)$ will be proved in [Section 2](#) (see in particular [Propositions 5.15](#) and [5.1](#)).

3.3 Selection of supervision points and budget

We evaluate [\(4.8\)](#) at a relatively small set of *Feynman–Kac supervision points* $\{x_k^{\text{FK}}\}_{k=1}^{N_{\text{FK}}} \subset \Omega$. These points can be selected by simple generic strategies, such as uniform sampling over Ω , low-discrepancy sequences, or an adaptive second pass that concentrates points in regions where an initial PINN exhibits large residuals.

Given a fixed computational budget, there is a natural trade-off between the number of supervision points N_{FK} and the number of trajectories per point N_{MC} : increasing N_{FK} improves spatial coverage, whereas increasing N_{MC} reduces the Monte Carlo variance at each point. Concrete parameter choices and additional implementation details for our numerical experiments

are deferred to Section 2 in Chapter 6.

4 Feynman–Kac-preconditioned PINNs (FK-PINNs)

4.1 Augmented loss with FK supervision

Given FK supervision points $\{x_k^{\text{FK}}\}_{k=1}^{N_{\text{FK}}} \subset \Omega$ and corresponding Monte Carlo estimates $\hat{u}_\beta^{\text{MC}}(x_k^{\text{FK}})$ of the Feynman–Kac functional (4.7), we define the FK supervision loss

$$\mathcal{R}_{\text{FK}}(\theta) = \frac{1}{N_{\text{FK}}} \sum_{k=1}^{N_{\text{FK}}} (u_\theta(x_k^{\text{FK}}) - \hat{u}_\beta^{\text{MC}}(x_k^{\text{FK}}))^2. \quad (4.9)$$

Here β denotes the Monte Carlo parameters (number of trajectories, time step, truncation time, etc.), as described in Algorithm 1.

The full FK-PINN objective augments the standard PINN loss with (4.9):

$$\begin{aligned} \mathcal{R}_{\text{FK-PINN}}(\theta, \phi) &= \lambda_{\text{PDE}}(\phi) \mathcal{R}_{\text{PDE}}(\theta) + \lambda_{\partial\Omega}(\phi) \mathcal{R}_{\partial\Omega}(\theta) \\ &\quad + \lambda_{\text{FK}}(\phi) \mathcal{R}_{\text{FK}}(\theta), \end{aligned} \quad (4.10)$$

where $\mathcal{R}_{\text{PDE}}$ and $\mathcal{R}_{\partial\Omega}$ are the empirical PDE residual and boundary losses defined in Section 2.2, and ϕ denotes the task-weighting parameters introduced below. When $\lambda_{\text{FK}}(\phi) = 0$ and the remaining weights are fixed, (4.10) reduces to the standard PINN objective.

4.2 Adaptive weighting via homoscedastic uncertainty

The objective (4.10) couples three losses with different noise levels: $\mathcal{R}_{\text{PDE}}$, $\mathcal{R}_{\partial\Omega}$, and $\mathcal{R}_{\text{FK}}$. We adopt the homoscedastic uncertainty weighting scheme of Kendall et al. (2018), in the same form used for PINNs by Niu et al. (2025), and interpret the weights as task-dependent noise variances. In their formulation, for task losses $\{\mathcal{R}_j\}$ with homoscedastic variances $\{\sigma_j^2\}$, the joint objective is

$$\mathcal{R}_{\text{multi}} = \sum_j \left(\frac{1}{2\sigma_j^2} \mathcal{R}_j + \log \sigma_j \right). \quad (4.11)$$

Motivated by this, for our three tasks $j \in \{\text{PDE}, \partial\Omega, \text{FK}\}$ we introduce log-variances $s_j := \log \sigma_j^2$ and define trainable weights $\lambda_j(\phi) := \exp(-s_j)$ with $\phi := \{s_j\}_j$. The three scalars in (4.10), $\lambda_{\text{PDE}}(\phi)$, $\lambda_{\partial\Omega}(\phi)$, and $\lambda_{\text{FK}}(\phi)$,

Algorithm 2 : Training a Feynman–Kac-Preconditioned PINN (FK-PINN)

Require: Domain Ω and boundary $\partial\Omega$; PDE operator $\mathcal{L}$ and data (f, g) ; neural network u_θ ; task-weight parameters ϕ ; FK Monte Carlo parameters β ; number of FK points N_{FK} ; numbers of interior and boundary collocation points.

- 1: **Stage 1: Offline FK supervision generation**
- 2: Choose FK supervision points $\{x_k^{\text{FK}}\}_{k=1}^{N_{\text{FK}}} \subset \Omega$ (e.g., uniform or low-discrepancy sampling).
- 3: **for** $k = 1, \dots, N_{\text{FK}}$ **do**
- 4: Approximate $u^*(x_k^{\text{FK}})$ by a Monte Carlo FK estimate $\hat{u}_\beta^{\text{MC}}(x_k^{\text{FK}})$ using Algorithm 1 with parameters β .
- 5: **end for**
- 6: Form the FK dataset $\mathcal{D}_{\text{FK}} = \{(x_k^{\text{FK}}, \hat{u}_\beta^{\text{MC}}(x_k^{\text{FK}}))\}_{k=1}^{N_{\text{FK}}}$.
- 7: **Stage 2: PINN training with FK supervision**
- 8: **while** stopping criterion not met **do**
- 9: Sample interior collocation points in Ω and boundary points on $\partial\Omega$.
- 10: Compute $\mathcal{R}_{\text{PDE}}(\theta)$ and $\mathcal{R}_{\partial\Omega}(\theta)$ from the current collocation batches.
- 11: Sample a minibatch from $\mathcal{D}_{\text{FK}}$ and compute $\mathcal{R}_{\text{FK}}(\theta)$ via (4.9).
- 12: Evaluate the total loss $\mathcal{R}_{\text{FK-PINN}}(\theta, \phi)$ in (4.10).
- 13: Update (θ, ϕ) using (stochastic) gradient descent.
- 14: **end while**
- 15: **return** trained parameters (θ, ϕ) .

are obtained in this way and are optimized jointly with θ . In practice, we initialize the s_j so that the initial weights are of comparable magnitude and clip s_j to a bounded interval to avoid degenerate solutions in which one loss (especially the noisy FK term) is effectively ignored.

4.3 Training algorithm

FK-PINN training consists of an offline stage that generates FK supervision and an online stage that trains the network with the augmented loss (4.10). The full procedure is summarized in Algorithm 2.

5 Theoretical guarantees for FK-based operator preconditioning

We now give an overview of our theoretical framework for FK-based operator preconditioning. Our presentation follows the PL* framework of [Liu et al. \(2022\)](#) and its adaptation to finite-width PINNs by [Rathore et al. \(2024\)](#). Full assumptions and proofs are deferred to Section 7.

5.1 Setup and PL* framework

We analyze gradient descent on the FK-PINN loss with fixed task weights. Throughout this section we fix constants $\lambda_{\text{PDE}} > 0$, $\lambda_{\partial\Omega} > 0$, and $\lambda_{\text{FK}} \geq 0$, and we consider the objectives

$$\begin{aligned}\mathcal{R}_{\text{PINN}}(\theta) &:= \lambda_{\text{PDE}} \mathcal{R}_{\text{PDE}}(\theta) + \lambda_{\partial\Omega} \mathcal{R}_{\partial\Omega}(\theta), \\ \mathcal{R}_{\text{FK-PINN}}(\theta) &:= \mathcal{R}_{\text{PINN}}(\theta) + \lambda_{\text{FK}} \mathcal{R}_{\text{FK}}(\theta),\end{aligned}$$

where $\mathcal{R}_{\text{PDE}}$ and $\mathcal{R}_{\partial\Omega}$ are defined in Section 2.2, and $\mathcal{R}_{\text{FK}}$ is the FK supervision loss in (4.9). The standard PINN corresponds to $\lambda_{\text{FK}} = 0$.

In Section 7, we will prove the conditioning statement used below via bounds on the empirical Gauss–Newton matrix.

Assumption 4.1 (Interpolation regime). Both $\mathcal{R}_{\text{PINN}}$ and $\mathcal{R}_{\text{FK}}$ are optimized in an interpolation regime: each loss has minimum value 0, and every global minimizer attains zero loss.

We stress here that Assumption 4.1 does *not* say that we assume our the PINNs trained with gradient descent to achieve zero loss (which would be meaningless), but rather that the network is sufficiently overparametrized that there exists at least one choice of weights and biases for which it will achieve zero loss. This is the standard interpolation setting used in [Liu et al. \(2022\)](#); [Rathore et al. \(2024\)](#), and has been shown to hold even for moderately over-parametrized networks ([Vardi et al., 2022](#); [Madden, 2024](#)).

Definition 4.2 (PL* condition ([Karimi et al., 2016](#))). Let $\mathcal{J}$ be a loss function and let θ^* be any global minimizer of $\mathcal{J}$. We say that $\mathcal{J}$ satisfies the PL* condition in a neighborhood $\mathcal{S}$ of θ^* with constant $\mu > 0$ if

$$\mathcal{J}(\theta) - \mathcal{J}(\theta^*) \leq \frac{1}{2\mu} \|\nabla \mathcal{J}(\theta)\|^2 \quad \text{for all } \theta \in \mathcal{S}.$$

This is the Polyak–Łojasiewicz (PL) inequality ([Polyak, 1963](#)), adapted to possibly non-isolated minimizers as in [Karimi et al. \(2016\)](#); [Liu et al. \(2022\)](#).

Under Assumption 4.1 we have $\mathcal{J}(\theta^*) = 0$, so the PL^* condition directly controls the loss in terms of the gradient norm.

Definition 4.3 (Condition number for PL^* functions). *Let $\mathcal{J}$ be differentiable and L -smooth on a set $\mathcal{S}$, and assume that $\mathcal{J}$ satisfies a PL^* condition on $\mathcal{S}$ with constant $\mu > 0$. We define the PL -based condition number of $\mathcal{J}$ on $\mathcal{S}$ as*

$$\kappa_{\text{PL}}(\mathcal{J}; \mathcal{S}) := \frac{L}{\mu}.$$

Here L -smooth means that the gradient is L -Lipschitz on $\mathcal{S}$: $\|\nabla \mathcal{J}(\theta) - \nabla \mathcal{J}(\theta')\| \leq L\|\theta - \theta'\|$.

For the two objectives of interest, we fix a neighborhood $\mathcal{S}$ of the interpolation manifold and write $\kappa_{\text{PINN}} := \kappa_{\text{PL}}(\mathcal{R}_{\text{PINN}}; \mathcal{S})$ and $\kappa_{\text{FK}} := \kappa_{\text{PL}}(\mathcal{R}_{\text{FK-PINN}}; \mathcal{S})$, omitting the dependence on $\mathcal{S}$ when clear from context.

For the losses considered here, the constants L and the PL^* constant μ are controlled by the largest and smallest nonzero eigenvalues of the linearized Gauss–Newton matrix, so κ_{PL} is comparable to the usual spectral condition number. As in Liu et al. (2022); Rathore et al. (2024), we assume that the empirical PINN objectives $\mathcal{R}_{\text{PINN}}$ and $\mathcal{R}_{\text{FK}}$ are L -smooth and satisfy a PL^* condition on a neighborhood $\mathcal{S}$ of the interpolation manifold.

Theorem 4.4 (Rathore et al. (2024), Theorem 8.4). *Consider the standard PINN loss $\mathcal{R}_{\text{PINN}}$ for a linear PDE and an interpolating minimizer θ^* as in Assumption 4.1. Let $\mathcal{T}_K(\theta^*)$ be the associated kernel integral operator with eigenvalues $(\lambda_j)_{j \geq 1}$ satisfying $\lambda_j \leq Cj^{-\beta}$ for some $C > 0$ and $\beta > 1$. Let κ_{PINN} denote the PL -based condition number of Definition 4.3 on a PL^* neighborhood of θ^* . Then there exists a constant $c > 0$ such that, with high probability for all sufficiently large number N of collocation points,*

$$\kappa_{\text{PINN}} \geq c N^{\beta/2}.$$

Thus, in the PDE-only case the PL -based condition number of the PINN loss deteriorates polynomially as the residual (collocation) mesh is refined.

5.2 Effect of FK supervision on conditioning

FK supervision augments the loss with a data term that, at the operator level, acts like a preconditioner and prevents the conditioning from degrading as the residual mesh is refined. The following theorem makes this effect precise in terms of the local PL^* conditioning of the FK-augmented loss. The uniform conditioning statement also requires a compatibility condition that prevents

the FK term from being blind to directions that are relevant for the PDE constraints.

Assumption 4.5 (No FK blind spots). Near the parameters of interest, there is no direction in parameter space that changes the PDE residual or boundary outputs on their training points while leaving the FK outputs unchanged on the FK supervision points. Equivalently, any small (first-order) parameter change that produces no change in FK predictions at the FK points also produces no change in the PDE residual and boundary predictions at their training points.

A precise formulation is given in Section 7 as Assumption 4.10. We can now formulate our main result of this section:

Theorem 4.6 (Informal). *Assume the setting of Theorem 4.4, the regularity assumptions on FK supervision stated in Section 7, and the compatibility condition of Assumption 4.10. Then there exist constants $\mu_0 > 0$, $L_0 < \infty$, and $C < \infty$, independent of the number N of PDE collocation points, such that, with high probability for all sufficiently large N , the FK-augmented objective $\mathcal{R}_{\text{FK-PINN}}$ is L_0 -smooth and satisfies a PL^* condition with constant μ_0 on a neighborhood $\mathcal{S}$ of the interpolation manifold. In particular,*

$$\kappa_{\text{FK}} = \kappa_{\text{PL}}(\mathcal{R}_{\text{FK-PINN}}; \mathcal{S}) \leq C$$

uniformly in N .

A detailed statement and proof are given in Section 7 (see in particular Theorem 4.15).

Combining Theorems 4.4 and 4.6 yields a sharp contrast between the two objectives: while for the standard PINN loss, the PL-based condition number κ_{PINN} deteriorates polynomially with the number of collocation points, the FK-augmented loss' condition number κ_{FK} remains uniformly bounded as the mesh is refined, provided the FK supervision satisfies the non-degeneracy and compatibility requirements stated in Section 7. This shows that the FK-supervision term can effectively enable PINNs to overcome training pathologies due to fine meshes.

Implications for gradient descent. Under the PL^* framework, Liu et al. (2022) show that gradient descent on an L -smooth loss $\mathcal{J}$ satisfying a PL^* condition with constant μ converges linearly to a global minimizer, with iteration complexity

$$T_{\mathcal{J}}(\varepsilon) = \mathcal{O}(\kappa_{\text{PL}}(\mathcal{J}; \mathcal{S}) \log(1/\varepsilon)), \quad \kappa_{\text{PL}}(\mathcal{J}; \mathcal{S}) = \frac{L(\mathcal{J}; \mathcal{S})}{\mu},$$

where $L(\mathcal{J}; \mathcal{S})$ is the Lipschitz constant of $\nabla \mathcal{J}$ on $\mathcal{S}$. For the standard PINN loss this gives $T_{\text{PINN}}(\varepsilon) = \mathcal{O}(\kappa_{\text{PINN}} \log(1/\varepsilon))$, and Theorem 4.4 suggests that $T_{\text{PINN}}(\varepsilon)$ grows like $N^{\beta/2} \log(1/\varepsilon)$ as N increases. In contrast, Theorem 4.6 shows that κ_{FK} stays bounded in N , so $T_{\text{FK}}(\varepsilon) = \mathcal{O}(\log(1/\varepsilon))$, i.e., reaching error ε costs about a factor $N^{\beta/2}$ more iterations for the PINN than for the FK-PINN.

6 An Illustrative Example

We now illustrate our theoretical results by comparing the FK-PINNs against the standard PINN baseline on a representative PDE: a steady-state Schrödinger-type equation. Further extensive numerical experiments on a wider range of problems, together with implementation details are provided in Chapter 6.

For a stationary system defined on a bounded domain $\Omega \subset \mathbb{R}^d$, we consider the following non-homogeneous *Schrödinger-type* equation:

$$\begin{aligned} -\frac{1}{2}\Delta\psi(x) + V(x)\psi(x) &= f(x) & x \in \Omega, \\ \psi(x) &= 0 & x \in \partial\Omega, \end{aligned} \quad (4.12)$$

where $\psi(x)$ represents the response wavefunction, $V(x)$ denotes a prescribed potential function, and $f(x)$ constitutes a given source term. The multi-scale nature of this Schrödinger-type equation, notably for oscillatory, high-frequency potentials, poses a significant challenge for conventional PINNs.

We set the domain Ω as $\Omega = [-1, 1] \times [-1, 1]$, while the potential $V(x)$ is that of a periodic crystalline lattice, defined as

$$V(x) = -V_0[\cos(kx_1) + \cos(kx_2)], \quad (4.13)$$

where $V_0 = 5$ is the potential depth and $k = 2\pi/a$ with lattice constant $a = 0.5$. This periodic structure induces strong local confinement and sharp gradients in the wavefunction $\psi(x)$. We further introduce a high-frequency external forcing term:

$$f(x) = A \sin(Kx_1) \sin(Kx_2), \quad (4.14)$$

where $A = 50$ and $K = 5$.

Figure 4.1 illustrates the potential V , forcing term f and corresponding wavefunction ψ . We report respectively in Figures 4.2 and 4.3 the wavefunctions

learned by a standard PINN, and an FK-PINN, with both identical architecture training parameters and schedule, after $30000 + 15000$ steps of Adam + L-BFGS. As the results clearly illustrate, while the standard PINN completely fails at learning the solution structure, due to the high-frequency oscillations making the loss landscape intractable, the proposed FK-PINNs effectively mitigates the underlying stiffness of the PDE and successfully recovers the intended solution profile.

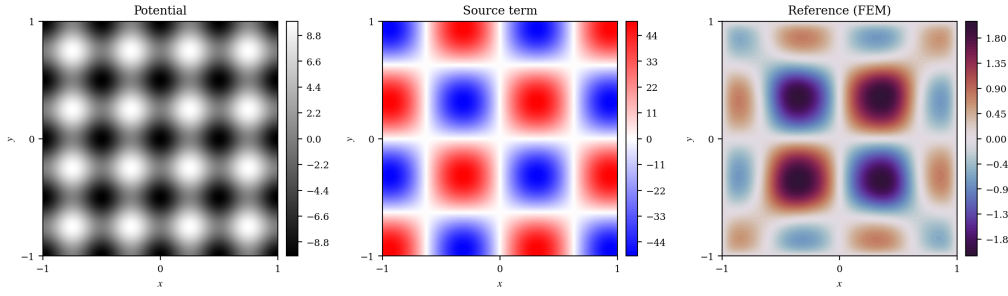


Figure 4.1: Data for the Schrödinger-type equation (4.12). **From left to right:** potential $V(\mathbf{x})$, source $f(\mathbf{x})$, and FEM reference ψ .

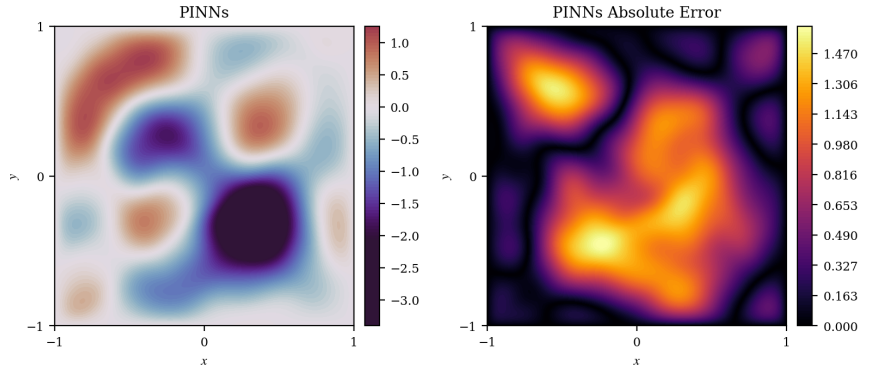


Figure 4.2: Numerical results for the standard PINN on the Schrödinger-type equation (4.12). **Left side:** PINN prediction **Right side:** absolute errors $|\psi - u_\theta|$

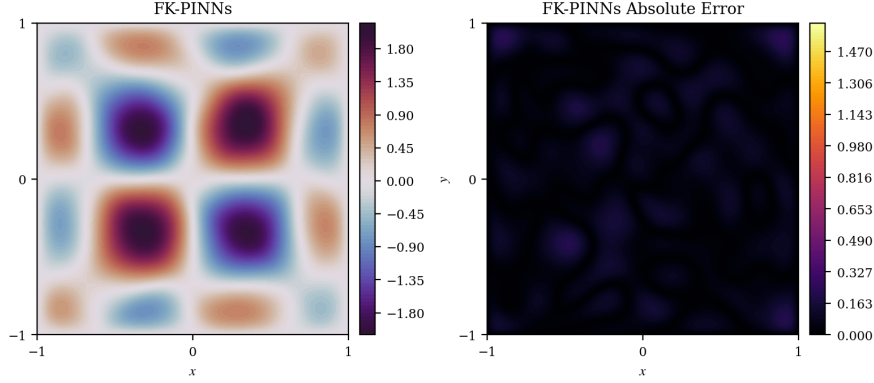


Figure 4.3: Numerical results for the FK-PINN on the Schrödinger-type equation (4.12). **Left side:** FK-PINN prediction **Right side:** absolute errors $|\psi - u_\theta|$

7 Proof of preconditioning effect of the FK data term

We now prove the conditioning result stated informally as Theorem 4.6. The idea of the proof is to compare the empirical Gauss–Newton matrix of the PDE-only objective with that of the FK-augmented objective, and to show that the supervised FK term prevents the smallest nonzero eigenvalues from collapsing as the residual mesh is refined.

We write $L_{\text{PDE+BC}}(\theta)$ for the empirical PDE residual plus Dirichlet boundary loss and $L_{\text{data}}(\theta)$ for a supervised data loss, and we consider

$$L_\lambda(\theta) := L_{\text{PDE+BC}}(\theta) + \lambda L_{\text{data}}(\theta).$$

This is the same structure as the fixed-weight FK-PINN objective in Section 5, with L_{data} playing the role of the FK supervision term and λ corresponding to λ_{FK} .

The proof combines the residual ill-conditioning mechanism in (De Ryck et al., 2024; Rathore et al., 2024) described by Proposition 4.8, with the local PL^* control for supervised least squares from (Liu et al., 2022). In an overparameterized regime, the data Gauss–Newton matrix is typically low rank, so adding it does not automatically bound the smallest eigenvalue of the sum. We therefore impose a compatibility condition, Assumption 4.10, which ensures that every direction relevant to the PDE and boundary terms is also seen by the data term. Under this condition, the smallest nonzero

eigenvalues of the combined Gauss–Newton matrix are uniformly bounded below, which yields the PL^* constant and the condition-number bound in Theorem 4.15.

7.1 PINN objective with noisy data

We consider a linear PDE with Dirichlet boundary conditions on a bounded domain $\Omega \subset \mathbb{R}^d$,

$$\begin{aligned}\mathcal{L}[u](x) &= f(x), \quad x \in \Omega, \\ u(x) &= g(x), \quad x \in \partial\Omega,\end{aligned}\tag{4.15}$$

where $\mathcal{L}$ is a linear differential operator, and f and g are given functions. We assume that (4.15) is well-posed in the sense that there exists a unique solution u^* in an appropriate function space, as we will make precise in Section 3.1.

Let $\theta \in \mathbb{R}^p$ denote the PINN parameter vector, and let $u(\cdot; \theta)$ be a neural network parameterization of u . We distinguish three types of training points:

- *Residual points* $\{x_i^{\text{res}}\}_{i=1}^{n_{\text{res}}} \subset \Omega$.
- *Boundary points* $\{x_j^{\text{bc}}\}_{j=1}^{n_{\text{bc}}} \subset \partial\Omega$.
- *Data points* $\{x_i^{\text{d}}\}_{i=1}^{n_{\text{d}}} \subset \Omega$.

Residual and boundary terms

The empirical residual and boundary losses are defined by

$$L_{\text{PDE+BC}}(\theta) := \frac{1}{2n_{\text{res}}} \sum_{i=1}^{n_{\text{res}}} (\mathcal{L}[u(x_i^{\text{res}}; \theta)] - f(x_i^{\text{res}}))^2 + \frac{1}{2n_{\text{bc}}} \sum_{j=1}^{n_{\text{bc}}} (u(x_j^{\text{bc}}; \theta) - g(x_j^{\text{bc}}))^2.\tag{4.16}$$

Noisy data term

We additionally observe (possibly noisy) function values at the data points,

$$y_i = u^*(x_i^{\text{d}}) + \xi_i, \quad i = 1, \dots, n_{\text{d}},$$

where the noise variables ξ_i are independent across i and satisfy $\mathbb{E}[\xi_i] = 0$.

The data-supervision term with weight $\lambda \geq 0$ is

$$L_{\text{data}}(\theta) := \frac{1}{2n_{\text{d}}} \sum_{i=1}^{n_{\text{d}}} (u(x_i^{\text{d}}; \theta) - y_i)^2.\tag{4.17}$$

The full PINN objective is then

$$L_\lambda(\theta) := L_{\text{PDE+BC}}(\theta) + \lambda L_{\text{data}}(\theta). \quad (4.18)$$

We denote by Θ_λ^* the set of global minimizers of L_λ :

$$\Theta_\lambda^* := \arg \min_{\theta \in \mathbb{R}^p} L_\lambda(\theta).$$

Stacked feature map and Gauss–Newton matrix.

To streamline notation, define the residual, boundary, and data feature maps

$$\begin{aligned} F_{\text{res}}(\theta) &:= \frac{1}{\sqrt{n_{\text{res}}}} (\mathcal{L}[u(x_1^{\text{res}}; \theta)] - f(x_1^{\text{res}}), \dots, \mathcal{L}[u(x_{n_{\text{res}}}^{\text{res}}; \theta)] - f(x_{n_{\text{res}}}^{\text{res}}))^{\top}, \\ F_{\text{bc}}(\theta) &:= \frac{1}{\sqrt{n_{\text{bc}}}} (u(x_1^{\text{bc}}; \theta) - g(x_1^{\text{bc}}), \dots, u(x_{n_{\text{bc}}}^{\text{bc}}; \theta) - g(x_{n_{\text{bc}}}^{\text{bc}}))^{\top}, \\ F_{\text{data}}(\theta) &:= \frac{1}{\sqrt{n_{\text{d}}}} (u(x_1^{\text{d}}; \theta) - y_1, \dots, u(x_{n_{\text{d}}}^{\text{d}}; \theta) - y_{n_{\text{d}}})^{\top}. \end{aligned}$$

Then

$$L_{\text{PDE+BC}}(\theta) = \frac{1}{2} \|F_{\text{res}}(\theta)\|_2^2 + \frac{1}{2} \|F_{\text{bc}}(\theta)\|_2^2, \quad L_{\text{data}}(\theta) = \frac{1}{2} \|F_{\text{data}}(\theta)\|_2^2.$$

We introduce the stacked feature map

$$F_\lambda(\theta) := \begin{bmatrix} F_{\text{res}}(\theta) \\ F_{\text{bc}}(\theta) \\ \sqrt{\lambda} F_{\text{data}}(\theta) \end{bmatrix} \in \mathbb{R}^{n_\lambda}, \quad n_\lambda := n_{\text{res}} + n_{\text{bc}} + n_{\text{d}}, \quad (4.19)$$

so that $L_\lambda(\theta) = \frac{1}{2} \|F_\lambda(\theta)\|_2^2$. Let $J_{F_\lambda}(\theta) \in \mathbb{R}^{n_\lambda \times p}$ denote the Jacobian of F_λ at θ . The associated empirical Gauss–Newton matrix is

$$G_\lambda(\theta) := J_{F_\lambda}(\theta)^{\top} J_{F_\lambda}(\theta) \in \mathbb{R}^{p \times p}.$$

Its eigenvalues describe the local curvature of L_λ in parameter space, and will provide the link between operator conditioning and PL-based optimization guarantees. From this point on, the conditioning question is reduced to bounding the smallest and largest eigenvalues of $G_\lambda(\theta)$ in the region where the PL* argument is applied.

Definition 4.7. Let $A \in \mathbb{R}^{p \times p}$ be symmetric and positive semidefinite. We write $\lambda_{\min}^+(A)$ for the smallest strictly positive eigenvalue of A .

7.2 Tangent function space and population Gauss–Newton (PDE-only)

We now recall the operator-theoretic interpretation of the PDE-only population Gauss–Newton matrix from [De Ryck et al. \(2024\)](#). This section mostly summarizes their construction in the finite-width setting, to make our later use of their results self-contained. The only reason to introduce the tangent space $\mathcal{H}(\theta)$, the map $T(\theta)$, and the operator $\mathcal{K}_\infty(\theta)$ is to express Gauss–Newton curvature in a form where the differential operator $\mathcal{L}$ appears as $\mathcal{A} = \mathcal{L}^* \mathcal{L}$. This will allow us to directly see which “directions” are poorly controlled by residual sampling and how a supervised term changes those directions.

Let $\theta \in \mathbb{R}^p$ be a parameter vector. For $j = 1, \dots, p$, we define the tangent functions

$$\phi_j(x; \theta) := \frac{\partial}{\partial \theta_j} u(x; \theta), \quad x \in \Omega.$$

The $\phi_j(\cdot; \theta)$ ’s span the corresponding tangent function space

$$\mathcal{H}(\theta) := \text{span}\{\phi_1(\cdot; \theta), \dots, \phi_p(\cdot; \theta)\} \subset L^2(\Omega, \mu),$$

where μ is the reference sampling measure on Ω (i.e., the distribution used to draw residual and data points).

We then introduce the linear map

$$T(\theta) : \mathbb{R}^p \rightarrow \mathcal{H}(\theta), \quad T(\theta)v := \sum_{j=1}^p v_j \phi_j(\cdot; \theta),$$

and its adjoint

$$T(\theta)^* : L^2(\Omega, \mu) \rightarrow \mathbb{R}^p, \quad (T(\theta)^* f)_j := \int_{\Omega} f(x) \phi_j(x; \theta) d\mu(x).$$

We then define the naturally associated (population) kernel integral operator as:

$$\mathcal{K}_\infty(\theta) := T(\theta)T(\theta)^* : L^2(\Omega, \mu) \rightarrow \mathcal{H}(\theta). \quad (4.20)$$

Here $\mathcal{H}(\theta)$ is the tangent function space spanned by $\{\phi_j(\cdot; \theta)\}$, and $\mathcal{K}_\infty(\theta) = T(\theta)T(\theta)^*$ is the corresponding (finite-width) neural tangent kernel operator in the sense of [De Ryck et al. \(2024\)](#); [Rathore et al. \(2024\)](#).

Population PDE objective and Gauss–Newton matrix.

For simplicity, we ignore the boundary term in this subsection. It contributes an additional positive semidefinite Gauss–Newton term, so keeping it separate can only strengthen the spectral lower bounds used later. With this simplification, the population counterpart of the residual loss is given by:

$$L_{\infty, \text{PDE}}(\theta) := \frac{1}{2} \int_{\Omega} (\mathcal{L}[u(x; \theta)] - f(x))^2 d\mu(x). \quad (4.21)$$

Define $\mathcal{A} := \mathcal{L}^* \mathcal{L}$, where $\mathcal{L}^*$ denotes the adjoint of $\mathcal{L}$ with respect to the $L^2(\Omega, \mu)$ inner product.

The population Gauss–Newton matrix for (4.21) is

$$G_{\infty, \text{PDE}}(\theta) := \int_{\Omega} \mathcal{L}[\nabla_{\theta} u(x; \theta)] \mathcal{L}[\nabla_{\theta} u(x; \theta)]^{\top} d\mu(x),$$

where $\nabla_{\theta} u(x; \theta) \in \mathbb{R}^p$ is the gradient of u with respect to θ . A direct computation shows that

$$G_{\infty, \text{PDE}}(\theta) = T(\theta)^* \mathcal{A} T(\theta). \quad (4.22)$$

By (De Ryck et al., 2024), the finite-width population Gauss–Newton matrix $G_{\infty, \text{PDE}}(\theta)$ has the same non-zero eigenvalues as the compact, self-adjoint operator

$$\mathcal{A} \circ \mathcal{K}_{\infty}(\theta) : \mathcal{H}(\theta) \rightarrow \mathcal{H}(\theta), \quad (\mathcal{A} \circ \mathcal{K}_{\infty}(\theta))f := \mathcal{A}(T(\theta)T(\theta)^* f),$$

and hence

$$\lambda_j(G_{\infty, \text{PDE}}(\theta)) = \lambda_j(\mathcal{A} \circ \mathcal{K}_{\infty}(\theta)), \quad \kappa(G_{\infty, \text{PDE}}(\theta)) = \kappa(\mathcal{A} \circ \mathcal{K}_{\infty}(\theta)), \quad (4.23)$$

for $j = 1, \dots, p$. We refer to De Ryck et al. (2024, Thm. 2.4) for the derivation, which is a finite-dimensional analogue of the fact that $S^* S$ and SS^* have the same non-zero eigenvalues for any bounded operator S between Hilbert spaces.

7.3 Population Gauss–Newton with FK data

We now incorporate the Feynman–Kac supervision term into the population analysis. For simplicity, we assume that residual and data points are sampled from the same measure μ on Ω , but the more general case follows from the same argument.

The population counterpart of the full loss (4.18) is

$$L_{\infty,\lambda}(\theta) := \frac{1}{2} \int_{\Omega} (\mathcal{L}[u(x;\theta)] - f(x))^2 d\mu(x) + \frac{\lambda}{2} \int_{\Omega} (u(x;\theta) - u^*(x))^2 d\mu(x). \quad (4.24)$$

The first term yields the Gauss–Newton matrix $T(\theta)^* \mathcal{A} T(\theta)$ as in (4.22). The data term contributes

$$G_{\infty,\text{data}}(\theta) := \lambda \int_{\Omega} \nabla_{\theta} u(x;\theta) \nabla_{\theta} u(x;\theta)^{\top} d\mu(x) = \lambda T(\theta)^* T(\theta),$$

since $T(\theta)v = \sum_{j=1}^p v_j \phi_j(\cdot;\theta)$ and $\nabla_{\theta} u(x;\theta) = (\phi_1(x;\theta), \dots, \phi_p(x;\theta))^{\top}$.

Thus the combined population Gauss–Newton matrix for $L_{\infty,\lambda}$ is

$$G_{\infty,\lambda}(\theta) := G_{\infty,\text{PDE}}(\theta) + G_{\infty,\text{data}}(\theta) = T(\theta)^* (\mathcal{A} + \lambda I) T(\theta). \quad (4.25)$$

Exactly as in the PDE-only case, $G_{\infty,\lambda}(\theta)$ is the matrix representation (in the basis $\{\phi_j(\cdot;\theta)\}_{j=1}^p$) of the compact, self-adjoint operator

$$(\mathcal{A} + \lambda I) \circ \mathcal{K}_{\infty}(\theta) : \mathcal{H}(\theta) \rightarrow \mathcal{H}(\theta),$$

and we obtain from De Ryck et al. (2024, Thm. 2.4) that their non-zero spectra and condition numbers coincide:

$$\lambda_j(G_{\infty,\lambda}(\theta)) = \lambda_j((\mathcal{A} + \lambda I) \circ \mathcal{K}_{\infty}(\theta)), \quad \kappa(G_{\infty,\lambda}(\theta)) = \kappa((\mathcal{A} + \lambda I) \circ \mathcal{K}_{\infty}(\theta)). \quad (4.26)$$

Intuitively, adding the data term simply replaces $\mathcal{A}$ by $\mathcal{A} + \lambda I$ in the operator-theoretic framework of De Ryck et al. (2024), thereby lifting the small eigenvalues associated with ill-conditioned PDE directions. This is the population analogue of Tikhonov regularization or ridge regression, but applied at the level of the PDE operator $\mathcal{A}$ composed with the tangent kernel $\mathcal{K}_{\infty}(\theta)$. This population calculation motivates what we look for empirically: if the data term contributes a matrix with eigenvalues bounded below independently of the number of residual points, then the sum $G_{\lambda} = G_{\text{res}} + G_{\text{bc}} + \lambda G_{\text{data}}$ should inherit a uniform lower bound once $\lambda > 0$ is fixed.

7.4 Finite-sample Gauss–Newton matrices and effective conditioning

We now move from the population operators in (4.22) and (4.25) to their empirical counterparts, and study how the data term affects the spectrum of G_{λ} built from finitely many residual, boundary, and data points.

Empirical Gauss–Newton matrices

The empirical residual and data Gauss–Newton matrices at a parameter θ are given by

$$G_{\text{res}}(\theta) := \frac{1}{n_{\text{res}}} \sum_{i=1}^{n_{\text{res}}} \mathcal{L}[\nabla_{\theta} u(x_i^{\text{res}}; \theta)] \mathcal{L}[\nabla_{\theta} u(x_i^{\text{res}}; \theta)]^{\top}, \quad (4.27)$$

$$G_{\text{data}}(\theta) := \frac{1}{n_{\text{d}}} \sum_{i=1}^{n_{\text{d}}} \nabla_{\theta} u(x_i^{\text{d}}; \theta) \nabla_{\theta} u(x_i^{\text{d}}; \theta)^{\top}. \quad (4.28)$$

The boundary Gauss–Newton term $G_{\text{bc}}(\theta)$ is defined analogously to $G_{\text{res}}(\theta)$, replacing $\mathcal{L}[u] - f$ by $u - g$ on $\partial\Omega$.

The full empirical Gauss–Newton matrix for L_{λ} at θ is then

$$G_{\lambda}(\theta) = G_{\text{res}}(\theta) + G_{\text{bc}}(\theta) + \lambda G_{\text{data}}(\theta). \quad (4.29)$$

In what follows, we focus on the behavior of $G_{\lambda}(\theta)$ at a global minimizer $\theta_{\lambda}^* \in \Theta_{\lambda}^*$, although the analysis applies more generally to any parameter θ in a region of interest.

Polynomial spectral decay for the PDE part

We assume that at the parameter of interest θ_{λ}^* the population operator $\mathcal{A} \circ \mathcal{K}_{\infty}(\theta_{\lambda}^*)$ has polynomially decaying eigenvalues:

$$\lambda_j(\mathcal{A} \circ \mathcal{K}_{\infty}(\theta_{\lambda}^*)) \leq C j^{-2\alpha}, \quad j = 1, 2, \dots, \quad (4.30)$$

for some constants $C > 0$ and $\alpha > 1/2$. This type of decay is typical for elliptic operators on smooth domains, possibly combined with smoothness properties of the tangent kernel $\mathcal{K}_{\infty}(\theta_{\lambda}^*)$ (see (De Ryck et al., 2024) for some examples).

Theorem 4.4 in the main text is proved in (Rathore et al., 2024, Theorem 8.4) in the same linear PDE and interpolation setting considered here. We will use the following finite-sample spectral scaling for the residual Gauss–Newton matrix at θ_{λ}^* , stated in the notation of Section 7.4.

Proposition 4.8. *There exist constants $c_{\text{res}} > 0$, $\Lambda_{\text{PDE}} < \infty$ and $\alpha > 1/2$ such that, for every $\delta \in (0, 1)$, there is $n_{\text{res}}^{\min}(\delta) \in \mathbb{N}$ with the following property: for all $n_{\text{res}} \geq n_{\text{res}}^{\min}(\delta)$, with probability at least $1 - \delta$ over the sampling of the residual points,*

$$\lambda_{\min}(G_{\text{res}}(\theta_{\lambda}^*)) \geq c_{\text{res}} n_{\text{res}}^{-\alpha}, \quad \lambda_{\max}(G_{\text{res}}(\theta_{\lambda}^*)) \leq \Lambda_{\text{PDE}}.$$

Proof. This is a restatement of (Rathore et al., 2024, Theorem 8.4) in the notation of (4.27), with $N = n_{\text{res}}$. $\square$

Proposition 4.8 captures the deterioration of the residual contribution under mesh refinement, as established in (Rathore et al., 2024). We now turn to the supervised term and its effect on the combined spectrum.

Data PL* condition and finite-sample concentration

The beneficial effect of the data term on conditioning requires that the tangent features at the data points are sufficiently rich and non-degenerate. Following the PL* viewpoint of (Liu et al., 2022), we encode this via a finite-width PL* result for the data term, Proposition 4.9, which yields a uniform spectral bound on the data tangent kernel and its associated empirical Gauss–Newton matrix. Informally, this proposition states that for a sufficiently wide network, the empirical data tangent kernel stays uniformly well conditioned in a ball around initialization, so that the supervised data loss satisfies a uniform PL* inequality with constants independent of n_d .

Proposition 4.9 (Data PL* via Liu et al. (2022)). *Consider the fully-connected network, initialization scheme, and fixed data points $\{x_i^d\}_{i=1}^{n_d} \subset \Omega$ in the setting of Liu et al. (2022, Section 4), and define the rescaled data feature map*

$$\mathcal{F}_{\text{data}}(\theta) := \frac{1}{\sqrt{n_d}}(u(x_1^d; \theta) - y_1, \dots, u(x_{n_d}^d; \theta) - y_{n_d}) \in \mathbb{R}^{n_d}.$$

Let θ_0 be the random initialization and

$$K_{\text{data}}(\theta) := D\mathcal{F}_{\text{data}}(\theta) D\mathcal{F}_{\text{data}}(\theta)^\top \in \mathbb{R}^{n_d \times n_d}$$

the associated data tangent kernel, and denote $\lambda_0 := \lambda_{\min}(K_{\text{data}}(\theta_0))$. Fix any radius $R > 0$, target $\mu_{\text{data}} \in (0, \lambda_0)$, and confidence level $\delta \in (0, 1)$. Then there exists a universal constant $C > 0$ such that, if

$$m \geq C \frac{n_d R^{6L+2}}{(\lambda_0 - \mu_{\text{data}})^2} \log\left(\frac{2n_d}{\delta}\right), \quad (4.31)$$

the following holds with probability at least $1 - \delta$ over the random initialization, simultaneously for all $\theta \in B(\theta_0, R)$:

$$\mu_{\text{data}} \leq \lambda_{\min}(K_{\text{data}}(\theta)) \leq \lambda_{\max}(K_{\text{data}}(\theta)) \leq M_{\text{data}}, \quad (4.32)$$

for some $M_{\text{data}} < \infty$ independent of n_{d} . In particular, the empirical data loss $L_{\text{data}}(\theta) = \frac{1}{2} \|\mathcal{F}_{\text{data}}(\theta)\|_2^2$ satisfies a μ_{data} -PL* inequality on $B(\theta_0, R)$. Moreover, $G_{\text{data}}(\theta) = D\mathcal{F}_{\text{data}}(\theta)^\top D\mathcal{F}_{\text{data}}(\theta)$ and $K_{\text{data}}(\theta)$ have the same nonzero eigenvalues, hence the nonzero eigenvalues of $G_{\text{data}}(\theta)$ lie in $[\mu_{\text{data}}, M_{\text{data}}]$. Equivalently, $\lambda_{\min}^+(G_{\text{data}}(\theta)) \geq \mu_{\text{data}}$ and $\lambda_{\max}(G_{\text{data}}(\theta)) \leq M_{\text{data}}$.

Proof sketch. This is the specialization of Liu et al. (2022, Theorem 4) to $\mathcal{F} = \mathcal{F}_{\text{data}}$, together with the identification of the PL* constant with $\lambda_{\min}(K_{\text{data}}(\theta))$ (Liu et al., 2022, Theorem 1) and the fact that $G_{\text{data}}(\theta)$ and $K_{\text{data}}(\theta)$ have the same non-zero eigenvalues. $\square$

Fix a confidence level $\delta_{\text{init}} \in (0, 1)$ and choose the width m so that (4.31) holds with $\delta = \delta_{\text{init}}$. Let $\mathcal{E}_{\text{init}}$ denote the corresponding high-probability initialization event. By Proposition 4.9, on $\mathcal{E}_{\text{init}}$ the following uniform spectral bounds hold on $B(\theta_0, R)$:

$$\lambda_{\min}^+(G_{\text{data}}(\theta)) \geq \mu_{\text{data}} \text{ and } \lambda_{\max}(G_{\text{data}}(\theta)) \leq M_{\text{data}} \text{ for all } \theta \in B(\theta_0, R). \quad (4.33)$$

When $p > n_{\text{d}}$, one typically has $\lambda_{\min}(G_{\text{data}}(\theta)) = 0$, so the relevant notion is the conditioning of $G_{\text{data}}(\theta)$ restricted to its range, quantified by $\lambda_{\min}^+(G_{\text{data}}(\theta))$.

At this stage we have a uniform lower bound on the nonzero curvature provided by the data term through (4.33). To conclude that the FK term stabilizes the smallest nonzero eigenvalues of the combined Gauss–Newton matrix, one needs an additional assumption that excludes parameter directions that influence the PDE residual while remaining invisible to the supervised data term.

The obstruction is geometric. In an overparameterized regime, $G_{\text{data}}(\theta)$ is typically low rank, so it cannot lift PDE curvature in parameter directions that are invisible to the data Jacobian. We therefore assume a compatibility condition that rules out such directions on the neighborhood where the PL* argument is applied.

Assumption 4.10. Fix a set $\mathcal{S} \subset B(\theta_0, R)$. For every $\theta \in \mathcal{S}$,

$$\text{Null}(G_{\text{data}}(\theta)) \subseteq \text{Null}(G_{\text{res}}(\theta) + G_{\text{bc}}(\theta)).$$

Assumption 4.10 states that any parameter direction that is invisible to the data term is also invisible to the PDE and boundary terms. Since the matrices are symmetric and positive semidefinite, this nullspace inclusion is equivalent to the range inclusion $\text{Range}(G_{\text{res}}(\theta) + G_{\text{bc}}(\theta)) \subseteq \text{Range}(G_{\text{data}}(\theta))$.

A simple way to see why such an assumption is needed is to consider a linear-in-parameters model $u(x; \theta) = \sum_{j=1}^p \theta_j \psi_j(x)$. If the FK points happen to satisfy $\sum_{j=1}^p v_j \psi_j(x_i^{\text{d}}) = 0$ for all FK points x_i^{d} for some nonzero v , then the FK loss has zero curvature in direction v . If at the same time $\mathcal{L}[\sum_{j=1}^p v_j \psi_j](x_k^{\text{res}}) \neq 0$ at some residual point x_k^{res} , then the PDE term still varies in direction v , and adding the FK term cannot prevent ill-conditioning along that direction.

We now make explicit the implication of Assumption 4.10 on the spectrum. It shows that, on the parameter region where the data tangent features remain non-degenerate, the FK term provides a uniform lower bound on the smallest nonzero Gauss–Newton eigenvalue.

Proposition 4.11. *Assume that Proposition 4.9 holds with radius $R > 0$, and let $\mathcal{S} \subset B(\theta_0, R)$ be a set on the corresponding initialization event. Assume also that Assumption 4.10 holds on $\mathcal{S}$. Then, on that initialization event, for every $\theta \in \mathcal{S}$,*

$$\lambda_{\min}^+(G_{\lambda}(\theta)) \geq \lambda \mu_{\text{data}}. \quad (4.34)$$

Proof. Work on the initialization event on which (4.33) holds on $\mathcal{S}$. Fix $\theta \in \mathcal{S}$. By Assumption 4.10 and the fact that $\text{Range}(A) = \text{Null}(A)^\perp$ for symmetric matrices, one has $\text{Range}(G_{\lambda}(\theta)) \subseteq \text{Range}(G_{\text{data}}(\theta))$. Let $v \in \text{Range}(G_{\lambda}(\theta))$ with $\|v\|_2 = 1$. Then $v \in \text{Range}(G_{\text{data}}(\theta))$, hence $v^\top G_{\text{data}}(\theta) v \geq \lambda_{\min}^+(G_{\text{data}}(\theta))$. Using positive semidefiniteness of $G_{\text{res}}(\theta)$ and $G_{\text{bc}}(\theta)$,

$$v^\top G_{\lambda}(\theta) v \geq \lambda v^\top G_{\text{data}}(\theta) v \geq \lambda \lambda_{\min}^+(G_{\text{data}}(\theta)) \geq \lambda \mu_{\text{data}},$$

where the last step uses (4.33). Taking the minimum over such v gives (4.34). $\square$

Remark 4.12. In the Dirichlet case the boundary loss has the same least-squares form as a supervised term, so its Gauss–Newton contribution $G_{\text{bc}}(\theta)$ is positive semidefinite. In Proposition 4.11 and Theorem 4.15 we only use this monotonicity, together with a uniform operator-norm bound on $G_{\text{bc}}(\theta_\lambda^*)$. Any additional non-degeneracy of the boundary sampling can only improve the spectral lower bounds, but we do not use it.

7.5 PL constant, condition number, and gradient descent

We now translate the spectral bounds from Proposition 4.11 into a Polyak–Łojasiewicz (PL) constant and condition number for L_λ , and derive implications for gradient descent.

PL inequality and PL-based condition number

Let $\mathcal{S} \subset \mathbb{R}^p$ be a set containing θ_λ^* . We say that L_λ satisfies a PL inequality with constant $\mu_\lambda > 0$ on $\mathcal{S}$ if

$$\frac{\|\nabla L_\lambda(\theta)\|_2^2}{2\mu_\lambda} \geq L_\lambda(\theta) - L_\lambda(\theta_\lambda^*), \quad \forall \theta \in \mathcal{S}. \quad (4.35)$$

Let β_{L_λ} denote a smoothness constant on $\mathcal{S}$, i.e. an upper bound on the operator norm of the Hessian,

$$\beta_{L_\lambda} \geq \sup_{\theta \in \mathcal{S}} \|H_{L_\lambda}(\theta)\|.$$

The PL-based condition number of L_λ on $\mathcal{S}$ is then defined by

$$\kappa_{L_\lambda}(\mathcal{S}) := \frac{\beta_{L_\lambda}}{\mu_\lambda}. \quad (4.36)$$

We will use the basic least-squares implication that underlies the PL* framework, as stated and used in (Liu et al., 2022, Theorem 1).

Lemma 4.13. *Let $L(\theta) = \frac{1}{2}\|F(\theta)\|_2^2$, where $F : \mathbb{R}^p \rightarrow \mathbb{R}^n$ is differentiable on a set $\mathcal{S}$. Suppose there exists $\theta^* \in \mathcal{S}$ with $F(\theta^*) = 0$. Define the sample tangent kernel $K_F(\theta) := J_F(\theta) J_F(\theta)^\top$. If there exists $\mu > 0$ such that $\lambda_{\min}(K_F(\theta)) \geq \mu$ holds for every $\theta \in \mathcal{S}$, then L satisfies the PL* inequality on $\mathcal{S}$ with constant μ .*

Proof. For $\theta \in \mathcal{S}$, $\nabla L(\theta) = J_F(\theta)^\top F(\theta)$, hence $\|\nabla L(\theta)\|_2^2 = F(\theta)^\top K_F(\theta) F(\theta)$. The assumed bound gives $\|\nabla L(\theta)\|_2^2 \geq \mu \|F(\theta)\|_2^2 = 2\mu L(\theta)$. Since $L(\theta^*) = 0$, this is the PL* inequality. This is the calculation used in (Liu et al., 2022, Theorem 1). $\square$

Lemma 4.14. *Let $F : \mathbb{R}^p \rightarrow \mathbb{R}^n$ be differentiable at θ , with Jacobian $J_F(\theta)$. Define $G_F(\theta) := J_F(\theta)^\top J_F(\theta)$ and $K_F(\theta) := J_F(\theta) J_F(\theta)^\top$. Then $G_F(\theta)$ and $K_F(\theta)$ have the same nonzero eigenvalues. If $J_F(\theta)$ has full row rank, then $K_F(\theta)$ is positive definite and $\lambda_{\min}(K_F(\theta)) = \lambda_{\min}^+(G_F(\theta))$.*

Applying Lemma 4.13 to F_λ shows that it suffices to control the smallest eigenvalue of the stacked sample kernel $K_{F_\lambda}(\theta)$ on $\mathcal{S}$.

Bounding the PL constant and condition number

Combining this relation with Proposition 4.11 yields the following result.

Theorem 4.15. *Let $\mathcal{S} \subset \mathbb{R}^p$ be a set containing a global minimizer θ_λ^* of L_λ . Assume that the architecture, initialization, and width satisfy Proposition 4.9 with radius $R > 0$ in such a way that $\mathcal{S} \subset B(\theta_0, R)$ on the corresponding initialization event. Let μ_{data} and M_{data} be the constants from (4.32). Assume in addition that $J_{F_\lambda}(\theta)$ has full row rank for every $\theta \in \mathcal{S}$, and that there exist constants $\beta_L < \infty$ and $C'' > 0$, independent of n_{res} , n_{d} , and λ , such that*

$$\sup_{\theta \in \mathcal{S}} \|H_{L_\lambda}(\theta)\| \leq \beta_L + C'' \lambda M_{\text{data}}. \quad (4.37)$$

Then, with high probability on the initialization event of Proposition 4.9, the loss L_λ satisfies a PL inequality on $\mathcal{S}$ with constant μ_λ obeying*

$$\mu_\lambda \geq \lambda \mu_{\text{data}}. \quad (4.38)$$

Moreover, L_λ is β_{L_λ} -smooth on $\mathcal{S}$ with $\beta_{L_\lambda} \leq \beta_L + C'' \lambda M_{\text{data}}$, and its PL-based condition number on $\mathcal{S}$ satisfies

$$\kappa_{L_\lambda}(\mathcal{S}) \leq \frac{\beta_L + C'' \lambda M_{\text{data}}}{\lambda \mu_{\text{data}}}. \quad (4.39)$$

Proof sketch. Work on the initialization event from Proposition 4.9, so that (4.33) holds on $\mathcal{S}$. Proposition 4.11 gives the uniform lower bound (4.34) on $\lambda_{\min}^+(G_\lambda(\theta))$ for every $\theta \in \mathcal{S}$. Since $J_{F_\lambda}(\theta)$ has full row rank on $\mathcal{S}$, Lemma 4.14 yields $\lambda_{\min}(K_{F_\lambda}(\theta)) = \lambda_{\min}^+(G_\lambda(\theta))$ on $\mathcal{S}$. Lemma 4.13 then gives a PL* inequality on $\mathcal{S}$ with constant μ_λ satisfying (4.38). The smoothness bound (4.37) yields $\beta_{L_\lambda} \leq \beta_L + C'' \lambda M_{\text{data}}$, and combining these bounds gives (4.39). $\square$

In the PDE-only case $\lambda = 0$, Assumption 4.8 implies $\mu_0 \asymp n_{\text{res}}^{-\alpha}$, so the PL based condition number $\kappa_{L_0}(\mathcal{S})$ grows like n_{res}^α . In contrast, (4.38)–(4.39) show that if $\lambda > 0$ is fixed independently of n_{res} , then μ_λ is bounded away from zero and $\kappa_{L_\lambda}(\mathcal{S})$ remains bounded as more residual points are added.

Gradient descent convergence

Consider gradient descent on L_λ ,

$$\theta_{k+1} = \theta_k - \eta \nabla L_\lambda(\theta_k),$$

with a fixed step size $\eta \in (0, 1/\beta_{L_\lambda}]$, and assume that the iterates stay in $\mathcal{S}$. Standard analysis of gradient descent under PL and smoothness assumptions gives

$$L_\lambda(\theta_{k+1}) - L_\lambda(\theta_\lambda^*) \leq (1 - \eta\mu_\lambda) (L_\lambda(\theta_k) - L_\lambda(\theta_\lambda^*)).$$

Choosing $\eta = 1/\beta_{L_\lambda}$ and using (4.36)–(4.39), we obtain

$$L_\lambda(\theta_{k+1}) - L_\lambda(\theta_\lambda^*) \leq \left(1 - \frac{1}{\kappa_{L_\lambda}(\mathcal{S})}\right) (L_\lambda(\theta_k) - L_\lambda(\theta_\lambda^*)).$$

Thus the number of iterations required to reach $L_\lambda(\theta_k) - L_\lambda(\theta_\lambda^*) \leq \varepsilon$ is

$$\mathcal{O}(\kappa_{L_\lambda}(\mathcal{S}) \log(1/\varepsilon)).$$

Using Theorem 4.15, this iteration complexity is bounded uniformly in n_{res} when $\lambda > 0$ is fixed.

8 Conclusion and discussion

We have introduced FK-PINNs, an easily implementable add-on to the standard PINN loss based on a Monte Carlo approximation of the Feynman–Kac representation of certain elliptic and parabolic PDEs. Our spectral analysis of the associated kernel operator shows that this term acts as an operator-level *preconditioner* for the loss, reducing the condition number of a natural Hessian-like operator and thereby providing quantifiable convergence guarantees for first-order optimization algorithms used to train these FK-PINNs. We stress again that although the presentation focused on linear elliptic Dirichlet problems for simplicity, the only role that the operator $\mathcal{L}$ plays is in the Feynman-Kac representation formula. Concretely, this means that our methodology extends to *any PDE for which a stochastic representation is available*, and there are many such examples, as we discuss in Appendix E. Before ending, we now briefly mention some natural questions raised by the current chapter:

- **More general families of PDEs.** A limitation of our approach is the reliance on the existence of a Feynman-Kac representation formula for the PDE of interest. While, as we show in Appendix E, this already constitutes a rather broad class, it turns out that this requirement can in fact be discarded altogether, as we discussed in the next item.

- **Other sources of supervised data.** As we just discussed, our analysis appears as if it were valid for PDEs with Feynman-Kac representation only. In fact, a careful read through the arguments presented in this Section 7 reveals that the “operator preconditioning” effect of FK supervision is *completely agnostic* to the “data” source. In practice, this means that the method applies for **any PDE**, as long as supervision data of reasonable quality is available. Such data could be obtained by external simulations, coarse FEM simulations, wet-lab measurements or any other source, and the conclusion would still apply.
- **Reliance on technical conditions.** A last important consideration worth mentioning here is the reliance on two theoretical assumptions whose soundness is debatable. Concerning the PL^* assumption on the PINN loss objective, despite there being a growing body of evidence suggesting that such an assumption is reasonable for mildly overparametrized neural networks (Liu et al., 2022; Taheri and Thrampoulidis, 2024), it would be an important contribution to the scientific machine learning community to establish rigorously whether such an assumption could reasonably hold for PINNs in the way they are commonly used by practitioners. Likewise, Assumption 4.5 on the well-alignment of the FK data with the PDE objective is hard to characterize in a meaningful way. Although it is clear to see that similar conclusions hold with weakened versions of this Assumption, having a clear characterization and understanding of its applicability would be of tremendous help for practitioners.

Chapter 5

End-to-end Error Guarantees for FK-PINNs Trained with Gradient Descent

In Chapter 4, we have introduced FK-PINNs, a variant of the standard PINN objective, and have shown that these enjoy a uniformly bounded condition number. In this Chapter, we will make use of this uniform control on the condition number to deduce convergence guarantees for gradient descent ran on FK-PINNs. Furthermore, we will combine these optimization guarantees with the learning-theoretic machinery from Chapter 2 and establish non-asymptotic *end-to-end* a priori error bounds for FK-PINNs trained with finitely many steps of (full-batch) gradient descent to solve linear elliptic PDEs, with explicit quantification of the error in terms of stepsize, gradient descent steps, network architecture, or PDE regularity. This chapter thus bridges the gap between the “ideal world” of ERM and the “real-world” of rigorous guarantees for SciML practitioners. The content of this chapter is, again, largely based off of (Tepakbong et al., 2026b).

1 End-to-end error analysis of FK-PINNs

We complement the operator-level PL^* analysis from Section 5 with a learning-theoretic error analysis of FK-PINNs. We first briefly recall the setup from Chapter 4: as per Section 2 therein, we consider the hypothesis class $\mathcal{F}$ of fully connected networks $u_{\mathbf{m}\theta}$ with depth L , layer-wise widths $(m_\ell)_{\ell=0}^L$ and tanh activation in all layers except for a linear output neuron. As described in Algorithm 2, the neural networks in $\mathcal{F}$ are trained to minimize the loss

$\mathcal{R}_{\text{FK-PINN}}$ (4.9), which serves as an empirical risk.

1.1 Setting and regime

We sample N_{FK} FK points x_i^{FK} , N_{int} interior points x_i^{int} , and $N_{\partial\Omega}$ boundary points $x_i^{\partial\Omega}$ independently and uniformly on their respective domains. At each FK point x_i^{FK} , a Monte Carlo approximation of the FK functional provides a supervision label $\hat{u}^{\text{MC}}(x_i^{\text{FK}})$, as described in Algorithm 1. Under the assumptions stated in Section 3, we establish non-asymptotic $L^2(\Omega)$ error bounds for trained FK-PINNs via an approximation–estimation–optimization decomposition. To our knowledge, these are the first *end-to-end* and *non-asymptotic* learning-error guarantees for PINNs with tanh activation—a widely used choice in practice—contrasting with earlier *asymptotic* convergence results in similar settings (e.g. Doumèche et al., 2025) and with non-asymptotic results restricted to piecewise polynomial activations (Jiao et al., 2022; Lu et al., 2022; Lei et al., 2025).

1.2 Non-asymptotic error bound

Our first result establishes that, with the Monte Carlo estimator described by Algorithm 1, each fidelity label in $\mathcal{R}_{\text{FK-PINN}}$ is a pointwise approximation of u^* with a deterministic bias of order $\sqrt{\Delta t}$ plus an exponentially small term in T_{max} and with sub-exponential Monte Carlo noise.

Proposition 5.1 (Sub-exponential FK noise). *Suppose that the FK data in $\mathcal{R}_{\text{FK-PINN}}$ is generated by simulating the diffusion with the Euler–Maruyama scheme described in Algorithm 1, with time step $0 < \Delta t \leq \Delta t_0$, truncation horizon $T_{\text{max}} \in (0, \infty]$ and a fixed number of Monte Carlo paths N_{MC} per point, where $\Delta t_0 > 0$ is a constant depending only on the problem data (as in Section 2). Then each FK datum admits a decomposition*

$$Y_i^{\text{FK}} = u^*(X_i^{\text{FK}}) + b(X_i^{\text{FK}}) + \zeta_i,$$

where $\{(X_i^{\text{FK}}, Y_i^{\text{FK}})\}_{i=1}^{N_{\text{FK}}}$ are i.i.d., the bias satisfies

$$|b(x)| \leq C_{\text{bias}}\sqrt{\Delta t} + C_T e^{-\kappa T_{\text{max}}} \quad \text{for all } x \in \Omega,$$

for constants $C_{\text{bias}}, C_T, \kappa > 0$ independent of Δt , N_{FK} and N_{MC} , with the convention $e^{-\kappa T_{\text{max}}} = 0$ when $T_{\text{max}} = \infty$, and, conditional on X_i^{FK} , the fluctuation ζ_i is mean-zero and sub-exponential with parameters that do not depend on N_{FK} or N_{MC} . A proof is given in Appendix 2.

We can now state our main learning-theoretic estimates.

Theorem 5.2 (Error analysis for depth-3 FK-PINNs). *Assume the setting of Section 1 and the PDE, regularity and PL^* conditions of Section 3. Let $s \geq 3$ and suppose that $u^* \in W^{s,\infty}(\Omega)$ and set $\beta := (s-2)/(3d+2(s-2))$. Consider FK-PINNs with depth $L = 3$ and width m , trained, according to Algorithm 2, by T steps of gradient descent on $\mathcal{R}_{\text{FK-PINN}}$ with sufficiently small step size. Suppose that the max layer width $m \asymp N_{\text{FK}}^{\frac{d}{2(3d+2(s-2))}}$, that $N_{\text{int}}, N_{\partial\Omega} \gtrsim N_{\text{FK}}$, and that the FK time step Δt and truncation horizon T_{max} are chosen so that the bias in Proposition 5.1 is at most of order $N_{\text{FK}}^{-\beta}$. Then, for any $\delta \in (0, 1)$, there exist constants $C, c_{\text{opt}} > 0$, independent of $(N_{\text{int}}, N_{\partial\Omega}, N_{\text{FK}})$ and T , such that with probability at least $1 - \delta$,*

$$\|u_{\mathbf{m}\theta_T} - u^*\|_{L^2(\Omega)} \leq C(N_{\text{FK}}^{-\beta} + e^{-c_{\text{opt}}T}), \quad (5.1)$$

up to logarithmic factors in N_{FK} and in the sample sizes.

Corollary 5.3 (Rate for the FK-PINN algorithm). *In the setting of Theorem 5.2, choose the width m and FK parameters $(\Delta t, T_{\text{max}})$ as in the theorem, and run gradient descent for*

$$T \gtrsim \log N_{\text{FK}}$$

iterations. Let $\hat{u}$ denote the resulting depth-3 FK-PINN. Then, up to logarithmic factors in N_{FK} and in the sample sizes, there exists a constant $C > 0$, independent of N_{FK} , such that with probability at least $1 - N_{\text{FK}}^{-10}$,

$$\|\hat{u} - u^*\|_{L^2(\Omega)} \leq C N_{\text{FK}}^{-\beta}, \quad (5.2)$$

where $\beta := (s-2)/(3d+2(s-2))$. See Appendix 3, Corollary 5.33 for proofs.

This result shows that, when N_{FK} grows and the width is tuned as in Theorem 5.2, FK-PINNs achieve the usual Sobolev rate in terms of N_{FK} , in line with the convergence theory of PINNs with dense physics sampling (Doumèche et al., 2025). In contrast, in the genuinely sparse FK regime where N_{FK} is fixed while N_{int} and $N_{\partial\Omega}$ increase, the error bound is saturated by the FK term and cannot vanish uniformly, which quantifies the statistical “cost” paid for the improved conditioning brought by FK supervision.

Remark 5.4. We bring to the reader’s attention here that the error bounds are given in $L^2(\Omega)$. This is due to the fact that, following the setup of Chapter 4, we are not working with boundary-adapted PINNs. As explained in Chapter 3, we can therefore not hope for estimates in any norm stronger than $H^{1/2}(\Omega)$ in this case. It is however clear that all the machinery developed in this chapter extends straightforwardly to the hypothesis space of boundary-adapted PINNs.

2 Proof of Proposition 5.1: influence of Monte Carlo Feynman–Kac label noise

This section provides the probabilistic error analysis for the Monte Carlo Feynman–Kac estimator used to construct the FK supervision labels in $\mathcal{R}_{\text{FK-PINN}}$. Our goal is to prove Proposition 5.1, which states that each FK datum can be written as a noisy point evaluation of the true solution u^* with sub-exponential noise whose parameters do not depend on the number of FK points.

Throughout this section, we fix a spatial point $x \in \Omega$ and study the error of the Monte Carlo Feynman–Kac estimator at this point. All constants may depend on the model data (domain, coefficients, source and boundary terms) and, when a truncation horizon $T_{\max}$ is present, on $T_{\max}$. They do *not* depend on the Monte Carlo sample size N_{MC} and they do not depend on the time step Δt once Δt is restricted to be smaller than a suitable threshold.

2.1 Setting and numerical scheme

We recall the diffusion setting that underlies the Feynman–Kac representation.

Let $\Omega \subset \mathbb{R}^d$ be a bounded domain with C^2 boundary $\partial\Omega$. Let $b : \mathbb{R}^d \rightarrow \mathbb{R}^d$ and $\sigma : \mathbb{R}^d \rightarrow \mathbb{R}^{d \times m}$ be globally Lipschitz functions. We assume that σ is uniformly elliptic, i.e. there exist constants $0 < \underline{\lambda} \leq \bar{\lambda} < \infty$ such that

$$\underline{\lambda} |v|^2 \leq v^\top (\sigma \sigma^\top)(x) v \leq \bar{\lambda} |v|^2, \quad \forall x, v \in \mathbb{R}^d.$$

Let $(W_t)_{t \geq 0}$ be an m -dimensional Brownian motion on some filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$.

The diffusion process $X = (X_t)_{t \geq 0}$ is the unique strong solution of

$$dX_t = b(X_t) dt + \sigma(X_t) dW_t, \quad X_0 = x \in \Omega. \quad (5.3)$$

We denote by

$$\tau := \inf\{t \geq 0 : X_t \notin \Omega\}$$

the first exit time of the diffusion from Ω , and assume $\tau < \infty$ almost surely for all starting points $x \in \Omega$.

Let $f : \bar{\Omega} \rightarrow \mathbb{R}$, $g : \partial\Omega \rightarrow \mathbb{R}$ and $V : \bar{\Omega} \rightarrow \mathbb{R}$ be bounded measurable functions. In the main text these correspond to the source term, boundary

data and zeroth-order potential of the elliptic boundary value problem. The associated (continuous-time) Feynman–Kac payoff is

$$Y := \int_0^\tau \exp\left(-\int_0^t V(X_s) ds\right) f(X_t) dt + \exp\left(-\int_0^\tau V(X_s) ds\right) g(X_\tau). \quad (5.4)$$

Under the regularity and well-posedness assumptions in Appendix D.1 of the main text, the elliptic boundary value problem admits a unique solution u^* , and one has the Feynman–Kac representation

$$u^*(x) = \mathbb{E}^x[Y],$$

where $\mathbb{E}^x$ denotes expectation for the diffusion (5.3) started at $X_0 = x$.

For the numerical approximation we fix a time step $\Delta t > 0$ and define the Euler–Maruyama scheme $(X_k^{\Delta t})_{k \in \mathbb{N}_0}$ by

$$X_0^{\Delta t} = x, \quad X_{k+1}^{\Delta t} = X_k^{\Delta t} + b(X_k^{\Delta t}) \Delta t + \sigma(X_k^{\Delta t}) \Delta W_k, \quad (5.5)$$

where $\Delta W_k := W_{(k+1)\Delta t} - W_{k\Delta t}$. The corresponding discrete exit time is

$$\tau_{\Delta t} := \inf\{k\Delta t : X_k^{\Delta t} \notin \Omega\}, \quad (5.6)$$

with the convention $\inf \emptyset = \infty$. Writing

$$N_{\Delta t} := \inf\{k \in \mathbb{N}_0 : X_k^{\Delta t} \notin \Omega\},$$

we have $\tau_{\Delta t} = N_{\Delta t} \Delta t$.

The discretized payoff functional is

$$Y^{\Delta t} := \sum_{k=0}^{N_{\Delta t}-1} \exp\left(-\Delta t \sum_{j=0}^{k-1} V(X_j^{\Delta t})\right) f(X_k^{\Delta t}) \Delta t + \exp\left(-\Delta t \sum_{j=0}^{N_{\Delta t}-1} V(X_j^{\Delta t})\right) g(X_{N_{\Delta t}}^{\Delta t}). \quad (5.7)$$

The associated time-discretized value at x is

$$u_{\Delta t}(x) := \mathbb{E}^x[Y^{\Delta t}].$$

Given $N_{\text{MC}} \in \mathbb{N}$, we consider i.i.d. copies $(Y^{\Delta t, (i)})_{i=1}^{N_{\text{MC}}}$ of $Y^{\Delta t}$ and define the Monte Carlo estimator (for fixed x and Δt)

$$\hat{u}^{\text{MC}}(x) := \frac{1}{N_{\text{MC}}} \sum_{i=1}^{N_{\text{MC}}} Y^{\Delta t, (i)}. \quad (5.8)$$

In practice [Algorithm 1](#) simulates trajectories until they exit Ω or until a truncation horizon $T_{\max} \in (0, \infty]$ is reached. For $T_{\max} = \infty$ this reduces to the infinite horizon estimator $\hat{u}^{\text{MC}}(x)$ defined above. For a finite truncation horizon we set $N_T := \lfloor T_{\max}/\Delta t \rfloor$ and define the truncated discrete stopping index

$$\kappa_{\Delta t}^{T_{\max}} := \min\{N_{\Delta t}, N_T\},$$

together with the truncated payoff

$$\begin{aligned} Y^{\Delta t, T_{\max}} := & \sum_{k=0}^{\kappa_{\Delta t}^{T_{\max}}-1} \exp\left(-\Delta t \sum_{j=0}^{k-1} V(X_j^{\Delta t})\right) f(X_k^{\Delta t}) \Delta t \\ & + \exp\left(-\Delta t \sum_{j=0}^{\kappa_{\Delta t}^{T_{\max}}-1} V(X_j^{\Delta t})\right) g(X_{\kappa_{\Delta t}^{T_{\max}}}^{\Delta t}). \end{aligned} \quad (5.9)$$

We write $u_{\Delta t}^{T_{\max}}(x) := \mathbb{E}^x[Y^{\Delta t, T_{\max}}]$ for the corresponding truncated value. For $T_{\max} = \infty$ we adopt the convention $Y^{\Delta t, \infty} := Y^{\Delta t}$ and $u_{\Delta t}^{\infty}(x) := u_{\Delta t}(x)$ so that the notation covers both finite and infinite horizons. Our aim is to control the tails of FK estimators of the form $\hat{u}^{\text{MC}}(x)$ and of their truncated counterparts, and hence show that the resulting FK labels behave like noisy point evaluations of u^* with sub-exponential noise.

2.2 Exit time estimates and exponential moments

We first establish exponential moments for the continuous exit time τ and the discrete exit time $\tau_{\Delta t}$. These are the key inputs for controlling the tails of the payoffs Y and $Y^{\Delta t}$ and hence the Monte Carlo error.

Continuous exit time

Theorem 5.5 (Exponential moments of the continuous exit time). *Suppose that $\Omega \subset \mathbb{R}^d$ is a bounded C^2 domain and that the diffusion X defined by (5.3) has globally Lipschitz coefficients and is uniformly elliptic. Then there exist constants $\lambda_c > 0$ and $C_c < \infty$ such that*

$$\sup_{y \in \bar{\Omega}} \mathbb{E}^y[e^{\lambda_c \tau}] \leq C_c.$$

In particular, the law of τ has exponential tails uniformly over starting points $y \in \bar{\Omega}$.

Proof. Under these assumptions, the generator $\mathcal{L}$ of X with homogeneous Dirichlet boundary conditions on $\partial\Omega$ admits a discrete spectrum with strictly positive first eigenvalue $\lambda_1 > 0$ (Pinsky, 1995). The killed semigroup $(P_t^\Omega)_{t \geq 0}$ therefore satisfies

$$\|P_t^\Omega\|_{L^\infty(\Omega) \rightarrow L^\infty(\Omega)} \leq Ce^{-\lambda_1 t}, \quad t \geq 0$$

for some constant $C < \infty$. It follows that

$$\sup_{y \in \bar{\Omega}} \mathbb{P}^y(\tau > t) = \sup_{y \in \bar{\Omega}} P_t^\Omega \mathbf{1}(y) \leq Ce^{-\lambda_1 t}, \quad t \geq 0.$$

Fix any $\lambda_c \in (0, \lambda_1)$. Then, integrating the tail bound,

$$\mathbb{E}^y[e^{\lambda_c \tau}] = 1 + \int_0^\infty \lambda_c e^{\lambda_c t} \mathbb{P}^y(\tau > t) dt \leq 1 + C\lambda_c \int_0^\infty e^{-(\lambda_1 - \lambda_c)t} dt < \infty,$$

with the bound uniform in $y \in \bar{\Omega}$. This proves the claim for a suitable constant C_c . $\square$

Discrete exit time

We next recall moment estimates for the discrete exit time $\tau_{\Delta t}$. We will rely on the results of Bouchard et al. (2017):

Theorem 5.6 (Moments of continuous and discrete exit times). *Assume that $\Omega \subset \mathbb{R}^d$ is a bounded C^2 domain with non-characteristic boundary, and that the coefficients b, σ in (5.3) are globally Lipschitz and of linear growth. Let τ be as above and $\tau_{\Delta t}$ be defined by (5.6). Then, for each integer $p \geq 1$, there exist finite constants $C_p < \infty$ and $\Delta t_p > 0$ such that for all $0 < \Delta t \leq \Delta t_p$,*

$$\begin{aligned} \sup_{y \in \bar{\Omega}} \mathbb{E}^y[\tau^p] &\leq C_p, \\ \sup_{y \in \bar{\Omega}} \mathbb{E}^y[\tau_{\Delta t}^p] &\leq C_p, \\ \sup_{y \in \bar{\Omega}} \mathbb{E}^y[|\tau_{\Delta t} - \tau|^p] &\leq C_p \Delta t^{p/2}. \end{aligned}$$

Proof. This is a specialization of the main results in Bouchard et al. (2017). See in particular Theorems 3.1 there, taken in the special case of globally Lipschitz coefficients and a bounded C^2 domain. $\square$

Combining Theorems 5.5 and 5.6, one can show that $\tau_{\Delta t}$ admits exponential moments uniformly over Δt small enough, by an elementary argument which we sketch in the next two lemmas.

Lemma 5.7 (Uniform exit in finite time). *Under the standing assumptions, there exist constants $T_* > 0$ and $p_0 \in (0, 1)$ such that*

$$\inf_{y \in \bar{\Omega}} \mathbb{P}^y(\tau \leq T_*) \geq p_0.$$

Moreover, there exist $\Delta t_0 > 0$ and $p \in (0, 1)$ such that for all $0 < \Delta t \leq \Delta t_0$,

$$\inf_{y \in \bar{\Omega}} \mathbb{P}^y(\tau_{\Delta t} \leq T_*) \geq p.$$

Proof sketch. For the continuous exit time, Theorem 5.5 yields $\sup_{y \in \bar{\Omega}} \mathbb{P}^y(\tau > t) \leq C_c e^{-\lambda_c t}$, hence, choosing T_* large enough that $C_c e^{-\lambda_c T_*} \leq 1/2$ gives

$$\inf_{y \in \bar{\Omega}} \mathbb{P}^y(\tau \leq T_*) \geq 1/2 =: p_0.$$

By Theorem 5.6, for some $p \geq 1$ we have $\sup_{y \in \bar{\Omega}} \mathbb{E}^y[|\tau_{\Delta t} - \tau|^p] \rightarrow 0$ as $\Delta t \rightarrow 0$, hence, for any $\varepsilon > 0$,

$$\sup_{y \in \bar{\Omega}} \mathbb{P}^y(|\tau_{\Delta t} - \tau| > \varepsilon) \leq \varepsilon^{-p} \sup_y \mathbb{E}^y[|\tau_{\Delta t} - \tau|^p] \rightarrow 0,$$

so $\tau_{\Delta t} \rightarrow \tau$ in probability, uniformly in y . Since the law of τ has a density (see, e.g., Pinsky (1995)), we have $\mathbb{P}^y(\tau = T_*) = 0$ for all y , and thus $\mathbb{P}^y(\tau_{\Delta t} \leq T_*) \rightarrow \mathbb{P}^y(\tau \leq T_*)$ uniformly in $y \in \bar{\Omega}$ as $\Delta t \rightarrow 0$. Hence for Δt small enough we still have

$$\inf_{y \in \bar{\Omega}} \mathbb{P}^y(\tau_{\Delta t} \leq T_*) \geq p_0/2 =: p \in (0, 1).$$

□

Proposition 5.8 (Exponential moments of the discrete exit time). *Under the standing assumptions, there exist constants $\lambda_d > 0$, $C_d < \infty$ and $\Delta t_0 > 0$ such that*

$$\sup_{y \in \bar{\Omega}, 0 < \Delta t \leq \Delta t_0} \mathbb{E}^y[e^{\lambda_d \tau_{\Delta t}}] \leq C_d.$$

Proof sketch. Let $T_* > 0$, $p \in (0, 1)$ and $\Delta t_0 > 0$ be as in Lemma 5.7. In particular, $\inf_y \mathbb{P}^y(\tau_{\Delta t} \leq T_*) \geq p$, so $\sup_y \mathbb{P}^y(\tau_{\Delta t} > T_*) \leq 1 - p$ for all $0 < \Delta t \leq \Delta t_0$.

Consider the continuous-time interpolation of the Euler scheme, which is a time-homogeneous Markov process. By the (strong) Markov property at times kT_* ,

$$\mathbb{P}^y(\tau_{\Delta t} > (k+1)T_*) = \mathbb{E}^y[\mathbf{1}_{\{\tau_{\Delta t} > kT_*\}} \mathbb{P}^{X_{kT_*}^{\Delta t}}(\tau_{\Delta t} > T_*)] \leq (1-p) \mathbb{P}^y(\tau_{\Delta t} > kT_*).$$

Iterating this inequality gives $\sup_{y \in \bar{\Omega}} \mathbb{P}^y(\tau_{\Delta t} > kT_*) \leq (1-p)^k$ for all $k \in \mathbb{N}$, $0 < \Delta t \leq \Delta t_0$. For any $t \geq 0$ with $kT_* \leq t < (k+1)T_*$, we have $\{\tau_{\Delta t} > t\} \subset \{\tau_{\Delta t} > kT_*\}$, so the same bound holds with t in place of kT_* .

Proceeding as in the proof of Theorem 5.5, we use the representation $\mathbb{E}^y[e^{\lambda \tau_{\Delta t}}] = 1 + \int_0^\infty \lambda e^{\lambda t} \mathbb{P}^y(\tau_{\Delta t} > t) dt$, and split the integral over the intervals $[kT_*, (k+1)T_*)$, yielding a geometric series upper bound by the previous paragraph.

Choosing $\lambda_d > 0$ so small that $e^{\lambda_d T_*}(1-p) < 1$, the series in question converges and its sum is finite and independent of y and $\Delta t \in (0, \Delta t_0]$. This gives the claimed bound with some finite constant C_d . $\square$

2.3 Exponential moments and tails for the payoff

We now transfer the exponential-moment bounds for τ and $\tau_{\Delta t}$ to the payoffs Y and $Y^{\Delta t}$.

Lemma 5.9 (Deterministic bound for $Y^{\Delta t}$). *Assume that f and g are bounded and that $V \geq 0$ on $\bar{\Omega}$. Then there exist constants $C_f, C_g \geq 0$ such that*

$$|f(x)| \leq C_f \quad \forall x \in \bar{\Omega}, \quad |g(z)| \leq C_g \quad \forall z \in \partial\Omega,$$

and

$$|Y^{\Delta t}| \leq C_f \tau_{\Delta t} + C_g \quad \text{almost surely.}$$

Proof. By boundedness of f and g there exist $C_f, C_g \geq 0$ with the stated properties. Since $V \geq 0$, all exponential factors in (5.7) are bounded by one. Thus

$$\begin{aligned} |Y^{\Delta t}| &\leq \sum_{k=0}^{N_{\Delta t}-1} |f(X_k^{\Delta t})| \Delta t + |g(X_{N_{\Delta t}}^{\Delta t})| \\ &\leq C_f N_{\Delta t} \Delta t + C_g = C_f \tau_{\Delta t} + C_g. \end{aligned}$$

$\square$

A similar argument yields a deterministic bound for Y in terms of τ .

Lemma 5.10 (Deterministic bound for Y). *Under the same assumptions as in Lemma 5.9, one has*

$$|Y| \leq C_f \tau + C_g \quad \text{almost surely.}$$

Combining these bounds with Theorem 5.5 and Proposition 5.8, we obtain exponential moments for the payoffs.

Proposition 5.11 (Exponential moments for Y and $Y^{\Delta t}$). *Under the standing assumptions and assuming in addition that $V \geq 0$ on $\bar{\Omega}$, there exist constants $\lambda_Y > 0$, $C_Y < \infty$ and $\Delta t_0 > 0$ such that*

$$\sup_{y \in \bar{\Omega}} \mathbb{E}^y[e^{\lambda_Y |Y|}] \leq C_Y,$$

and

$$\sup_{y \in \bar{\Omega}, 0 < \Delta t \leq \Delta t_0} \mathbb{E}^y[e^{\lambda_Y |Y^{\Delta t}|}] \leq C_Y.$$

Proof. By Lemma 5.9,

$$e^{\lambda |Y^{\Delta t}|} \leq e^{\lambda C_g} e^{\lambda C_f \tau \Delta t},$$

and by Lemma 5.10,

$$e^{\lambda |Y|} \leq e^{\lambda C_g} e^{\lambda C_f \tau}.$$

Using Theorem 5.5 and Proposition 5.8, choose $\lambda_Y > 0$ small enough that $\lambda_Y C_f < \lambda_c$ and $\lambda_Y C_f < \lambda_d$. Then the stated bounds follow immediately, with

$$C_Y := e^{\lambda_Y C_g} \max\{C_c, C_d\}.$$

□

Remark 5.12. A real-valued random variable Z is sub-exponential if there exists $\alpha > 0$ with $\mathbb{E}[e^{\lambda Z}] < \infty$ for all $|\lambda| < \alpha$. Proposition 5.11 shows that Y and $Y^{\Delta t}$ are sub-exponential, with sub-exponential norms bounded uniformly in Δt for $\Delta t \leq \Delta t_0$.

2.4 Concentration for the Monte Carlo estimator at fixed Δt

We now derive non-asymptotic tail bounds for the Monte Carlo estimator $\hat{u}^{\text{MC}}(x)$ at a fixed time step $\Delta t \leq \Delta t_0$. The key tool is a Bernstein-type inequality for sub-exponential random variables.

A Bernstein inequality for sub-exponential variables

Theorem 5.13 (Bernstein inequality for sub-exponential variables). *Let $Z_1, \dots, Z_N$ be i.i.d. real-valued random variables with $\mathbb{E}[Z_i] = 0$ and suppose there exist constants $\nu^2, b > 0$ satisfying*

$$\mathbb{E}[e^{\lambda Z_i}] \leq \exp\left(\frac{\nu^2 \lambda^2}{2}\right) \quad \text{for all } |\lambda| < 1/b.$$

Then there exists a universal constant $c > 0$ such that, for all $t > 0$,

$$\mathbb{P}\left(\left|\frac{1}{N} \sum_{i=1}^N Z_i\right| > t\right) \leq 2 \exp\left(-cN \min\left(\frac{t^2}{\nu^2}, \frac{t}{b}\right)\right).$$

Proof. See, for example, [Vershynin \(2018, Theorem 2.8.1\)](#). $\square$

Application to the Euler payoff

We apply Theorem 5.13 to the centered Euler payoff $Y^{\Delta t} - u_{\Delta t}(x)$.

Lemma 5.14 (Sub-exponential parameters for $Y^{\Delta t} - u_{\Delta t}(x)$). *Under the standing assumptions, there exist constants $\nu > 0$, $b > 0$ and $\Delta t_0 > 0$ such that, for all $\Delta t \leq \Delta t_0$, the centered random variable $Z^{\Delta t} := Y^{\Delta t} - u_{\Delta t}(x)$ satisfies*

$$\mathbb{E}[e^{\lambda Z^{\Delta t}}] \leq \exp\left(\frac{\nu^2 \lambda^2}{2}\right) \quad \text{for all } |\lambda| < 1/b.$$

The constants ν and b depend only on the model data and can be chosen independent of $\Delta t \leq \Delta t_0$.

Proof. Proposition 5.11 guarantees that there exist $\lambda_Y > 0$ and $C_Y < \infty$ such that

$$\sup_{0 < \Delta t \leq \Delta t_0} \mathbb{E}[e^{\lambda_Y |Y^{\Delta t}|}] \leq C_Y.$$

This implies that the moment generating function $\mathbb{E}[e^{\lambda Y^{\Delta t}}]$ is finite on $(-\lambda_Y, \lambda_Y)$ uniformly in $\Delta t \leq \Delta t_0$. Since $Z^{\Delta t} = Y^{\Delta t} - u_{\Delta t}(x)$ differs from $Y^{\Delta t}$ by a constant shift, the same holds for $Z^{\Delta t}$.

A standard argument (see [Vershynin \(2018, Section 2.7\)](#)) shows that there exist parameters (ν, b) depending only on the exponential-moment bounds of $Y^{\Delta t}$ such that

$$\mathbb{E}[e^{\lambda Z^{\Delta t}}] \leq \exp\left(\frac{\nu^2 \lambda^2}{2}\right) \quad \text{for all } |\lambda| < 1/b,$$

uniformly over $\Delta t \leq \Delta t_0$. We will omit the technical details here. $\square$

Combining Lemma 5.14 with Theorem 5.13 yields the following concentration inequality.

Proposition 5.15 (Concentration for the Monte Carlo error at fixed Δt). *Under the standing assumptions, there exist constants $c > 0$, $\nu > 0$, $b > 0$ and $\Delta t_0 > 0$ such that for all $\Delta t \leq \Delta t_0$, all $N_{\text{MC}} \in \mathbb{N}$ and all $t > 0$,*

$$\mathbb{P}\left(|\hat{u}^{\text{MC}}(x) - u_{\Delta t}(x)| > t\right) \leq 2 \exp\left(-cN_{\text{MC}} \min\left(\frac{t^2}{\nu^2}, \frac{t}{b}\right)\right).$$

Proof. For fixed x , Δt and N_{MC} , define $Z_i^{\Delta t} := Y^{\Delta t, (i)} - u_{\Delta t}(x)$ for $i = 1, \dots, N_{\text{MC}}$. These variables are i.i.d. with zero mean, and by Lemma 5.14 they satisfy the moment generating function condition of Theorem 5.13 with parameters (ν, b) independent of $\Delta t \leq \Delta t_0$. Since

$$\hat{u}^{\text{MC}}(x) - u_{\Delta t}(x) = \frac{1}{N_{\text{MC}}} \sum_{i=1}^{N_{\text{MC}}} Z_i^{\Delta t},$$

Theorem 5.13 yields the stated inequality. $\square$

Remark 5.16 (sub-Gaussian vs exponential regimes). The bound in Proposition 5.15 has two regimes. For deviations $0 < t \leq \nu^2/b$, the quadratic term dominates and the tails behave essentially like $\exp(-cN_{\text{MC}}t^2/\nu^2)$. For larger deviations, the linear term dominates and the decay is of order $\exp(-cN_{\text{MC}}t/b)$. In both cases the constants are independent of N_{MC} and of Δt for $\Delta t \leq \Delta t_0$.

2.5 Discretization bias

We next bound the bias $u_{\Delta t}(x) - u^*(x)$ of the time-discretized FK value.

Theorem 5.17 (Weak error for killed Euler schemes). *Assume that $\Omega \subset \mathbb{R}^d$ is a bounded C^2 domain with non-characteristic boundary, that the coefficients b, σ in (5.3) are C_b^4 with bounded derivatives up to order 4, and that f, g, V are bounded with bounded derivatives up to the necessary order. Then there exist constants $C_{\text{bias}} < \infty$ and $\Delta t_0 > 0$ such that, for all $0 < \Delta t \leq \Delta t_0$,*

$$|u_{\Delta t}(x) - u^*(x)| \leq C_{\text{bias}} \sqrt{\Delta t}.$$

Proof. This is a specialization of the weak error estimates for killed Euler schemes in Gobet (2000); Gobet and Menozzi (2010). Our assumptions on the regularity of b, σ, f, g, V and on the geometry of Ω are stronger than those in the cited papers, so their hypotheses are satisfied. The functional $Y^{\Delta t}$ in (5.7) matches the discretization considered there. The stated bound follows, with a constant C_{bias} independent of Δt for Δt small enough. $\square$

Lemma 5.18 (Bias due to time truncation). *Assume the standing assumptions of this appendix and those of Lemma 5.9. Then there exist constants $\tilde{\lambda} > 0$, $C_T < \infty$ and $\Delta t_0 > 0$ such that, for all $x \in \Omega$, all $0 < \Delta t \leq \Delta t_0$ and all $T_{\text{max}} > 0$,*

$$|u_{\Delta t}^{T_{\text{max}}}(x) - u_{\Delta t}(x)| \leq C_T e^{-\tilde{\lambda} T_{\text{max}}}.$$

Proof. On the event $\{\tau_{\Delta t} \leq T_{\max}\}$ the discrete trajectory exits Ω before the truncation horizon and $\kappa_{\Delta t}^{T_{\max}} = N_{\Delta t}$, hence $Y^{\Delta t, T_{\max}} = Y^{\Delta t}$. On the complement $\{\tau_{\Delta t} > T_{\max}\}$ the payoffs $Y^{\Delta t}$ and $Y^{\Delta t, T_{\max}}$ differ only through the contribution of the running functional f after time $T_{\max}$ and through the terminal term that involves g . Using the representation (5.7) and the definition (5.9), together with boundedness of f and g and the fact that all exponential factors are at most one, we obtain

$$|Y^{\Delta t} - Y^{\Delta t, T_{\max}}| \leq C_f \tau_{\Delta t} \mathbf{1}_{\{\tau_{\Delta t} > T_{\max}\}} + 2C_g \mathbf{1}_{\{\tau_{\Delta t} > T_{\max}\}},$$

with the constants C_f and C_g from Lemma 5.9. Taking expectations and using Proposition 5.8, for any $\lambda \in (0, \lambda_d)$,

$$\mathbb{P}^x(\tau_{\Delta t} > t) \leq e^{-\lambda t} \mathbb{E}^x[e^{\lambda \tau_{\Delta t}}] \leq C_d e^{-\lambda t},$$

and therefore

$$\mathbb{E}^x[\tau_{\Delta t} \mathbf{1}_{\{\tau_{\Delta t} > T_{\max}\}}] = \int_{T_{\max}}^{\infty} \mathbb{P}^x(\tau_{\Delta t} > u) du \leq \frac{C_d}{\lambda} e^{-\lambda T_{\max}},$$

as well as

$$\mathbb{P}^x(\tau_{\Delta t} > T_{\max}) \leq C_d e^{-\lambda T_{\max}}.$$

Combining these bounds yields

$$|u_{\Delta t}^{T_{\max}}(x) - u_{\Delta t}(x)| = |\mathbb{E}^x[Y^{\Delta t, T_{\max}} - Y^{\Delta t}]| \leq \left(\frac{C_f C_d}{\lambda} + 2C_g C_d \right) e^{-\lambda T_{\max}}.$$

The claim follows by setting $\tilde{\lambda} := \lambda$ and $C_T := \frac{C_f C_d}{\lambda} + 2C_g C_d$, and by choosing Δt_0 as in Proposition 5.8. $\square$

These bounds summarize the non-asymptotic behavior of the infinite horizon Monte Carlo FK estimator at a fixed point x . For the learning-theoretic analysis we only need that, for each fixed x , the FK label produced by Algorithm 1 behaves like a noisy point evaluation of $u^*(x)$ with sub-exponential noise whose parameters do not depend on the number of FK labels or on the Monte Carlo budget. This is precisely the content of Proposition 5.1 in the main text, whose proof we now provide.

Proof of Proposition 5.1. We are now ready to finish. Fix $x \in \Omega$ and consider the FK estimator produced at this spatial point by Algorithm 1, with time step $0 < \Delta t \leq \Delta t_0$, truncation horizon $T_{\max} \in (0, \infty]$ and Monte Carlo budget N_{MC} . For each trajectory we simulate a payoff $Y^{\Delta t, T_{\max}}$ as

in (5.9). The FK label at x is the empirical average of N_{MC} independent copies of $Y^{\Delta t, T_{\text{max}}}$, that is

$$\hat{u}^{T_{\text{max}}}(x) := \frac{1}{N_{\text{MC}}} \sum_{i=1}^{N_{\text{MC}}} Y^{\Delta t, T_{\text{max}}, (i)}.$$

Step 1: sub-exponential Monte Carlo fluctuations.

We distinguish the cases $T_{\text{max}} < \infty$ and $T_{\text{max}} = \infty$.

If $T_{\text{max}} < \infty$, then at most $N_T = \lfloor T_{\text{max}}/\Delta t \rfloor$ time steps are accumulated before truncation. Boundedness of f and g and nonnegativity of V imply (as in Lemma 5.9) the deterministic bound

$$|Y^{\Delta t, T_{\text{max}}}| \leq C_f T_{\text{max}} + C_g \quad \text{almost surely,}$$

uniformly in x and $0 < \Delta t \leq \Delta t_0$. Hence, for each fixed x , the centered variable

$$Z^{\Delta t, T_{\text{max}}} := Y^{\Delta t, T_{\text{max}}} - u_{\Delta t}^{T_{\text{max}}}(x)$$

is bounded and therefore sub-exponential, with parameters that depend only on C_f , C_g and T_{max} and do not depend on N_{MC} or on the number of FK points. Standard properties of sub-exponential variables (see, e.g., [Vershynin \(2018, Section 2.7\)](#)) imply that empirical averages of i.i.d. sub-exponential variables are again sub-exponential, and the sub-exponential parameters can be chosen independently of the sample size. Thus, conditional on $X_0 = x$,

$$\hat{u}^{T_{\text{max}}}(x) - u_{\Delta t}^{T_{\text{max}}}(x) = \frac{1}{N_{\text{MC}}} \sum_{i=1}^{N_{\text{MC}}} Z^{\Delta t, T_{\text{max}}, (i)}$$

is mean-zero and sub-exponential with parameters depending only on the problem data and on T_{max} , and not on N_{MC} .

If $T_{\text{max}} = \infty$, then $Y^{\Delta t, \infty} = Y^{\Delta t}$ and $u_{\Delta t}^{\infty}(x) = u_{\Delta t}(x)$. Proposition 5.11 yields constants $\lambda_Y > 0$, $C_Y < \infty$ and $\Delta t_0 > 0$ such that

$$\sup_{y \in \bar{\Omega}, 0 < \Delta t \leq \Delta t_0} \mathbb{E}^y[e^{\lambda_Y |Y^{\Delta t}|}] \leq C_Y.$$

In particular, for each fixed x and $0 < \Delta t \leq \Delta t_0$, the centered variable

$$Z^{\Delta t} := Y^{\Delta t} - u_{\Delta t}(x)$$

is sub-exponential with parameters depending only on C_Y and λ_Y and independent of Δt , N_{MC} and the number of FK points; cf. Lemma 5.14. As in the

finite-horizon case, empirical averages of i.i.d. copies of $Z^{\Delta t}$ are again sub-exponential, with parameters that can be chosen independent of N_{MC} . Thus $\hat{u}^\infty(x) - u_{\Delta t}(x)$ is mean-zero and sub-exponential with parameters depending only on the model data.

In both cases we may therefore write

$$\hat{u}^{T_{\max}}(x) = u_{\Delta t}^{T_{\max}}(x) + \zeta(x),$$

where, conditional on $X_0 = x$, the fluctuation $\zeta(x)$ is mean-zero and sub-exponential with parameters that do not depend on N_{MC} or on the number of FK points (and are uniform over $0 < \Delta t \leq \Delta t_0$).

Step 2: discretization and truncation bias.

We now decompose the bias of the (possibly truncated) FK value. Adding and subtracting $u_{\Delta t}(x)$ and $u^*(x)$ yields

$$\hat{u}^{T_{\max}}(x) = u^*(x) + (u_{\Delta t}(x) - u^*(x)) + (u_{\Delta t}^{T_{\max}}(x) - u_{\Delta t}(x)) + \zeta(x).$$

By Theorem 5.17 the discretization bias satisfies

$$|u_{\Delta t}(x) - u^*(x)| \leq C_{\text{bias}} \sqrt{\Delta t} \quad \text{for all } 0 < \Delta t \leq \Delta t_0,$$

while Lemma 5.18 gives

$$|u_{\Delta t}^{T_{\max}}(x) - u_{\Delta t}(x)| \leq C_T e^{-\tilde{\lambda} T_{\max}} \quad \text{for all } T_{\max} > 0,$$

with the convention $e^{-\tilde{\lambda} T_{\max}} = 0$ when $T_{\max} = \infty$. Defining

$$b(x) := (u_{\Delta t}(x) - u^*(x)) + (u_{\Delta t}^{T_{\max}}(x) - u_{\Delta t}(x)),$$

we obtain the decomposition

$$\hat{u}^{T_{\max}}(x) = u^*(x) + b(x) + \zeta(x),$$

with

$$|b(x)| \leq C_{\text{bias}} \sqrt{\Delta t} + C_T e^{-\tilde{\lambda} T_{\max}},$$

and with $\zeta(x)$ mean-zero and sub-exponential with parameters independent of N_{MC} and of the number of FK points, uniformly over $0 < \Delta t \leq \Delta t_0$.

Step 3: random FK locations. In the FK-PINN construction we draw i.i.d. supervision locations $X_i^{\text{FK}} \in \Omega$, and at each such point we compute an independent estimator $\hat{u}^{T_{\max}}(X_i^{\text{FK}})$ using a fresh set of N_{MC} trajectories. Setting

$$Y_i^{\text{FK}} := \hat{u}^{T_{\max}}(X_i^{\text{FK}}), \quad b(X_i^{\text{FK}}) := b(X_i^{\text{FK}}), \quad \zeta_i := \zeta(X_i^{\text{FK}}),$$

the pairs $\{(X_i^{\text{FK}}, Y_i^{\text{FK}})\}_{i=1}^{N_{\text{FK}}}$ are i.i.d., and, conditional on $X_i^{\text{FK}} = x$, each Y_i^{FK} admits the decomposition

$$Y_i^{\text{FK}} = u^*(x) + b(x) + \zeta_i,$$

with $|b(x)| \leq C_{\text{bias}}\sqrt{\Delta t} + C_T e^{-\tilde{\lambda}T_{\text{max}}}$ and with ζ_i mean-zero and sub-exponential with parameters that do not depend on N_{FK} or N_{MC} . This is exactly the statement in Proposition 5.1, thus we are done.

3 End-to-end error analysis for FK-PINNs

The goal of this section is to prove Theorem 5.2. To this end, we will develop a comprehensive error analysis, based on a decomposition of the $L^2(\Omega)$ error in three terms, which we will control separately through learning-theoretic arguments, as outlined in Chapter 2. The reason we restrict ourselves to $L^2(\Omega)$ estimates only is merely for ease of exposition, as it is clear that all the arguments of Chapter 3 could also be applied (using boundary-adapted PINNs) to get estimates in $H^2(\Omega)$ instead. Likewise, the same arguments could be straightforwardly extended to parabolic PDEs, as discussed in Appendix C.

3.1 Preliminaries

We begin by recalling the setting in which we will be working together with notations and technical assumptions that we will be working under for the remainder of this section.

Notations

We work on a bounded Lipschitz domain $\Omega \subset [0, 1]^d$ with boundary $\partial\Omega$ and outward unit normal ν . Integration with respect to Lebesgue measure on Ω is written as $\int_{\Omega} \cdot dx$, while $\int_{\partial\Omega} \cdot ds$ denotes integration with respect to surface measure on $\partial\Omega$.

For $1 \leq p \leq \infty$ we write $L^p(\Omega)$ and $L^p(\partial\Omega)$ for the usual Lebesgue spaces, with norms

$$\|u\|_{L^p(\Omega)} := \begin{cases} (\int_{\Omega} |u(x)|^p dx)^{1/p}, & 1 \leq p < \infty, \\ \text{ess sup}_{x \in \Omega} |u(x)|, & p = \infty, \end{cases}$$

and similarly on $\partial\Omega$. We write $C(\bar{\Omega})$ for the space of continuous functions on the closure of Ω with norm $\|u\|_{C(\bar{\Omega})} := \sup_{x \in \bar{\Omega}} |u(x)|$.

For $s \in \mathbb{N}$ and $1 \leq p \leq \infty$, the Sobolev space $W^{s,p}(\Omega)$ consists of functions in $L^p(\Omega)$ whose weak derivatives up to order s lie in $L^p(\Omega)$, endowed with norm

$$\|u\|_{W^{s,p}(\Omega)} := \left(\sum_{|\alpha| \leq s} \|D^\alpha u\|_{L^p(\Omega)}^p \right)^{1/p} \quad \text{for } p < \infty,$$

and with the obvious modification for $p = \infty$. We use $H^s(\Omega) := W^{s,2}(\Omega)$. Fractional Sobolev spaces $H^s(\Omega)$ for $s \notin \mathbb{N}$ are defined in the standard way, for instance via real interpolation or via the Gagliardo seminorm, see [Evans \(2022\)](#). We denote the associated norms by $\|\cdot\|_{H^s(\Omega)}$.

Throughout, $\langle \cdot, \cdot \rangle$ denotes the $L^2(\Omega)$ inner product, and $\|\cdot\|_{L^2(\Omega)}$ is abbreviated to $\|\cdot\|_2$ when the domain is clear from the context. For random variables we write $\mathbb{E}[\cdot]$ for expectation and $\mathbb{P}(\cdot)$ for probabilities.

Setting and assumptions

We recall the PDE from [Section 2.1](#). Throughout this appendix we work in the elliptic setting for clarity. All statements are formulated and proved for the second order elliptic problem in (4.1)–(4.2). The same learning theoretic arguments extend to parabolic problems that admit a Feynman–Kac representation, once the time variable has been discretized and the resulting time discretization error is controlled separately (see [Appendix C](#)).

Assumption 5.19 (PDE well posedness and FK representation). The operator $\mathcal{L}$ in (4.1) is uniformly elliptic with bounded measurable coefficients and satisfies the structural conditions stated in [Section 2.1](#). The data f and g are such that (4.1)–(4.2) admit a unique strong solution

$$u^* \in H^2(\Omega) \cap C(\bar{\Omega}),$$

with $u^*|_{\partial\Omega} = g$. Moreover, there exists a diffusion process $(X_t)_{t \geq 0}$ and an exit time τ such that u^* admits a Feynman–Kac representation of the form

$$u^*(x) = \mathbb{E}_x \left[\int_0^\tau r(X_t) dt + h(X_\tau) \right], \quad x \in \Omega,$$

for suitable running and terminal functionals r and h , as described in [Section 3](#) and [Appendix E](#).

We follow the sampling scheme described in [Algorithm 1](#) and [Algorithm 2](#). Let

$$X^{\text{int}} \sim U(\Omega), \quad X^{\partial\Omega} \sim U(\partial\Omega),$$

be interior and boundary points drawn independently from the uniform distributions on Ω and $\partial\Omega$. The corresponding population PDE and boundary risks are

$$\mathcal{R}_{\text{PDE}}(u) := |\Omega| \mathbb{E}[|\mathcal{L}u(X^{\text{int}}) - f(X^{\text{int}})|^2], \quad (5.10)$$

$$\mathcal{R}_{\partial\Omega}(u) := |\partial\Omega| \mathbb{E}[|u(X^{\partial\Omega}) - g(X^{\partial\Omega})|^2]. \quad (5.11)$$

In the FK term we work in a data scarce regime. We sample FK points from the same underlying geometry as the interior points, but we keep their number small. Concretely, we assume

$$X^{\text{FK}} \sim U(\Omega),$$

independently of X^{int} and $X^{\partial\Omega}$, while the sample size N_{FK} is allowed to remain bounded or grow much more slowly than N_{int} and $N_{\partial\Omega}$. At each such point we generate a Monte Carlo FK estimate $\hat{u}^{\text{MC}}(X^{\text{FK}})$ as in [Algorithm 1](#). [Proposition 5.1](#) shows that these labels admit the decomposition

$$\hat{u}^{\text{MC}}(X^{\text{FK}}) = u^*(X^{\text{FK}}) + b(X^{\text{FK}}) + \zeta,$$

where the deterministic bias $b : \Omega \rightarrow \mathbb{R}$ satisfies

$$|b(x)| \leq C_{\text{bias}}\sqrt{\Delta t} + C_T e^{-\kappa T_{\text{max}}} \quad \text{for all } x \in \Omega,$$

for constants $C_{\text{bias}}, C_T, \kappa > 0$ independent of N_{FK} and of the number of FK trajectories, and where, conditional on X^{FK} , the fluctuation ζ is mean-zero and sub-exponential with parameters that do not depend on N_{FK} or on the number of trajectories. The corresponding population FK risk is

$$\mathcal{R}_{\text{FK}}(u) := \mathbb{E}[|u(X^{\text{FK}}) - u^*(X^{\text{FK}})|^2]. \quad (5.12)$$

For fixed nonnegative weights λ_{PDE} , $\lambda_{\partial\Omega}$ and λ_{FK} we define the population FK-PINN risk

$$\mathcal{R}(u) := \lambda_{\text{PDE}}\mathcal{R}_{\text{PDE}}(u) + \lambda_{\partial\Omega}\mathcal{R}_{\partial\Omega}(u) + \lambda_{\text{FK}}\mathcal{R}_{\text{FK}}(u). \quad (5.13)$$

In what follows we assume that the PDE and boundary weights $\lambda_{\text{PDE}}, \lambda_{\partial\Omega}$, and that the FK weight is non-negative. For convenience, we will use the abbreviation

$$\lambda_{\min} := \min\{\lambda_{\text{PDE}}, \lambda_{\partial\Omega}\}.$$

Note that rescaling the three weights only changes the constants in the bounds below, so this choice entails no loss of generality. The empirical FK-PINN risk is obtained by Monte Carlo approximation of the expectations in [\(5.10\)–\(5.12\)](#). Given i.i.d. samples

$$\{x_i^{\text{int}}\}_{i=1}^{N_{\text{int}}} \subset \Omega, \quad \{x_j^{\partial\Omega}\}_{j=1}^{N_{\partial\Omega}} \subset \partial\Omega, \quad \{x_k^{\text{FK}}\}_{k=1}^{N_{\text{FK}}} \subset \Omega,$$

and the corresponding FK labels $\{\hat{u}^{\text{MC}}(x_k^{\text{FK}})\}_{k=1}^{N_{\text{FK}}}$, we set

$$\hat{\mathcal{R}}_{\text{PDE}}(u) := |\Omega| \frac{1}{N_{\text{int}}} \sum_{i=1}^{N_{\text{int}}} |\mathcal{L}u(x_i^{\text{int}}) - f(x_i^{\text{int}})|^2, \quad (5.14)$$

$$\hat{\mathcal{R}}_{\partial\Omega}(u) := |\partial\Omega| \frac{1}{N_{\partial\Omega}} \sum_{j=1}^{N_{\partial\Omega}} |u(x_j^{\partial\Omega}) - g(x_j^{\partial\Omega})|^2, \quad (5.15)$$

$$\hat{\mathcal{R}}_{\text{FK}}(u) := \frac{1}{N_{\text{FK}}} \sum_{k=1}^{N_{\text{FK}}} |u(x_k^{\text{FK}}) - \hat{u}^{\text{MC}}(x_k^{\text{FK}})|^2, \quad (5.16)$$

and

$$\hat{\mathcal{R}}(u) := \lambda_{\text{PDE}} \hat{\mathcal{R}}_{\text{PDE}}(u) + \lambda_{\partial\Omega} \hat{\mathcal{R}}_{\partial\Omega}(u) + \lambda_{\text{FK}} \hat{\mathcal{R}}_{\text{FK}}(u). \quad (5.17)$$

Up to the adaptive task weights used in Equation (4.11), the empirical loss $\hat{\mathcal{R}}(u_\theta)$ coincides with $\mathcal{R}_{\text{FK-PINN}}(\theta, \varphi)$.

We will make use of the following classical PDE estimate of Agmon et al. (1959), as we already did in Chapter 3.

Lemma 5.20 (PDE stability). *Under Assumption 5.19, there exists a constant $C_{\text{stab}} > 0$ such that for every $u \in H^2(\Omega) \cap C(\bar{\Omega})$ with $u|_{\partial\Omega} \in L^2(\partial\Omega)$,*

$$\|u - u^*\|_{L^2(\Omega)}^2 \leq C_{\text{stab}} \left(\mathcal{R}_{\text{PDE}}(u) + \mathcal{R}_{\partial\Omega}(u) \right). \quad (5.18)$$

We will work with a hypothesis class $\mathcal{U}$ of candidate solutions, specified in the next subsection. For any $u \in \mathcal{U}$ we define

$$\mathcal{E}_{\text{app}} := \inf_{v \in \mathcal{U}} \|v - u^*\|_{L^2(\Omega)}, \text{ and } \mathcal{E}_{\text{stat}} := \sup_{v \in \mathcal{U}} |\mathcal{R}(v) - \hat{\mathcal{R}}(v)|. \quad (5.19)$$

Later we also introduce an optimization error $\mathcal{E}_{\text{opt}}$ that measures the gap between the empirical minimizer and the iterate produced by gradient descent. These three terms will form the building blocks of the final non-asymptotic error bound.

Neural network architecture

We consider fully connected feedforward neural networks with tanh activations in the hidden layers and a linear output. For an integer depth $L \geq 1$ and a fixed width vector $(m_\ell)_{\ell=0}^L$ with $m_0 = d$ and $m_L = 1$, a parameter vector

$$\mathbf{m}\theta := \{W_\ell, b_\ell\}_{\ell=1}^L, \quad W_\ell \in \mathbb{R}^{m_\ell \times m_{\ell-1}}, \quad b_\ell \in \mathbb{R}^{m_\ell},$$

defines a function $u_{\mathbf{m}\theta} : \Omega \rightarrow \mathbb{R}$ by

$$z_0(x) := x \in \mathbb{R}^d, \quad (5.20)$$

$$z_\ell(x) := \tanh(W_\ell z_{\ell-1}(x) + b_\ell), \quad \ell = 1, \dots, L-1, \quad (5.21)$$

$$u_{\mathbf{m}\theta}(x) := W_L z_{L-1}(x) + b_L \in \mathbb{R}. \quad (5.22)$$

We write $P := P(L, (m_\ell)_\ell)$ for the total number of free parameters in $\mathbf{m}\theta$ and call L the depth and $m := \max_\ell m_\ell$ the width of the network.

For later use we note a simple upper bound on the number of free parameters. By definition,

$$P = \sum_{\ell=1}^L (m_\ell m_{\ell-1} + m_\ell), \quad m_0 = d, \quad m_L = 1 \quad \text{and} \quad m_\ell \leq m \quad \text{for all } 0 \leq \ell \leq L.$$

This implies that $P \leq C_{\text{arch}}(d) L m^2$ for a constant $C_{\text{arch}}(d) > 0$ depending only on the input dimension d . In all subsequent bounds we may therefore replace P by $C_{\text{arch}}(d) L m^2$ if we wish to make the dependence on the architectural hyperparameters explicit.

For the learning-theoretic analysis we restrict attention to networks whose outputs and low order derivatives are uniformly bounded on $\overline{\Omega}$. For fixed L , $(m_\ell)_\ell$ and $B > 0$ we first define the parameter set

$$\Theta_{L,m} := \left\{ \mathbf{m}\theta = (W_\ell, b_\ell)_{\ell=1}^L : W_\ell \in \mathbb{R}^{m_\ell \times m_{\ell-1}}, \quad b_\ell \in \mathbb{R}^{m_\ell} \right\}.$$

The associated hypothesis class is

$$\mathcal{U}_{L,m,B} := \{u_{\mathbf{m}\theta} : \mathbf{m}\theta \in \Theta_{L,m}, \quad \|u_{\mathbf{m}\theta}\|_{C^2(\overline{\Omega})} \leq B\}, \quad (5.23)$$

where $\|\cdot\|_{C^2(\overline{\Omega})}$ denotes the usual C^2 norm that controls the function and its first and second order partial derivatives on $\overline{\Omega}$. The class $\mathcal{U}_{L,m,B}$ is nonempty for every B large enough, since constant functions belong to it.

To lighten notation we often write $\mathcal{U}$ for $\mathcal{U}_{L,m,B}$ and denote by P the maximal number of parameters of networks in $\mathcal{U}$. All constants in the sequel may depend on (L, m, B, d) but never on the sample sizes $(N_{\text{int}}, N_{\partial\Omega}, N_{\text{FK}})$.

3.2 Approximation error bound

We first bound the approximation error in (5.19) in terms of the size of the network. To this end, we recall the approximation bound for Sobolev functions due to De Ryck et al. (2021), presented in a simplified format.

Theorem 5.21 (Approximation by two hidden layer tanh networks, simplified). *Let $d \in \mathbb{N}$, let $s \in \mathbb{N}$ with $s \geq 3$ and let $f \in W^{s,\infty}([0,1]^d)$. There exist constants $C_{\text{approx}}, C_{\text{width}} > 0$, depending only on $(d, s, \|f\|_{W^{s,\infty}})$, such that for every integer N large enough there is a tanh network $\hat{f}_N$ with two hidden layers whose total number of parameters satisfies*

$$P_N \leq C_{\text{width}} N^d$$

and such that, for all multi indices α with $|\alpha| \leq 2$,

$$\|D^\alpha f - D^\alpha \hat{f}_N\|_{L^\infty([0,1]^d)} \leq C_{\text{approx}} N^{-(s-|\alpha|)}. \quad (5.24)$$

We now apply this result to the hypothesis class $\mathcal{U}$ defined in (5.23) and to the true solution u^* .

Corollary 5.22 (Approximation error for $\mathcal{U}_{L,m,B}$). *Suppose that Theorem 5.19 holds and that $u^* \in W^{s,\infty}(\Omega)$ for some integer $s \geq 3$. Fix any depth $L \geq 3$. Then there exist constants $C_{\text{app},1}, C_{\text{app},2} > 0$ and $m_0 \in \mathbb{N}$ such that for every width $m \geq m_0$ the hypothesis class $\mathcal{U}_{L,m,B}$ contains a network u^{approx} with*

$$\|u^* - u^{\text{approx}}\|_{C^2(\bar{\Omega})} \leq C_{\text{app},1} P(L, m)^{-\gamma}, \quad \gamma := \frac{s-2}{d}, \quad (5.25)$$

where $P(L, m)$ denotes the total number of free parameters of networks in $\mathcal{U}_{L,m,B}$. In particular,

$$\mathcal{E}_{\text{app}} = \inf_{u \in \mathcal{U}_{L,m,B}} \|u - u^*\|_{L^2(\Omega)} \leq C_{\text{app},2} P(L, m)^{-\gamma}. \quad (5.26)$$

Proof. Extend u^* to a function in $W^{s,\infty}([0,1]^d)$ with comparable norm using a standard extension operator. Apply Theorem 5.21 with this extension. For each integer N large enough we obtain a two-hidden-layer tanh network $\hat{u}_N$ with depth $L = 3$, widths of order N^d in each hidden layer, total number of parameters $P_N \leq C_{\text{width}} N^d$, and

$$\|D^\alpha u^* - D^\alpha \hat{u}_N\|_{L^\infty([0,1]^d)} \leq C_{\text{approx}} N^{-(s-|\alpha|)}, \quad |\alpha| \leq 2.$$

In particular,

$$\|u^* - \hat{u}_N\|_{C^2(\bar{\Omega})} \leq C N^{-(s-2)}$$

for a constant C depending on $(d, s, \|u^*\|_{W^{s,\infty}})$.

Any such two-hidden-layer network can be realized as a special case of our architecture with depth $L \geq 3$ and width m as soon as m is larger than the maximal width of $\hat{u}_N$, since the extra layers and neurons can be made

inactive by setting their weights to zero or to identities. Thus, for every $L \geq 3$ there exists m_0 such that for all $m \geq m_0$ the network $\hat{u}_N$ belongs to $\mathcal{U}_{L,m,B}$ provided B is chosen large enough.

Let $N_{\min}$ be the (fixed) threshold required in Theorem 5.21 and choose

$$N := \max \left\{ N_{\min}, \lfloor (P(L, m)/C_{\text{width}})^{1/d} \rfloor \right\}.$$

Then $P_N \leq C_{\text{width}} N^d \leq P(L, m)$ and therefore

$$\|u^* - \hat{u}_N\|_{C^2(\bar{\Omega})} \leq C N^{-(s-2)} \leq C' P(L, m)^{-\gamma}, \quad \gamma = \frac{s-2}{d},$$

for a constant $C' > 0$. Taking $u^{\text{approx}} = \hat{u}_N$ yields (5.25) with $C_{\text{app},1} = CC'$. Since Ω is bounded, $\|v\|_{L^2(\Omega)} \leq C_{\Omega} \|v\|_{C(\bar{\Omega})}$ for every continuous v , and (5.26) follows from (5.25) by absorbing C_{Ω} into $C_{\text{app},2}$. $\square$

As a consequence of these approximation error bounds, we can control the corresponding summand in the error decomposition.

Lemma 5.23 (Control of the population risk by C^2 approximation). *Under Theorem 5.19 and with the weights $(\lambda_{\text{PDE}}, \lambda_{\partial\Omega}, \lambda_{\text{FK}})$ as in (5.13), there exists a constant $C_{\text{risk}} > 0$, depending only on the PDE coefficients, the domain, and the weights, such that for every $u \in C^2(\bar{\Omega})$ with*

$$\|u - u^*\|_{C^2(\bar{\Omega})} \leq \varepsilon,$$

one has

$$\mathcal{R}(u) \leq C_{\text{risk}} \varepsilon^2. \quad (5.27)$$

In particular, for the approximating network u^{approx} from Corollary 5.22 we have

$$\mathcal{R}(u^{\text{approx}}) \leq \tilde{C}_{\text{risk}} P^{-2\gamma}, \quad (5.28)$$

for a constant $\tilde{C}_{\text{risk}} > 0$ independent of P .

Proof. Since $\mathcal{L}$ is a second order operator with bounded measurable coefficients, there exists a constant $C_{\mathcal{L}} > 0$ such that

$$\sup_{x \in \bar{\Omega}} |\mathcal{L}(u - u^*)(x)| \leq C_{\mathcal{L}} \|u - u^*\|_{C^2(\bar{\Omega})}.$$

Therefore

$$\mathcal{R}_{\text{PDE}}(u) = \int_{\Omega} (\mathcal{L}u(x) - f(x))^2 dx = \int_{\Omega} (\mathcal{L}(u - u^*)(x))^2 dx \leq |\Omega| C_{\mathcal{L}}^2 \varepsilon^2.$$

On the boundary we have

$$\sup_{x \in \partial\Omega} |u(x) - g(x)| = \sup_{x \in \partial\Omega} |u(x) - u^*(x)| \leq \|u - u^*\|_{C(\bar{\Omega})} \leq \varepsilon,$$

which gives

$$\mathcal{R}_{\partial\Omega}(u) = \int_{\partial\Omega} (u(x) - g(x))^2 ds \leq |\partial\Omega| \varepsilon^2.$$

Finally, for the FK term we use that u^* is itself the target in (5.12), hence

$$\mathcal{R}_{\text{FK}}(u) = \mathbb{E}[|u(X^{\text{FK}}) - u^*(X^{\text{FK}})|^2] \leq \|u - u^*\|_{C(\bar{\Omega})}^2 \leq \varepsilon^2.$$

Collecting these three bounds and inserting the weights gives (5.27) with a constant C_{risk} that depends only on the coefficients of $\mathcal{L}$, the domain, and the weights. The bound (5.28) then follows from (5.25). $\square$

3.3 Optimization error bound

We next control the optimization error incurred by terminating gradient descent after a finite number of iterations. Throughout this subsection we fix the sample set and view the empirical FK-PINN loss

$$\ell(\mathbf{m}\theta) := \hat{\mathcal{R}}(u_{\mathbf{m}\theta})$$

as a deterministic function of the parameters.

We rely on the Polyak–Łojasiewicz framework of Karimi et al. (2016); Liu et al. (2022). A function $\ell : \mathbb{R}^P \rightarrow \mathbb{R}$ satisfies a local PL* condition on a set $S \subset \mathbb{R}^P$ with constant $\mu > 0$ if

$$\ell(\mathbf{m}\theta) - \ell(\mathbf{m}\theta^*) \leq \frac{1}{2\mu} \|\nabla \ell(\mathbf{m}\theta)\|_2^2, \quad \forall \mathbf{m}\theta \in S, \quad (5.29)$$

for some global minimizer $\mathbf{m}\theta^* \in S$ of ℓ . If, in addition, ℓ is β -smooth on S (that is, $\|\nabla \ell(\mathbf{m}\theta) - \nabla \ell(\mathbf{m}\theta')\|_2 \leq \beta \|\mathbf{m}\theta - \mathbf{m}\theta'\|_2$ for all $\mathbf{m}\theta, \mathbf{m}\theta' \in S$), the ratio

$$\kappa_\ell(S) := \frac{\beta}{\mu}$$

plays the role of a condition number.

The following result is a specialization of Liu et al. (2022, Theorem 6).

Theorem 5.24 (Gradient descent under a local PL* condition). *Let ℓ be differentiable and β -smooth on a convex set $S \subset \mathbb{R}^P$, and assume that ℓ*

satisfies the PL^* condition (5.29) on S with constant $\mu > 0$. Let $\mathbf{m}\theta_0 \in S$ and consider gradient descent

$$\mathbf{m}\theta_{t+1} = \mathbf{m}\theta_t - \eta \nabla \ell(\mathbf{m}\theta_t),$$

with a fixed step size $\eta \in (0, 2/\beta]$. Suppose that all iterates remain in S . Then for all $t \geq 0$,

$$\ell(\mathbf{m}\theta_t) - \ell(\mathbf{m}\theta^*) \leq (1 - \eta\mu)^t (\ell(\mathbf{m}\theta_0) - \ell(\mathbf{m}\theta^*)). \quad (5.30)$$

In Theorem 4.6 we proved that the FK-augmented loss satisfies a PL^* inequality with a condition number that stays uniformly bounded as the number of collocation points increases. We now make this consequence explicit for the empirical loss.

Corollary 5.25 (Uniform PL^* geometry for FK-PINNs). *Under the assumptions of Theorem 4.6 there exist a convex set $S \subset \mathbb{R}^P$, constants $\mu_{\text{FK}} > 0$, $\beta_{\text{FK}} < \infty$ and an initialization $\mathbf{m}\theta_0 \in S$ such that the empirical loss $\ell(\mathbf{m}\theta) = \hat{\mathcal{R}}(u_{\mathbf{m}\theta})$ satisfies the PL^* condition (5.29) on S with constant μ_{FK} , is β_{FK} -smooth on S , and the associated condition number*

$$\kappa_{\text{FK}} := \frac{\beta_{\text{FK}}}{\mu_{\text{FK}}}$$

is bounded by a constant that does not depend on $(N_{\text{int}}, N_{\partial\Omega}, N_{\text{FK}})$.

Under Theorem 5.25, gradient descent with a suitable step size converges linearly to an empirical minimizer. We quantify the corresponding optimization error in terms of the number of iterations.

Proposition 5.26 (Optimization error bound). *Let $\{\mathbf{m}\theta_t\}_{t \geq 0}$ be the gradient descent iterates for $\ell(\mathbf{m}\theta) = \hat{\mathcal{R}}(u_{\mathbf{m}\theta})$ with step size $\eta \in (0, 2/\beta_{\text{FK}}]$, initialized at $\mathbf{m}\theta_0$ as in Theorem 5.25. Let*

$$\hat{u} \in \underset{u \in \mathcal{U}}{\operatorname{argmin}} \hat{\mathcal{R}}(u)$$

be an empirical minimizer and write $\mathbf{m}\theta^$ for a parameter vector with $u_{\mathbf{m}\theta^*} = \hat{u}$. Define the optimization error*

$$\mathcal{E}_{\text{opt}}(t) := \hat{\mathcal{R}}(u_{\mathbf{m}\theta_t}) - \hat{\mathcal{R}}(\hat{u}). \quad (5.31)$$

Then, under Corollary 5.25, there exist constants $C_{\text{opt}}, c_{\text{opt}} > 0$ independent of the sample sizes and of t such that

$$\mathcal{E}_{\text{opt}}(t) \leq C_{\text{opt}} \exp(-c_{\text{opt}} t). \quad (5.32)$$

In particular, for every $t \geq 0$ the network $u_{\mathbf{m}\theta_t}$ is an $\mathcal{E}_{\text{opt}}(t)$ -approximate empirical minimizer of $\hat{\mathcal{R}}$ in the sense that

$$\hat{\mathcal{R}}(u_{\mathbf{m}\theta_t}) \leq \inf_{u \in \mathcal{U}} \hat{\mathcal{R}}(u) + \mathcal{E}_{\text{opt}}(t).$$

Proof. Inequality (5.32) follows directly from Theorem 5.24 with $\mu = \mu_{\text{FK}}$ and $\beta = \beta_{\text{FK}}$, taking $\mathbf{m}\theta^*$ to be any parameter vector that realizes an empirical minimizer $\hat{u}$ of $\hat{\mathcal{R}}$. $\square$

3.4 Statistical error bound

We now turn to the statistical error $\mathcal{E}_{\text{stat}}$ in (5.19), which captures the deviation between the population risk and its empirical approximation. Our goal is to obtain non-asymptotic high probability bounds on

$$\mathcal{E}_{\text{stat}} = \sup_{u \in \mathcal{U}} |\mathcal{R}(u) - \hat{\mathcal{R}}(u)|$$

in terms of the number of collocation points and the complexity of the hypothesis class. The analysis follows the now standard Rademacher complexity approach in statistical learning theory, but it has two distinctive features.

First, we operate in a data scarce regime for the FK term: N_{FK} is allowed to remain bounded while N_{int} and $N_{\partial\Omega}$ grow. Second, we must control the complexity not only of the network outputs $u_{\mathbf{m}\theta}$ but also of their first and second order derivatives, since the PDE residual involves $\partial^\alpha u_{\mathbf{m}\theta}$ for $|\alpha| \leq 2$. To carry our analysis, we will naturally heavily rely on the concepts and results presented in Chapters 2 and 3.

We will now bound $\mathcal{E}_{\text{stat}}$ using the VC-dimension estimates from Theorem 3.18, which we recall apply for pseudo-dimension as well in virtue of Lemma 3.22. Recall that we have defined

$$\mathcal{R}(u) = \lambda_{\text{PDE}} \mathcal{R}_{\text{PDE}}(u) + \lambda_{\partial\Omega} \mathcal{R}_{\partial\Omega}(u) + \lambda_{\text{FK}} \mathcal{R}_{\text{FK}}(u),$$

and that $\hat{\mathcal{R}}$ is defined in (5.17). We first separate the contributions of the three components.

Lemma 5.27 (Decomposition of the statistical error). *For every hypothesis class $\mathcal{U}$,*

$$\mathcal{E}_{\text{stat}} \leq \lambda_{\text{PDE}} \mathcal{E}_{\text{stat}}^{\text{PDE}} + \lambda_{\partial\Omega} \mathcal{E}_{\text{stat}}^{\partial\Omega} + \lambda_{\text{FK}} \mathcal{E}_{\text{stat}}^{\text{FK}},$$

where

$$\begin{aligned}\mathcal{E}_{\text{stat}}^{\text{PDE}} &:= \sup_{u \in \mathcal{U}} \left| \mathcal{R}_{\text{PDE}}(u) - \hat{\mathcal{R}}_{\text{PDE}}(u) \right|, \\ \mathcal{E}_{\text{stat}}^{\partial\Omega} &:= \sup_{u \in \mathcal{U}} \left| \mathcal{R}_{\partial\Omega}(u) - \hat{\mathcal{R}}_{\partial\Omega}(u) \right|, \\ \mathcal{E}_{\text{stat}}^{\text{FK}} &:= \sup_{u \in \mathcal{U}} \left| \mathcal{R}_{\text{FK}}(u) - \hat{\mathcal{R}}_{\text{FK}}(u) \right|.\end{aligned}$$

Proof. This is an immediate consequence of the triangle inequality and the linearity of $\mathcal{R}$ and $\hat{\mathcal{R}}$ in their three components. $\square$

We now bound the pseudodimension of classes described by parameter-independent multiplications and sums.

Lemma 5.28 (Pseudo-dimension and fixed multipliers / finite sums). *Let $\mathcal{F} \subset \{f : \mathcal{Z} \rightarrow \mathbb{R}\}$ be a function class with $\text{Pdim}(\mathcal{F}) = d < \infty$ and let $g : \mathcal{Z} \rightarrow \mathbb{R}$ be a fixed (non-random) function. Then*

$$\text{Pdim}(g\mathcal{F}) := \text{Pdim}(\{z \mapsto g(z)f(z) : f \in \mathcal{F}\}) \leq d.$$

Moreover, if $\mathcal{F}_1, \dots, \mathcal{F}_m$ are function classes with pseudo-dimensions $d_j := \text{Pdim}(\mathcal{F}_j)$, and

$$\mathcal{F}_{\text{sum}} := \left\{ z \mapsto \sum_{j=1}^m f_j(z) : f_j \in \mathcal{F}_j \right\},$$

then there exists a constant $C_m > 0$, depending only on m , such that

$$\text{Pdim}(\mathcal{F}_{\text{sum}}) \leq C_m \sum_{j=1}^m d_j.$$

Proof. For the first claim, fix a finite sample $z_1, \dots, z_n$ and thresholds $r_1, \dots, r_n \in \mathbb{R}$. If we replace the class $\mathcal{F}$ by $g\mathcal{F}$, the sign of $g(z_i)f(z_i) - r_i$ at each point z_i is determined by the sign of $f(z_i) - r_i/g(z_i)$ whenever $g(z_i) \neq 0$, while points with $g(z_i) = 0$ contribute no additional sign variability. Thus the number of sign patterns achievable by $g\mathcal{F}$ on any finite sample is at most the number achievable by $\mathcal{F}$, so $\text{Pdim}(g\mathcal{F}) \leq \text{Pdim}(\mathcal{F})$.

The bound for finite sums is a special case of the general pseudo-dimension bound for linear combinations of real-valued function classes as shown by [Bartlett and Anthony \(1999, Chapter 11\)](#). Since the number m of summands is fixed in our setting, its contribution can be absorbed into the constant C_m . $\square$

We treat each term in Lemma 5.27 with Theorem 2.29 applied to an appropriate function class. We illustrate the argument for $\mathcal{E}_{\text{stat}}^{\text{PDE}}$ only, as the remaining terms follow in the same way.

Define

$$\mathcal{F}_{\text{PDE}} := \left\{ x \mapsto |\Omega| (\mathcal{L}u(x) - f(x))^2 : u \in \mathcal{U} \right\}.$$

Define the (unsquared) residual class

$$\mathcal{G}_{\text{PDE}} := \left\{ x \mapsto \mathcal{L}u(x) - f(x) : u \in \mathcal{U} \right\}.$$

By the bounds above we have

$$\sup_{g \in \mathcal{G}_{\text{PDE}}} \sup_{x \in \Omega} |g(x)| \leq C_{\text{PDE}}.$$

The mapping $u \mapsto \mathcal{L}u$ is linear in u and involves only first and second order derivatives. In particular, every $g \in \mathcal{G}_{\text{PDE}}$ can be written as a finite linear combination (with fixed coefficient functions given by the coefficients of $\mathcal{L}$ and by $-f$) of functions from the derivative classes $\mathcal{F}^{(0)}, \mathcal{F}^{(1)}, \mathcal{F}^{(2)}$, defined in the obvious way. Note that for every $g \in \mathcal{G}_{\text{PDE}}$ the function

$$x \mapsto |\Omega| g(x)^2$$

belongs to $\mathcal{F}_{\text{PDE}}$, and the mapping $t \mapsto |\Omega| t^2$ is $2|\Omega|C_{\text{PDE}}$ -Lipschitz on $[-C_{\text{PDE}}, C_{\text{PDE}}]$. Hence, by the contraction inequality in Lemma 2.31,

$$\mathfrak{R}_{N_{\text{int}}}(\mathcal{F}_{\text{PDE}}) \leq 2|\Omega|C_{\text{PDE}} \mathfrak{R}_{N_{\text{int}}}(\mathcal{G}_{\text{PDE}}).$$

To bound $\mathfrak{R}_{N_{\text{int}}}(\mathcal{G}_{\text{PDE}})$ it suffices to control the Rademacher complexities of the derivative classes $\mathcal{F}^{(0)}, \mathcal{F}^{(1)}, \mathcal{F}^{(2)}$. Indeed, writing $\mathcal{L}$ in coordinates as

$$\mathcal{L}u(x) = \sum_{|\alpha| \leq 2} c_{\alpha}(x) D^{\alpha}u(x),$$

with bounded coefficient functions c_{α} , we have for any fixed sample $x_1, \dots, x_{N_{\text{int}}}$,

$$\mathfrak{R}_{N_{\text{int}}}(\mathcal{G}_{\text{PDE}} \mid x_{1:N_{\text{int}}}) \leq \sum_{|\alpha| \leq 2} \|c_{\alpha}\|_{L^{\infty}(\Omega)} \mathfrak{R}_{N_{\text{int}}}(\mathcal{F}^{(|\alpha|)} \mid x_{1:N_{\text{int}}}),$$

and the constant shift $-f$ does not affect Rademacher complexity. By the pseudo-dimension bounds from Theorem 3.18, the Dudley integral inequality

Theorem 2.35 and the covering number bound from Proposition 2.38, this yields

$$\mathfrak{R}_{N_{\text{int}}}(\mathcal{G}_{\text{PDE}}) \lesssim (PN_2) \sqrt{\frac{\log(PN_2)}{N_{\text{int}}}},$$

up to logarithmic factors and constants depending only on (d, L) and the coefficients of $\mathcal{L}$.

Applying Theorem 2.29, Theorem 2.35 and Proposition 2.38 to $\mathcal{F}_{\text{PDE}}$, together with the bound $\mathfrak{R}_{N_{\text{int}}}(\mathcal{G}_{\text{PDE}}) \lesssim (PN_2) \sqrt{\log(PN_2)/N_{\text{int}}}$, yields the following bound:

Lemma 5.29 (Statistical error for the PDE residual). *There exist constants $C_{\text{PDE},1}, C_{\text{PDE},2} > 0$, depending only on (d, L, m, B) and on the coefficients of $\mathcal{L}$, such that for every $\delta \in (0, 1)$, with probability at least $1 - \delta$,*

$$\begin{aligned} \mathcal{E}_{\text{stat}}^{\text{PDE}} &\leq C_{\text{PDE},1} |\Omega| C_{\text{PDE}}^2 \sqrt{\frac{(PN_2)^2 \log(PN_2) + \log(2/\delta)}{N_{\text{int}}}} \\ &\quad + C_{\text{PDE},2} |\Omega| C_{\text{PDE}}^2 \frac{(PN_2)^2 \log(PN_2) + \log(2/\delta)}{N_{\text{int}}}. \end{aligned} \quad (5.33)$$

The boundary term $\mathcal{E}_{\text{stat}}^{\partial\Omega}$ is handled analogously, using that the boundary loss involves only $u_{\mathbf{m}\theta}$ itself and no derivatives. In this case the relevant function class is generated by $\mathcal{F}^{(0)}$ and the same argument shows that

$$\mathcal{E}_{\text{stat}}^{\partial\Omega} \lesssim |\partial\Omega| C_{\partial\Omega}^2 (PN_0) \sqrt{\frac{\log(PN_0) + \log(1/\delta)}{N_{\partial\Omega}}}. \quad (5.34)$$

for a suitable constant $C_{\partial\Omega}$.

The FK term requires a small modification. Recall that $\mathcal{R}_{\text{FK}}(u)$ is defined using the true solution u^* in (5.12), while $\hat{\mathcal{R}}_{\text{FK}}(u)$ is defined with the noisy labels $\hat{u}^{\text{MC}}(x_k^{\text{FK}})$. Writing

$$\hat{u}^{\text{MC}}(x_k^{\text{FK}}) = u^*(x_k^{\text{FK}}) + b(x_k^{\text{FK}}) + \zeta_k,$$

with b and ζ_k as in Proposition 5.1, we deduce

$$\begin{aligned}
\mathcal{E}_{\text{stat}}^{\text{FK}} &= \sup_{u \in \mathcal{U}} \left| \mathbb{E}[(u(X^{\text{FK}}) - u^*(X^{\text{FK}}))^2] - \frac{1}{N_{\text{FK}}} \sum_{k=1}^{N_{\text{FK}}} (u(x_k^{\text{FK}}) - \hat{u}^{\text{MC}}(x_k^{\text{FK}}))^2 \right| \\
&\leq \sup_{u \in \mathcal{U}} \left| \mathbb{E}[(u(X^{\text{FK}}) - u^*(X^{\text{FK}}))^2] - \frac{1}{N_{\text{FK}}} \sum_{k=1}^{N_{\text{FK}}} (u(x_k^{\text{FK}}) - u^*(x_k^{\text{FK}}))^2 \right| \\
&\quad + \sup_{u \in \mathcal{U}} \left| \frac{1}{N_{\text{FK}}} \sum_{k=1}^{N_{\text{FK}}} [(u(x_k^{\text{FK}}) - u^*(x_k^{\text{FK}}))^2 - (u(x_k^{\text{FK}}) - \hat{u}^{\text{MC}}(x_k^{\text{FK}}))^2] \right| \\
&=: A_{\text{FK}} + B_{\text{FK}}.
\end{aligned}$$

The term A_{FK} can be bounded exactly as in Lemma 5.29, now using only the class $\mathcal{F}^{(0)}$ on Ω , since it is an empirical-process deviation for the squared error with respect to u^* . For B_{FK} we expand the difference of squares and use that

$$(u - u^*)^2 - (u - \hat{u}^{\text{MC}})^2 = 2(u - u^*)(b(x_k^{\text{FK}}) + \zeta_k) - (b(x_k^{\text{FK}}) + \zeta_k)^2.$$

Using $\|u - u^*\|_{C(\bar{\Omega})} \leq 2B$ and the uniform bound on b , the deterministic contribution of b to B_{FK} is at most of order

$$\sup_{x \in \Omega} |b(x)| + \sup_{x \in \Omega} |b(x)|^2.$$

In the regime of Theorem 5.2, we choose $(\Delta t, T_{\text{max}})$ as functions of N_{FK} so that $\sup_{x \in \Omega} |b(x)| \lesssim N_{\text{FK}}^{-\beta}$, with β as in the theorem. This bias term can then be absorbed into the approximation error and we do not track it separately in the statistical bounds below. The genuinely stochastic part, involving the ζ_k , is mean-zero and sub-exponential, and a standard argument yields a contribution of order $\mathcal{O}(N_{\text{FK}}^{-1/2})$.

Combining these estimates, we obtain the following finite-sample bound on the full statistical error.

Proposition 5.30 (Finite-sample statistical error bound). *Under Theorem 5.19 and the standing assumptions on $\mathcal{U}$, there exist constants $C_{\text{stat}}, C_{\text{FK}} > 0$ such that for every $\delta \in (0, 1)$, with probability at least $1 - \delta$,*

$$\begin{aligned}
\mathcal{E}_{\text{stat}} &\leq C_{\text{stat}} \left[(PN_2) \sqrt{\frac{\log(PN_2) + \log(6/\delta)}{N_{\text{int}}}} + (PN_2) \sqrt{\frac{\log(PN_2) + \log(6/\delta)}{N_{\partial\Omega}}} \right] \\
&\quad + C_{\text{FK}} \left[(PN_2) \sqrt{\frac{\log(PN_2) + \log(6/\delta)}{N_{\text{FK}}}} + \frac{\log(6/\delta)}{N_{\text{FK}}} \right]. \tag{5.35}
\end{aligned}$$

In particular, for fixed network size P and fixed confidence level, the first bracket in (5.35) tends to zero as $N_{\text{int}}, N_{\partial\Omega} \rightarrow \infty$, while the second bracket decays only as $N_{\text{FK}}^{-1/2}$ (up to logarithmic factors). Thus, when N_{FK} is small or remains bounded, the FK contribution controls the asymptotic order of $\mathcal{E}_{\text{stat}}$ and induces a nonvanishing statistical floor of order

$$\mathcal{E}_{\text{stat}} \gtrsim (PN_2) \sqrt{\frac{\log(PN_2)}{N_{\text{FK}}}}.$$

The dependence on $\log(6/\delta)$ in (5.35) comes from a simple union bound. We first apply Lemma 5.29 to control $\mathcal{E}_{\text{stat}}^{\text{PDE}}$ with failure probability at most $\delta/3$ by replacing $\log(2/\delta)$ there with $\log(6/\delta)$. The same argument applied to the boundary term $\mathcal{E}_{\text{stat}}^{\partial\Omega}$ yields an analogous bound, again with failure probability at most $\delta/3$. Finally, the FK contribution $\mathcal{E}_{\text{stat}}^{\text{FK}}$ is controlled in the same way, with its own failure probability bounded by $\delta/3$ and with the same replacement of $\log(2/\delta)$ by $\log(6/\delta)$. A union bound over these three events shows that, with probability at least $1 - \delta$, all three component bounds hold simultaneously, which implies (5.35) via Lemma 5.27.

3.5 Putting it all together: end-to-end non-asymptotic error bounds for FK-PINNs

We are now ready to combine the approximation, statistical and optimization bounds to prove the non-asymptotic error estimate announced in Theorem 5.2.

Let $\hat{u} \in \mathcal{U}$ be an empirical minimizer of $\hat{\mathcal{R}}$ and let $u_{\mathbf{m}\theta_T}$ be the output of T steps of gradient descent on $\ell(\mathbf{m}\theta) = \hat{\mathcal{R}}(u_{\mathbf{m}\theta})$ starting from $\mathbf{m}\theta_0$ as in Theorem 5.25. We write

$$\mathcal{E}_{\text{tot}}(T) := \|u_{\mathbf{m}\theta_T} - u^*\|_{L^2(\Omega)}.$$

Theorem 5.31 (Non-asymptotic error bound for FK-PINNs). *Suppose that Theorem 5.19 holds and that the assumptions of Corollary 5.25 are satisfied, and that $u^* \in W^{s,\infty}(\Omega)$ for some integer $s \geq 3$. Fix confidence level $\delta \in (0, 1)$. Then there exist constants $C_{\text{app}}, C_{\text{stat}}, C_{\text{FK}}, C_{\text{opt}}, c_{\text{opt}} > 0$, independent of $(N_{\text{int}}, N_{\partial\Omega}, N_{\text{FK}})$ and of T , such that with probability at least $1 - \delta$ over*

the sampling of the collocation and FK points,

$$\begin{aligned} \mathcal{E}_{\text{tot}}(T) \leq & C_{\text{app}} P^{-\gamma} + C_{\text{stat}}(PN_2) \sqrt{\frac{\log(PN_2) + \log(6/\delta)}{N_{\text{int}}}} + C_{\text{stat}}(PN_2) \sqrt{\frac{\log(PN_2) + \log(6/\delta)}{N_{\partial\Omega}}} \\ & + C_{\text{FK}}(PN_2) \sqrt{\frac{\log(PN_2) + \log(6/\delta)}{N_{\text{FK}}}} + C_{\text{opt}} \exp(-c_{\text{opt}}T), \end{aligned} \quad (5.36)$$

where $\gamma = (s - 2)/d$ and P is the maximal number of parameters of networks in $\mathcal{U}$. The constants $C_{\text{app}}, C_{\text{stat}}, C_{\text{FK}}, C_{\text{opt}}, c_{\text{opt}}$ may depend on the task weights $(\lambda_{\text{PDE}}, \lambda_{\partial\Omega}, \lambda_{\text{FK}})$ and on the PDE data, but not on the sample sizes $(N_{\text{int}}, N_{\partial\Omega}, N_{\text{FK}})$ nor on T .

Proof. Fix $\delta \in (0, 1)$ and work on the event where the statistical bound (5.35) of Proposition 5.30 holds. Set

$$u_T := u_{\mathbf{m}\theta_T},$$

let $\hat{u}$ be an empirical minimizer of $\hat{\mathcal{R}}$ as in Lemma 2.10, and let $u^{\text{approx}} \in \mathcal{U}$ be the approximating network from Corollary 5.22.

First we control the population risk of u_T . Adding and subtracting empirical risks gives

$$\begin{aligned} \mathcal{R}(u_T) - \mathcal{R}(u^*) &= [\mathcal{R}(u_T) - \hat{\mathcal{R}}(u_T)] + [\hat{\mathcal{R}}(u_T) - \hat{\mathcal{R}}(\hat{u})] \\ &\quad + [\hat{\mathcal{R}}(\hat{u}) - \mathcal{R}(\hat{u})] + [\mathcal{R}(\hat{u}) - \mathcal{R}(u^*)]. \end{aligned}$$

On the high probability event, the first and third brackets are bounded in absolute value by $\mathcal{E}_{\text{stat}}$. The second bracket is exactly $\mathcal{E}_{\text{opt}}(T)$ by (5.31). For the last bracket we apply the excess risk decomposition Lemma 2.10 with comparison function u^{approx} , which yields

$$\mathcal{R}(\hat{u}) - \mathcal{R}(u^*) \leq 2\mathcal{E}_{\text{stat}} + 2[\mathcal{R}(u^{\text{approx}}) - \mathcal{R}(u^*)].$$

Since $\mathcal{R}(u^*) = 0$ and $\mathcal{R}$ is nonnegative, this last term is at most $2\mathcal{R}(u^{\text{approx}})$. Combining these bounds we arrive at

$$\mathcal{R}(u_T) - \mathcal{R}(u^*) \leq 4\mathcal{E}_{\text{stat}} + \mathcal{E}_{\text{opt}}(T) + 2\mathcal{R}(u^{\text{approx}}). \quad (5.37)$$

By Lemma 5.23 we have $\mathcal{R}(u^{\text{approx}}) \leq \tilde{C}_{\text{risk}} P^{-2\gamma}$, so

$$\mathcal{R}(u_T) - \mathcal{R}(u^*) \leq 4\mathcal{E}_{\text{stat}} + \mathcal{E}_{\text{opt}}(T) + C_1 P^{-2\gamma} \quad (5.38)$$

for a constant $C_1 > 0$ independent of the sample sizes and of P .

Next we use PDE stability. From (5.13) and the nonnegativity of $\mathcal{R}_{\text{FK}}$ we have, for every u ,

$$\lambda_{\min}(\mathcal{R}_{\text{PDE}}(u) + \mathcal{R}_{\partial\Omega}(u)) \leq \mathcal{R}(u) - \mathcal{R}(u^*),$$

with $\lambda_{\min} = \min\{\lambda_{\text{PDE}}, \lambda_{\partial\Omega}\}$. Applying this with $u = u_T$ in (5.18) gives

$$\|u_T - u^*\|_{L^2(\Omega)}^2 \leq \frac{C_{\text{stab}}}{\lambda_{\min}} (\mathcal{R}(u_T) - \mathcal{R}(u^*)). \quad (5.39)$$

Substituting (5.38) into (5.39) we obtain

$$\|u_T - u^*\|_{L^2(\Omega)}^2 \leq C_2(\mathcal{E}_{\text{stat}} + P^{-2\gamma} + \mathcal{E}_{\text{opt}}(T)),$$

for a constant $C_2 > 0$ independent of $(N_{\text{int}}, N_{\partial\Omega}, N_{\text{FK}})$ and of T . Taking square roots and inserting the bounds (5.35) and (5.32), then absorbing numerical factors and the architectural bound $P \lesssim Lm^2$ into the constants, yields the claimed inequality (5.36). $\square$

Remark 5.32 (Error bound in terms of depth and width). Recall that for networks in $\mathcal{U}_{L,m,B}$ the number of parameters satisfies

$$P(L, m) \leq C_{\text{arch}}(d) L m^2.$$

Writing $\tilde{P} := C_{\text{arch}}(d) L m^2$ and using $\log P \lesssim \log \tilde{P} \lesssim \log(Lm)$, the bound (5.36) can be rewritten (up to a change of the constants) as

$$\begin{aligned} \mathcal{E}_{\text{tot}}(T) &\lesssim (L m^2)^{-\gamma} + L^2 m^3 \sqrt{\frac{\log(Lm) + \log(6/\delta)}{N_{\text{int}}}} + L^2 m^3 \sqrt{\frac{\log(Lm) + \log(6/\delta)}{N_{\partial\Omega}}} \\ &\quad + L^2 m^3 \sqrt{\frac{\log(Lm) + \log(6/\delta)}{N_{\text{FK}}}} + \exp(-c_{\text{opt}}T), \end{aligned}$$

with $\gamma = (s - 2)/d$. Moreover $N_{\text{hid}} = \sum_{\ell=1}^{L-1} m_\ell \leq (L - 1)m$, hence $N_2 \leq (C_{d,2}(L - 1)m + 3) \lesssim_d Lm$. Therefore $(PN_2) \lesssim_d L^2 m^3$.

Corollary 5.33 (Depth-3 FK-PINN: optimal width and rate). *Assume the setting of Theorem 5.31 and Corollary 5.22, with $s \geq 3$ and $\gamma = (s - 2)/d$. Fix the depth to $L = 3$. Using $P(L, m) \lesssim m^2$ in the master bound and suppressing logarithmic factors in the sample sizes and in m , we may write, for any training time $T \geq 0$,*

$$\mathcal{E}_{\text{tot}}(T) \lesssim m^{-2\gamma} + m^3 \left(N_{\text{int}}^{-1/2} + N_{\partial\Omega}^{-1/2} + N_{\text{FK}}^{-1/2} \right) + \exp(-c_{\text{opt}}T).$$

In the data-scarce FK regime (with N_{int} and $N_{\partial\Omega}$ large compared to N_{FK}), balance the approximation term $m^{-2\gamma}$ with the FK term $m^3 N_{\text{FK}}^{-1/2}$. This gives the width

$$m \asymp N_{\text{FK}}^{\frac{1}{2(3+2\gamma)}} = N_{\text{FK}}^{\frac{d}{2(3d+2(s-2))}}.$$

For this architecture there exists a constant $C > 0$ such that, for any $T \geq 0$ and with probability at least $1 - \delta$,

$$\mathcal{E}_{\text{tot}}(T) \leq C \left(N_{\text{FK}}^{-\frac{\gamma}{3+2\gamma}} + m^3 N_{\text{int}}^{-1/2} + m^3 N_{\partial\Omega}^{-1/2} + \exp(-c_{\text{opt}}T) \right), \quad m \asymp N_{\text{FK}}^{\frac{1}{2(3+2\gamma)}}.$$

where $m \asymp N_{\text{FK}}^{\frac{d}{2(3d+2(s-2))}}$. In particular, apart from the optimization term $\exp(-c_{\text{opt}}T)$ and up to logarithmic factors, a depth-3 FK-PINN with this width attains the statistical rate

$$\mathcal{E}_{\text{tot}}(T) = \mathcal{O} \left(N_{\text{FK}}^{-\frac{\gamma}{3+2\gamma}} \right) = \mathcal{O} \left(N_{\text{FK}}^{-\frac{s-2}{3d+2(s-2)}} \right) \quad (\text{up to logs}).$$

once N_{int} and $N_{\partial\Omega}$ are large enough for the terms $m^3 N_{\text{int}}^{-1/2}$ and $m^3 N_{\partial\Omega}^{-1/2}$ to be negligible. Finally, choosing $T \asymp \log N_{\text{FK}}$ so that $\exp(-c_{\text{opt}}T) \lesssim N_{\text{FK}}^{-\beta}$ with $\beta := \frac{\gamma}{3+2\gamma}$, taking $\delta := N_{\text{FK}}^{-10}$ in Theorem 5.31 (and hence in the bound above), and defining $\hat{u} := u_{\mathbf{m}\theta_T}$ yields exactly Corollary 5.3 (up to the logarithmic factors already suppressed here).

Corollary 5.33 shows that, if the number of FK trajectories N_{FK} were allowed to grow, a depth-3 FK-PINN with optimally chosen width would achieve the usual Sobolev rate $N_{\text{FK}}^{-(s-2)/(3d+2(s-2))}$ (up to logs). This matches the asymptotic behaviour proved by Doumèche et al. (2025) for PINNs with dense physics sampling: in that ideal regime, FK supervision does not worsen the leading statistical exponent.

Our focus, however, is the genuinely sparse FK regime, where N_{FK} is fixed while N_{int} and $N_{\partial\Omega}$ may grow. In this setting the FK term behaves as an irreducible noise floor: even as $N_{\text{int}}, N_{\partial\Omega} \rightarrow \infty$ and $T \rightarrow \infty$, the excess risk cannot be made uniformly small, because it is pinned down by the few noisy FK labels. This loss of uniform guarantees for the excess risk is the real “price” of FK supervision. We accept it in exchange for the PL* geometry and uniform conditioning provided by the FK term (Corollary 5.25), which yield fast optimization (Proposition 5.26). How to exploit a fixed FK budget (placement of FK points, choice of λ_{FK} , etc.) so as to mitigate this statistical saturation while retaining the optimization benefits is an interesting avenue for future work.

4 Conclusion and discussion

In this Chapter, we have leveraged the results from Chapter 4 concerning the improved conditioning of FK-PINNs loss landscape to derive rigorous end-to-end error bounds for FK-PINNs trained with finitely many steps of gradient descent, with all error components explicit in the training parameters and neural network architecture, making therefore a concrete step towards provably accurate and reliable neural-based solvers for numerical PDEs. The analysis is performed by precisely controlling the error incurred by Monte Carlo estimates of the Feynman-Kac supervision labels, together with a learning-theoretic decomposition of the excess risk, as developed in Chapters 2 and 3. Before the numerical experiments which we will show in the next chapter, we close with some relevant remarks and open questions.

- **Error bounds in stronger norms.** For the sake of sticking closely with the framework as introduced in Chapter 4, which is intended for arbitrary PDEs, we did not include the boundary-adapted factor in our analysis. This means that the best norm in which we can obtain estimate is the $H^{1/2}(\Omega)$, as explained in Chapter 3 (we have presented the bound in $L^2(\Omega)$ for simplicity). It is however clear that the analysis performed in this chapter extends verbatim to boundary-adapted PINNs, and thus the same theoretical guarantees apply in $H^2(\Omega)$ for boundary-adapted FK-PINNs.
- **More general PDE families.** Due to its reliance on techniques similar to the ones presented in Chapters 2 and 3, the end-to-end analysis performed in this chapter is therefore in principle limited to the same kind of limitations. Namely, the end-to-end a priori error bounds we obtained only hold for linear elliptic and parabolic equations, and extending the approach to more general PDE families would crucially require the existence of a “strong-convexity estimate” akin to the one from (Agmon et al., 1959) in the elliptic case.
- **Guarantees for variants of gradient descent.** Our end-to-end analysis is performed for PINNs trained with full-batch gradient descent. Though significant, this does not quite align with common practice, where other algorithms such as SGD, Adam, L-BFGS and others are preferred and reported to perform well (Han et al., 2018; Rathore et al., 2024; Müller and Zeinhofer, 2023). It would thus be of value to the community to investigate how such similar error guarantees could be obtained for these algorithms which are commonly used in practice.

Chapter 6

Numerical Experiments

We now illustrate the theoretical results from Chapters 3 and 4 with a comprehensive collection of numerical experiments representing a range of problems of important practical interest. Such problems include mean escape time problems, Poisson equations, Schrödinger equation or the committor problem. For each of those, we try to specifically highlight how our proposed solutions can effectively circumvent issues regarding boundary conditions enforcement or training pathologies, and compare those with other existing approaches in the literature. The results presented in this chapter are in large part taken from (Tepakbong et al., 2026a,b).

1 Influence of distance-to-boundary function for PINNs solving Dirichlet problems

We begin by illustrating the theoretical results from Chapter 3 by considering two Mean Escape Time problems: a two-dimensional Ornstein–Uhlenbeck process in a ball, and a random particle in a double-well potential field. As per Theorem 3.1, finding the MET for these processes is equivalent to solving PDEs of the form

$$\mathcal{L}\tau(x) = -1 \text{ for } x \in \Omega, \quad \tau(x) = 0 \text{ for } x \in \partial\Omega,$$

with $\mathcal{L}$ a second-order elliptic operator whose coefficients depend on the stochastic process under consideration.

In all experiments we approximate τ by neural network ansatz functions $f_{\theta}: \Omega \rightarrow \mathbb{R}$, which are then trained using different PINN formulations. We

consider the Boundary-Adapted PINN as per Definition 3.7, a classical strong-form PINN with an explicit boundary penalty term, and a variational PINN (VPINN) based on a weak formulation with Nitsche-type boundary terms (Liao and Ming, 2021). In addition, we introduce two modified variants of the Boundary Adapted architecture, obtained by multiplying the neural network by $\rho^{3/2}$ and $\rho^{1/2}$ instead of ρ , where ρ denotes an $(r + 2)$ -smooth distance approximation as per Definition 3.6. We refer to these as “over-enforced” and “under-enforced” PINNs. These variants still satisfy the homogeneous Dirichlet boundary condition and are included to assess the importance of the “distance-like” behaviour of ρ near $\partial\Omega$. Before reporting the numerical results, we briefly describe these architectures and the associated training methodology.

1.1 Computation of normalized distance function

As made clear throughout Chapter 3, choosing ρ as normalized distance function to the boundary $\partial\Omega$, according to Definition 3.6 is of crucial importance. Depending on how Ω is defined and available numerically, there are a number of ways to construct such functions ρ , as shown in (Sukumar and Srivastava, 2022) and references within. In the examples we give below, we are in a situation where $\Omega = \{x \in \mathbb{R}^d : \varphi(x) \leq 0\}$ for some sufficiently smooth function $\phi : \mathbb{R}^d \rightarrow \mathbb{R}$. We can therefore choose our normalized distance function as

$$\rho : x \mapsto \frac{\varphi(x)}{\sqrt{\varphi(x)^2 + \|\nabla\varphi(x)\|^2}},$$

which we can check is indeed a distance function normalized to the first order in the sense of Definition 3.6. Numerically, we find this choice to already yield notable improvements. It would however be possible (and advisable, as the theory shows), to choose ρ normalized to higher order $r + 2$. We describe in Appendix D a practical algorithm to construct such functions.

1.2 Training losses and evaluation metrics

All methods share the same fully connected feedforward network N_{θ} , and differ only in how the boundary condition is enforced and in the resulting training losses.

PDE residual, boundary, and data-fidelity losses. For all formulations that use a strong-form residual, we define the empirical PDE loss

$$\hat{J}_{\text{pde}}(\theta) := \frac{1}{n} \sum_{i=1}^n |\mathcal{L}(f_{\theta})(x_i) + 1|^2, \quad x_i \in \Omega. \quad (6.1)$$

The homogeneous Dirichlet boundary condition $\tau = 0$ is enforced in the standard PINN by the empirical $L^2(\partial\Omega)$ boundary loss

$$\hat{J}_{\text{bc}}(\boldsymbol{\theta}) := \frac{1}{n_{\text{bc}}} \sum_{k=1}^{n_{\text{bc}}} |f_{\boldsymbol{\theta}}(x_k^{\text{bc}})|^2, \quad x_k^{\text{bc}} \in \partial\Omega. \quad (6.2)$$

To tame the typically badly-behaved loss-landscape of the objective (6.1) and avoid convergence to spurious local minima, all architectures are further equipped with a data-fidelity loss based on Monte Carlo approximations of τ at interior points. Namely, we proceed as explained in Chapter 4: for a small number n_{data} of uniformly sampled $\{x_j^{\text{data}}\}_{j=1}^{n_{\text{data}}} \subset \Omega$, and precomputed estimates $\tau(x_j^{\text{data}})$, we set

$$\hat{J}_{\text{data}}(\boldsymbol{\theta}) := \frac{1}{n_{\text{data}}} \sum_{j=1}^{n_{\text{data}}} |f_{\boldsymbol{\theta}}(x_j^{\text{data}}) - \tau(x_j^{\text{data}})|^2. \quad (6.3)$$

Boundary Adapted, over-enforced, and under-enforced PINNs. The Boundary Adapted architecture incorporates the zero Dirichlet boundary condition into the ansatz via the smooth distance approximation ρ , which vanishes on $\partial\Omega$ and is strictly positive in Ω . Writing $N_{\boldsymbol{\theta}}$ for a standard feedforward network, we define

$$f_{\boldsymbol{\theta}}^{\text{adapted}}(x) := \rho(x) N_{\boldsymbol{\theta}}(x), f_{\boldsymbol{\theta}}^{\text{over}}(x) := \rho(x)^{3/2} N_{\boldsymbol{\theta}}(x), \text{ and } f_{\boldsymbol{\theta}}^{\text{under}}(x) := \rho(x)^{1/2} N_{\boldsymbol{\theta}}(x), \quad (6.4)$$

which we refer to as the boundary *adapted*, *over-enforced*, and *under-enforced* PINNs, respectively. In all three cases the boundary condition is satisfied by construction, so only the PDE residual and data-fidelity losses are used, obtained from (6.1)–(6.3) by replacing $f_{\boldsymbol{\theta}}$ with the corresponding $f_{\boldsymbol{\theta}}^{\text{arch}}$.

Standard PINN and variational PINN. The standard PINN uses the strong-form PDE residual (6.1), the boundary loss (6.2), and the data-fidelity loss (6.3). For the VPINN we instead adopt a deep Nitsche formulation (Liao and Ming, 2021): we fix in advance a finite collection of smooth test functions $\{\phi_k\}_{k=1}^{n_{\text{test}}} \subset H^1(\Omega)$. Let $a_{\text{N}}(\cdot, \cdot)$ and $l(\cdot)$ denote the Nitsche bilinear and linear forms obtained from $\mathcal{L}$ by integration by parts together with Nitsche-type boundary terms; in particular, the last term in a_{N} corresponds to the weak imposition of the Dirichlet condition. The exact solution u then satisfies

$$a_{\text{N}}(u, v) = l(v) \quad \text{for all } v \in H^1(\Omega).$$

Given a neural approximation f_{θ} , we evaluate, for each test function ϕ_k ,

$$R_k(f_{\theta}) := a_N(f_{\theta}, \phi_k) - l(\phi_k), \text{ for } k = 1, \dots, n_{\text{test}},$$

where all domain and boundary integrals in a_N and l are approximated by Monte Carlo quadrature. We decompose $R_k(f_{\theta})$ into its interior-plus-flux contribution and its Nitsche boundary-penalty contribution and define the associated empirical losses as the average squared residuals of these two parts, denoted (with a mild abuse of notation) by $\hat{J}_{\text{pde}}(\theta)$ and $\hat{J}_{\text{bc}}(\theta)$, respectively. The VPINN also uses the same data-fidelity loss (6.3), but does not employ the pointwise boundary loss (6.2).

Adaptive weighting. Rather than fixing the loss weights by hand, we learn them adaptively following the uncertainty-based multi-task weighting proposed in (Kendall et al., 2018), which we have found to consistently improve the performance of all our models. For a given architecture, let $\{\hat{J}_i(\theta)\}_{i \in \mathcal{I}}$ denote its active loss components (PDE, boundary, data). We introduce one trainable positive parameter λ_i per component and define

$$\hat{J}(\theta, \{\lambda_i\}_{i \in \mathcal{I}}) := \sum_{i \in \mathcal{I}} \frac{1}{2\lambda_i^2} \hat{J}_i(\theta) + \sum_{i \in \mathcal{I}} \log \lambda_i. \quad (6.5)$$

In particular, the standard PINN and the VPINN use three adaptive weights $\lambda_{\text{pde}}, \lambda_{\text{bc}}, \lambda_{\text{data}}$ (for the VPINN, λ_{bc} is associated with the Nitsche boundary term), while the Boundary Adapted, over-enforced, and under-enforced PINNs use two weights $\lambda_{\text{pde}}, \lambda_{\text{data}}$. All network and weighting parameters are learned jointly by minimizing the objective (6.5).

Error metrics. In addition to the training losses $\hat{J}_i$, we evaluate each trained model using relative L^2 , H^1 , and H^2 errors with respect to the exact MET solution τ . For a given approximation f_{θ} and $X \in \{L^2, H^1, H^2\}$, we define

$$X_{\text{rel}} \equiv X_{\text{rel}}(f_{\theta}) := \frac{\|f_{\theta} - \tau\|_{X(\Omega)}}{\|\tau\|_{X(\Omega)}}. \quad (6.6)$$

1.3 Ornstein-Uhlenbeck process in a ball

The first mean escape time problem we consider is that of a two-dimensional Ornstein-Uhlenbeck process in a domain $\Omega := \{x \in \mathbb{R}^2 : \|x\| < R\}$. In this case, the process $(X_t)_{t \geq 0}$ initiated at $x \in \Omega$ solves the SDE

$$dX_t = -\theta X_t dt + \varsigma dW_t, \quad X_0 = x,$$

where we set $R = 2, \theta = 3, \varsigma^2 = 3$ as fixed parameters. The infinitesimal generator of this process is given by

$$\mathcal{L}u(x) = -\theta x \cdot \nabla u(x) + \frac{\varsigma^2}{2} \Delta u(x).$$

All PINN variants in this example share the same fully connected feedforward network with tanh activations, three hidden layers, and 30 neurons per hidden layer. For the Boundary Adapted, over-enforced, and under-enforced architectures we use the smooth distance approximation

$$\rho(x) = \frac{R^2 - \|x\|^2}{2R},$$

which one can verify satisfies the conditions in Definition 3.6. The strong-form residual and boundary terms are approximated using $n_{\text{pde}} = 8192$ interior collocation points and $n_{\text{bc}} = 2048$ boundary collocation points.

For the variational PINN we additionally fix $n_{\text{test}} = 50$ randomly generated test functions of the form $\phi_k(x) = \exp\left(-\frac{\|x - c_k\|^2}{2s_{\text{test}}^2}\right)$, with centers c_k sampled uniformly in Ω , and fixed length scale $s_{\text{test}} = R/5$. Lastly, the data-fidelity loss uses $n_{\text{data}} = 256$ interior data points with Monte Carlo estimates of $\tau(x_j^{\text{data}})$.

All models are trained for 10000 epochs of Adam on the same decaying schedule, followed by L-BFGS with the same parameters and stopping criteria. The reported error metrics are computed with respect to the exact mean escape time τ , whose exact expression has been derived in (Kersting et al., 2023).

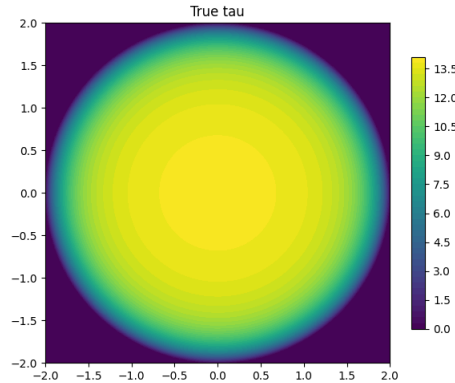


Figure 6.1: Reference solution τ for the 2D Ornstein-Uhlenbeck problem.

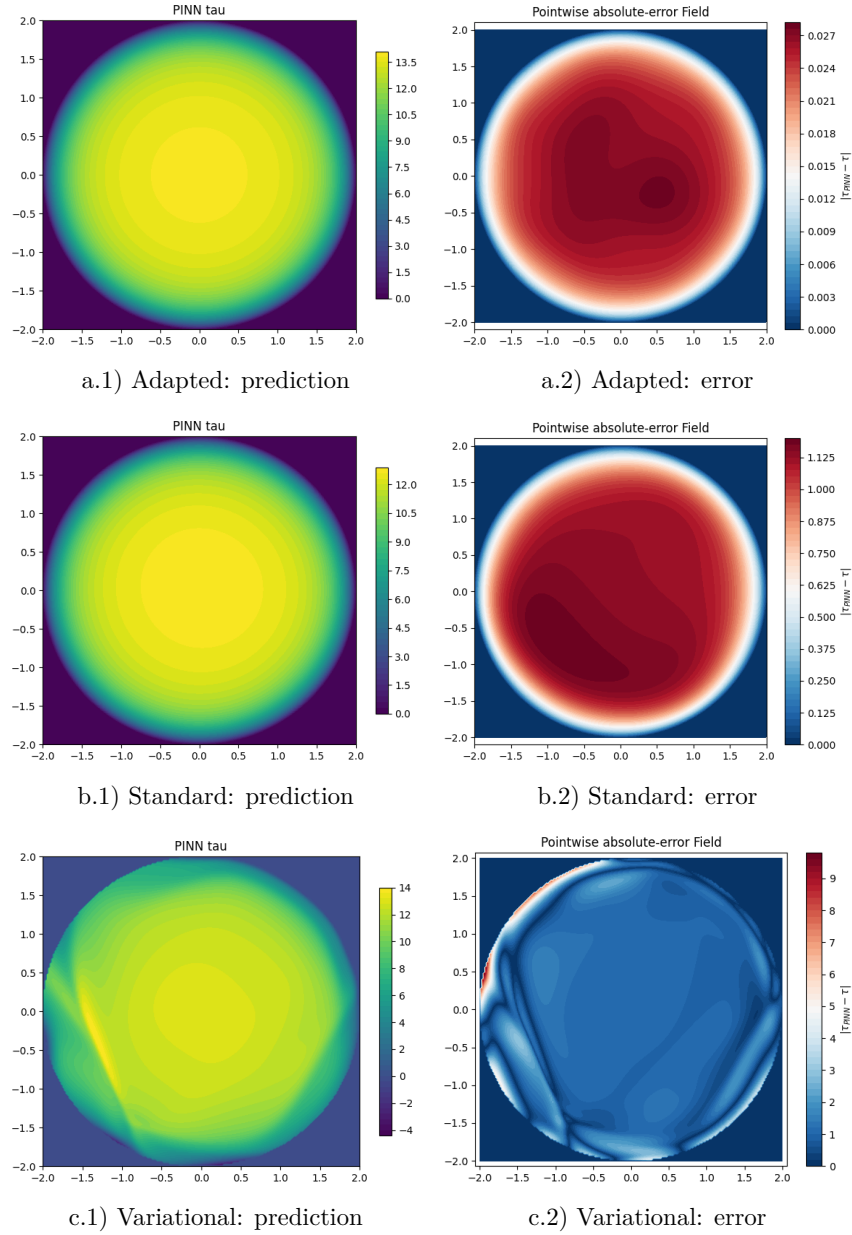


Figure 6.2: Predicted solutions (left column) and pointwise error fields (right column) for the adapted, standard, and variational PINNs on the 2D Ornstein-Uhlenbeck MET problem, using the reference solution τ in Figure 6.1.

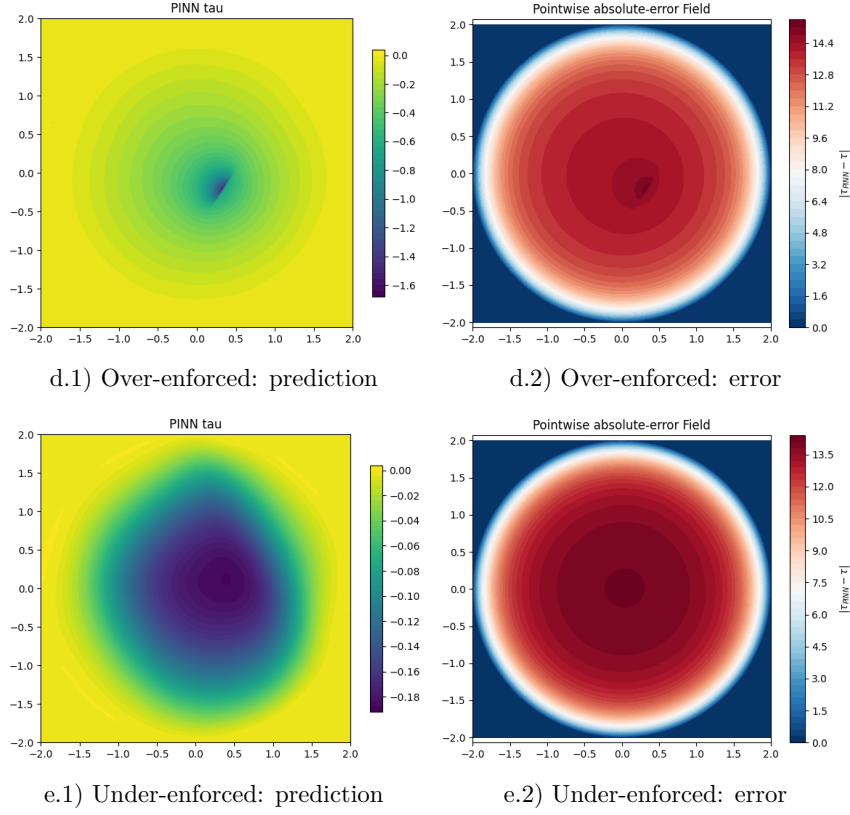


Figure 6.3: Predicted solutions (left column) and pointwise error fields (right column) for the over- and under-enforced PINNs on the 2D Ornstein-Uhlenbeck MET problem, using the reference solution τ in Figure 6.1.

Method	$L^2_{\text{rel}}(\ \tau\ _{L^2} = 1.155 \times 10^1)$	$H^1_{\text{rel}}(\ \tau\ _{H^1} = 2.266 \times 10^1)$	$H^2_{\text{rel}}(\ \tau\ _{H^2} = 1.841 \times 10^2)$
Standard	$4.887 \times 10^{-2} \pm 3.872 \times 10^{-2}$	$4.994 \times 10^{-2} \pm 3.926 \times 10^{-2}$	$5.041 \times 10^{-2} \pm 3.964 \times 10^{-2}$
Variational	$1.169 \times 10^{-1} \pm 5.697 \times 10^{-2}$	$6.491 \times 10^{-1} \pm 6.106 \times 10^{-1}$	3.269 ± 4.211
Over-enforced	$1.013 \pm 5.774 \times 10^{-4}$	1.006 ± 0.000	$1.001 \pm 5.774 \times 10^{-4}$
Under-enforced	$1.009 \pm 1.732 \times 10^{-3}$	$1.005 \pm 5.774 \times 10^{-4}$	1.000 ± 0.000
Adapted	$2.006 \times 10^{-3} \pm 3.014 \times 10^{-4}$	$1.998 \times 10^{-3} \pm 2.981 \times 10^{-4}$	$1.990 \times 10^{-3} \pm 2.946 \times 10^{-4}$

Table 6.1: Error metrics (mean $\pm$ standard deviation over 4 runs) for different PINN variants solving the Ornstein-Uhlenbeck MET problem.

Method	PDE loss	Data loss	BC loss
Standard	$5.229 \times 10^{-1} \pm 6.802 \times 10^{-1}$	$\mathbf{6.221} \times 10^{-1} \pm \mathbf{8.254} \times 10^{-2}$	$2.170 \times 10^{-1} \pm 3.759 \times 10^{-1}$
Variational	$8.532 \times 10^{-4} \pm 1.397 \times 10^{-3}$	$2.239 \pm 8.159 \times 10^{-1}$	4.920 ± 8.522
Over-enforced	$5.450 \times 10^{-2} \pm 4.157 \times 10^{-2}$	$1.254 \times 10^2 \pm 2.623$	—
Under-enforced	$4.332 \times 10^{-1} \pm 9.777 \times 10^{-2}$	$1.254 \times 10^2 \pm 1.401$	—
Adapted	$\mathbf{5.639} \times 10^{-5} \pm \mathbf{3.821} \times 10^{-6}$	$6.829 \times 10^{-1} \pm 1.580 \times 10^{-2}$	—

Table 6.2: Training losses (mean $\pm$ standard deviation over 4 runs) for different PINN variants solving the Ornstein-Uhlenbeck MET problem.

1.4 Particle in a double-well potential field

The second mean escape time problem we consider is that of a two-dimensional diffusion in a double-well potential on the elliptical domain $\Omega := \{(x, y) \in \mathbb{R}^2 : (x/a)^2 + (y/b)^2 < 1\}$. In this case, the process $(X_t)_{t \geq 0}$ initiated at $x \in \Omega$ solves the SDE

$$dX_t = b(X_t) dt + \varsigma(X_t) dW_t, \quad X_0 = x,$$

in which the drift is given by the negative gradient of the double-well potential

$$\Phi(x, y) = A \left(1 - \exp \left(- \frac{(x-1)^2 + (y-0.5)^2}{\nu^2} \right) \right) \left(1 - \exp \left(- \frac{(x+1)^2 + (y+0.5)^2}{\nu^2} \right) \right),$$

where we set $a = 2$, $b = 1.5$, $A = 5$, $\nu = 0.1$, and $\epsilon = 1.2$ as fixed parameters. We thus have $b(x, y) = -\nabla \Phi(x, y)$, and the diffusion matrix is chosen constant, $\varsigma = \text{diag}(\sqrt{2\epsilon}, \sqrt{\epsilon})$. The infinitesimal generator of this process is given by

$$\mathcal{L}u(x, y) = \epsilon u_{xx}(x, y) + \frac{\epsilon}{2} u_{yy}(x, y) + b_1(x, y) u_x(x, y) + b_2(x, y) u_y(x, y).$$

Again, all PINN variants in this example share the same fully connected feed-forward network with tanh activations, three hidden layers, and 30 neurons per hidden layer. For the Boundary Adapted, over-enforced, and under-enforced architectures we follow (Sukumar and Srivastava, 2022) and use the smooth distance approximation

$$\rho(x, y) = \frac{\omega(x, y)}{\sqrt{\omega(x, y)^2 + \|\nabla \omega(x, y)\|^2}}, \quad \omega(x, y) := 1 - (x/a)^2 - (y/b)^2,$$

which one can verify satisfies the conditions in Definition 3.6. The strong-form residual and boundary terms are approximated using $n_{\text{pde}} = 8192$ interior collocation points and $n_{\text{bc}} = 2048$ boundary collocation points.

For the variational PINN we additionally fix $n_{\text{test}} = 50$ randomly generated test functions of the form $\phi_k(x, y) = \exp(-\|(x, y) - c_k\|^2 / (2s_{\text{test}}^2))$, with centers c_k sampled uniformly in Ω , and fixed length scale $s_{\text{test}} = \pi ab / 10$. Lastly, the data-fidelity loss uses $n_{\text{data}} = 256$ interior data points with Monte Carlo estimates of $\tau(x_j^{\text{data}})$.

All models are trained for 5,000 epochs of Adam on the same decaying schedule, followed by L-BFGS with the same parameters and stopping criteria. The reported error metrics are computed with respect to a reference solution τ_{ref} , obtained with the finite element method (FEM).

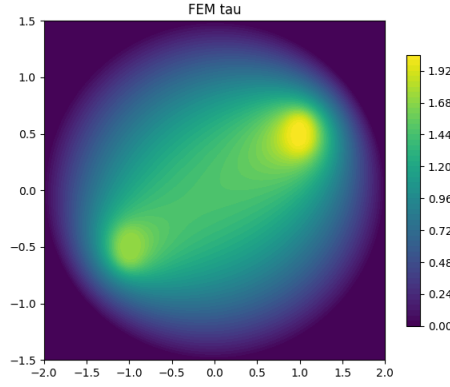


Figure 6.4: FEM solution τ_{ref} for the 2D double-well MET problem.

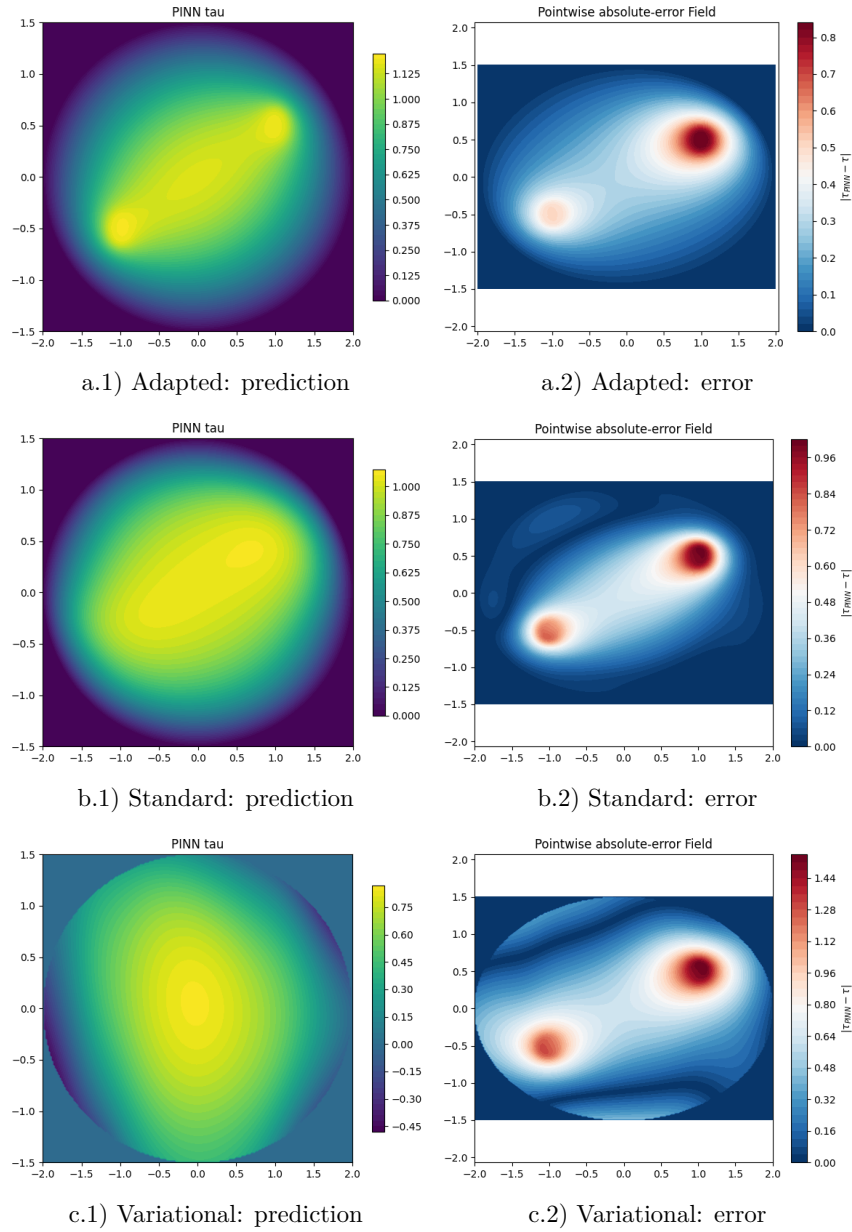


Figure 6.5: Predicted solutions (left column) and pointwise error fields (right column) for the adapted, standard, and variational PINNs on the double-well MET problem, using the FEM solution τ_{ref} in Figure 6.4 as reference.

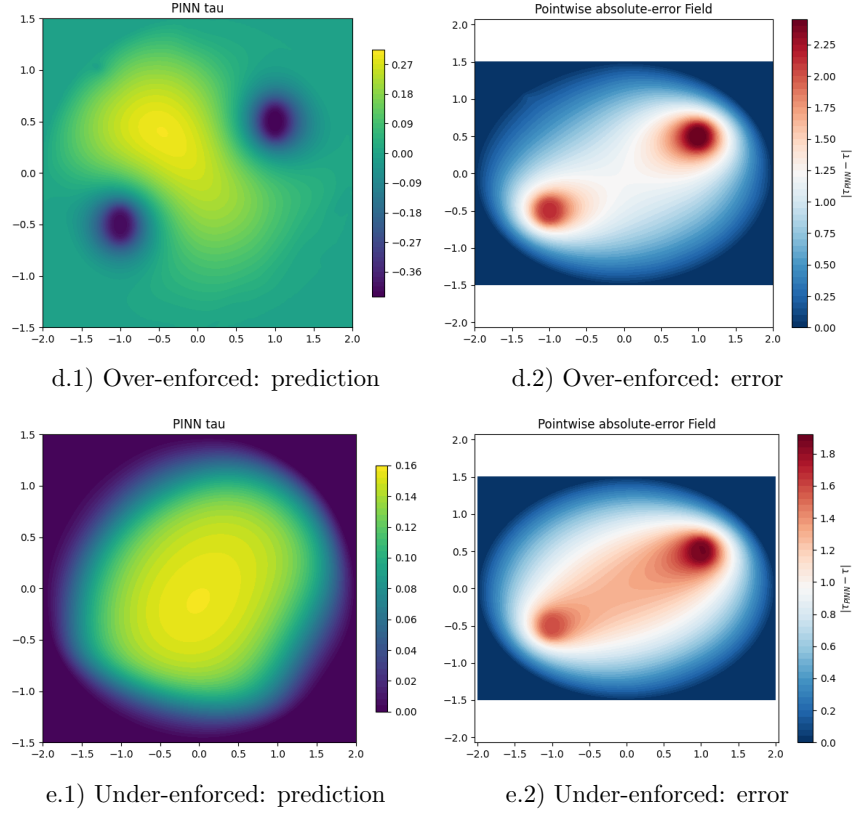


Figure 6.6: Predicted solutions (left column) and pointwise error fields (right column) for the over- and under-enforced PINNs on the double-well MET problem, using the FEM solution τ_{ref} in Figure 6.4 as reference.

Method	$L^2_{\text{rel}}(\ \tau\ _{L^2} = 9.538 \times 10^{-1})$	$H^1_{\text{rel}}(\ \tau\ _{H^1} = 1.834)$	$H^2_{\text{rel}}(\ \tau\ _{H^2} = 8.577)$
Standard	$2.953 \times 10^{-1} \pm 1.200 \times 10^{-2}$	$4.499 \times 10^{-1} \pm 1.393 \times 10^{-2}$	$6.530 \times 10^{-1} \pm 2.366 \times 10^{-2}$
Variational	$5.639 \times 10^{-1} \pm 1.100 \times 10^{-2}$	$1.173 \pm 9.637 \times 10^{-2}$	$1.197 \times 10^1 \pm 1.703$
Over-enforced	$4.857 \times 10^{-1} \pm 3.981 \times 10^{-1}$	$5.865 \times 10^{-1} \pm 4.187 \times 10^{-1}$	$8.943 \times 10^{-1} \pm 2.767 \times 10^{-1}$
Under-enforced	$9.047 \times 10^{-1} \pm 8.524 \times 10^{-3}$	$9.202 \times 10^{-1} \pm 9.625 \times 10^{-3}$	$9.625 \times 10^{-1} \pm 8.305 \times 10^{-3}$
Adapted	$2.643 \times 10^{-1} \pm 3.512 \times 10^{-4}$	$3.233 \times 10^{-1} \pm 8.386 \times 10^{-4}$	$4.031 \times 10^{-1} \pm 1.300 \times 10^{-3}$

Table 6.3: Error metrics (mean $\pm$ standard deviation over 4 runs) for different PINN variants solving the Double-Well MET problem.

Method	PDE loss	Data loss	BC loss
Standard	$3.580 \pm 5.859 \times 10^{-1}$	$9.612 \times 10^{-3} \pm 2.935 \times 10^{-3}$	$1.466 \times 10^{-11} \pm 8.607 \times 10^{-12}$
Variational	$1.059 \times 10^{-6} \pm 6.009 \times 10^{-7}$	$1.050 \times 10^{-1} \pm 9.377 \times 10^{-3}$	$8.653 \times 10^{-18} \pm 9.463 \times 10^{-18}$
Over-enforced	6.912 ± 5.865	$1.565 \times 10^{-1} \pm 2.674 \times 10^{-1}$	—
Under-enforced	$1.088 \pm 2.784 \times 10^{-2}$	$4.127 \times 10^{-1} \pm 1.489 \times 10^{-2}$	—
Adapted	$2.316 \times 10^{-4} \pm 7.213 \times 10^{-6}$	$2.734 \times 10^{-3} \pm 2.190 \times 10^{-4}$	—

Table 6.4: Training losses (mean $\pm$ standard deviation over 4 runs) for different PINN variants solving the Double-Well MET problem.

Overall, the numerical experiments confirm that the adapted PINN formulation is the most accurate and robust across all test cases, consistently achieving the lowest relative errors and the smallest variability over independent runs. By contrast, the baseline and unadapted variants either yield significantly larger errors or exhibit pronounced run-to-run fluctuations, indicating a lack of stability. Most notably, the over- and under-enforced PINNs frequently fail *catastrophically* to approximate the PDE solution, in agreement with Proposition 3.9, which shows that inappropriate choices of the multiplication function destroy the smoothness of the effective approximation target and thereby invalidate the convergence guarantees to the true solution.

2 Influence of Feynman-Kac supervision on the training dynamics

We now move to consider the effect of Feynman-Kac supervision on the training dynamics, as predicted by the results in Chapter 4.

2.1 Equations considered

We begin by specifying the concrete equations used to benchmark the proposed FK-PINNs framework. As discussed earlier, our focus is on regimes where traditional PINNs often become inaccurate or unstable, due to difficulties such as multiscale coupling, singular perturbations, high frequencies etc. in the PDE data, leading to extremely stiff loss landscapes. Besides the Schrödinger equation introduced in the main text, we also consider the three following representative classes of problems:

Poisson equation

We consider a prototypical elliptic bounded value problem on a bounded domain $\Omega \subset \mathbb{R}^d$ with boundary $\partial\Omega$:

$$-\Delta u(x) = f(x) \quad x \in \Omega, \quad (6.7)$$

with Dirichlet boundary conditions

$$u(x) = g(x) \quad x \in \partial\Omega.$$

Here $u(x)$ denotes the stationary potential or steady-state field generated by a source term $f(x)$. In multiscale settings, $f(x)$ and/or the solution $u(x)$ can exhibit sharp layers and small-scale structures, which pose challenges for standard PINNs due to spectral bias.

Mean Escape Time

Consider an overdamped diffusion process in $\mathbb{R}^d$,

$$dX_t = b(X_t) dt + \sigma(X_t) dW_t, \quad (6.8)$$

and a bounded domain $\Omega \subset \mathbb{R}^d$ with boundary $\partial\Omega$. The mean first exit time (or mean escape time) $\tau(x)$ from Ω solves the elliptic PDE

$$\mathcal{L} \tau(x) = -1, \quad x \in \Omega, \quad (6.9)$$

with boundary condition

$$\tau(x) = 0, \quad x \in \partial\Omega,$$

where $\mathcal{L}$ is the infinitesimal generator

$$\mathcal{L}\phi = -\nabla V \cdot \nabla \phi + \beta^{-1} \Delta \phi \quad (6.10)$$

where β is inverse temperature. The function $\tau(x)$ quantifies how long, on average, trajectories starting from x remain inside Ω . In metastable or small-noise regimes, $\tau(x)$ can vary over exponentially large scales, leading to stiff elliptic problems characterized by sharp boundary layers and high-contrast magnitudes that are challenging for standard PINNs.

Committor function

For the same diffusion process

$$dX_t = b(X_t) dt + \sigma(X_t) dW_t, \quad (6.11)$$

let $A, B \subset \mathbb{R}^d$ be two disjoint sets, and define $\Omega = \mathbb{R}^d \setminus (A \cup B)$. The committor function $q(x)$ is the probability that a trajectory starting at x reaches B before A . It is characterized as the solution of the elliptic boundary value problem

$$\mathcal{L} q(x) = 0 \quad x \in \Omega, \quad (6.12)$$

with boundary conditions

$$q(x) = 0 \quad x \in A, \quad q(x) = 1 \quad x \in B,$$

where $\mathcal{L}$ is the same generator as above. The committor encodes the effective reaction coordinate in transition path theory and large deviation theory, typically exhibiting sharp interfaces between metastable basins. Such boundary layers and rare-event structures are difficult to resolve with conventional PINNs, especially under small-noise conditions.

2.2 Evaluation of FK-PINNs against baseline PINNs

In this section, we evaluate our proposed FK-PINNs by testing them on the aforementioned PDEs, and comparing how they perform against standard PINNs as baseline. This choice is to isolate the effect of the Feynman-Kac supervision terms on the training dynamics.

Implementation details

Neural Network Architectures. All neural networks (baseline and FK-PINNs) used for these experiments are fully connected tanh networks with 4 hidden layers, and 128 neurons per layer.

Training Schedule. All neural networks (baseline and FK-PINNs) used for these experiments are trained using the Adam optimizer for 30,000 iterations, followed by 15,000 iterations of L-BFGS to fine-tune the solution. For the Adam phase, we implement a step-based exponential decay schedule, where the initial learning rate of 10^{-3} is decayed every k iterations by a factor of $\gamma = \exp(\ln(10^{-3})/(N/k))$, reaching a final learning rate of 10^{-6} .

Feynman-Kac data supervision. To generate the FK “data points”, we use [Algorithm 1](#) and set $\Delta t = 10^{-3}$, $N_{\text{MC}} = 500$, and $T_{\text{max}} = \infty$ (we simulate

the trajectories until escape). As for the number of FK points, we first choose a number N_{coll} of collocation points, and fix a proportion $p_{\text{data}} \in (0, 1)$ of randomly selected points for which we compute the FK label. We thus define

$$N_{\text{FK}} := \lfloor p_{\text{data}} N_{\text{coll}} \rfloor, \quad N_{\text{int}} := N_{\text{coll}} - N_{\text{FK}}. \quad (6.13)$$

We fix $p_{\text{data}} = 0.02$ for all experiments in 2.2. We will further explore the effects of different N_{MC} , Δt , and p_{data} on the experimental results in 2.5.

Metrics. To evaluate the performance of the models, we use the L^2 and H^1 error metrics in both absolute and relative forms.

$$L^2 \text{ absolute error: } L_{\text{abs}}^2(f_\theta) = \sqrt{\frac{1}{N} \sum_{n=1}^N |f_\theta(x_n) - u_{\text{ref}}(x_n)|^2} \quad (6.14)$$

$$L^2 \text{ relative error: } L_{\text{rel}}^2(f_\theta) = \frac{\sqrt{\frac{1}{N} \sum_{n=1}^N |f_\theta(x_n) - u_{\text{ref}}(x_n)|^2}}{\sqrt{\frac{1}{N} \sum_{n=1}^N |u_{\text{ref}}(x_n)|^2}} \quad (6.15)$$

$$H^1 \text{ absolute error: } H_{\text{abs}}^1(f_\theta) = \sqrt{\frac{1}{N} \sum_{n=1}^N |f_\theta(x_n) - u_{\text{ref}}(x_n)|^2 + \|\nabla f_\theta(x) - \nabla u_{\text{ref}}(x)\|^2} \quad (6.16)$$

$$H^1 \text{ relative error: } H_{\text{rel}}^1(f_\theta) = \frac{\sqrt{\frac{1}{N} \sum_{n=1}^N |f_\theta(x_n) - u_{\text{ref}}(x_n)|^2 + \|\nabla f_\theta(x) - \nabla u_{\text{ref}}(x)\|^2}}{\sqrt{\frac{1}{N} \sum_{n=1}^N |u_{\text{ref}}(x_n)|^2 + \|\nabla u_{\text{ref}}(x)\|^2}} \quad (6.17)$$

Poisson equation

In this example, we solve the Poisson equation on the square $\Omega = [0, 1] \times [0, 1]$, with source term given by

$$f(x_1, x_2) = \pi^2 [51 \sin(\pi x_1) \sin(4\pi x_2) + 156 \sin(5\pi x_1) \sin(\pi x_2) \\ + 24 \sin(2\pi x_1) \sin(2\pi x_2) + 244 \sin(5\pi x_1) \sin(6\pi x_2)],$$

and exact solution u^* :

$$u^*(x_1, x_2) = 3 \sin(\pi x_1) \sin(4\pi x_2) + 6 \sin(5\pi x_1) \sin(\pi x_2) \\ + 3 \sin(2\pi x_1) \sin(2\pi x_2) + 4 \sin(5\pi x_1) \sin(6\pi x_2).$$

We set $N_{\text{coll}} = 10000$ interior collocation points, and ($N_{\text{bc}} = 400$) boundary collocation points.

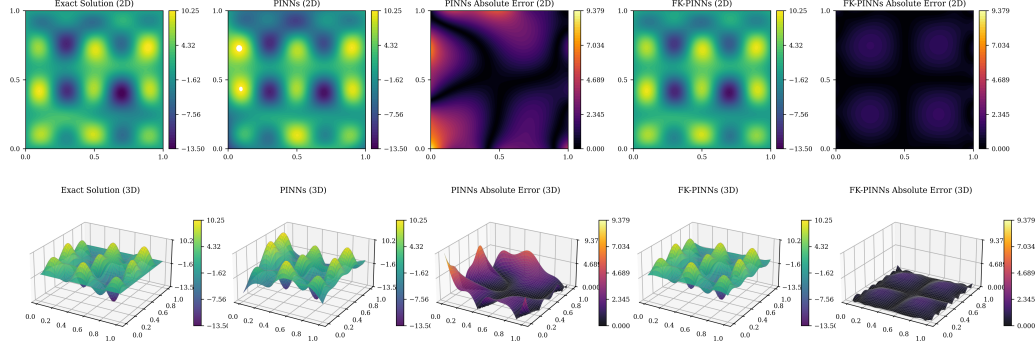


Figure 6.7: The ground truth solution (*Col.1*), predicted 2D,3D results by PINNs (*Col.2*), 2D,3D absolute error by PINNs (*Col.3*), predicted 2D,3D results by FK-PINNs (*Col.4*), 2D,3D absolute error by FK-PINNs (*Col.5*) on Poisson equations.

Figure 6.7 compares the performance of PINNs and FK-PINNs on this Poisson problem with high-frequency oscillations. While both models capture the global solution profile, PINNs exhibit significant absolute errors (up to 9.0), particularly near the oscillatory peaks and boundaries. In contrast, by leveraging the “extra data”, FK-PINNs effectively mitigate spectral bias, achieving a much finer reconstruction with absolute errors nearly an order of magnitude lower.

Figure 6.8 likewise demonstrates that FK-PINNs achieve superior convergence stability and a significantly lower PDE loss compared to standard PINNs, effectively mitigating the optimization stagnation observed in the vanilla baseline.

Mean Escape Time

We set the domain Ω as the regular hexagon centered at the origin with circumradius $R = 2$. We set V as a double-well potential function:

$$V(x_1, x_2) = \frac{1}{4}(x_1^2 - 1)^2 + \frac{\alpha}{2}x_2^2, \quad (6.18)$$

where $\alpha = 1$. The corresponding Mean Escape Time PDE is then given by:

$$-\nabla V \cdot \nabla \tau + \beta^{-1} \Delta \tau = -1, \quad (6.19)$$

with inverse temperature $\beta = 5$ and zero Dirichlet boundary condition.

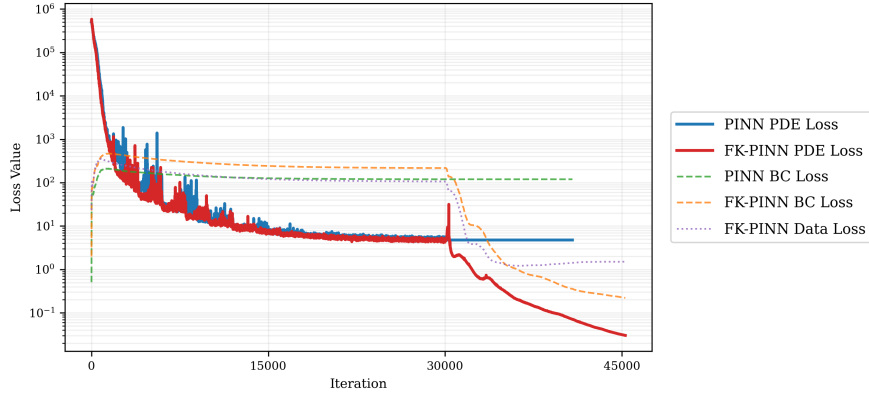


Figure 6.8: Comparison of PDE loss, BC loss of PINNs versus PDE loss, BC loss, Data loss of FK-PINNs for Poisson equation

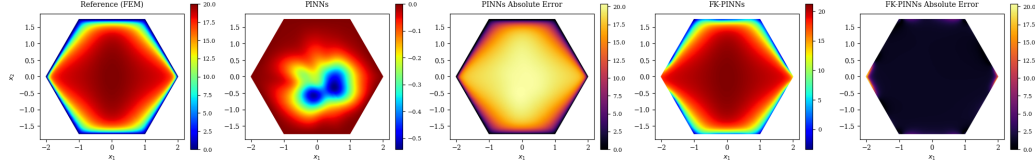


Figure 6.9: Comparison of Mean Escape Time PDE solutions learned by PINNs (left) and FK-PINNs(right)

Figure 6.9 shows the solutions learned both by PINNs and FK-PINNs. We see that PINNs (Cols. 2–3) exhibit *catastrophic failure* on this task, and completely fail to produce a meaningful solution. FK-PINNs (Cols. 4–5), on the other hand, successfully learn the solution and manage to capture the transition layers near the boundary, despite the corner singularities.

The training loss evolutions in Figure 6.10 reveal that standard PINNs encounter significant optimization stagnation after the initial phase, seemingly getting stuck around a spurious local minimum for the PDE loss. We see on the other hand that FK-PINNs manage to break through the poorly conditioned loss landscape and converge to the true solution.

Committer function

We now report the performance of FK-PINNs on the committor function problem on the domain $\Omega = [-1.5, 1.2] \times [-0.2, 2]$ with Müller-Brown potential, which is a standard benchmark in transition path studies (E and Vanden-Eijnden, 2006; Vanden-Eijnden et al., 2010). It is defined as the sum

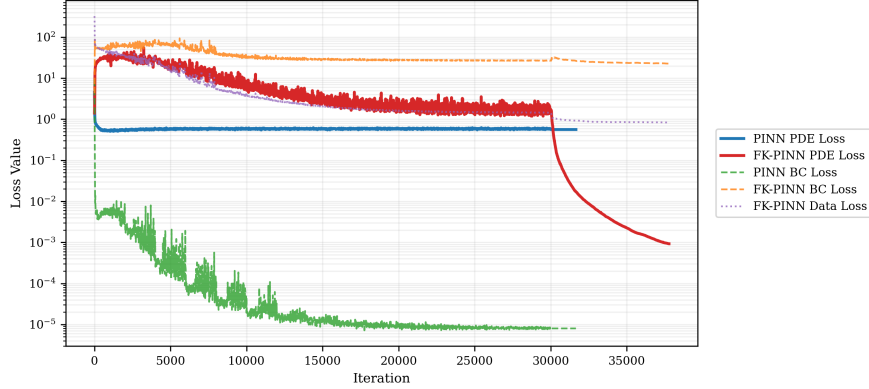


Figure 6.10: Comparing the evolution of loss components for solving the Mean Escape Time problem between PINNs and FK-PINNs

of four exponentials:

$$V(x, y) = \sum_{k=1}^4 A_k \exp [a_k(x - x_k^0)^2 + b_k(x - x_k^0)(y - y_k^0) + c_k(y - y_k^0)^2] \quad (6.20)$$

The specific coefficients used to define the potential energy surface in this work are provided in Table 6.5. These parameters ensure the presence of the characteristic “three-well” structure often used to benchmark transition path sampling and committor function approximations.

We further define two metastable states, A and B , which correspond to the reactant and product regions, respectively. These states are represented as circular boundary regions $\bar{\Omega}_A$ and $\bar{\Omega}_B$ with a fixed radius of $r = 0.2$. The centers of these disks, $\mathbf{c}_A$ and $\mathbf{c}_B$, are positioned at the primary local minima of the potential:

- **State A (Reactant):** $\mathbf{c}_A = (-0.5582, 1.4417)$, satisfying the boundary condition $q(\mathbf{x}) = 0$.
- **State B (Product):** $\mathbf{c}_B = (0.6235, 0.0281)$, satisfying the boundary condition $q(\mathbf{x}) = 1$.

The domain for the committor equation is thus defined as $\Omega' := \Omega \setminus (\bar{\Omega}_A \cup \bar{\Omega}_B)$

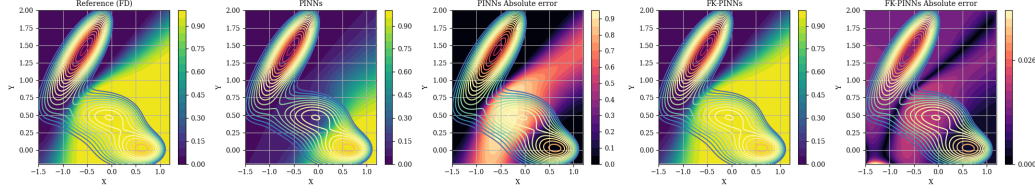


Figure 6.11: Comparison of Committor Function under Müller-Brown potential learned by PINNs (left) and FK-PINNs (right)

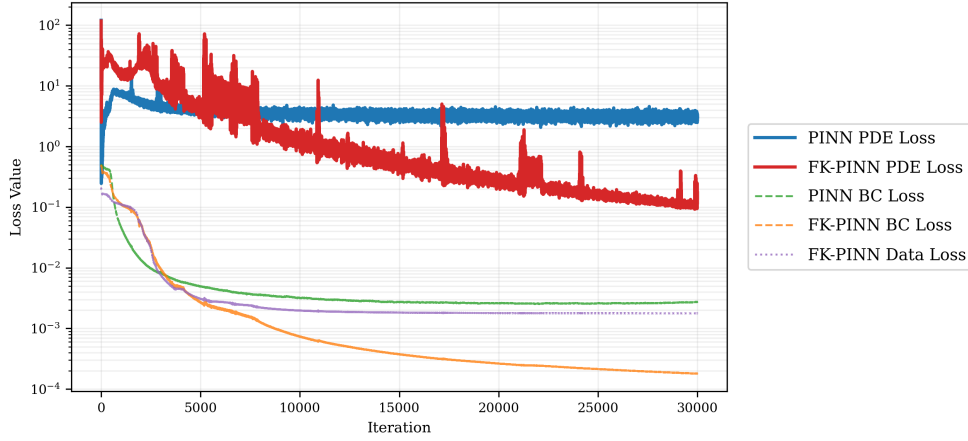


Figure 6.12: Comparison of the convergence behavior of individual loss components in PINNs and FK-PINNs trained with Adam optimizer only

Table 6.5: Parameters for the Müller-Brown Potential Energy Surface.

Index (k)	A_k	a_k	b_k	c_k	x_k^0	y_k^0
1	-200	-1	0	-10	1.0	0.0
2	-100	-1	0	-10	0.0	0.5
3	-170	-6.5	11	-6.5	-0.5	1.5
4	15	0.7	0.6	0.7	-1.0	1.0

As highlighted in Figure 6.11, the PINN solution fails to meaningfully capture the transition regions and the dynamics induced by the loss landscape, unlike FK-PINNs which recover a fully satisfactory solution. The improved training dynamics are also apparent in Figure 6.12.

Model	Metric	Poisson (2.2)	Schrödinger-type (6)	Mean Escape Time (2.2)	Committor (2.2)
PINNs	L^2 Abs Err	1.333 ± 0.616	0.475 ± 0.148	16.56 ± 0.055	1.028 ± 0.283
	L^2 Rel Err	0.322 ± 0.149	0.624 ± 0.195	1.007 ± 0.003	0.839 ± 0.661
	H^1 Abs Err	12.42 ± 4.345	2.893 ± 0.415	43.55 ± 0.039	3.154 ± 0.794
	H^1 Rel Err	0.171 ± 0.061	0.512 ± 0.073	1.002 ± 0.001	0.547 ± 0.074
FK-PINNs	L^2 Abs Err	0.761 ± 0.011	0.073 ± 0.008	1.755 ± 0.093	0.791 ± 0.039
	L^2 Rel Err	0.1184 ± 0.003	0.096 ± 0.010	0.107 ± 0.006	0.030 ± 0.008
	H^1 Abs Err	7.822 ± 0.412	1.043 ± 0.083	1.755 ± 0.093	1.495 ± 0.105
	H^1 Rel Err	0.108 ± 0.006	0.185 ± 0.015	0.395 ± 0.051	0.153 ± 0.049

Table 6.6: Performance comparison between standard PINNs and FK-PINNs on various problems. All results are averaged over 5 independent runs with different random seeds.

2.3 FK supervision improves the loss landscape

We now collect both qualitative and quantitative evidence supporting the improvement of the loss landscape as predicted by [Theorem 4.6](#).

Hessian condition number

To provide quantitative justification for the superior convergence of FK-Enhanced PINNs, we analyze the local curvature of the loss landscape near a minimizer through the lens of the empirical Hessian matrix, $H = \nabla_{\theta}^2 \mathcal{L}(\theta)$. As discussed in [Section 5](#), the conditioning of the Hessian governs the stability and speed of gradient-based optimization.

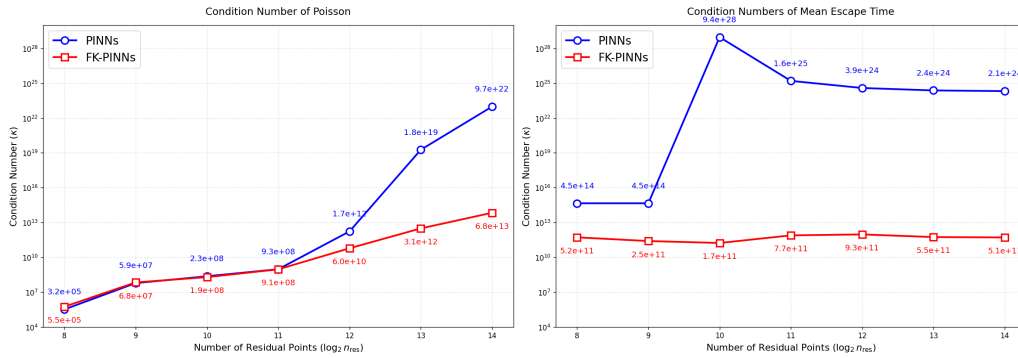


Figure 6.13: Condition numbers near a local minimum for PINNs and FK-PINNs for Poisson equation (left) and the Mean Escape Time Problem (right).

The evolution of the Hessian condition number as the number of collocation points increases plotted in [Figure 6.13](#), suggests, as predicted by [Theorem 4.4](#), that for the standard PINN, the condition number grows polynomially fast

with sampling density. The condition number of FK-PINNs however, does not appear to follow such a growth rate, and remains for most sample sizes orders of magnitude below that of the standard PINN, in accordance with [Theorem 4.6](#).

Visualizing the loss landscape geometry

We now turn to a more qualitative understanding of the training difficulty for standard PINNs and FK-PINNs. To do so, we decide to directly visualize the loss landscape itself around a minimizer, by projecting the loss onto a 2D parameter subspace, and visually observe the result ([Li et al., 2018](#); [Krishnapriyan et al., 2021](#); [Zhao et al., 2024](#)). The resulting landscapes for the Mean Escape Time PDE, as seen in [Figure 6.14](#), confirm what the theory and the other numerical results so far suggest: the PINN loss landscape exhibits a highly non-convex, jagged geometry, with many oscillations. This explains the failure of PINNs on this challenging PDE. On the other hand, the loss landscape of FK-PINNs appears much more regular and smooth, with a unique, well localized minimizer. This thus provides a vivid illustration of our theory.

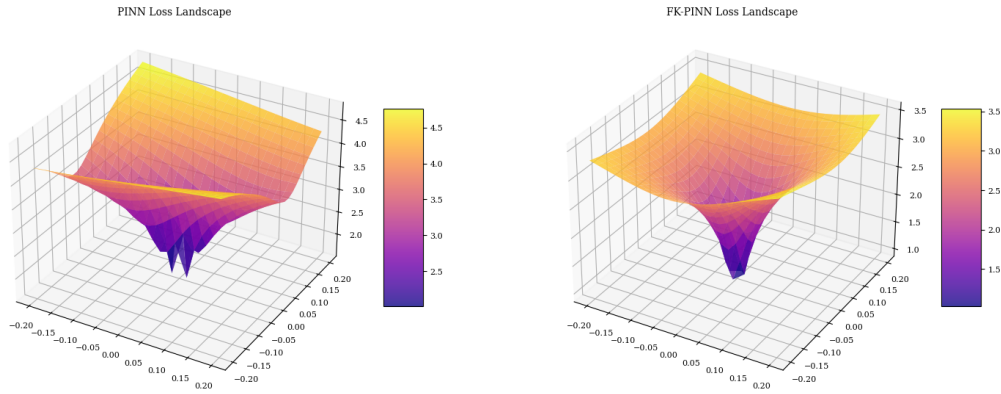


Figure 6.14: Loss landscape of Standard PINN (left) and FK-PINN (right) trained on the Mean Escape Time PDE, next to a minimizer.

2.4 Comparison with other operator preconditioning techniques

Now that the improvements of FK-PINNs over the baseline PINNs is clear, we compare the FK-PINNs with two other “pre-conditioning inspired” approaches to train them. The first one we consider is the Energy Natural Gradient Descent algorithm (ENGd), proposed by [Müller and Zeinhofer \(2023\)](#),

and which consists in performing gradient descent in the function space induced by the PDE. The second one is the NysNewtown-Conjugate Gradient Descent Algorithm (NNCG), proposed by Rathore et al. (2024), and which we run after Adam and L-BFGS, as the authors of said paper do to save computational cost.

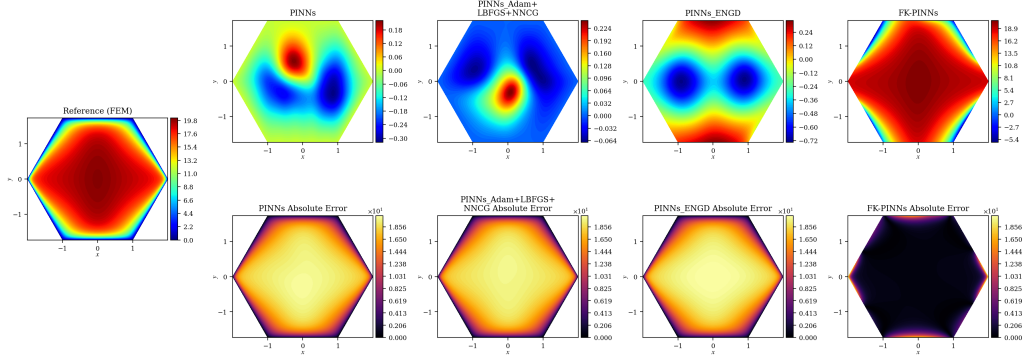


Figure 6.15: **Comparison of different PINNs solving the Mean Escape Time PDE** trained with (from left to right) Standard PINN (Adam + L-BFGS), Adam + L-BFGS + NNCG, ENG, FK-PINN method

Model	L2 Error	H1 Error
PINNs (Adam+L-BFGS)	1.002 ± 0.002	1.001 ± 0.001
PINNs (Adam+L-BFGS+NNCG)	0.986 ± 0.011	1.006 ± 0.001
PINNs (ENG)	1.016 ± 0.001	1.006 ± 0.014
FK-PINNs (Adam+L-BFGS)	0.174 ± 0.004	0.699 ± 0.008

Table 6.7: Relative L^2 and H^1 errors for each of the methods in Figure 6.15

As per the results reported in Figure 6.15 and Table 6.7, the other methods can not handle the stiff loss landscape of the Mean Escape Time PDE and fail just as much as the standard PINN does, despite their non-negligible computational overhead.

2.5 Influence of the Monte Carlo budget

Finally, we report how the “Monte Carlo budget”, i.e., the number N_{FK} of FK data points we allow ourselves to use relative to the number of collocation points, the number N_{MC} of paths we use to compute the Monte Carlo average,

and the timestep Δt we use to discretize the SDE influence the performance of FK-PINNs. We take $T_{\max} = \infty$ in all of the experiments presented below.

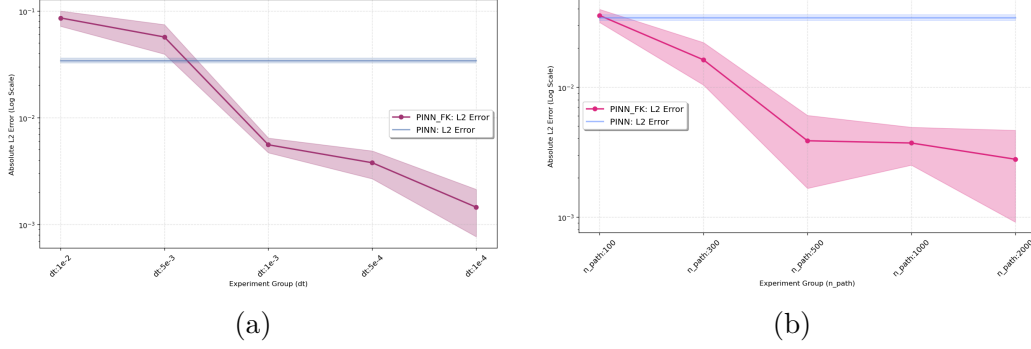


Figure 6.16: Sensitivity Analysis of Key Parameters in Feynman-Kac Monte Carlo Supervision. Figure (a): Fixed $N_{MC}=500$, influence of Δt on solution accuracy. Figure (b): Fixed $\Delta t = 1e - 3$, influence of N_{MC} on solution accuracy

As shown in Figure 6.16, and as one could expect, too coarse discretization Δt of the SDE leads to solutions which perform worse than the baseline PINN, which could be explained by the large amount of noise in the labels shifting the objective away from the true PDE solution. However we observe clear solution improvement for any discretization finer than $\Delta t = 10^{-3}$, which is well within the standard discretization sizes used to simulate SDE solutions. We likewise observe that for $\Delta t = 10^{-3}$, as even as little as $N_{MC} = 300$ provides clear improvement over the baseline, suggesting that even modest Monte Carlo computational budgets are already sufficient to “lift” the poorly conditioned PINN landscape with our method.

P_data	L2 Error	H1 Error	Time (s)
p_data=0	1.003 ± 0.004	1.001 ± 0.001	319.7 ± 5.5
p_data=0.001	0.189 ± 0.037	0.594 ± 0.069	456.3 ± 93.1
p_data=0.01	0.120 ± 0.025	0.431 ± 0.143	665.7 ± 14.1
p_data=0.02	0.107 ± 0.007	0.353 ± 0.086	807.4 ± 22.1
p_data=0.05	0.111 ± 0.011	0.224 ± 0.072	1309.7 ± 19.6
p_data=0.1	0.098 ± 0.007	0.299 ± 0.081	2198.2 ± 7.7
p_data=0.2	0.106 ± 0.010	0.382 ± 0.081	3968.9 ± 25.1

Table 6.8: Influence of the choice of p_{data} on the time overhead and the relative L^2 and H^1 error metrics. The PINNs here are trained to solve the Mean Escape Time PDE, and we’ve taken $\Delta t = 10^{-3}$, $N_{\text{MC}} = 500$

We finally show in Table 6.8 how the proportion p_{data} of points used as FK supervision influences the solution quality. Perhaps contrary as to what one would expect, we find that, for the Mean Escape Time at least, there are diminishing returns when adding FK supervision data: while the solution quality improves as p_{data} slowly increases from zero, we quickly find that there is a threshold where there is no clear improvement in the solution quality, while the time needed to simulate all the $N_{\text{MC}} \times N_{\text{FK}}$ paths until escape becomes prohibitively large, as the mean escape time is by design costly to estimate by Monte Carlo averages.

These observations suggest that it would be sensible, depending on the PDE being solved, to first fix Δt , and then preemptively simulate some Euler-Maruyama trajectories of the underlying stochastic process X_t until escape, in order to estimate roughly the amount of time needed to fully simulate a path. One can then tune p_{data} , N_{MC} , and possibly T_{max} according to both the time and computational budget one has at disposition.

3 Conclusion and discussion

Our numerical experiments confirm that boundary conditions enforcement and extraneous supervision, when done properly, can alleviate or even eliminate PINN failure modes. We note however that the mathematical theory that we’ve built is not yet able to exactly quantify the error incurred by PINNs, as the underlying constants in the technical assumptions are unknown. More effort remains thus needed, especially on the optimization analysis, to have a truly reliable and interpretable end-to-end error pipeline for PINNs.

Chapter 7

Conclusion and Outlook

In this thesis, we have made explicit the theoretical foundations on which the establishment of rigorous mathematical guarantees for scientific machine learning lie on. Based upon these, we have then provided for physics-informed neural networks (PINNs) trained to solve second-order elliptic Dirichlet boundary value problems, new error guarantees, which improve on the known state-of-the-art. These improved error guarantees take the following form:

- **Approximation error bounds for shallow ReLU² networks in higher-order Sobolev norms.** By extending on the variation space-based approximation framework originally proposed by [Barron \(2002\)](#), and improved upon for ReLU networks by [Bach \(2017\)](#), we’ve provided for the first time higher-order Sobolev norm approximation error bounds for shallow ReLU² networks. This extends non-trivially L^∞ approximation results for shallow ReLU^k networks due to [Yang and Zhou \(2025\)](#).
- **VC dimension and pseudo-dimension bounds for derivatives of fully-connected tanh and ReLU² networks.** We’ve given, for the first time, bounds on complexity measures for hypothesis spaces consisting of tanh and ReLU² networks, and their higher-order derivatives. This builds and improves upon a variety of classical ([Bartlett and Anthony, 1999](#); [Karpinski and Macintyre, 1997](#)) and recent ([D’Inverno et al., 2025](#); [Yang et al., 2023](#)) results.
- **Provably consistent scheme for enforcement of Dirichlet boundary conditions in PINNs.** We’ve shown the inconsistency of the standard “soft-enforcement” approach of imposing Dirichlet boundary conditions on PINNs by penalizing the $L^2(\partial\Omega)$ norm of the input when

aiming for bounds on the $H^2(\Omega)$ a priori error. We’ve therefore introduced, based on [Sukumar and Srivastava \(2022\)](#)’s results, a provably consistent way to “hard-enforce” those conditions. The algorithm is detailed in Appendix D.

- **A priori $H^2(\Omega)$ -error bounds for PINNs solving elliptic Dirichlet boundary value problems.** Building on those results, we’ve derived for the first time $H^2(\Omega)$ a priori error bounds for shallow *boundary-adapted PINNs* using ReLU^2 or $\tanh$ as activation. As it is common folklore that shallow architectures are preferred in applications, and that both of these activations are among the most popular in scientific machine learning, our results significantly improve our understanding in the possible successes of these methods.

Furthermore, we’ve moved out of the “ideal world” of empirical risk minimization, and taken a step towards guarantees for actual neural networks trained with finitely many steps of (stochastic) gradient descent as practitioners care about, by providing a simple regularization scheme for PINNs which we prove regularizes the loss landscape and ensures convergence of gradient descent. We call PINNs regularized in this way *FK-PINNs* and described the mechanism together with its benefits in Chapter 4. Leveraging these results, we’ve then provided end-to-end error bounds for FK-PINNs trained with finitely many steps of gradient descent, making a concrete bridge between theoretical and practical guarantees for PINNs, and scientific machine learning in general. We note that all the results in this thesis could be extended in a more or less direct fashion to parabolic PDEs, as we explain in Appendix C.

We now close this thesis by discussing natural questions arising from our work, and possible related future directions to explore.

1 Extension to operator learning

Throughout this thesis, we have kept our focus on physics-informed neural networks, which learn to solve a specific PDE instance on a given domain by directly approximating the solution. Another related approach which has garnered tremendous interest in recent years consists in teaching a neural network to learn the *parameter-to-solution operator* of a parametric PDE, rather than the solution for a specific PDE instance ([Lu et al., 2021a](#); [Kovachki et al., 2023](#)). This problem is of course much more challenging, as it now involves learning a mapping between infinite-dimensional function

spaces. However, due to its equally important practical implications, it is of as much interest to obtain rigorous theoretical guarantees for this family of algorithms. Naturally, one may then wonder how much of the framework and results presented in this thesis apply to *neural operators* too.

Although the general ideas explained in Chapter 2 do in fact translate more or less directly to the neural operator framework (or in fact, to pretty much any such parametric learning framework, for that matter), a number of subtleties make the establishment of systematic and unified rigorous error guarantees for neural operators quite challenging. Just to name a few:

- **Characterization of smoothness for solution operators.** As made clear throughout this thesis, a clear, quantitative characterization of regularity for the mapping of interest is essential in order to obtain any kind of approximation guarantees for neural networks. Unfortunately, such characterization for infinite-dimensional mappings is highly non-trivial and can typically only be obtained under a number of restrictions on the underlying PDE setup. While for the case of learning *functionals* (*function-to-scalar* mappings), regularity can be reasonably encoded through moduli of smoothness with corresponding guarantees Song et al. (2023); Shi et al. (2025), for operator learning these quantitative guarantees require enhanced control on the regularity of the solution operator (Lipschitz, C^k ...) (Kovachki et al., 2021) which are not realistic in most PDE setups, and even then are relatively pessimistic regarding the algorithm’s performance (Lanthaler, 2023; Lanthaler and Stuart, 2026). Altogether, the existing literature suggests that a generic ERM-based approach may not suffice to establish guarantees for operator learning in the way practitioners hope for.
- **Finite-dimensional representations.** A critical aspect of operator learning is that, while the operator of interest acts on infinite-dimensional spaces, computers can only represent finite-dimensional objects. This forces the user to choose *encoding* and *decoding* operators to respectively associate the input and output spaces to finite dimensional representations (Kovachki et al., 2023). While there are many ways to achieve this, for one to get error guarantees, one needs to understand how well these encoding/decoding choices accurately represent the function spaces of interest with respect to the underlying operator. This is where, once again, the existing results suggest that blindly choosing “data-only” representations can not yield strong guarantees (Lanthaler, 2023). Thus, choosing a basis that is adapted to the

PDE at hand is the option that is most likely to yield positive results. Unfortunately, efficiently computing a representing such a basis is typically an issue most likely to be successfully handled in PDE-specific cases rather than generically.

- **Dependence on meshing and resolution.** Related to the previous bullet point is the dependence on meshing/resolution of the operator network, which is a new component specific to this framework (Li et al., 2021; Kovachki et al., 2023). This dependence needs to be accounted for either directly through the architecture, or through the analysis (Zhang et al., 2023).
- **Restriction to compact subsets.** Another important element to consider in the error analysis of neural operators is that guarantees are typically only possible when approximating over *compact* subsets. This may be too restrictive for many PDE applications where we can't reasonably assume the input to come from (or be mapped to) compact subsets. To work around this restriction, authors have considered working over L^p spaces weighted by a finite-measure (Lu et al., 2021a; Kovachki et al., 2023). While this takes care of the compactness issue, this further limits the applicability and meaningfulness of these bounds for practical applications.

Therefore, the general ideas of this thesis, though applicable to neural operators, still require further work in order to be turned into complete, rigorous end-to-end guarantees for operator learning, which is still a growing field of research.

2 More general PDEs and boundary conditions

A general theme we've seen throughout this thesis is the following: *one needs to take the PDE structure into account in order to get strong mathematical guarantees.* We have mostly concerned ourselves to elliptic/parabolic PDEs with favourable boundary conditions, as it is a natural setting in which this structure, which we typically express in terms of regularity and near-boundary behavior, is easiest to describe. It is also a setting where strong-convexity estimates (see Chapter 3), which are necessary for the error analysis, are available. Moving away from elliptic and parabolic problems however, characterizing this structure, and coming up with architectures tailored to it becomes much more challenging. We list some possible work in

this direction.

2.1 Capturing more general boundary behavior

- **Different boundary conditions.** While we have focused on and provided results for PINNs adapted to solve PDEs with Dirichlet boundary conditions, this class represents a relatively small (and nice) subset of boundary conditions which arise naturally in applications. Understanding how to adapt the approach proposed in Chapter 3 to PDEs with Neumann, Robin or a mixture of different boundary conditions would be an important and practically meaningful extension of the work we’ve carried out in this thesis. We note that at the core of any approach in such direction would most likely be the algorithm proposed by [Sukumar and Srivastava \(2022\)](#), which proposes a way to construct neural networks which naturally satisfy such conditions. On the theoretical side however, understanding how well such modified neural networks can approximate smooth solutions remains largely open.
- **Adaptation to boundary layers.** As we’ve discussed in Chapter 3, for many problems of interest in engineering and the natural sciences, the solution is best described in terms of a *boundary layer*: there exists a small neighborhood Ω_ε of the boundary $\partial\Omega$, such that the solution u^* grows exponentially fast in Ω_ε , and behaves like a quasi-constant away from it ([Freidlin and Wentzell, 2012](#)). Building neural networks which can effectively and provably capture this structural condition would be an immensely valuable contribution for numerous applications.
- **Boundaries with kinks and corners.** Even for linear elliptic problems with smooth data, polygonal/polyhedral domains generically induce corner/edge singularities, so the solution is not well-described by a globally smooth near-boundary ansatz ([Gilbarg and Trudinger, 1977](#)). In particular, boundary-enforcing schemes based on distance-to-boundary functions can interact poorly with corner geometry, since derivatives of the boundary factor may become large or ill-behaved near vertices. A promising direction is to replace distance-function constructions by polygon-aware maps built from smooth generalized barycentric coordinates: the Wachspress-based transfinite formulation of [Sukumar and Roy \(2026\)](#) for instance exactly enforces Dirichlet conditions on convex polygons while retaining a bounded Laplacian of the trial function, and is thus a natural direction to explore going forward.

2.2 Other classes of regularity for PDE solutions

While we have considered PDEs for which solutions in $W^{r,\infty}$ or H^r exist, it is well-known that for many practically important PDEs, existence in such nice function spaces does not hold necessarily, and may require neural architectures adapted to those regimes. To illustrate what such examples could be, and the need for a corresponding error analysis, we give two examples below:

- **Rough coefficients and multiscale structure.** When coefficients are highly oscillatory or only weakly regular, the solution may have limited global smoothness and the strong-form residual can become poorly conditioned. A practically meaningful extension of the present framework would be to incorporate *scale separation* or *coarse-fine* structure into the PINN design and training procedure, for instance via homogenization-inspired pipelines such as NH-PINN (Leung et al., 2022) or by learning effective (homogenized) coefficients from data as in Park and Zhu (2022). On the analysis side, a key open point would be to understand which notion of regularity (and which norms) control approximation and training stability in such rough regimes.
- **BV/entropy solutions (shocks and discontinuities).** For non-linear hyperbolic problems, the physically relevant solution concept is typically an entropy solution in L^1/BV , and discontinuities are unavoidable (Dafermos, 2005). In such settings, minimizing a pointwise strong-form residual is structurally misaligned near shocks. A natural direction is to build the admissibility mechanism into the training objective (weak/finite-volume formulations and entropy inequalities) and to enforce conservation through interface fluxes or domain decomposition, as advocated for instance in conservative PINN formulations (Jagtap et al., 2020). Obtaining error bounds for this type of formulation would likewise be extremely meaningful.

3 Efficient, interpretable and provably convergent training algorithms for PINNs

As we’ve discussed throughout this thesis, obtaining error guarantees based on the empirical risk minimization framework laid out in Chapter 2 is not sufficient. To turn these results into actual guarantees for practical algorithms, a comprehensive convergence theory for PINNs trained using first and second order optimization algorithms is needed.

In this direction, a number of algorithms have been proposed that treat PINN training as a structured nonlinear least-squares problem, aiming to control the optimization error in a more principled manner than generic SGD training. Such approaches often consist in a first-order warm start followed by full-batch quasi-Newton refinement, and benchmarks indicate that such second-order phases can be decisive for high accuracy (Urbán et al., 2025). This has motivated PINN-tailored Gauss–Newton methods, including Levenberg–Marquardt for shallow architectures (Shahab et al., 2026) and energy natural gradient descent (Müller and Zeinhofer, 2023), as well as scalable matrix-free Newton–CG variants with randomized preconditioning and related dual residual-space implementations (Mckay et al., 2025; Bioli et al., 2025; Rathore et al., 2024; Jnini and Vella, 2025).

What remains missing for end-to-end error guarantees is therefore optimizer-specific convergence theory that turns these practical stopping rules into quantitative optimization-error term with constants expressed in PDE-conditioned quantities. Initial results in this direction exist for gradient descent and natural-gradient dynamics in simplified or over-parameterized PINN models (Gao et al., 2023; Xu et al., 2024), and extending them to the second-order solvers that dominate successful PINN training is a natural next step.

More fundamentally, having a clearer understanding of the underlying dynamics of first and second order algorithms used to train PINNs would be extremely valuable as well, in order to better grasp which PDEs are better suited for PINNs, and possibly design PDE-dependent regularization schemes taking this information into account. While there do exist results in the literature explaining the spectral bias of neural networks trained with gradient descent (Soudry et al., 2018; Cao et al., 2021; Jin and Montúfar, 2023), these works generally apply for classical classification and regression tasks, and do not directly apply to PINNs or physics-informed machine learning in general.

4 Looking at the bigger picture: theoretical guarantees for modern AI algorithms

Going further, it would be interesting to understand how the mathematical theory developed in this thesis extends beyond scientific computing and PDE-centric problems to modern machine learning, especially in light of the recent jump in capability of large-scale architectures such as Transformers (Vaswani et al., 2017) and score-based diffusion models (Ho et al., 2020). Indeed, one may notice that many of the analytical issues emphasized throughout this

thesis, such as approximation of objects living in infinite-dimensional spaces, training dynamics and stability, and quantification of generalization ability reappear in modern AI under different names.

An illustrative example is given by score-based generation, where sampling is formulated as the numerical approximation of a reverse-time stochastic dynamics (Song et al., 2021). In one common continuous-time formulation, a forward diffusion is defined by $dX_t = f(X_t, t) dt + g(t) dW_t$, and generation corresponds to integrating the reverse-time SDE

$$d\bar{X}_t = (f(\bar{X}_t, t) - g(t)^2 \nabla_x \log p_t(\bar{X}_t)) dt + g(t) d\bar{W}_t,$$

where the unknown score $\nabla_x \log p_t$ is approximated by a neural network. This viewpoint naturally leads to the same type of end-to-end decomposition used in this thesis, namely a separation between approximation error in the learned score field and numerical error from time discretization of the reverse dynamics, with stability estimates acting as the mechanism that turns local errors into global guarantees.

A second connection appears in recent mathematical work that models Transformers as maps acting on probability measures, and interprets self-attention as an interacting particle system whose continuum limit is governed by a mean-field PDE (Geshkovski et al., 2025; Rigollet, 2025). In such models, the evolution of the token distribution μ_t can be written at the level of measures as a continuity equation of the form

$$\partial_t \mu_t + \nabla \cdot (\mu_t v_\theta[\mu_t]) = 0,$$

for a nonlocal velocity field v_θ induced by attention. This reframing suggests that tools developed for rigorous SciML, including stability theory for measure-valued evolutions and error bounds for learned solution operators, may also contribute to a mathematical understanding of modern AI models that are, at their core, learned dynamical systems on distributions.

Appendix A

Complement on learning theory

This appendix is dedicated to complementary proofs and results from empirical process theory, which complete the discussion in Chapter 2.

1 Proof of Talagrand contraction lemma

1.1 Real-valued contraction inequality

We now give the proof of Lemma 2.31, which is none other than the celebrated Ledoux-Talagrand contraction principle, a more general version thereof can be found in (Ledoux and Talagrand, 2013).

Lemma A.1 (Contraction lemma). *Fix a sample $Z_1, \dots, Z_n$. Let $\sigma_1, \dots, \sigma_n$ be independent Rademacher random variables, independent of the sample. Let $\mathcal{G}$ be a class of real-valued functions on the sample space of the Z_i . For $i = 1, \dots, n$, let $\Phi_i : \mathbb{R} \rightarrow \mathbb{R}$ be L -Lipschitz and satisfy $\Phi_i(0) = 0$. Then*

$$\mathbb{E}_\sigma \left[\sup_{g \in \mathcal{G}} \frac{1}{n} \sum_{i=1}^n \sigma_i \Phi_i(g(Z_i)) \right] \leq L \mathbb{E}_\sigma \left[\sup_{g \in \mathcal{G}} \frac{1}{n} \sum_{i=1}^n \sigma_i g(Z_i) \right],$$

where $\mathbb{E}_\sigma$ denotes expectation with respect to $(\sigma_1, \dots, \sigma_n)$, conditional on $Z_1, \dots, Z_n$.

Proof. Throughout the proof, the sample $Z_1, \dots, Z_n$ is fixed. For $m \in \{0, 1, \dots, n\}$, define

$$U_m(g) := \sum_{i=1}^m \sigma_i \Phi_i(g(Z_i)).$$

We first prove the one-step inequality obtained by conditioning on the last sign, and then iterate it over $i = n, n-1, \dots, 1$.

Fix $\sigma_1, \dots, \sigma_{n-1}$ and consider the conditional expectation in σ_n . Using that σ_n takes values ± 1 with probability $1/2$,

$$\mathbb{E}_{\sigma_n} \left[\sup_{g \in \mathcal{G}} (U_{n-1}(g) + \sigma_n \Phi_n(g(Z_n))) \right] = \frac{1}{2}A + \frac{1}{2}B,$$

where

$$A := \sup_{g \in \mathcal{G}} (U_{n-1}(g) + \Phi_n(g(Z_n))), \text{ and } B := \sup_{g \in \mathcal{G}} (U_{n-1}(g) - \Phi_n(g(Z_n))).$$

Since the suprema need not be attained, fix $\eta > 0$ and choose $g_+, g_- \in \mathcal{G}$ such that

$$U_{n-1}(g_+) + \Phi_n(g_+(Z_n)) \geq A - \eta, \text{ and } U_{n-1}(g_-) - \Phi_n(g_-(Z_n)) \geq B - \eta.$$

Averaging these inequalities gives

$$\frac{1}{2}A + \frac{1}{2}B \leq \frac{1}{2}(U_{n-1}(g_+) + U_{n-1}(g_-)) + \frac{1}{2}(\Phi_n(g_+(Z_n)) - \Phi_n(g_-(Z_n))) + \eta.$$

By L -Lipschitz continuity of Φ_n ,

$$\Phi_n(g_+(Z_n)) - \Phi_n(g_-(Z_n)) \leq |\Phi_n(g_+(Z_n)) - \Phi_n(g_-(Z_n))| \leq L |g_+(Z_n) - g_-(Z_n)|.$$

Now note that for any real numbers a_+, a_-, b_+, b_- ,

$$\frac{1}{2}(a_+ + a_-) + \frac{L}{2}|b_+ - b_-| \leq \frac{1}{2} \sup_{g \in \mathcal{G}} (U_{n-1}(g) + L g(Z_n)) + \frac{1}{2} \sup_{g \in \mathcal{G}} (U_{n-1}(g) - L g(Z_n)),$$

because the right-hand side upper bounds both choices obtained by replacing $|b_+ - b_-|$ with $\pm(b_+ - b_-)$. Applying this with $a_{\pm} = U_{n-1}(g_{\pm})$ and $b_{\pm} = g_{\pm}(Z_n)$, we obtain

$$\mathbb{E}_{\sigma_n} \left[\sup_{g \in \mathcal{G}} (U_{n-1}(g) + \sigma_n \Phi_n(g(Z_n))) \right] \leq \mathbb{E}_{\sigma_n} \left[\sup_{g \in \mathcal{G}} (U_{n-1}(g) + L \sigma_n g(Z_n)) \right] + \eta.$$

Since $\eta > 0$ was arbitrary, the same inequality holds without η . Finally, repeat the same conditioning argument for $\sigma_{n-1}, \sigma_{n-2}, \dots, \sigma_1$. This yields

$$\mathbb{E}_{\sigma} \left[\sup_{g \in \mathcal{G}} \sum_{i=1}^n \sigma_i \Phi_i(g(Z_i)) \right] \leq L \mathbb{E}_{\sigma} \left[\sup_{g \in \mathcal{G}} \sum_{i=1}^n \sigma_i g(Z_i) \right],$$

and dividing by n gives the claim. $\square$

Lemma A.2 (Contraction for the absolute-value Rademacher averages). *Fix a sample $Z_1, \dots, Z_n$. Let $\sigma_1, \dots, \sigma_n$ be independent Rademacher random variables, independent of the sample. Let $\psi : \mathbb{R} \rightarrow \mathbb{R}$ be L -Lipschitz and satisfy $\psi(0) = 0$. Then*

$$\mathbb{E}_\sigma \left[\sup_{g \in \mathcal{G}} \left| \frac{1}{n} \sum_{i=1}^n \sigma_i \psi(g(Z_i)) \right| \right] \leq 2L \mathbb{E}_\sigma \left[\sup_{g \in \mathcal{G}} \left| \frac{1}{n} \sum_{i=1}^n \sigma_i g(Z_i) \right| \right].$$

Proof. Fix $\sigma = (\sigma_1, \dots, \sigma_n)$. Using $|x| \leq x^+ + (-x)^+$ in the form $|x| \leq x + (-x)$ after taking suprema,

$$\sup_{g \in \mathcal{G}} \left| \frac{1}{n} \sum_{i=1}^n \sigma_i \psi(g(Z_i)) \right| \leq \sup_{g \in \mathcal{G}} \frac{1}{n} \sum_{i=1}^n \sigma_i \psi(g(Z_i)) + \sup_{g \in \mathcal{G}} \frac{1}{n} \sum_{i=1}^n (-\sigma_i) \psi(g(Z_i)).$$

Taking $\mathbb{E}_\sigma$ and using that $(-\sigma_1, \dots, -\sigma_n)$ has the same distribution as $(\sigma_1, \dots, \sigma_n)$ gives

$$\mathbb{E}_\sigma \left[\sup_{g \in \mathcal{G}} \left| \frac{1}{n} \sum_{i=1}^n \sigma_i \psi(g(Z_i)) \right| \right] \leq 2 \mathbb{E}_\sigma \left[\sup_{g \in \mathcal{G}} \frac{1}{n} \sum_{i=1}^n \sigma_i \psi(g(Z_i)) \right].$$

But we are now done, since applying Lemma A.1 with $\Phi_i = \psi$ for $i = 1, \dots, n$ to bound the right-hand side by $2L \mathbb{E}_\sigma [\sup_{g \in \mathcal{G}} \frac{1}{n} \sum_{i=1}^n \sigma_i g(Z_i)]$, and then using $\sup_g \sum_i \sigma_i g(Z_i) \leq \sup_g |\sum_i \sigma_i g(Z_i)|$ yields the desired inequality. $\square$

1.2 A Banach-valued contraction lemma

We extend Lemma A.1 from real-valued classes to Banach-valued classes. This yields Lemma 2.32 by specializing to the Banach space ℓ_1^m .

Lemma A.3. *Fix a sample $Z_1, \dots, Z_n$. Let $\sigma_1, \dots, \sigma_n$ be independent Rademacher random variables, independent of the sample. Let $(\mathbb{B}, \|\cdot\|_{\mathbb{B}})$ be a Banach space, and let $\mathcal{F}$ be a class of measurable maps $f : \mathcal{Z} \rightarrow \mathbb{B}$. For $i = 1, \dots, n$, let $\Phi_i : \mathbb{B} \rightarrow \mathbb{R}$ be L -Lipschitz with $\Phi_i(0) = 0$. Then*

$$\mathbb{E}_\sigma \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \sigma_i \Phi_i(f(Z_i)) \right] \leq L \mathbb{E}_\sigma \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \left\| \sum_{i=1}^n \sigma_i f(Z_i) \right\|_{\mathbb{B}} \right].$$

Proof. Throughout the proof, the sample $Z_1, \dots, Z_n$ is fixed. For $m \in \{0, 1, \dots, n\}$, define

$$U_m(f) := \sum_{i=1}^m \sigma_i \Phi_i(f(Z_i)).$$

We prove the one-step inequality obtained by conditioning on σ_n , and then iterate it over $i = n, n-1, \dots, 1$.

Fix $\sigma_1, \dots, \sigma_{n-1}$ and consider the conditional expectation in σ_n . Using that σ_n takes values ± 1 with probability $1/2$,

$$\mathbb{E}_{\sigma_n} \left[\sup_{f \in \mathcal{F}} (U_{n-1}(f) + \sigma_n \Phi_n(f(Z_n))) \right] = \frac{1}{2}A + \frac{1}{2}B,$$

where $A := \sup_{f \in \mathcal{F}} (U_{n-1}(f) + \Phi_n(f(Z_n)))$ and $B := \sup_{f \in \mathcal{F}} (U_{n-1}(f) - \Phi_n(f(Z_n)))$. Since the suprema need not be attained, fix $\eta > 0$ and choose $f_+, f_- \in \mathcal{F}$ such that

$$U_{n-1}(f_+) + \Phi_n(f_+(Z_n)) \geq A - \eta, \text{ and } U_{n-1}(f_-) - \Phi_n(f_-(Z_n)) \geq B - \eta.$$

Averaging gives

$$\frac{1}{2}A + \frac{1}{2}B \leq \frac{1}{2}(U_{n-1}(f_+) + U_{n-1}(f_-)) + \frac{1}{2}(\Phi_n(f_+(Z_n)) - \Phi_n(f_-(Z_n))) + \eta.$$

By the L -Lipschitz property of Φ_n ,

$$\Phi_n(f_+(Z_n)) - \Phi_n(f_-(Z_n)) \leq L \|f_+(Z_n) - f_-(Z_n)\|_{\mathbb{B}}.$$

By the definition of the norm and Hahn–Banach, for every $\delta > 0$ there exists a linear functional $\ell \in \mathbb{B}^*$ with $\|\ell\|_{\mathbb{B}^*} \leq 1$ such that

$$\|f_+(Z_n) - f_-(Z_n)\|_{\mathbb{B}} \leq \ell(f_+(Z_n) - f_-(Z_n)) + \delta.$$

Using this bound and the same symmetrization in the last sign as in the proof of Lemma A.1 yields

$$\mathbb{E}_{\sigma_n} \left[\sup_{f \in \mathcal{F}} (U_{n-1}(f) + \sigma_n \Phi_n(f(Z_n))) \right] \leq \mathbb{E}_{\sigma_n} \left[\sup_{f \in \mathcal{F}} (U_{n-1}(f) + L \sigma_n \ell(f(Z_n))) \right] + \eta + \frac{L\delta}{2}.$$

Since $\eta > 0$ and $\delta > 0$ were arbitrary, we can drop them. Iterating the same conditioning argument for $\sigma_{n-1}, \dots, \sigma_1$ gives

$$\mathbb{E}_{\sigma} \left[\sup_{f \in \mathcal{F}} \sum_{i=1}^n \sigma_i \Phi_i(f(Z_i)) \right] \leq L \mathbb{E}_{\sigma} \left[\sup_{f \in \mathcal{F}} \sup_{\|\ell\|_{\mathbb{B}^*} \leq 1} \sum_{i=1}^n \sigma_i \ell(f(Z_i)) \right].$$

Finally, $\sup_{\|\ell\|_{\mathbb{B}^*} \leq 1} \sum_{i=1}^n \sigma_i \ell(f(Z_i)) = \|\sum_{i=1}^n \sigma_i f(Z_i)\|_{\mathbb{B}}$ by duality, and dividing by n yields the claim. $\square$

Proof of Lemma 2.32. Apply Lemma A.3 with $\mathbb{B} = \ell_1^m$ and $f_f(z) = (g_{f,1}(z), \dots, g_{f,m}(z))$. The Lipschitz assumption on ψ_i is exactly Lipschitz continuity with respect to $\|\cdot\|_{\ell_1^m}$. We obtain

$$\mathbb{E}_{\sigma} \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \sigma_i \psi_i(g_{f,1}(Z_i), \dots, g_{f,m}(Z_i)) \right] \leq L \mathbb{E}_{\sigma} \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \left\| \sum_{i=1}^n \sigma_i f_f(Z_i) \right\|_{\ell_1^m} \right].$$

For a fixed f , $\|\sum_{i=1}^n \sigma_i f_f(Z_i)\|_{\ell_1^m} = \sum_{j=1}^m |\sum_{i=1}^n \sigma_i g_{f,j}(Z_i)|$, hence

$$\sup_{f \in \mathcal{F}} \left\| \sum_{i=1}^n \sigma_i f_f(Z_i) \right\|_{\ell_1^m} \leq \sum_{j=1}^m \sup_{f \in \mathcal{F}} \left| \sum_{i=1}^n \sigma_i g_{f,j}(Z_i) \right|.$$

For each j , the coordinate functions $g_{f,j}$ belong to $\mathcal{G}_j$, so

$$\mathbb{E}_\sigma \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \left| \sum_{i=1}^n \sigma_i g_{f,j}(Z_i) \right| \right] \leq \hat{\mathfrak{R}}_n(\mathcal{G}_j).$$

Combining these inequalities yields the claim. $\square$

2 Covering numbers and chaining

In this Section, we will give a proof for the “single-scale” covering bound stated in Theorem 2.34, and then extend it to the celebrated Dudley’s entropy integral bound given in Theorem 2.35. To achieve this, we will first need a bound on Rademacher complexity of finite classes.

2.1 A maximal inequality for finite classes

The next lemma gives a logarithmic bound for the Rademacher complexity over a finite class, essentially by using the subgaussianity of bounded random variables together with the well-known Chernoff bound. It is often referred to in the literature as *Massart’s lemma* (Mohri et al., 2018).

Lemma A.4 (Massart’s lemma). *Let $V \subset \mathbb{R}^n$ be finite and set $B := \max_{v \in V} (\sum_{i=1}^n v_i^2)^{1/2}$. Then*

$$\mathbb{E}_\sigma \left[\max_{v \in V} \left| \frac{1}{n} \sum_{i=1}^n \sigma_i v_i \right| \right] \leq \frac{B}{n} \sqrt{2 \log(2|V|)}.$$

Proof. Let $V^\pm := V \cup (-V)$, so $\max_{v \in V} |\sum_{i=1}^n \sigma_i v_i| = \max_{v \in V^\pm} \sum_{i=1}^n \sigma_i v_i$. Fix $s > 0$. By Jensen’s inequality,

$$\exp \left(s \mathbb{E}_\sigma \left[\max_{v \in V^\pm} \sum_{i=1}^n \sigma_i v_i \right] \right) \leq \mathbb{E}_\sigma \left[\exp \left(s \max_{v \in V^\pm} \sum_{i=1}^n \sigma_i v_i \right) \right] \leq \sum_{v \in V^\pm} \mathbb{E}_\sigma \left[\exp \left(s \sum_{i=1}^n \sigma_i v_i \right) \right].$$

By independence of the σ_i , we get

$$\mathbb{E}_\sigma \left[\exp \left(s \sum_{i=1}^n \sigma_i v_i \right) \right] = \prod_{i=1}^n \mathbb{E}_\sigma [\exp(s \sigma_i v_i)].$$

Using $\mathbb{E}_\sigma[\exp(s\sigma_i v_i)] = \cosh(sv_i) \leq \exp(s^2 v_i^2/2)$, we thus get

$$\mathbb{E}_\sigma \left[\exp \left(s \sum_{i=1}^n \sigma_i v_i \right) \right] \leq \exp \left(\frac{s^2}{2} \sum_{i=1}^n v_i^2 \right) \leq \exp \left(\frac{s^2 B^2}{2} \right).$$

Therefore $\exp(s \mathbb{E}_\sigma[\max_{v \in V^\pm} \sum \sigma_i v_i]) \leq |V^\pm| \exp(s^2 B^2/2)$, so

$$\mathbb{E}_\sigma \left[\max_{v \in V^\pm} \sum_{i=1}^n \sigma_i v_i \right] \leq \frac{\log |V^\pm|}{s} + \frac{s B^2}{2}.$$

Optimizing over s yields $\mathbb{E}_\sigma[\max_{v \in V^\pm} \sum \sigma_i v_i] \leq B \sqrt{2 \log |V^\pm|}$. Dividing by n and using $|V^\pm| \leq 2|V|$ concludes the proof. $\square$

2.2 Proof of Theorem 2.34

We now give a proof of Theorem 2.34, which is the classical covering number-based bound on Rademacher complexity. The idea here is simply to approximate $\mathcal{G}$ by a finite ε -net in $\|\cdot\|_{L_2(P_n)}$, and then apply Lemma A.4 to the induced finite set of evaluation vectors.

Proof of Theorem 2.34. Fix $\varepsilon > 0$ and let $\mathcal{C} \subset \mathcal{G}$ be a minimal ε -cover of $\mathcal{G}$ in $\|\cdot\|_{L_2(P_n)}$. For each $g \in \mathcal{G}$, pick $\pi(g) \in \mathcal{C}$ with $\|g - \pi(g)\|_{L_2(P_n)} \leq \varepsilon$. Then for every σ ,

$$\sup_{g \in \mathcal{G}} \left| \frac{1}{n} \sum_{i=1}^n \sigma_i g(Z_i) \right| \leq \sup_{g \in \mathcal{G}} \left| \frac{1}{n} \sum_{i=1}^n \sigma_i (g(Z_i) - \pi(g)(Z_i)) \right| + \sup_{h \in \mathcal{C}} \left| \frac{1}{n} \sum_{i=1}^n \sigma_i h(Z_i) \right|.$$

For the first term, Cauchy–Schwarz gives $P_n|g - \pi(g)| \leq \|g - \pi(g)\|_{L_2(P_n)} \leq \varepsilon$, hence the first term is at most ε . Taking $\mathbb{E}_\sigma$ yields

$$\hat{\mathfrak{R}}_n(\mathcal{G}) \leq \varepsilon + \mathbb{E}_\sigma \left[\sup_{h \in \mathcal{C}} \left| \frac{1}{n} \sum_{i=1}^n \sigma_i h(Z_i) \right| \right].$$

Now we apply Lemma A.4 to the finite set $V := \{(h(Z_1), \dots, h(Z_n)) : h \in \mathcal{C}\}$. Since $\|h\|_{L_2(P_n)} \leq c$, we have $\sum_{i=1}^n h(Z_i)^2 \leq nc^2$, so $B \leq \sqrt{n} c$. Furthermore, recall that $|V| = |\mathcal{C}| = N(\varepsilon, \mathcal{G}, \|\cdot\|_{L_2(P_n)})$, hence the first claimed bound in Theorem 2.34 holds, while the second one is obtained by taking the infimum over ε . $\square$

2.3 Proof of Theorem 2.35

We are now ready to prove the Dudley entropy integral bound from Theorem 2.35. The main idea of the proof relies on the now classical chaining decomposition $g = (g - g_N) + \sum_{j=1}^N (g_j - g_{j-1})$, where g_j are successive approximations from finer and finer covers (Sen, 2018).

Proof of Theorem 2.35. Fix the sample $Z_1, \dots, Z_n$ and denote $d_n(g, h) := \|g - h\|_{L_2(P_n)}$. Likewise, define $\varepsilon_0 := 1$, and for $j \in \mathbb{N}$, set $\varepsilon_j := 2^{-j}$. Finally, for each $j \geq 1$, let $\mathcal{C}_j \subseteq \mathcal{G}$ be an ε_j -cover of $(\mathcal{G}, d_n)$ with cardinality $|\mathcal{C}_j| = N(\varepsilon_j, \mathcal{G}, \|\cdot\|_{L_2(P_n)})$. Also set $\mathcal{C}_0 := \{0\}$, which is an ε_0 -cover because $|g| \leq 1$ implies $d_n(g, 0) \leq 1$.

For each $g \in \mathcal{G}$ and each $j \geq 0$, choose a representative $g_j \in \mathcal{C}_j$ such that $d_n(g, g_j) \leq \varepsilon_j$. Then, for every $N \in \mathbb{N}$, one has

$$g = (g - g_N) + \sum_{j=1}^N (g_j - g_{j-1}).$$

Using subadditivity of $|\cdot|$ and the definition of $\hat{\mathfrak{R}}_n(\mathcal{G})$, it thus follows that $\hat{\mathfrak{R}}_n(\mathcal{G}) \leq T_0 + \sum_{j=1}^N T_j$, where

$$T_0 := \mathbb{E}_\sigma \left[\sup_{g \in \mathcal{G}} \left| \frac{1}{n} \sum_{i=1}^n \sigma_i(g - g_N)(Z_i) \right| \right], \text{ and } T_j := \mathbb{E}_\sigma \left[\sup_{g \in \mathcal{G}} \left| \frac{1}{n} \sum_{i=1}^n \sigma_i(g_j - g_{j-1})(Z_i) \right| \right].$$

We now bound each summand separately: for T_0 , applying Khintchine inequality in the form $\mathbb{E}_\sigma |\sum_{i=1}^n \sigma_i a_i| \leq (\sum_{i=1}^n a_i^2)^{1/2}$ (Boucheron et al., 2013) yields

$$T_0 \leq \sup_{g \in \mathcal{G}} \frac{1}{n} \left(\sum_{i=1}^n (g - g_N)^2(Z_i) \right)^{1/2} = \sup_{g \in \mathcal{G}} d_n(g, g_N) \leq \varepsilon_N.$$

We now fix $j \in \{1, \dots, N\}$ and want to control T_j : first note that the increment $g_j - g_{j-1}$ depends on g only through the pair $(g_j, g_{j-1}) \in \mathcal{C}_j \times \mathcal{C}_{j-1}$, hence the set of possible increments has cardinality at most $|\mathcal{C}_j| |\mathcal{C}_{j-1}|$. Moreover, by the triangle inequality and the defining approximation property, we have

$$d_n(g_j, g_{j-1}) \leq d_n(g_j, g) + d_n(g, g_{j-1}) \leq \varepsilon_j + \varepsilon_{j-1} = 3\varepsilon_j.$$

By the definition of d_n , this implies $\sum_{i=1}^n (g_j - g_{j-1})^2(Z_i) \leq 9n\varepsilon_j^2$.

Lemma A.4 can thus be applied to this finite class of increments, viewed as a subset of $\mathbb{R}^n$, with $B = 3\sqrt{n}\varepsilon_j$ and $|V| \leq |\mathcal{C}_j||\mathcal{C}_{j-1}|$, which after replacing yields

$$T_j \leq 3\varepsilon_j \sqrt{\frac{2\log(2|\mathcal{C}_j||\mathcal{C}_{j-1}|)}{n}} \leq 6\varepsilon_j \sqrt{\frac{\log(2|\mathcal{C}_j|)}{n}},$$

where we used $|\mathcal{C}_{j-1}| \leq |\mathcal{C}_j|$.

Combining the bounds for T_0 and T_j gives

$$\hat{\mathfrak{R}}_n(\mathcal{G}) \leq \varepsilon_N + \frac{6}{\sqrt{n}} \sum_{j=1}^N \varepsilon_j \sqrt{\log(2N(\varepsilon_j, \mathcal{G}, \|\cdot\|_{L_2(P_n)}))}.$$

Now define $H(\varepsilon) := \sqrt{\log(2N(\varepsilon, \mathcal{G}, \|\cdot\|_{L_2(P_n)}))}$. Since the function $\varepsilon \mapsto N(\varepsilon, \mathcal{G}, \|\cdot\|_{L_2(P_n)})$ is nonincreasing, so is H . Monotonicity of H together with the equality $\varepsilon_{j-1} = 2\varepsilon_j$ therefore yields

$$\varepsilon_j H(\varepsilon_j) = 2(\varepsilon_{j-1} - \varepsilon_j) H(\varepsilon_j) \leq 2 \int_{\varepsilon_j}^{\varepsilon_{j-1}} H(\varepsilon) d\varepsilon.$$

Summing over $j = 1, \dots, N$, we get $\sum_{j=1}^N \varepsilon_j H(\varepsilon_j) \leq 2 \int_{\varepsilon_N}^1 H(\varepsilon) d\varepsilon$, and therefore

$$\hat{\mathfrak{R}}_n(\mathcal{G}) \leq \varepsilon_N + \frac{12}{\sqrt{n}} \int_{\varepsilon_N}^1 \sqrt{\log(2N(\varepsilon, \mathcal{G}, \|\cdot\|_{L_2(P_n)}))} d\varepsilon.$$

Now fix $\alpha \in (0, 1]$. If $\alpha > 1/2$, then $\hat{\mathfrak{R}}_n(\mathcal{G}) \leq 1 \leq 4\alpha$, so the desired bound holds. Assume $\alpha \leq 1/2$, and let $N \in \mathbb{N}$ be such that $\varepsilon_N \leq 2\alpha < \varepsilon_{N-1}$. Then $\alpha < \varepsilon_N \leq 2\alpha$, and the above inequality implies

$$\hat{\mathfrak{R}}_n(\mathcal{G}) \leq 2\alpha + \frac{12}{\sqrt{n}} \int_{\alpha}^1 \sqrt{\log(2N(\varepsilon, \mathcal{G}, \|\cdot\|_{L_2(P_n)}))} d\varepsilon.$$

This bound holds for every $\alpha \in (0, 1]$, hence taking the infimum over α concludes the proof. $\square$

We now prove the localized variant from Lemma 2.36. The argument repeats the chaining construction above, but starts from radius γ instead of 1, and the entropy integral stops at γ .

Proof of Lemma 2.36. Fix the sample $Z_1, \dots, Z_n$ and set $d_n(g, h) := \|g - h\|_{L_2(P_n)}$. Define the localized class $\mathcal{G}_\gamma := \{g \in \mathcal{G} : d_n(g, 0) \leq \gamma\}$. For $j \in \mathbb{N}$, set $\varepsilon_j := \gamma 2^{-j}$, and let $\mathcal{C}_j \subseteq \mathcal{G}$ be an ε_j -cover of $(\mathcal{G}, d_n)$. Also set $\mathcal{C}_0 := \{0\}$, which is an ε_0 -cover of $\mathcal{G}_\gamma$.

For $g \in \mathcal{G}_\gamma$ and $j \geq 0$, choose $g_j \in \mathcal{C}_j$ such that $d_n(g, g_j) \leq \varepsilon_j$. Then $g = (g - g_N) + \sum_{j=1}^N (g_j - g_{j-1})$ and the same decomposition as in the proof of Theorem 2.35 yields

$$\hat{\mathfrak{R}}_n(\mathcal{G}_\gamma) \leq \varepsilon_N + \frac{6}{\sqrt{n}} \sum_{j=1}^N \varepsilon_j \sqrt{\log(2N(\varepsilon_j, \mathcal{G}, \|\cdot\|_{L_2(P_n)}))}.$$

Using the same monotonicity argument as before, we obtain

$$\hat{\mathfrak{R}}_n(\mathcal{G}_\gamma) \leq \varepsilon_N + \frac{12}{\sqrt{n}} \int_{\varepsilon_N}^{\gamma} \sqrt{\log(2N(\varepsilon, \mathcal{G}, \|\cdot\|_{L_2(P_n)}))} d\varepsilon.$$

Let $\beta \in (0, \gamma)$. Choose N such that $\varepsilon_N \leq 2\beta < \varepsilon_{N-1}$. Then $\beta < \varepsilon_N \leq 2\beta$, and the bound becomes

$$\hat{\mathfrak{R}}_n(\mathcal{G}_\gamma) \leq 2\beta + \frac{12}{\sqrt{n}} \int_{\beta}^{\gamma} \sqrt{\log(2N(\varepsilon, \mathcal{G}, \|\cdot\|_{L_2(P_n)}))} d\varepsilon.$$

Since $\beta \in (0, \gamma)$ was arbitrary, replacing 2β by 4β and taking the infimum over β yields the displayed form in Lemma 2.36. $\square$

2.4 Covering numbers from pseudo-dimension in the ℓ_∞ metric

We prove Proposition 2.38. The argument is based on discretization and a growth-function bound for threshold sets induced by a pseudo-dimension class.

Proof of Proposition 2.38. Fix $n \geq d$ and $Z_{1:n}$. Let $m := \lceil B/\varepsilon \rceil$. For $g \in \mathcal{G}$, define the discretized evaluation vector $v(g) \in \{0, \dots, m\}^n$ by $v(g)_i := \lfloor g(Z_i)/\varepsilon \rfloor$. If $v(g) = v(h)$, then $|g(Z_i) - h(Z_i)| \leq \varepsilon$ for every i , so choosing one representative g per distinct discretized vector yields an ε -cover of $\mathcal{G}|_{Z_{1:n}}$ in d_∞ . Therefore it suffices to bound the number of possible vectors $v(g)$.

Consider the class of threshold sets on $\mathcal{Z} \times \mathbb{R}$,

$$\mathcal{A} := \{ \{ (z, t) \in \mathcal{Z} \times \mathbb{R} : g(z) \geq t \} : g \in \mathcal{G} \}.$$

By the definition of pseudo-dimension, $\text{vc}(\mathcal{A}) \leq d$. Apply the Sauer–Shelah bound to $\mathcal{A}$ evaluated on the finite set $\{(Z_i, k\varepsilon) : 1 \leq i \leq n, 1 \leq k \leq m\}$, which has cardinality nm . This gives that the number of distinct incidence patterns of $\mathbf{1}_{\{g(Z_i) \geq k\varepsilon\}}$ over all $g \in \mathcal{G}$ is at most $(enm/d)^d$.

Since the discretized vector $v(g)$ is uniquely determined by the collection of comparisons $\{g(Z_i) \geq k\varepsilon\}_{i,k}$, the number of distinct vectors $v(g)$ is bounded by $(enm/d)^d$. Using $m \leq B/\varepsilon + 1 \leq 2B/\varepsilon$ and absorbing the factor 2 into constants yields

$$\mathcal{C}(\varepsilon, \mathcal{G}|_{Z_{1:n}}, d_\infty) \leq \left(\frac{eBn}{\varepsilon d}\right)^d.$$

Taking the supremum over $Z_{1:n}$ yields the stated bound on $\mathcal{C}_\infty(\varepsilon, \mathcal{G}, n)$. $\square$

3 Some learning-theoretic considerations of kernel methods

Although this thesis is almost entirely focused on theoretical guarantees for scientific machine learning algorithms based on hypotheses consisting of artificial neural networks trained with gradient descent, another powerful family of algorithms to solve machine learning tasks are the so-called *kernel methods*, based on the rich and comparatively much more mature theory of *Reproducing Kernel Hilbert Spaces* (RKHS) (Schölkopf and Smola, 2002; Steinwart and Christmann, 2008). In this section, we thus intend to briefly outline how the machinery developed in Chapter 2 plays out for this very important family of hypothesis spaces. In particular, we will discuss how kernel methods can be applied to SciML problems and how learning-theoretic arguments specific to these methods can be used to derive rigorous error guarantees for RKHS-based scientific machine learning.

3.1 RKHS: basic notions

We begin by introducing the formal definition of a positive definite kernel, together with the Moore-Aronszajn theorem, which are at the very basis of RKHS theory:

Definition A.5 (Positive definite kernels). *For a fixed (metric) space $\mathcal{X}$, a function $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ is said to be a (symmetric) positive definite kernel if $k(x, x') = k(x', x)$ and, for every $n \in \mathbb{N}$, every $x_1, \dots, x_n \in \mathcal{X}$, and every $a_1, \dots, a_n \in \mathbb{R}$, it holds $\sum_{i=1}^n \sum_{j=1}^n a_i a_j k(x_i, x_j) \geq 0$.*

Theorem A.6 (Moore-Aronszajn). *Every positive definite kernel k admits a unique Hilbert space $\mathcal{H}_k$ of functions $f : \mathcal{X} \rightarrow \mathbb{R}$, called the reproducing kernel Hilbert space of k , such that $k(x, \cdot) \in \mathcal{H}_k$ for every $x \in \mathcal{X}$ and the reproducing identity $f(x) = \langle f, k(x, \cdot) \rangle_{\mathcal{H}_k}$ holds for every $f \in \mathcal{H}_k$ and $x \in \mathcal{X}$.*

Theorem A.6 is well-known, and we refer the reader to, e.g., (Schölkopf and Smola, 2002) and references within for proofs and further discussion. The reproducing property implies that the point evaluation functional $\delta_x : f \mapsto f(x)$ is continuous on $\mathcal{H}_k$ and yields the bound $|f(x)| \leq \|f\|_{\mathcal{H}_k} \sqrt{k(x, x)}$. In particular, if $k(x, x) \leq \kappa^2$ for all $x \in \mathcal{X}$, then $|f(x)| \leq \kappa \|f\|_{\mathcal{H}_k}$.

It is often convenient to think of the kernel k through a so-called *feature map*, which “maps” the original “features” in $\mathcal{X}$ to a certain kernel-dependent Hilbert space, allowing for potentially more powerful representations. More specifically, it can be shown that there exists a Hilbert space $\mathcal{F}$ and a map $\varphi : \mathcal{X} \rightarrow \mathcal{F}$ such that the identity $k(x, x') = \langle \varphi(x), \varphi(x') \rangle_{\mathcal{F}}$ holds for all $(x, x') \in \mathcal{X} \times \mathcal{X}$. One may then want to choose the kernel k so that the mapped features can satisfactorily represent the phenomenon one wishes to learn. This point of view is at the origin of the celebrated kernel trick (or, equivalently, *Support Vector Machines*), lifting features to a higher dimensional space $\mathcal{F}$ where they are easily learnable (Steinwart and Christmann, 2008).

3.2 Examples of kernels and what they encode

As mentioned earlier, a kernel can be read as a similarity function, but positive definiteness is the property that matters: it guarantees that every Gram matrix $(k(x_i, x_j))_{i,j=1}^n$ is positive semidefinite, hence can be interpreted as an inner-product matrix in some (possibly infinite-dimensional) feature space. The induced RKHS $\mathcal{H}_k$ is then the hypothesis space obtained by taking linear prediction in that feature space and pulling it back to $\mathcal{X}$. We now briefly list some popular choices of kernels which have proven successful in numerous applications. For an extended discussion, we refer the reader to, e.g., (Schölkopf and Smola, 2002; Berlinet and Thomas-Agnan, 2011) and references within.

Linear kernel. On $\mathcal{X} = \mathbb{R}^d$, the linear kernel is defined as $k_{\text{lin}}(x, x') := \langle x, x' \rangle$. Its RKHS consists exactly of linear functionals $f_w(x) = \langle w, x \rangle$, and the RKHS norm coincides with the Euclidean norm of the weight vector in the sense that $\|f_w\|_{\mathcal{H}_{k_{\text{lin}}}} = \|w\|_2$. Thus, constraining $\|f\|_{\mathcal{H}_k}$ or penalizing it recovers the classical ℓ_2 geometry of ridge regression.

Polynomial kernels. Fix $p \in \mathbb{N}$ and $c \geq 0$. The degree- p polynomial kernel is defined as $k_{\text{poly}}(x, x') := (\langle x, x' \rangle + c)^p$. It corresponds to linear prediction in a feature space containing monomials of the input coordinates up to degree p , with weights determined by the choice of c and p . Increasing p expands the hypothesis class by allowing higher-order interactions, while

the kernel trick ensures that optimization and evaluation can still be carried out using only inner products $\langle x, x' \rangle$.

Gaussian (RBF) kernel. Fix a *scale parameter* $\sigma > 0$. The Gaussian kernel, also sometimes called *Radial Basis Function* (RBF) kernel, is defined as

$$k_{\text{rbf}}(x, x') := \exp(-\|x - x'\|_2^2 / (2\sigma^2)).$$

It is translation-invariant and radial, and thus measures similarity through Euclidean distance. The scale parameter σ controls how quickly similarity decays with distance: smaller σ makes the kernel more local and typically allows more rapidly varying fits, whereas larger σ couples observations over longer distances and biases toward smoother functions. A common expressivity statement is universality, meaning that $\mathcal{H}_{k_{\text{rbf}}}$ is dense in $C(\mathcal{X})$ with respect to the supremum norm when $\mathcal{X}$ is compact (Sriperumbudur et al., 2011). Universality is an approximation statement rather than a rate statement: quantitative learning rates depend on how the target aligns with the RKHS norm and on the spectrum of the associated kernel operator (Zhou, 2003).

Matérn kernels and explicit smoothness control. Fix $\nu > 0$ (smoothness), $\ell > 0$ (length scale), and optionally an amplitude $\sigma_k^2 > 0$. The stationary, isotropic Matérn kernel on $\mathbb{R}^d$ is given by

$$k_{\text{mat},\nu,\ell}(x, x') := \sigma_k^2 \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\frac{\sqrt{2\nu} \|x - x'\|_2}{\ell} \right)^\nu K_\nu \left(\frac{\sqrt{2\nu} \|x - x'\|_2}{\ell} \right),$$

where Γ is the Gamma function and K_ν is the modified Bessel function of the second kind (Williams and Rasmussen, 2006).

For half-integer $\nu = m + \frac{1}{2}$, $k_{\text{mat},\nu,\ell}$ simplifies to an exponential times a polynomial in $\|x - x'\|_2$. For example,

$$k_{\text{mat},3/2,\ell}(x, x') = \sigma_k^2 \left(1 + \sqrt{3} \frac{\|x - x'\|_2}{\ell} \right) \exp \left(-\sqrt{3} \frac{\|x - x'\|_2}{\ell} \right).$$

A complementary characterization of the Matérn kernel is through its spectral density, which behaves as

$$S_{\nu,\ell}(\omega) \propto \left(\frac{2\nu}{\ell^2} + \|\omega\|_2^2 \right)^{-(\nu+d/2)}.$$

On bounded domains and under standard regularity assumptions, the RKHS $\mathcal{H}_{k_{\text{mat}}, \nu, \ell}$ is norm-equivalent to a Sobolev space of order $\nu + d/2$, so ν directly controls the degree of smoothness encoded by the hypothesis space (Williams and Rasmussen, 2006; Steinwart and Christmann, 2008).

3.3 Representer theorem and its practical implications

Given inputs $x_1, \dots, x_n$, define the Gram matrix $K \in \mathbb{R}^{n \times n}$ by $K_{ij} = k(x_i, x_j)$. What made kernel methods so attractive and successful, beyond their solid theoretical grounding, is the widely acclaimed *representer theorem*, which shows that although $\mathcal{H}_k$ can be infinite-dimensional, regularized ERM problems depend on f only through the finite vector $(f(x_1), \dots, f(x_n))$ and the RKHS norm. The representer theorem states that any minimizer must lie in the span of $\{k(x_i, \cdot)\}_{i=1}^n$, so the problem reduces to optimizing over n coefficients and the Gram matrix K .

Theorem A.7 (Representer theorem). *Let $\mathcal{H}_k$ be an RKHS on $\mathcal{X}$ with kernel k . Fix inputs $x_1, \dots, x_n \in \mathcal{X}$, let $\Phi: \mathbb{R}^n \rightarrow \mathbb{R} \cup \{\infty\}$ be any function, and let $\Omega: [0, \infty) \rightarrow \mathbb{R}$ be strictly increasing. Consider the objective*

$$J(f) := \Phi(f(x_1), \dots, f(x_n)) + \Omega(\|f\|_{\mathcal{H}_k}).$$

If J attains its minimum over $\mathcal{H}_k$ at some f^ , then f^* admits a representation of the form $f^*(\cdot) = \sum_{i=1}^n c_i k(\cdot, x_i)$ for some coefficients $c_1, \dots, c_n \in \mathbb{R}$.*

Proof. Let $\mathcal{S} := \text{span}\{k(x_1, \cdot), \dots, k(x_n, \cdot)\} \subset \mathcal{H}_k$ and write $f = f_{\mathcal{S}} + f_{\perp}$ with $f_{\mathcal{S}} \in \mathcal{S}$ and $f_{\perp} \in \mathcal{S}^{\perp}$. For each $i \in \{1, \dots, n\}$,

$$f(x_i) = \langle f, k(x_i, \cdot) \rangle_{\mathcal{H}_k} = \langle f_{\mathcal{S}}, k(x_i, \cdot) \rangle_{\mathcal{H}_k} + \langle f_{\perp}, k(x_i, \cdot) \rangle_{\mathcal{H}_k} = f_{\mathcal{S}}(x_i),$$

since $k(x_i, \cdot) \in \mathcal{S}$ and $f_{\perp} \perp \mathcal{S}$. Therefore $\Phi(f(x_1), \dots, f(x_n)) = \Phi(f_{\mathcal{S}}(x_1), \dots, f_{\mathcal{S}}(x_n))$.

Moreover, $\|f\|_{\mathcal{H}_k}^2 = \|f_{\mathcal{S}}\|_{\mathcal{H}_k}^2 + \|f_{\perp}\|_{\mathcal{H}_k}^2$, so $\|f\|_{\mathcal{H}_k} \geq \|f_{\mathcal{S}}\|_{\mathcal{H}_k}$, with strict inequality if $f_{\perp} \neq 0$. Since Ω is strictly increasing, $J(f) \geq J(f_{\mathcal{S}})$, with strict inequality if $f_{\perp} \neq 0$. Hence any minimizer must satisfy $f_{\perp} = 0$, which means $f^* \in \mathcal{S}$ and has the stated expression. $\square$

As a concrete illustration, consider kernel ridge regression (KRR) (Steinwart and Christmann, 2008), which is the prototypical squared-loss, Tikhonov-regularized ERM over an RKHS (Bauer et al., 2007). Despite being a basic regression method, KRR is a perfect starting point to see the practical meaning of the representer theorem: one solves an infinite-dimensional optimization problem over functions by solving an $n \times n$ linear system involving the Gram matrix.

Corollary A.8. *Fix data $(x_i, y_i)_{i=1}^n$ and $\lambda > 0$. Consider the squared-loss, Tikhonov-regularized ERM problem*

$$\hat{f}_\lambda \in \arg \min_{f \in \mathcal{H}_k} \left(\frac{1}{n} \sum_{i=1}^n (f(x_i) - y_i)^2 + \lambda \|f\|_{\mathcal{H}_k}^2 \right).$$

Then $\hat{f}_\lambda(\cdot) = \sum_{i=1}^n c_i k(\cdot, x_i)$ for some $c \in \mathbb{R}^n$. Writing K for the Gram matrix and $y = (y_1, \dots, y_n)$, one common choice of coefficients is

$$c = (K + n\lambda I)^{-1}y,$$

so that $\hat{f}_\lambda(x) = k_x^\top (K + n\lambda I)^{-1}y$ for every $x \in \mathcal{X}$, where $k_x := (k(x, x_1), \dots, k(x, x_n))$.

Corollary A.8 exhibits the main computational implication (and appeal) of Theorem A.7: once K has been formed, training reduces to a linear algebra problem. If the number of observations is small enough, the system may be solved directly. However, it is more often than not the case that n is too large for direct solving to be tractable. To handle this case, many variants have been proposed, e.g. through low-rank approximations of K (Rudi and Rosasco, 2017), or by approximately solving the system using gradient descent (Yao et al., 2007; Chang et al., 2017). Of course, these approximations lead to a trade-off between computational cost and solution accuracy, and there is a rich and relatively mature literature on this issue (Berlinet and Thomas-Agnan, 2011; Bach, 2024).

3.4 Estimation bounds and kernel-dependent learning rates

We finally illustrate more concretely how to apply the learning-theoretic machinery to the specific class of kernel methods. We first begin by bounding the *statistical error*: a standard way to control the estimation error over an RKHS ball is to combine contraction with an RKHS Rademacher bound. We illustrate this in the below lemma, which quantifies the capacity of the hypothesis class $\{f \in \mathcal{H}_k : \|f\|_{\mathcal{H}_k} \leq \Lambda\}$ in a form that corresponds exactly to the Rademacher-based generalization bounds from Chapter 2.

Lemma A.9 (Rademacher complexity of an RKHS ball). *Let k be a reproducing kernel on $\mathcal{X}$ and let $\mathcal{H}_k$ be its RKHS. Fix $\Lambda > 0$ and define $\mathcal{G} := \{x \mapsto f(x) : f \in \mathcal{H}_k, \|f\|_{\mathcal{H}_k} \leq \Lambda\}$. Then for every sample $x_1, \dots, x_n$,*

$$\hat{\mathfrak{R}}_n(\mathcal{G}) \leq \frac{\Lambda}{n} \left(\sum_{i=1}^n k(x_i, x_i) \right)^{1/2}.$$

In particular, if $k(x, x) \leq \kappa^2$ for all $x \in \mathcal{X}$, then $\hat{\mathfrak{R}}_n(\mathcal{G}) \leq \Lambda\kappa/\sqrt{n}$.

Proof. By the reproducing property,

$$\sup_{\|f\|_{\mathcal{H}_k} \leq \Lambda} \left| \frac{1}{n} \sum_{i=1}^n \sigma_i f(x_i) \right| = \Lambda \left\| \frac{1}{n} \sum_{i=1}^n \sigma_i k(x_i, \cdot) \right\|_{\mathcal{H}_k}.$$

Taking $\mathbb{E}_\sigma$ and using Jensen's inequality,

$$\hat{\mathfrak{R}}_n(\mathcal{G}) \leq \Lambda \left(\mathbb{E}_\sigma \left\| \frac{1}{n} \sum_{i=1}^n \sigma_i k(x_i, \cdot) \right\|_{\mathcal{H}_k}^2 \right)^{1/2} = \Lambda \left(\frac{1}{n^2} \sum_{i=1}^n k(x_i, x_i) \right)^{1/2}.$$

□

The bound in Lemma A.9 controls estimation over an RKHS ball. To obtain kernel-dependent learning rates for a concrete estimator, one also needs an approximation statement that quantifies how the target relates to the RKHS geometry. For squared-loss regression, this is often expressed through the kernel integral operator and so-called *source conditions* (Wu et al., 2006; Smale and Zhou, 2007).

Definition A.10 (Kernel integral operator and effective dimension). *Let P_X denote the marginal of P on $\mathcal{X}$. The kernel integral operator $T_k : L_2(P_X) \rightarrow L_2(P_X)$ is defined by*

$$(T_k f)(x) := \int_{\mathcal{X}} k(x, x') f(x') dP_X(x').$$

Writing $(\mu_j)_{j \geq 1}$ for the eigenvalues of T_k , we denote by $\mathcal{N}(\lambda)$ and call effective dimension at scale $\lambda > 0$ the quantity

$$\mathcal{N}(\lambda) := \text{Tr}(T_k(T_k + \lambda I)^{-1}) = \sum_{j \geq 1} \frac{\mu_j}{\mu_j + \lambda}.$$

Definition A.11 (Population and empirical KRR). *Assume $y = f_\rho(x) + \xi$ with $f_\rho(x) := \mathbb{E}[y | x]$, $\mathbb{E}[\xi | x] = 0$, and $\mathbb{E}[\xi^2 | x] \leq \sigma^2$. For $\lambda > 0$, define the population KRR solution*

$$f_\lambda \in \arg \min_{f \in \mathcal{H}_k} \left(\mathbb{E}(f(x) - y)^2 + \lambda \|f\|_{\mathcal{H}_k}^2 \right),$$

and the empirical KRR solution $\hat{f}_\lambda$ by replacing $\mathbb{E}(f(x) - y)^2$ with $\frac{1}{n} \sum_{i=1}^n (f(x_i) - y_i)^2$.

Proposition A.12 (Approximation via a source condition). *Suppose that $f_\rho \in L_2(P_X)$ satisfies the source condition $f_\rho = T_k^r g$ for some $r > 0$ and some $g \in L_2(P_X)$ with $\|g\|_{L_2(P_X)} \leq C$. Then the population regularization bias obeys*

$$\|f_\lambda - f_\rho\|_{L_2(P_X)}^2 \leq C^2 \lambda^{2r}.$$

Proposition A.12 makes the approximation role of λ explicit: smaller λ allows a closer approximation to f_ρ , and the exponent r quantifies how compatible the target is with the kernel. The case $r = 1/2$ corresponds to the canonical assumption that $f_\rho \in \mathcal{H}_k$, with $\|f_\rho\|_{\mathcal{H}_k} = \|g\|_{L_2(P_X)}$ in that representation (Wu et al., 2006; Steinwart and Christmann, 2008).

Proposition A.13 (Estimation in terms of effective dimension). *Assume $k(x, x) \leq \kappa^2$ for all $x \in \mathcal{X}$ and $\mathbb{E}[\xi^2 | x] \leq \sigma^2$. Then the estimation term of KRR can be controlled as*

$$\mathbb{E}\|\hat{f}_\lambda - f_\lambda\|_{L_2(P_X)}^2 \leq \frac{c_1 \sigma^2}{n} \mathcal{N}(\lambda),$$

where c_1 is a universal constant.

Proposition A.13 is a standard result in kernel regression theory. Precise high-probability forms and refinements under different noise assumptions can be found in (Smale and Zhou, 2007; Rudi and Rosasco, 2017; Steinwart and Christmann, 2008). Combining these results, we can finally get error guarantees for KRR under the stated assumptions.

Theorem A.14 (Kernel-dependent learning rates for KRR from approximation plus estimation). *Suppose the assumptions of Propositions A.12 and A.13 hold, and assume an effective-dimension bound $\mathcal{N}(\lambda) \leq Q\lambda^{-\alpha}$ for some $\alpha > 0$, all $\lambda \in (0, 1]$, and some $Q > 0$. Then*

$$\mathbb{E}\|\hat{f}_\lambda - f_\rho\|_{L_2(P_X)}^2 \leq \frac{2c_1 \sigma^2}{n} \mathcal{N}(\lambda) + 2C^2 \lambda^{2r}.$$

In particular, choosing $\lambda \asymp n^{-1/(2r+\alpha)}$ yields the rate $\mathbb{E}\|\hat{f}_\lambda - f_\rho\|_{L_2(P_X)}^2 \lesssim n^{-2r/(2r+\alpha)}$, with an implicit constant depending on C, σ^2, Q , and fixed kernel parameters.

Proof. By the triangle inequality and the inequality $\|u+v\|^2 \leq 2\|u\|^2 + 2\|v\|^2$,

$$\mathbb{E}\|\hat{f}_\lambda - f_\rho\|_{L_2(P_X)}^2 \leq 2\mathbb{E}\|\hat{f}_\lambda - f_\lambda\|_{L_2(P_X)}^2 + 2\|f_\lambda - f_\rho\|_{L_2(P_X)}^2.$$

Apply Proposition A.13 to the first term and Proposition A.12 to the second. Under $\mathcal{N}(\lambda) \leq Q\lambda^{-\alpha}$, optimize the upper bound $\lambda \mapsto n^{-1}\lambda^{-\alpha} + \lambda^{2r}$ by

balancing the two contributions, which gives $\lambda \asymp n^{-1/(2r+\alpha)}$ and the claimed rate. $\square$

We briefly comment the quantitative decay assumption on $\mathcal{N}(\lambda)$ and how it relates to the choice of kernel: the exponent α is controlled by the eigenvalue decay of T_k , hence by the kernel and the input distribution. The parameter r is an approximation condition on the target relative to the same operator. For many Sobolev-type kernels (including Matérn kernels on bounded domains under standard conditions), eigenvalues decay polynomially, leading to $\mathcal{N}(\lambda) \leq Q\lambda^{-\alpha}$ for an α that depends on the dimension and the kernel smoothness, see (Steinwart and Christmann, 2008). For very smooth kernels such as the Gaussian kernel, the eigenvalues can decay super-polynomially, and $\mathcal{N}(\lambda)$ can be bounded by polylogarithmic functions of $1/\lambda$ under additional assumptions, leading to correspondingly faster rates up to logarithmic factors (Smale and Zhou, 2007).

3.5 Applications to SciML problems

To conclude, we now discuss how the learning-theoretic analysis of kernels as discussed so far can be applied to SciML problems. As we’ve seen in the KRR analysis above, the rate comes from an estimation-approximation decomposition of the error: Proposition A.12 controls the approximation (regularization bias) under a source condition, while Proposition A.13 controls the estimation term via the effective dimension. Theorem A.14 is then obtained by balancing these two contributions through the choice of λ . Physics-informed losses are amenable the same style of analysis, except that the loss will be built from more general linear measurements of u —for example $(\mathcal{L}u)(x)$, $(\mathcal{B}u)(z)$ —rather than only point evaluations $u(x)$.

To illustrate how the picture change for SciML problems, let $\mathcal{H}_k$ be an RKHS on $\mathcal{X}$, and let $\ell_1, \dots, \ell_m$ be a set of bounded linear functionals on $\mathcal{H}_k$. Consider objectives of the form

$$J(u) := \Phi(\ell_1(u), \dots, \ell_m(u)) + \Omega(\|u\|_{\mathcal{H}_k}),$$

where $\Phi : \mathbb{R}^m \rightarrow \mathbb{R} \cup \{\infty\}$ and $\Omega : [0, \infty) \rightarrow \mathbb{R}$ is strictly increasing. This includes KRR as the special case $\ell_r(u) = u(x_r)$, and it also covers collocation-type PDE solvers where the ℓ_r encode interior residual and boundary errors. In this setting, the same orthogonal-decomposition argument as in the proof of Theorem A.7 applies, yielding a generalized representer theorem with $k(x_i, \cdot)$ replaced by the Riesz representers $r_{\ell_i} \in \mathcal{H}_k$ defined by $\ell_i(u) = \langle u, r_{\ell_i} \rangle_{\mathcal{H}_k}$.

More concretely, let $\Omega \subset \mathbb{R}^d$ and consider a linear boundary value problem $\mathcal{L}u = f$ in Ω with $\mathcal{B}u = g$ on $\partial\Omega$. A natural approach to solve this PDE over a RKHS hypothesis would be to directly adopt a *physics-informed kernel learning* approach (Doumèche et al., 2024): selecting interior points $x_1, \dots, x_{n_\Omega} \in \Omega$ and boundary points $z_1, \dots, z_{n_{\partial\Omega}} \in \partial\Omega$, and solving the Tikhonov-regularized residual fit

$$\hat{u}_\lambda \in \arg \min_{u \in \mathcal{H}_k} \left(\frac{1}{n_\Omega} \sum_{i=1}^{n_\Omega} ((\mathcal{L}u)(x_i) - f(x_i))^2 + \frac{\beta}{n_{\partial\Omega}} \sum_{j=1}^{n_{\partial\Omega}} ((\mathcal{B}u)(z_j) - g(z_j))^2 + \lambda \|u\|_{\mathcal{H}_k}^2 \right),$$

for $\beta > 0$ and $\lambda > 0$. When the maps $u \mapsto (\mathcal{L}u)(x)$ and $u \mapsto (\mathcal{B}u)(z)$ are bounded linear functionals on $\mathcal{H}_k$, the representer theorem implies that $\hat{u}_\lambda$ is expressible in terms of kernels composed with differential operators, of the form

$$\hat{u}_\lambda(\cdot) = \sum_{i=1}^{n_\Omega} a_i (\mathcal{L}_1 k)(x_i, \cdot) + \sum_{j=1}^{n_{\partial\Omega}} b_j (\mathcal{B}_1 k)(z_j, \cdot),$$

and the corresponding normal equations involve quantities such as $(\mathcal{L}_1 \mathcal{L}_2 k)(x_i, x_{i'})$, $(\mathcal{B}_1 \mathcal{B}_2 k)(z_j, z_{j'})$, and $(\mathcal{L}_1 \mathcal{B}_2 k)(x_i, z_j)$, akin to the Gram-matrix structure in Corollary A.8.

From the learning-theoretic viewpoint, nothing essentially changes relative to Section 3.4: one is still fitting a linear model in an RKHS, except that the data-fit now acts on u through operators $u \mapsto (\mathcal{L}u)(x_i)$ and $u \mapsto (\mathcal{B}u)(z_j)$. In particular, the same type of approximation-estimation decomposition as in Theorem A.14 can be carried out once one specifies an approximation condition for the solution relative to $\mathcal{H}_k$ and controls the effective dimension of the relevant operator-applied kernel operator. For a rigorous mathematical treatment of this “physics-informed kernel” approach, we refer the interested reader to (Doumèche et al., 2024). For a further generalization of kernel methods to the learning of *operators* between infinite-dimensional function spaces where such a framework is applied, we refer the reader to (Batlle et al., 2024, 2025).

Of course, the error analysis of the optimal RKHS solution does not necessarily tell the full story, and as we mentioned earlier, the cost of inverting the linear system given by the representer theorem is prohibitively large. But the theory of RKHS can be used to design efficient approximate solutions or optimizers to mitigate these issues (Schwencke and Furtlehner, 2025; Xu et al., 2025). The development of tractable kernel-based algorithms to efficiently solve SciML problems remains an active and rapidly growing area of research.

Appendix B

Approximation error bounds for deep ReQU networks

We now give a proof of Theorem 3.16 for the approximation abilities of deep ReQU networks based on the “standard machinery” first introduced in (Yarotsky, 2017): we partition the domain into small cubes, on which τ can be approximated by Taylor polynomials of appropriate order, and we use a partition of unity to extend each local estimate to a global one. While our approach is similar to earlier related works (Gühring and Raslan, 2021; De Ryck et al., 2021), the main difference is that we make use of the representation power of ReQU networks (He et al., 2023; Bo et al., 2020) to construct exact partitions of unity, multiplication operator and local Taylor polynomials, leading to a fully explicit construction of the Neural Networks architecture and weights, with explicit bounds on their magnitude in terms of the problem’s parameters. We also note that the advantage of Theorem 3.16, when compared with similar results, is that *the depth does not need to increase with the approximation accuracy*, and can remain bounded logarithmically in dimension and smoothness.

1 Realization of the multiplication operator

Proposition B.1. *For any $m \in \mathbb{N}^+$, there exists a ReQU FCNN $\hat{\Pi}_m$ of depth at most $2 \log_2 2m$, width at most $8m$, and with at most $48m$ parameters all bounded by 1 such that for any $x_1, \dots, x_m \in \mathbb{R}^m$, we have*

$$\hat{\Pi}_m(x_1, \dots, x_m) = \prod_{i=1}^m x_i$$

Proof. Denote by $\sigma : x \mapsto \max(0, x)^2$ the ReQU activation, and observe that for any $x, y \in \mathbb{R}$, we have

$$xy = c_1 \cdot \sigma(A_1[x \ y]^T),$$

where $A_1 = \begin{bmatrix} 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 \end{bmatrix}^T$ and $c_1 = \frac{1}{4} [1 \ 1 \ 1 \ 1]^T$. And likewise, for any $x \in \mathbb{R}$, we have

$$x = c_2 \cdot \sigma(A_2x + b_2),$$

where $A_2 = [1 \ -1 \ 1 \ -1]^T$, $b_2 = [1 \ -1 \ -1 \ 1]^T$ and $c_2 = \frac{1}{4} [1 \ 1 \ 1 \ 1]^T$.

This shows that both the identity $x \mapsto x$ and the multiplication operator $(x, y) \mapsto xy$ can be realized by shallow ReQU networks of width 4 with a total of 12 parameters all bounded by 1. Accordingly, we will respectively denote by Id and $\hat{\times}$ the ReQU neural networks as described above which implements these functions. To realize the multiplication of m numbers, we now iterate the above construction in a binary tree-like fashion, as described e.g. in (Opschoor et al., 2019):

We will define a finite sequence of mappings $R^1, R^2, \dots, R^L$ as follows: for any even $k \in \mathbb{N}^+$, let

$$R^1(y_1, \dots, y_k) := (y_1 \times y_2, \dots, y_{k-1} \times y_k) \in \mathbb{R}^{k/2},$$

and for any $\ell \geq 1$, let $R^{\ell+1} = R \circ R^\ell$, which acts on input vectors half the size of those on which R^ℓ acts (note that the definition of R depends on the input vector size). The last index L is the one for which the input size is equal to 2, such that R^L can be seen as a map from $\mathbb{R}^{2^L}$ to $\mathbb{R}$.

Given m numbers $x_1, \dots, x_m$ we want to multiply, we thus define $\tilde{m} := \inf\{2^k : k \in \mathbb{N}^+, 2^k \geq m\}$, and if $\tilde{m} > m$, we let $x_{m+1} = \dots = x_{\tilde{m}} = 1$ (which can be implemented by using an appropriately sized bias vector in the input). The pairwise multiplications can be exactly realized by $\hat{\times}$, while the successive compositions $R \circ R^\ell \equiv f \circ g$, where f and g are FCNNs with matching input and output size can be realized by adding a layer $\hat{\text{Id}}_{\mathbb{R}^k}$ between f and g , where k is the input (resp. output) dimension of f (resp. g), and $\hat{\text{Id}}_{\mathbb{R}^k}$ the width-wise concatenation of k $\hat{\text{Id}}$ networks.

The ReQU FCNN implementation of the m -multiplication operator is then given by

$$\hat{\Pi}_m(x_1, \dots, x_m) := \hat{R}^{\log_2 \tilde{m}} \circ \hat{\text{Id}}_{\mathbb{R}^2} \circ \dots \circ \hat{\text{Id}}_{\mathbb{R}^{\tilde{m}/2}} \circ \hat{R}^1(x_1, \dots, x_{\tilde{m}}).$$

Where $\hat{R}$ is defined analogously as R , with the pairwise multiplication implemented by $\hat{\times}$. From the way it is defined, the identity $\hat{\Pi}_m(x_1, \dots, x_m) = \prod_{i=1}^m x_i$ is immediate. By construction, we see that we respectively have $\log_2 \tilde{m}$ multiplication and identity layers, each of which is implemented by the concatenation of $\tilde{m}/2^\ell$ respectively $\hat{\times}$ and $\hat{\text{Id}}$ networks. Since $\tilde{m} \leq 2m$, this implies that $\hat{\Pi}_m$ is a ReQU FCNN with parameters all bounded by 1, of depth at most $2\log_2 2m$, of width at most $8m$, and with a number of parameters bounded by

$$\sum_{\ell=1}^{\log_2 \tilde{m}} \tilde{m}/2^\ell (12 + 12) \leq 48m.$$

This is exactly the desired claim, and the proof is thus complete $\square$

Remark B.2. Note that (Petersen and Voigtlaender, 2018) have shown that sparse realizations of the identity mapping are possible for ReLU-activated FCNNs, which can lead to a decrease of the network size. This suggests that a sparsification of the identity mapping for ReQU network is also likely possible, and could lead to better bounds on the number of parameters than those given by Proposition B.1.

2 Realization of a partition of unity

The next main ingredient of our approximation result is the realization of an exact partition of unity by ReQU FCNNs. To avoid the technicalities induced by the potentially complex geometry of Ω , we recall that $\Omega \subseteq [0, 1]^d$, and we thus build a partition of unity for the unit hypercube instead. The validity of this approach is justified by Lemma 3.10, which guarantees that if the target function, defined on Ω is smooth enough, then it can be extended to the hypercube $[0, 1]^d$.

Recall that $\sigma : x \mapsto \max(0, x)^2$ denotes the ReQU activation function, and define the “sigmoid-like” function

$$\psi_0 : x \mapsto 2\sigma(x) - 4\sigma(x - 1/2) + 2\sigma(x - 1),$$

with the following piecewise closed-form expression:

$$\begin{cases} \psi_0(x) = 0, x \leq 0 \\ \psi_0(x) = 2x^2, 0 \leq x \leq 1/2 \\ \psi_0(x) = 2x^2 - 4(x - 1/2)^2, 1/2 \leq x \leq 1 \\ \psi_0(x) = 1, x \geq 1. \end{cases}$$

From ψ_0 we can construct the following “bump function”

$$\psi : x \mapsto \psi_0(x+1) - \psi_0(x)$$

with the following properties:

$$\begin{cases} \psi \in W^{2,\infty}(\mathbb{R}), \\ \text{supp}(\psi) = (-1, 1), \\ \|\psi\|_{L^\infty(\mathbb{R})} = 1, \\ \|\psi'\|_{L^\infty(\mathbb{R})} = 2, \\ \|\psi''\|_{L^\infty(\mathbb{R})} = 4. \end{cases} \quad (\text{B.1})$$

Where ψ' and ψ'' respectively denote the first and second weak derivatives of ψ . Furthermore, we note that ψ satisfies the following identity for all $x \in \mathbb{R}$:

$$\psi(x) = c \cdot \sigma(Ax + b)$$

where $A := [1 \ 1 \ 1 \ 1 \ 1]$, $b := [1 \ 1/2 \ 0 \ -1/2 \ -1]$, and $c := [2 \ -4 \ -4 \ 4 \ 2]^T$. We thus see that ϕ can be realized by a shallow ReQU network of width 5 and weights bounded by 4. Define for $N \in \mathbb{N}^+$ to be chosen later and $\mathbf{j} = (j_1, \dots, j_d) \in \{0, 1, \dots, N\}^d$ the functions $\phi_{\mathbf{j}} : \mathbb{R}^d \rightarrow \mathbb{R}$, given by the following expression:

$$\phi_{\mathbf{j}} : x \mapsto \prod_{k=1}^d \psi \left(N \left(x_k - \frac{j_k}{N} \right) \right) \quad (\text{B.2})$$

Proposition B.3. *The family of functions $\{\phi_{\mathbf{j}} : \mathbf{j} \in \{0, 1, \dots, N\}^d\}$ forms an exact partition of unity of the unit hypercube $[0, 1]^d$, in the sense that it satisfies the following properties:*

(i) for all $\mathbf{j} \in \{0, 1, \dots, N\}^d$,

$$\text{supp}(\phi_{\mathbf{j}}) \subseteq \left\{ x = (x_1, \dots, x_d) \in \mathbb{R}^d : \left| x_k - \frac{j_k}{N} \right| < \frac{1}{N} \ \forall \ 1 \leq k \leq d \right\}$$

(ii) for all $x \in [0, 1]^d$,

$$\sum_{\mathbf{j} \in \{0, 1, \dots, N\}^d} \phi_{\mathbf{j}}(x) = 1$$

Proof. Item (i) follows immediately from the definition of ψ , we thus only need to prove that the ϕ_j sum to 1 on the unit hypercube. We proceed by induction over the dimension d :

For the base case $d = 1$, we show by induction on $N \in \mathbb{N}^+$ that the $\{\phi_j : j \in \{0, 1, \dots, N\}\}$ form a partition of unity of the unit interval. The case $N = 1$ is a direct computation, while for any $N \geq 2$, fix $x \in [0, 1]$ and observe that because each ϕ_j is only supported on $[(j-1)/N, (j+1)/N]$, the only indices j for which $\phi_j(x) \neq 0$ will only be those such that $Nx - 1 < j < Nx + 1$. In particular, there can only be two such indices, which we denote w.l.o.g. j_x and $j_x + 1$. But now we can compute

$$\begin{aligned} \sum_{j=0}^N \phi_j(x) &= \phi_{j_x}(x) + \phi_{j_x+1}(x) \\ &= \psi\left(N\left(x - \frac{j_x}{N}\right)\right) + \psi\left(N\left(x - \frac{j_x}{N}\right) - 1\right) \\ &= 1 \end{aligned}$$

where the last equality follows from the case $N = 1$ applied to $y \equiv N\left(x - \frac{j_x}{N}\right) \in [0, 1]$. This proves that $\{\phi_j : j \in \{0, 1, \dots, N\}\}$ is an exact partition of unity, as desired.

Now, for $d \geq 2$, assume that both $\{\phi_j : j \in \{0, 1, \dots, N\}\}$ and $\{\phi_{\mathbf{j}} : \mathbf{j} \in \{0, 1, \dots, N\}^{d-1}\}$ respectively form partitions of unity for the unit (hyper)cubes of appropriate dimensions. By distributing the product, we get for all $x = (x_1, \dots, x_d) \in [0, 1]^d$:

$$\begin{aligned} 1 &= \sum_{\ell=0}^N \psi\left(N\left(x_d - \frac{\ell}{N}\right)\right) \times \sum_{\mathbf{j} \in \{0, 1, \dots, N\}^{d-1}} \prod_{k=1}^{d-1} \psi\left(N\left(x_k - \frac{j_k}{N}\right)\right) \\ &= \sum_{\mathbf{j} \in \{0, 1, \dots, N\}^d} \prod_{k=1}^d \psi\left(N\left(x_k - \frac{j_k}{N}\right)\right), \end{aligned}$$

as desired. Hence the proof is complete. $\square$

Observe that ψ is given by

$$\psi : x \mapsto 2\sigma(x+1) - 2\sigma(x-1) + 4\sigma(x-1/2) - 4\sigma(x+1/2) = c \cdot \sigma(Ax + b)$$

where $A = \begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix}^T$, $b = \begin{bmatrix} 1 & -1 & -1/2 & 1/2 \end{bmatrix}^T$ and $c = \begin{bmatrix} 2 & -2 & 4 & -4 \end{bmatrix}^T$. Since the multiplication operator can also be represented by a ReQU FCNN

thanks to Proposition B.1, this shows that our partition of unity can be exactly realized by appropriately scaled ReQU networks, whose size and parameter’s magnitude can be explicitly bounded.

Lastly, we note the following fact regarding the “disjointness” of the respective supports of each $\phi_{\mathbf{j}}$:

Lemma B.4. *For any $x \in [0, 1]^d$, the set $\{\mathbf{j} : \phi_{\mathbf{j}}(x) \neq 0\} \subseteq \{0, 1, \dots, N\}^d$ has at most 2^d elements.*

Proof. Let us first fix $1 \leq k \leq d$, and consider x_k , the k th component of x . The k th factor in the expression of $\phi_{\mathbf{j}}$ does not vanish if and only if j_k/N lies in the interval $(x_k - 1/N, x_k + 1/N)$, which contains at most two grid points. Since this condition needs to be satisfied for all the d components of $\mathbf{j}$, the conclusion follows directly. $\square$

3 Realization of localized Taylor polynomials

The last element we need to approximate our target function is to build local Taylor polynomial estimators. We will make use of a property specific to ReLU^k -activated Neural Networks, which is their ability to exactly represent any polynomial. This representation property was first shown in (Li et al., 2020; Bo et al., 2020) through a constructive argument, and later in (Chen et al., 2022) by exploiting the connection between ReLU^k DNNs and B-splines. These representation results were used to derive approximation rates in higher-smoothness spaces by applying results from the classical approximation theory literature. More recently, (He et al., 2023) have come up with another constructive proof of this representation result, which allows to control not only the number of parameters, but also their magnitude. We will thus be relying the following theorem:

Theorem B.5 (Adapted from Theorem 1 in (He et al., 2023)). *Let $L \in \mathbb{N}^+$, $B > 0$ and $m_1 = \dots = m_L = 2(2^L + 1)^d$. Then any real polynomial of degree at most 2^L*

$$P : x \mapsto \sum_{\alpha \in \mathbb{N}^d : \|\alpha\|_1 \leq 2^L} a_{\alpha} x^{\alpha}$$

can be exactly realized by a ReQU architecture $\Sigma_{\mathbf{m}_1:L, M}^{\sigma}(B)$ with

$$M = B^{-2(2^L-1)} (2^{L-1} + 1)^{2(d+1)+2^L} \left(\sum_{\alpha \in \mathbb{N}^d : \|\alpha\|_1 \leq 2^L} |a_{\alpha}| \right).$$

Furthermore, the architecture implementing this polynomial can be chosen to have at most $2(2L + d)(2^L + 1)^d$ non-zero parameters.

Now that we have a way to represent polynomials of any degree, we just need to find a family of polynomials that can approximate our target C^r -smooth function in $W^{r,\infty}$ norm on each sub-hypercube which partitions the domain. A natural such candidate is the family of local Taylor polynomials of order $r - 1$. In the following, for $N \in \mathbb{N}^+$ and $\mathbf{j} \in \{0, 1, \dots, N\}^d$, we write

$$\Omega_{\mathbf{j}} := \frac{1}{N}[j_1 - 1, j_1] \times \dots \times \frac{1}{N}[j_d - 1, j_d] \quad (\text{B.3})$$

for notational convenience. We have the following estimate on the Sobolev distance between a smooth function and its Taylor polynomial approximation on $\Omega_{\mathbf{j}}$:

Proposition B.6. *Let $N \in \mathbb{N}^+$, $f \in C^r(\Omega_{\mathbf{j}})$, $\mathbf{j} \in \{0, 1, \dots, N\}^d$ and $\hat{f}$ be its $r - 1$ st order Taylor approximation at $x_{0,\mathbf{j}} := \frac{1}{N}(j_1 - 1/2, \dots, j_d - 1/2)$:*

$$\hat{f} : x \mapsto \sum_{\alpha \in \mathbb{N}^d, \|\alpha\|_1 \leq r-1} \frac{\partial^\alpha f(x_{0,\mathbf{j}})}{\alpha!} (x - x_{0,\mathbf{j}})^\alpha.$$

Then for all integers $0 \leq k \leq r - 1$, we have the estimate

$$\|f - \hat{f}\|_{W^{k,\infty}(\Omega_{\mathbf{j}})} \leq \frac{(d/2)^{r-k}}{N^{r-k}(r-k)!} \|f\|_{W^{r,\infty}(\Omega_{\mathbf{j}})}.$$

Proof. The result follows from a slight modification of the proof of (De Ryck et al., 2021, Lemma A.9), but since the argument is short, we give it here for completeness: let $\gamma \in \mathbb{N}^d$ with $\|\gamma\|_1 \leq r - 1$ and recall that by Taylor's theorem with exact remainder applied to $\partial^\gamma f \in C^{r-\|\gamma\|_1}$, we have that for all $x \in \Omega_{\mathbf{j}}$, there exists a $c \in [0, 1]$ such that

$$\begin{aligned} \partial^\gamma f(x) &= \sum_{\alpha \in \mathbb{N}^d, \|\alpha\|_1 \leq r-1-\|\gamma\|_1} \frac{\partial^\alpha \partial^\gamma f(x_{0,\mathbf{j}})}{\alpha!} (x - x_{0,\mathbf{j}})^\alpha \\ &\quad + \sum_{\beta \in \mathbb{N}^d, \|\beta\|_1 = r-\|\gamma\|_1} \frac{\partial^\beta \partial^\gamma f(c(x - x_{0,\mathbf{j}}) + x_{0,\mathbf{j}})}{\beta!} (x - x_{0,\mathbf{j}})^\beta \end{aligned} \quad (\text{B.4})$$

and likewise, we can directly compute

$$\begin{aligned} \partial^\gamma \hat{f}(x) &= \sum_{\|\alpha\|_1 \leq r-1, \alpha \geq \beta} \frac{\partial^\alpha f(x_{0,\mathbf{j}})}{\alpha!} \frac{\alpha!}{(\alpha - \gamma)!} (x - x_{0,\mathbf{j}})^{\alpha - \gamma} \\ &= \sum_{\alpha \in \mathbb{N}^d, \|\alpha\|_1 \leq r-1-\|\gamma\|_1} \frac{\partial^\alpha \partial^\gamma f(x_{0,\mathbf{j}})}{\alpha!} (x - x_{0,\mathbf{j}})^\alpha. \end{aligned} \quad (\text{B.5})$$

We thus get the estimate

$$\begin{aligned} \|\partial^\gamma f - \partial^\gamma \hat{f}\|_{L^\infty(\Omega_{\mathbf{j}})} &\leq \|f\|_{W^{r,\infty}(\Omega_{\mathbf{j}})} \sum_{\beta \in \mathbb{N}^d: \|\beta\|_1 = r - \|\gamma\|_1} \frac{1}{(2N)^{\|\beta\|_1} \beta!} \\ &= \|f\|_{W^{r,\infty}(\Omega_{\mathbf{j}})} \frac{(d/2)^{r - \|\gamma\|_1}}{N^{r - \|\gamma\|_1} (r - \|\gamma\|_1)!} \end{aligned} \quad (\text{B.6})$$

where we used the fact that $\|x - x_{0,\mathbf{j}}\|_\infty \leq 1/2N$ whenever $x \in \Omega_{\mathbf{j}}$ for the first inequality, and the last equality follows from the multinomial theorem applied in the form

$$\sum_{\beta \in \mathbb{N}^d: \|\beta\|_1 = r - \|\gamma\|_1} \frac{(r - \|\gamma\|_1)!}{\beta!} = \sum_{\beta \in \mathbb{N}^d: \|\beta\|_1 = r - \|\gamma\|_1} \binom{r - \|\gamma\|_1}{\beta} = d^{r - \|\gamma\|_1}.$$

□

Remark B.7. In a setting like ours where the target function is C^k smooth, Taylor polynomials can be used for approximation in $W^{k,\infty}$ -norm. However, from a purely approximation-theoretic standpoint, C^k regularity is not required: indeed, if the target function merely lies in $W^{k,\infty}$, in which case its derivatives may not be defined pointwise, we can get an equivalent result by using averaged Taylor polynomials instead, which is a form of the Bramble-Hilbert lemma (Brenner and Scott, 2007; De Ryck et al., 2021).

4 Final approximation error bound

Proof of Theorem 3.13. Let $N \geq 2$ be an integer, and set our neural network approximation of $u \in C^r(\mathbb{R}^d)$ as

$$\hat{f}_{NN} : x \mapsto \sum_{\mathbf{j} \in \{0,1,\dots,N\}^d} \hat{\Pi}_2(\phi_{\mathbf{j}}(x), p_{\mathbf{j}}(x)) = \sum_{\mathbf{j} \in \{0,1,\dots,N\}^d} \phi_{\mathbf{j}}(x) \times p_{\mathbf{j}}(x) \quad (\text{B.7})$$

where $\hat{\Pi}_2$ is the multiplication operator, ϕ_j is defined in (B.2), and p_j is the $r - 1$ st order Taylor approximation of u centered around the center of the hypercube $\Omega_{\mathbf{j}}$ defined in (B.3).

Recall the following estimates for the size and number of parameters of $\hat{\Pi}_2$, $\phi_{\mathbf{j}}$ and $p_{\mathbf{j}}$ respectively, we have that:

- $\hat{\Pi}_2$ is realized by a shallow ReQU network of width 4, with 12 non-zero parameters all bounded by 1,

- for all $\mathbf{j} \in \{0, 1, \dots, N\}^d$, $\phi_{\mathbf{j}}$ is realized by a deep ReQU network of depth $2\log_2(2d) + 1$, width $5d$, and at most $74d$ non-zero parameters all bounded by N ,
- for all $\mathbf{j} \in \{0, 1, \dots, N\}^d$, $p_{\mathbf{j}}$ is realized by a deep ReQU network of depth at most $\lceil \log_2(r-1) \rceil$, width $2r^d$, and at most $2r^d(d + 2\lceil \log_2(r-1) \rceil)$ non-zero parameters all bounded according to Theorem B.5, by

$$M = \sup_{x_0 \in [0,1]^d} B^{-2r} \left(\frac{r+1}{2} \right)^{2d+r+1} \left(\sum_{\alpha \in \mathbb{N}^d: \|\alpha\|_1 \leq r-1} \left| \frac{\partial^\alpha f(x_0)}{\alpha!} \right| \right). \quad (\text{B.8})$$

Lastly, note that for any collection of non-negative real numbers $y_1, \dots, y_m \geq 0$, we have the identity

$$\sum_{i=1}^m y_i = \frac{1}{2} \left(1 + \sum_{i=1}^m y_i \right)^2 - \frac{1}{2} \left(\sum_{i=1}^m y_i \right)^2 - \frac{1}{2} = c \cdot \sigma(A\vec{y}),$$

where $\vec{y} = [y_1 \ y_2 \ \dots \ y_m \ 1]^T \in \mathbb{R}^{m+1}$,

$$A = \begin{bmatrix} 1 & \dots & 1 & 1 \\ 1 & \dots & 1 & 0 \\ 0 & \dots & 0 & 1/\sqrt{2} \end{bmatrix} \in \mathbb{R}^{3 \times (m+1)},$$

and $c = [1/2 \ -1/2 \ -1/2]^T \in \mathbb{R}^3$. Hence the summation of each of the N^d subnetworks outputs can be realized by a shallow ReQU network of width $N^d + 1$ with $2N^d + 5$ non-zero parameters all bounded by 1. It thus follows by construction that $\hat{f}_{NN}$ as defined in (B.7) can be realized by a deep ReQU Network of depth $3 + 2\log_2(2d) + \lceil \log_2(r-1) \rceil$, width $\max\{2(rN)^d, dN^d\}$, and at most $(2r^d(d + 2\log_2(2d) + \lceil \log_2(r-1) \rceil) + 74d + 12)N^d$ non-zero parameters all bounded by $\max\{N, M\}$, where M is defined in (B.8).

Using the triangle inequality, we can estimate the $W^{2,\infty}(\Omega)$ approximation error as follows:

$$\begin{aligned} \left\| \hat{f}_{NN} - u \right\|_{W^{2,\infty}([0,1]^d)} &= \left\| \sum_{\mathbf{j} \in \{0,1,\dots,N\}^d} \phi_{\mathbf{j}} \times (p_{\mathbf{j}} - u) \right\|_{W^{2,\infty}([0,1]^d)} \\ &\leq \sum_{\mathbf{j} \in \{0,1,\dots,N\}^d} \left\| \phi_{\mathbf{j}} \times (p_{\mathbf{j}} - u) \right\|_{W^{2,\infty}(\Omega_{\mathbf{j}})} \\ &= \sum_{\mathbf{j} \in \{0,1,\dots,N\}^d} \max_{\alpha \in \mathbb{N}^d: \|\alpha\|_1 \leq 2} \left\| \partial^\alpha \left(\phi_{\mathbf{j}} \times (p_{\mathbf{j}} - u) \right) \right\|_{L^\infty(\Omega_{\mathbf{j}})}. \end{aligned}$$

We now apply the Sobolev-Leibniz rule to bound each summand: for all $\mathbf{j} \in \{0, 1, \dots, N\}^d$ and $\|\alpha\|_1 \leq 2$, we have

$$\begin{aligned}
\left\| \partial^\alpha \left(\phi_{\mathbf{j}} \times (p_{\mathbf{j}} - u) \right) \right\|_{L^\infty(\Omega_{\mathbf{j}})} &= \left\| \sum_{\beta: \beta \leq \alpha} \binom{\alpha}{\beta} \partial^\beta \phi_{\mathbf{j}} \times \partial^{\alpha-\beta} (p_{\mathbf{j}} - u) \right\|_{L^\infty(\Omega_{\mathbf{j}})} \\
&\leq \sum_{\beta: \beta \leq \alpha} \left\| \binom{\alpha}{\beta} \partial^\beta \phi_{\mathbf{j}} \times \partial^{\alpha-\beta} (p_{\mathbf{j}} - u) \right\|_{L^\infty(\Omega_{\mathbf{j}})} \\
&\leq 2 \sum_{\beta: \beta \leq \alpha} \left\| \partial^\beta \phi_{\mathbf{j}} \right\|_{L^\infty(\Omega_{\mathbf{j}})} \left\| \partial^{\alpha-\beta} (p_{\mathbf{j}} - u) \right\|_{L^\infty(\Omega_{\mathbf{j}})} \\
&\leq 2 \sum_{\beta: \beta \leq \alpha} \left\| \phi_{\mathbf{j}} \right\|_{W^{\|\beta\|_1, \infty}(\Omega_{\mathbf{j}})} \left\| p_{\mathbf{j}} - u \right\|_{W^{\|\alpha-\beta\|_1, \infty}(\Omega_{\mathbf{j}})} \\
&\leq 2 \sum_{k=0}^2 \sum_{\substack{\beta: \beta \leq \alpha \\ \|\beta\|_1 = k}} \left\| \phi_{\mathbf{j}} \right\|_{W^{k, \infty}(\Omega_{\mathbf{j}})} \left\| p_{\mathbf{j}} - u \right\|_{W^{2-k, \infty}(\Omega_{\mathbf{j}})},
\end{aligned}$$

where we used the fact that $\binom{\alpha}{\beta} \leq 2$ whenever $\|\alpha\|_1 \leq 2$. Since $|\{\beta \leq \alpha\}| \leq |\{\|\beta\|_1 \leq \|\alpha\|_1\}| \leq \binom{2+d}{2}$, we deduce that for all $\mathbf{j} \in \{0, 1, \dots, N\}^d$,

$$\max_{\alpha \in \mathbb{N}^d: \|\alpha\|_1 \leq 2} \left\| \partial^\alpha \left(\phi_{\mathbf{j}} \times (p_{\mathbf{j}} - u) \right) \right\|_{L^\infty(\Omega_{\mathbf{j}})} \leq 2 \binom{2+d}{2} \sum_{k=0}^2 \left\| \phi_{\mathbf{j}} \right\|_{W^{k, \infty}(\Omega_{\mathbf{j}})} \left\| p_{\mathbf{j}} - u \right\|_{W^{2-k, \infty}(\Omega_{\mathbf{j}})}. \quad (\text{B.9})$$

Fix $\mathbf{j} \in \{0, 1, \dots, N\}^d$ and recall the expression of $\phi_{\mathbf{j}}$ given in (B.2). For $\gamma = (1, 0, \dots, 0)^T \in \mathbb{N}^d$ and all $x \in [0, 1]^d$, we can compute

$$\partial^\gamma \phi_{\mathbf{j}}(x) = N \psi' \left(N \left(x_1 - \frac{j_1}{N} \right) \right) \prod_{k=2}^d \psi \left(N \left(x_k - \frac{j_k}{N} \right) \right).$$

And likewise, for $\gamma = (2, 0, \dots, 0)^T \in \mathbb{N}^d$ and all $x \in [0, 1]^d$, we have

$$\partial^\gamma \phi_{\mathbf{j}}(x) = N^2 \psi'' \left(N \left(x_1 - \frac{j_1}{N} \right) \right) \prod_{k=2}^d \psi \left(N \left(x_k - \frac{j_k}{N} \right) \right).$$

In light of (B.1), we can thus deduce by analogous computations that

$$\left\| \phi_{\mathbf{j}} \right\|_{W^{k, \infty}(\Omega_{\mathbf{j}})} = \begin{cases} 1, & k = 0 \\ 2N, & k = 1 \\ 4N^2, & k = 2. \end{cases}$$

Likewise recall that by Proposition B.6, we have for $0 \leq k \leq 2$:

$$\|p_{\mathbf{j}} - u\|_{W^{k,\infty}(\Omega_{\mathbf{j}})} \leq \frac{(d/2)^{r-k}}{N^{r-k}(r-k)!} \|u\|_{W^{r,\infty}(\Omega_{\mathbf{j}})} \leq \frac{(d/2)^{r-k}}{N^{r-k}(r-k)!} \|u\|_{W^{r,\infty}(\mathbb{R}^d)}.$$

The above being true for all $\mathbf{j} \in \{0, 1, \dots, N\}^d$, we thus deduce that for all $x \in [0, 1]^d$ and $\gamma \in \mathbb{N}^d$ with $\|\gamma\|_1 \leq 2$,

$$\begin{aligned} \left| \partial^\gamma \left(\hat{f}_{NN}(x) - u(x) \right) \right| &= \left| \sum_{\mathbf{j} \in \{0, 1, \dots, N\}^d} \partial^\gamma \left(\phi_{\mathbf{j}}(x) \cdot [p_{\mathbf{j}}(x) - u(x)] \right) \right| \\ &= \left| \sum_{\substack{\mathbf{j} \in \{0, 1, \dots, N\}^d \\ x \in \text{supp}(\phi_{\mathbf{j}})}} \partial^\gamma \left(\phi_{\mathbf{j}}(x) \cdot [p_{\mathbf{j}}(x) - u(x)] \right) \right| \\ &\leq \sum_{\substack{\mathbf{j} \in \{0, 1, \dots, N\}^d \\ x \in \text{supp}(\phi_{\mathbf{j}})}} \left| \partial^\gamma \left(\phi_{\mathbf{j}}(x) \cdot [p_{\mathbf{j}}(x) - u(x)] \right) \right| \\ &\leq 2^d \max_{\alpha \in \mathbb{N}^d: \|\alpha\|_1 \leq 2} \|\partial^\alpha (\phi_{\mathbf{j}} \times (p_{\mathbf{j}} - u))\|_{L^\infty(\Omega_{\mathbf{j}})} \\ &\leq 2^{d+1} \binom{2+d}{2} \sum_{k=0}^2 \|\phi_{\mathbf{j}}\|_{W^{k,\infty}(\Omega_{\mathbf{j}})} \|p_{\mathbf{j}} - u\|_{W^{2-k,\infty}(\Omega_{\mathbf{j}})} \\ &\leq 2^{d+3} \binom{2+d}{2} \sum_{k=0}^2 N^k \frac{(d/2)^{r+k-2}}{N^{r+k-2}(r+k-2)!} \|u\|_{W^{r,\infty}(\mathbb{R}^d)} \\ &\leq 2^{d+3} M \binom{2+d}{2} N^{2-r} \|u\|_{W^{r,\infty}(\mathbb{R}^d)} \end{aligned}$$

whenever $N \geq 2$, where we used inequality (B.9), Lemma B.4, and wrote

$$M := 3 \max_{0 \leq k \leq 2} \frac{(d/2)^{r+k-2}}{(r+k-2)!}.$$

We finish the proof by taking the essential supremum over $x \in [0, 1]^d$ and the maximum over $\{\gamma \in \mathbb{N}^d : \|\gamma\|_1 \leq 2\}$. $\square$

Appendix C

Theoretical guarantees for PINNs solving linear parabolic PDEs

In this Appendix, we show how the framework developed in Chapter 3 to establish theoretical guarantees for PINNs solving elliptic PDEs can be relatively straightforwardly applied analogously to parabolic PDEs. We here aim to clearly explain what the analogous framework becomes, along with what the key theoretical and computational challenges would be in that case.

1 Linear parabolic initial-boundary value problem with Dirichlet boundary conditions

Let $T > 0$ and $Q_T := (0, T) \times \Omega$, with parabolic boundary $\partial_p Q_T := (\{0\} \times \bar{\Omega}) \cup ([0, T] \times \partial\Omega)$. We consider linear initial-boundary value problems of the form (Lieberman, 1996):

$$\begin{cases} u_t(t, x) - \mathcal{L}_t u(t, x) = f(t, x), & (t, x) \in Q_T, \\ u(t, x) = g(t, x), & (t, x) \in (0, T) \times \partial\Omega, \\ u(0, x) = u_0(x), & x \in \Omega, \end{cases} \quad (\text{C.1})$$

where $\mathcal{L}_t$ is a uniformly elliptic second-order operator acting on the space variables x , for example in nondivergence form

$$\mathcal{L}_t = c(t, x) + \sum_{i=1}^d b_i(t, x) \frac{\partial}{\partial x_i} + \frac{1}{2} \sum_{i,j=1}^d \Sigma_{ij}(t, x) \frac{\partial^2}{\partial x_i \partial x_j}. \quad (\text{C.2})$$

The homogeneous case $g \equiv 0$ is the direct parabolic counterpart of (3.3) in Chapter 3.

Problems of the form (C.1) naturally arise in the study of diffusion processes: to illustrate more specifically, let $(X_t^x)_{t \geq 0}$ be the solution of the SDE

$$dX_t^x = b(X_t^x) dt + \varsigma(X_t^x) dW_t, \quad X_0^x = x \in \Omega, \quad (\text{C.3})$$

where $b : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is the drift, $\varsigma : \mathbb{R}^d \rightarrow \mathbb{R}^{d \times d}$ is the diffusion coefficient, and W is a standard d -dimensional Brownian motion. Write $\Sigma := \varsigma \varsigma^T$, and define the first exit time

$$\tau_\Omega^x := \inf\{t \geq 0 : X_t^x \notin \Omega\},$$

and the survival function

$$S(t, x) := \mathbb{P}(\tau_\Omega^x > t),$$

which determines the exit-time distribution. Under standard regularity and uniform ellipticity assumptions, S is the classical solution of (C.1) with the choices

$$u = S, \quad f \equiv 0, \quad g \equiv 0, \quad \text{and } u_0 \equiv 1,$$

and with $\mathcal{L}_t \equiv \mathcal{L}$ equal to the generator of (C.3) (Pavliotis, 2014; Holmes-Cerfon, 2022).

Likewise, if X_0 has density φ_0 on Ω , then the law of X_t admits a density $\varphi(t, \cdot)$ evolving according to the *forward* (Fokker–Planck) equation, i.e. according to (C.1) with

$$u = \varphi, \quad f \equiv 0, \quad g \equiv 0, \quad \text{and } u_0 = \varphi_0,$$

but with the spatial operator replaced by the formal adjoint $\mathcal{L}^*$ of $\mathcal{L}$. This absorbing Dirichlet condition corresponds to “killing” the process at τ_Ω . This model has numerous applications, including reaction kinetics in chemistry and biology, reliability analysis in engineering, or barrier-crossing type events in transition path theory (Schuss et al., 2007; Redner, 2001; Holcman and Schuss, 2015).

2 Technical assumptions

We now introduce the technical conditions we impose on the PDE C.1. In many ways, these can be seen as parallel to Assumptions (A.1)–(A.3) in

Chapter 3, but now expressed on the space-time cylinder Q_T . We will write $a_{ij}(t, x) := \frac{1}{2}\Sigma_{ij}(t, x)$, so that

$$\mathcal{L}_t = c(t, x) + \sum_{i=1}^d b_i(t, x) \partial_{x_i} + \sum_{i,j=1}^d a_{ij}(t, x) \partial_{x_i x_j}.$$

We will assume that the following holds:

P.1 $\Omega \subset \mathbb{R}^d$ is bounded with $C^{1,1}$ boundary.

P.2 (Uniform parabolicity.) $a(t, x)$ is symmetric and uniformly elliptic on $[0, T] \times \bar{\Omega}$: there exists $\lambda > 0$ such that

$$\xi^T a(t, x) \xi \geq \lambda \|\xi\|_2^2 \text{ for all } (t, x) \in [0, T] \times \bar{\Omega}, \text{ and } \xi \in \mathbb{R}^d.$$

P.3 (Boundedness and spatial Lipschitz regularity.)

$$\xi^T a(t, x) \xi \leq \lambda \|\xi\|_2^2 \text{ for all } (t, x) \in [0, T] \times \bar{\Omega} \text{ and all } \xi \in \mathbb{R}^d.$$

P.4 (Higher Hölder regularity.) Fix $\alpha \in (0, 1)$ and an integer $m \in \mathbb{N}_0$, and set $\beta := m + \alpha$. Assume $\partial\Omega$ is $C^{2+\beta}$ and $a_{ij}, b_i, c \in C^{\beta/2, \beta}([0, T] \times \bar{\Omega})$, where $C^{\beta/2, \beta}$ denotes the standard parabolic Hölder class.

Under these assumptions, we have existence of global-in-time solutions, together with a “strong-convexity estimate” akin to Equation (3.6) for the elliptic case:

Theorem C.1. *Under Assumptions **P.1–P.3**, for every $f \in L^2(Q_T)$ there exists a unique*

$$w \in H^1(0, T; L^2(\Omega)) \cap L^2(0, T; H^2(\Omega) \cap H_0^1(\Omega))$$

solving

$$\begin{cases} w_t - \mathcal{L}_t w = f & \text{in } Q_T, \\ w = 0 & \text{on } (0, T) \times \partial\Omega, \\ w(0, \cdot) = 0 & \text{in } \Omega. \end{cases}$$

Moreover, there exists $C_{\text{par}} > 0$ (depending only on d, T, Ω, λ and the coefficient bounds in Assumption **P.3**) such that

$$\|w_t\|_{L^2(Q_T)} + \|w\|_{L^2(0, T; H^2(\Omega))} \leq C_{\text{par}} \|f\|_{L^2(Q_T)}. \quad (\text{C.4})$$

Likewise, if we assume higher regularity on the PDE data, we have higher regularity guarantees on the target function:

Theorem C.2 (Global Schauder estimate of order β). *Assume Assumptions P.2 and P.4. If $f \in C^{\beta/2, \beta}(Q_T)$, then there exists a unique solution*

$$w \in C^{1+\beta/2, 2+\beta}(Q_T) \cap C(\bar{Q}_T)$$

of

$$\begin{cases} w_t - \mathcal{L}_t w = f & \text{in } Q_T, \\ w = 0 & \text{on } (0, T) \times \partial\Omega, \\ w(0, \cdot) = 0 & \text{in } \Omega. \end{cases}$$

Moreover, there exists $C_{\text{sch}} > 0$ such that

$$\|w\|_{C^{1+\beta/2, 2+\beta}(Q_T)} \leq C_{\text{sch}} \|f\|_{C^{\beta/2, \beta}(Q_T)}.$$

Theorems C.1 and C.2 are well-known global L^2 and Schauder regularity results for linear uniformly parabolic equations in non-divergence form with Dirichlet boundary conditions (Lieberman, 1996, Ch. VII). Here, they are needed to ensure, akin to Theorem 3.2, that (a) the solution exists and lies in a smoothness space suitable to neural network approximation and (b) minimizing the PDE residual is indeed akin to approximating the PDE solution.

3 Boundary-adapted PINNs on the space-time cylinder

Similarly to what we saw Chapter 3, the central issue is enforcing the constraints on $\partial_p Q_T$ in a way that is consistent with the strong norm suggested by (C.4). One can indeed check that the simple approach of adding initial and boundary penalties to the loss function is inconsistent with strong norms, and thus a hard-enforcement of the boundary conditions is needed for stronger guarantees.

Fix an integer $r \in \mathbb{N}_0$. Let ρ be an $(r+2)$ -smooth distance approximation in the sense of Definition 3.6. Fix $\eta \in C^\infty([0, T])$ such that $\eta(0) = 0$ and $\eta(t) > 0$ for $t \in (0, T]$. We also fix a function $\ell : \bar{Q}_T \rightarrow \mathbb{R}$ that matches the data on $\partial_p Q_T$, that is, $\ell = g$ on $(0, T) \times \partial\Omega$ and $\ell(0, \cdot) = u_0$ in Ω . When a classical solution is sought, we assume the compatibility condition $u_0 = g(0, \cdot)$ on $\partial\Omega$.

Remark C.3 (Practical construction of boundary-fitting function). The existence of such a ρ and such an ℓ with the regularity needed for higher-order

estimates is not automatic and, in practice, is handled on a case-by-case basis. For example, if one has an extension operator E that maps $g(t, \cdot)$ on $\partial\Omega$ to a function $(Eg)(t, \cdot)$ on $\bar{\Omega}$ of the desired regularity, then one may set $\ell(t, x) := (Eg)(t, x) + \zeta(t)(u_0(x) - (Eg)(0, x))$ with $\zeta \in C^\infty([0, T])$ and $\zeta(0) = 1$. Finding an algorithm to systematically compute such functions in practice, as was done by [Sukumar and Srivastava \(2022\)](#) for the elliptic case, is an important problem to solve.

Definition C.4 (Boundary-adapted parabolic PINNs). *Let w_θ be a FCNN on $[0, T] \times \Omega$ in the sense of Definition 3.5, with input dimension $d + 1$ (that is, $m_0 = d + 1$). We define*

$$v_\theta(t, x) := \ell(t, x) + \eta(t)\rho(x)w_\theta(t, x), \text{ for } (t, x) \in [0, T] \times \bar{\Omega}. \quad (\text{C.5})$$

Then v_θ satisfies the boundary and initial conditions in (C.1) for every θ . If u denotes the solution of (C.1) and $e_\theta := v_\theta - u$, then e_θ satisfies homogeneous boundary and initial conditions, so Theorem C.1 applies once e_θ has the required regularity.

4 Learning-theoretic framework

Having defined the problem and hypothesis space we'll be working with, we can introduce the empirical risk minimization framework we'll be using for the error analysis. As in (3.13)–(3.14), we work with a strong-form residual loss, now on Q_T .

Definition C.5 (Parabolic residual risks). *For any sufficiently regular v on Q_T , define the population risk*

$$\mathcal{R}_T(v) := \|v_t - \mathcal{L}_t v - f\|_{L^2(Q_T)}^2.$$

Let $Z_i = (t_i, x_i)$ be i.i.d. samples from a probability measure μ on Q_T , for instance the uniform distribution on Q_T . The empirical risk is

$$\mathcal{R}_{T,n}(v) := \frac{1}{n} \sum_{i=1}^n (v_t(Z_i) - \mathcal{L}_{t_i} v(Z_i) - f(Z_i))^2. \quad (\text{C.6})$$

Theorem C.1 gives the parabolic analogue of the role played by Theorem 3.4 and (3.8) in Chapter 3.

Corollary C.6 (Residual risk controls a strong parabolic error norm). *Let u be the solution of (C.1), and let v satisfy the same initial and boundary conditions. Then*

$$\|(v - u)_t\|_{L^2(Q_T)}^2 + \|v - u\|_{L^2(0,T;H^2(\Omega))}^2 \leq C_{\text{par}} \mathcal{R}_T(v) - C_{\text{par}} \mathcal{R}_T(u), \quad (\text{C.7})$$

and, in particular, $\|(v - u)_t\|_{L^2(Q_T)}^2 + \|v - u\|_{L^2(0,T;H^2(\Omega))}^2 \leq C_{\text{par}} \mathcal{R}_T(v)$.

4.1 Extending the elliptic ERM analysis to the parabolic case

For the error analysis, the starting point is Corollary C.6, which is the parabolic analogue of the strong-convexity estimate used in Chapter 3. Since u solves (C.1), we have $u_t - \mathcal{L}_t u - f = 0$ in Q_T , hence $\mathcal{R}_T(u) = 0$. If v satisfies the same initial and boundary conditions as u , Corollary C.6 gives the stability estimate

$$\|(v - u)_t\|_{L^2(Q_T)}^2 + \|v - u\|_{L^2(0,T;H^2(\Omega))}^2 \leq C_{\text{par}} \mathcal{R}_T(v).$$

This is the motivation for working with the boundary-adapted ansatz of Definition C.4, which enforces the constraints on $\partial_p Q_T$ by construction, so that every v_θ falls within the scope of the estimate.

Once these stability estimates are established, the excess-risk decomposition from Chapter 3 applies verbatim with $\mathcal{R}$ and $\mathcal{R}_n$ replaced by $\mathcal{R}_T$ and $\mathcal{R}_{T,n}$ from Definition C.5. The two remaining elements to get precise error bounds are an approximation bound for the population minimizer over $\mathcal{F}_{T,n} := \{v_\theta : \theta \in \Theta_n\}$, and an oracle inequality controlling $\mathcal{R}_T(v_n) - \inf_{v \in \mathcal{F}_{T,n}} \mathcal{R}_T(v)$ when v_n minimizes $\mathcal{R}_{T,n}$.

Compared with the elliptic case, the statistical analysis involves the same kind of complexity control, but for the derivatives that appear in the parabolic residual. Writing $Z = (t, x)$, the quantity being learned is the class

$$\{Z \mapsto v_t(Z) - \mathcal{L}_t v(Z) : v \in \mathcal{F}_{T,n}\},$$

so one needs uniform bounds and VC or pseudo-dimension estimates for $\partial_t \mathcal{F}_{T,n}$ together with $\partial_x^\alpha \mathcal{F}_{T,n}$ for $\|\alpha\|_1 \leq 2$. The empirical process induced by the loss function is driven by the strong-form residual $v_t - \mathcal{L}_t v - f$, so the relevant complexity is that of the derivative families that appear in this expression. Concretely, one needs uniform control of the classes $\partial_t \mathcal{F}_T$, $\partial_{x_i} \mathcal{F}_T$, and $\partial_{x_i x_j} \mathcal{F}_T$ on Q_T , where $\mathcal{F}_T := \{v_\theta : \theta \in \Theta\}$. Since $v_\theta = \ell + \eta \rho w_\theta$, each of these derivatives is a linear combination of $\partial_t w_\theta$ and spatial derivatives of w_θ ,

with coefficients given by fixed smooth functions built from η , ρ , and ℓ . As a result, the VC and pseudo-dimension estimates from Chapter 3 extend to the parabolic residual after replacing the input dimension d by $d + 1$ and taking the supremum over the time derivative together with spatial derivatives up to order two.

For approximation, it is convenient to write the error in a factorized form. Setting $\tilde{u} := u - \ell$, one has $\tilde{u} = 0$ on $\partial_p Q_T$, and

$$v_\theta - u = \eta(t)\rho(x) \left(w_\theta(t, x) - \frac{\tilde{u}(t, x)}{\eta(t)\rho(x)} \right).$$

Thus the approximation task is to approximate the quotient $\tilde{u}/(\eta\rho)$ by the network w_θ in norms strong enough to control $\mathcal{R}_T(v_\theta)$. The spatial factor ρ plays the same role as in the elliptic framework of Chapter 2, since for each fixed t the quotient $\tilde{u}(t, \cdot)/\rho(\cdot)$ is controlled up to $\partial\Omega$ under the same structural assumptions on ρ .

Remark C.7 (Boundary-adaptedness in the parabolic case). There is however an additional issue in the parabolic case, which is similar to what we discussed in Proposition 3.9: the behaviour near $t = 0$, where $\eta(0) = 0$. To avoid a loss of regularity in $\tilde{u}/(\eta\rho)$, it is necessary for the parabolic “boundary-fitting” function ℓ to be chosen such that $\tilde{u}(t, \cdot)$ vanishes at $t = 0$ with the same order as $\eta(t)$. Finding a way to systematically construct such maps is an important open question.

4.2 Expected parabolic a priori bound in the maximal-regularity norm

The parabolic analogue of Theorem 3.24 combines the same three ingredients as in Chapter 3, namely stability, approximation, and an oracle inequality, but with the strong norm suggested by Theorem C.1. The below informal theorem thus shows the bound one expects after carrying out these steps:

Theorem C.8 (Parabolic a priori error bound, expected form). *Let u solve (C.1), let $\mathcal{F}_{T,n}$ be a boundary-adapted hypothesis space generated by Definition C.4, and let $v_n \in \operatorname{argmin}_{v \in \mathcal{F}_{T,n}} \mathcal{R}_{T,n}(v)$, where $\mathcal{R}_{T,n}$ is defined in (C.6). Assume the conditions of Theorem C.1, and assume the boundedness and complexity hypotheses needed to apply the localized oracle-inequality argument of Theorem 3.23 to the parabolic residual class induced by $\partial_t - \mathcal{L}_t$. Then there exists a constant $C > 0$, depending on C_{par} and on coefficient bounds from Assumption P.3, such that for all $t \geq 0$ one has with probability*

at least $1 - \exp(-t)$,

$$\|(v_n - u)_t\|_{L^2(Q_T)}^2 + \|v_n - u\|_{L^2(0,T;H^2(\Omega))}^2 \leq C \left(\inf_{v \in \mathcal{F}_{T,n}} \mathcal{R}_T(v) + \frac{\text{VC}_{T,n}}{n} \log n + \frac{t}{n} \right),$$

where $\text{VC}_{T,n}$ denotes a VC-dimension bound for the derivative classes of $\mathcal{F}_{T,n}$ that enter $v_t - \mathcal{L}_t v$, namely $\partial_t \mathcal{F}_{T,n}$ and $\partial_x^\alpha \mathcal{F}_{T,n}$ for $\|\alpha\|_1 \leq 2$.

Under the Schauder regime of Theorem C.2, one expects the approximation term to be controlled by a network approximation rate on Q_T applied to $(u - \ell)/(\eta\rho)$, proceeding similarly as in Chapter 3. Quantitatively, we expect the resulting rates to be the rates from Chapter 3 with the spatial dimension d replaced by the space-time dimension $d + 1$, and with the smoothness measured in the mixed derivatives needed to control v_t and the second-order spatial derivatives appearing in $\mathcal{L}_t$. The statistical term scales in the same way, with $\text{VC}_{T,n}$ replacing the elliptic complexity measure, and the maximal-regularity norm on the left-hand side is the natural parabolic substitute for the elliptic $H^2(\Omega)$ error controlled in Chapter 3.

Appendix D

More on Distance-to-Boundary Functions

In this Appendix, we provide some complementary results on distance-to-boundary functions, as per Definition 3.6. In particular, we will prove Proposition 3.9 regarding the necessity of neural network ansatz using *normalized distance functions* to consistently approximate the PDE solution. We will also describe, in Algorithm 3, an easy-to-implement way to compute such normalized distance functions in arbitrary domains Ω . We refer the reader to (Shapiro, 2007; Sukumar and Srivastava, 2022) for more results and practical considerations around these issues.

1 Proof of Proposition 3.9

To prove Proposition 3.9, we begin by showing that any zero-trace $C^{r+2}(\bar{\Omega})$ function can be expressed—within a tubular neighborhood of $\partial\Omega$ —as the product of the Euclidean distance-to-boundary function $d_{\partial\Omega}$ with a $C^r(\bar{\Omega})$ function. Although this result is a straightforward consequence of standard PDE and geometric regularity theory, we have been unable to locate a reference for it.

Lemma D.1. *Assume that Assumption (A.1) holds, so that Ω is a C^{r+2} domain, and denote by $d_{\partial\Omega}(x)$ the Euclidean distance between $x \in \mathbb{R}^d$ and $\partial\Omega$. Then for any function $u \in C^{r+2}(\bar{\Omega})$ which vanishes on the boundary, there exist a constant $\delta > 0$, an open tubular neighborhood $U := \{x \in \Omega : d_{\partial\Omega}(x) < \delta\}$ and a function $f \in C^r(\bar{U})$ such that*

$$u(x) = f(x) d_{\partial\Omega}(x) \quad \text{for all } x \in U.$$

Proof. Since $\partial\Omega$ is C^{r+2} , it follows from standard results that there exists $\delta > 0$ such that the signed distance function

$$d^*(x) := \begin{cases} d_{\partial\Omega}(x) & \text{if } x \in \Omega, \\ -d_{\partial\Omega}(x) & \text{if } x \in \mathbb{R}^d \setminus \Omega, \end{cases}$$

is of class C^{r+2} on the tubular neighborhood $\mathcal{N}_\delta := \{x : |d^*(x)| < \delta\}$, and the nearest-point projection

$$\pi : \mathcal{N}_\delta \rightarrow \partial\Omega, \quad x \mapsto \text{the unique point of } \partial\Omega \text{ closest to } x,$$

as well as the unit normal $\boldsymbol{\nu}$ on $\partial\Omega$, extend to C^{r+1} maps on $\mathcal{N}_\delta$ (Gilbarg and Trudinger, 1977, Lemma 14.16). In particular, on the one-sided neighborhood $U := \mathcal{N}_\delta \cap \Omega$, every x admits the representation

$$x = \Psi(q, t) := q + t\boldsymbol{\nu}(q), \quad q = \pi(x) \in \partial\Omega, \quad t = d(x) \in [0, \delta), \quad (\text{D.1})$$

and the map $\Psi : \partial\Omega \times [0, \delta) \rightarrow \bar{\Omega}$ is a C^{r+1} diffeomorphism onto its image.

Now for $u \in C^{r+2}(\bar{\Omega})$ and $(q, t) \in \partial\Omega \times [0, \delta)$, define $\tilde{u}(q, t) := u(\Psi(q, t)) = u(q + t\boldsymbol{\nu}(q))$. Because Ψ is C^{r+1} and $u \in C^{r+2}(\bar{\Omega})$, it follows that $\tilde{u} \in C^{r+1}(\partial\Omega \times [0, \delta))$. For $t > 0$, by the fundamental theorem of calculus in the t -variable we have

$$\tilde{u}(q, t) = \int_0^t \partial_t \tilde{u}(q, s) ds = t \int_0^1 \partial_t \tilde{u}(q, \theta t) d\theta,$$

and hence

$$\tilde{f}(q, t) := \frac{\tilde{u}(q, t)}{t} = \int_0^1 \partial_t \tilde{u}(q, \theta t) d\theta.$$

Define $\tilde{f}(q, 0) := \partial_t \tilde{u}(q, 0)$. Since $\partial_t \tilde{u} \in C^r(\partial\Omega \times [0, \delta))$, it follows that $\tilde{f} \in C^r(\partial\Omega \times [0, \delta))$.

Finally, define $f := \tilde{f} \circ \Psi^{-1}$ on the set $U = \mathcal{N}_\delta \cap \Omega$. Since Ψ is a C^{r+1} diffeomorphism, it follows that $f \in C^r(\bar{U})$. Lastly, we have by construction that $f(x) = u(x)/d_{\partial\Omega}(x)$ for all $x \in U$, hence the proof is complete. $\square$

Equipped with Lemma D.1, we can now proceed with the proof of Proposition 3.9.

Proof of Proposition 3.9. Under Assumptions (A.1)—(A.3), we have that $\tau \in C^{r+2}(\bar{\Omega})$, hence by Lemma D.1, there exist $\delta_1 > 0$ and $f_1 \in C^r(\bar{\Omega})$ such that $\tau(x) = f_1(x)d_{\partial\Omega}(x)$ for all $x \in \Omega \cap U_{\delta_1}$, where U_{δ_1} denotes the width

δ_1 one-sided tubular neighborhood of $\partial\Omega$. Likewise, because $\rho \in C^{r+2}(\bar{\Omega})$, we can find $\delta_2 > 0$ and $f_2 \in C^r(\bar{\Omega})$ such that $\rho(x) = f_2(x)d_{\partial\Omega}(x)$ for all $x \in \Omega \cap U_{\delta_2}$. Note also that ρ does not vanish anywhere in Ω by Definition 3.6, hence the ratio τ/ρ is always well-defined and C^{r+2} -smooth away from the boundary $\partial\Omega$. It is thus enough to prove our result to show that f_2 remains strictly positive in a tubular neighborhood of $\partial\Omega$.

To that end, we reproduce the change of coordinates argument from the proof of Lemma D.1: denote by $\Psi : \partial\Omega \times [0, \delta_2)$ the C^{r+1} diffeomorphism defined as in (D.1), and let $\tilde{\rho}(q, t) := \rho(\psi(q, t))$. Applying the fundamental theorem of calculus in the normal direction, we again define

$$\tilde{f}_2(q, t) := \begin{cases} \int_0^1 \partial_t \tilde{\rho}(q, \theta t) d\theta, & t > 0 \\ \partial_t \tilde{\rho}(q, 0), & t = 0, \end{cases}$$

which is a $C^r(\partial\Omega \times [0, \delta_2))$ function. Furthermore, we have by definition $\partial_t \tilde{\rho}(q, 0) = \partial_{\nu} \rho = 1$. Hence by continuity, and by shrinking δ_2 if necessary, we find that $\tilde{f}_2(x) \geq 1/2$ for all $x \in U_{\delta_2}$.

Finally, let $\delta := \delta_1 \wedge \delta_2$. We have shown that τ/ρ is C^{r+2} -smooth in Ω , and equal to f_1/f_2 on U_δ , where both f_1, f_2 lie in $C^r(\bar{\Omega})$, and f_2 does not vanish. It thus follows that τ/ρ lies in $C^r(\bar{\Omega})$, as desired. $\square$

2 Construction of higher-order normalized distance approximations

Definition 3.6 enforces normalization in the first normal derivative. In some constructions it is useful to match additional normal derivatives at the boundary. The procedure below is adapted from (Shapiro, 2007).

For a function u defined in a neighborhood of $\partial\Omega$ and $q \in \partial\Omega$, we write

$$\partial_{\nu}^k u(q) := \left. \frac{d^k}{dt^k} u(q + t\nu(q)) \right|_{t=0}, \quad \text{for } k \in \mathbb{N},$$

whenever the derivative exists.

We will say that u is normalized to order m if its normal derivatives agree with those of the distance-to-boundary function up to order m .

Definition D.2. Let $r \in \mathbb{N}^+$ and $m \in \{1, \dots, r\}$. A function $\rho : \bar{\Omega} \rightarrow \mathbb{R}$ is called an r -smooth approximate distance function normalized to order m if $\rho \in C^r(\bar{\Omega})$, $\rho(x) = 0$ for all $x \in \partial\Omega$, $\rho(x) > 0$ for all $x \in \Omega$, and

$$\partial_{\nu} \rho(q) = 1, \quad \text{and} \quad \partial_{\nu}^k \rho(q) = 0 \quad \text{for } k = 2, \dots, m, \quad \text{for all } q \in \partial\Omega.$$

Algorithm 3 : Shapiro normalization to order m

Require: A function $\phi \in C^{r+m}(\bar{\Omega})$ with $\phi(x) = 0$ for all $x \in \partial\Omega$, $\phi(x) > 0$ for all $x \in \Omega$, and $\nabla\phi(q) \neq 0$ for all $q \in \partial\Omega$. An integer $m \in \mathbb{N}^+$. A tubular neighborhood U of $\partial\Omega$ with nearest-point projection $\pi : U \rightarrow \partial\Omega$ as in Lemma D.1.

1: Define the first-order normalization

$$\rho_1(x) \leftarrow \frac{\phi(x)}{\sqrt{\phi(x)^2 + \|\nabla\phi(x)\|^2}}, \quad \text{for } x \in \bar{\Omega}.$$

2: **for** $k = 2, \dots, m$ **do**

3: Compute the boundary function $a_k(q) \leftarrow \partial_{\nu}^k \rho_{k-1}(q)$ for $q \in \partial\Omega$.

4: Extend a_k to U by $\tilde{a}_k(x) \leftarrow a_k(\pi(x))$ for $x \in U$.

5: Update on U by

$$\rho_k(x) \leftarrow \rho_{k-1}(x) - \frac{1}{k!} \rho_1(x)^k \tilde{a}_k(x), \quad \text{for } x \in U.$$

6: Set $\rho_k \leftarrow \rho_{k-1}$ on $\bar{\Omega} \setminus U$.

7: **end for**

8: **return** ρ_m .

Algorithm 3 starts from a smooth defining function and enforces the conditions in Definition D.2 by cancelling successive normal derivatives at the boundary.

Proposition D.3 shows that Algorithm 3 enforces the required boundary normalization.

Proposition D.3. *Let $r \in \mathbb{N}^+$ and $m \in \mathbb{N}^+$. Under the assumptions of Algorithm 3, the output ρ_m satisfies*

$$\rho_m(q) = 0, \quad \partial_{\nu} \rho_m(q) = 1, \quad \text{and} \quad \partial_{\nu}^k \rho_m(q) = 0 \quad \text{for } k = 2, \dots, m, \quad \text{for all } q \in \partial\Omega.$$

In particular, if ρ_m is positive in Ω , then ρ_m is an r -smooth approximate distance function normalized to order m in the sense of Definition D.2.

Proof. Fix $q \in \partial\Omega$ and define $\gamma(t) := q + t\nu(q)$ for $t \geq 0$ sufficiently small, and $\varphi_k(t) := \rho_k(\gamma(t))$.

We first check the base case $k = 1$. Since $\phi(q) = 0$ and $\nabla\phi(q) \neq 0$, the level set $\{\phi = 0\}$ has unit inward normal $\nu(q) = \nabla\phi(q)/\|\nabla\phi(q)\|$. Moreover,

$\rho_1(q) = 0$ and, by differentiating $\rho_1 = \phi(\phi^2 + \|\nabla\phi\|^2)^{-1/2}$ along ν and then evaluating at q ,

$$\partial_\nu \rho_1(q) = \frac{\partial_\nu \phi(q)}{\|\nabla\phi(q)\|} = \frac{\nabla\phi(q) \cdot \nu(q)}{\|\nabla\phi(q)\|} = 1.$$

Hence $\varphi_1(t) = t + O(t^2)$ as $t \rightarrow 0^+$, so $\varphi_1(t)^k = t^k + O(t^{k+1})$ for each fixed $k \in \mathbb{N}^+$.

We now prove by induction that for each $k \in \{1, \dots, m\}$,

$$\varphi'_k(0) = 1, \text{ and } \varphi_k^{(j)}(0) = 0 \text{ for } j = 2, \dots, k.$$

The statement holds for $k = 1$ by the argument above. Assume it holds for $k - 1$ with some $k \in \{2, \dots, m\}$. By the update in Algorithm 3, there is a neighborhood of $t = 0$ on which

$$\varphi_k(t) = \varphi_{k-1}(t) - \frac{1}{k!} \varphi_1(t)^k \varphi_{k-1}^{(k)}(0),$$

because $\tilde{a}_k$ is constant along the normal ray $t \mapsto \gamma(t)$. Differentiating $j < k$ times and evaluating at $t = 0$, every term involving $\varphi_1(t)^k$ contains a factor $\frac{d^\ell}{dt^\ell}(\varphi_1(t)^k)|_{t=0}$ with $\ell < k$, which vanishes since $\varphi_1(t)^k = t^k + O(t^{k+1})$. It follows that $\varphi_k^{(j)}(0) = \varphi_{k-1}^{(j)}(0)$ for $j = 1, \dots, k - 1$, hence $\varphi'_k(0) = 1$ and $\varphi_k^{(j)}(0) = 0$ for $j = 2, \dots, k - 1$ by the induction hypothesis.

Finally, differentiating k times and evaluating at $t = 0$, the only remaining term from the product rule is the one in which all k derivatives fall on $\varphi_1(t)^k$, yielding

$$\varphi_k^{(k)}(0) = \varphi_{k-1}^{(k)}(0) - \frac{1}{k!} (\varphi_1(t)^k)^{(k)}|_{t=0} \varphi_{k-1}^{(k)}(0) = \varphi_{k-1}^{(k)}(0) - \varphi_{k-1}^{(k)}(0) = 0,$$

since $(\varphi_1(t)^k)^{(k)}|_{t=0} = k!$.

Translating back to $\partial_\nu^j \rho_m(q) = \varphi_m^{(j)}(0)$ proves the claimed boundary identities. $\square$

Remark D.4. Algorithm 3 uses a tubular neighborhood U and the nearest-point projection π only to act as an extension of the boundary data a_k off $\partial\Omega$ that is constant along normal rays. For the update $\rho_k = \rho_{k-1} - \frac{1}{k!} \rho_1^k \tilde{a}_k$, one does not need closed-form access to π . It is enough to have a procedure that, given a_k on $\partial\Omega$, produces an (approximately) constant-along-normals extension in a neighborhood of the boundary.

Two common choices to construct such an approximation are the following:

- One can compute $\tilde{a}_k$ by propagating the boundary values along the normal direction into U . A convenient analytic description is the transport problem

$$\boldsymbol{\nu}(x) \cdot \nabla \tilde{a}_k(x) = 0 \quad \text{in } U, \quad \text{with } \tilde{a}_k|_{\partial\Omega} = a_k.$$

On a grid covering U , this can be approximated by one-sided finite differences for the directional derivative $\boldsymbol{\nu} \cdot \nabla \tilde{a}_k$, chosen so that values are taken from neighboring points that are closer to $\partial\Omega$ and then carried outward into U .

- One can approximate the closest-point map numerically and set $\tilde{a}_k(x) \approx a_k(\pi_h(x))$, for example by a closest-point mapping to a discrete surface representation, or by a few Newton iterations that project x onto the implicit surface $\{\phi = 0\}$ when $\partial\Omega$ is given by a level set.

In either case, the role of π in Proposition D.3 is conceptual, and the construction can be implemented using only local computations in a narrow band around $\partial\Omega$.

Appendix E

More on Feynman–Kac Representations and Probabilistic PDE Solutions

In this Appendix, we complete the discussion from Chapter 4 by gathering a comprehensive list of probabilistic representation formulas for linear and nonlinear parabolic and elliptic PDEs. We will present representative examples—diffusion and jump–diffusion generators, semilinear and fully nonlinear equations, branching diffusions, and stochastic control/HJB problems—to illustrate that PDEs admitting such *Feynman–Kac-like* representations form a large and mathematically rich class. We refer the interested readers to (Dynkin, 1965; Ethier and Kurtz, 2009; Karatzas and Shreve, 2014) and references within for further background on Markov processes and classical Feynman–Kac formulas.

1 Linear parabolic and elliptic equations with diffusion generators

Let B be a d -dimensional Brownian motion and consider the Itô SDE

$$dX_s = b(X_s) ds + \sigma(X_s) dB_s, \quad X_t = x \in \mathbb{R}^d, \quad (\text{E.1})$$

where $b : \mathbb{R}^d \rightarrow \mathbb{R}^d$ and $\sigma : \mathbb{R}^d \rightarrow \mathbb{R}^{d \times d}$ satisfy standard Lipschitz and linear growth conditions (Karatzas and Shreve, 2014). The associated generator $\mathcal{L}$ acts on smooth test functions φ as

$$(\mathcal{L}\varphi)(x) = b(x) \cdot \nabla \varphi(x) + \frac{1}{2} \text{Tr}(a(x) D^2 \varphi(x)), \quad a(x) := \sigma(x) \sigma(x)^\top. \quad (\text{E.2})$$

A prototypical linear parabolic Cauchy problem on $[0, T] \times \mathbb{R}^d$ is

$$\begin{cases} \partial_t u(t, x) + (\mathcal{L}u(t, \cdot))(x) - V(t, x)u(t, x) + f(t, x) = 0, & (t, x) \in [0, T) \times \mathbb{R}^d, \\ u(T, x) = g(x), & x \in \mathbb{R}^d, \end{cases} \quad (\text{E.3})$$

with bounded (or suitably integrable) data V, f, g . Under standard regularity assumptions on b, σ, V, f, g (e.g. [Dynkin, 1965](#); [Karatzas and Shreve, 2014](#)), the classical Feynman–Kac formula yields

$$u(t, x) = \mathbb{E}^{t, x} \left[e^{-\int_t^T V(s, X_s) ds} g(X_T) + \int_t^T e^{-\int_t^s V(r, X_r) dr} f(s, X_s) ds \right], \quad (\text{E.4})$$

where X solves (E.1) with initial condition $X_t = x$.

Elliptic boundary value problems such as Poisson, mean exit time, or committor equations arise by dropping the time derivative in (E.3). For a bounded C^2 domain $D \subset \mathbb{R}^d$, define the exit time $\tau_D := \inf\{s \geq 0 : X_{t+s} \notin D\}$. Under standard conditions, the Dirichlet problem

$$\begin{cases} (\mathcal{L}u)(x) - V(x)u(x) + f(x) = 0, & x \in D, \\ u(x) = g(x), & x \in \partial D, \end{cases} \quad (\text{E.5})$$

admits the representation

$$u(x) = \mathbb{E}^x \left[\int_0^{\tau_D} e^{-\int_0^s V(X_r) dr} f(X_s) ds + e^{-\int_0^{\tau_D} V(X_r) dr} g(X_{\tau_D}) \right], \quad (\text{E.6})$$

and analogous formulas hold for Neumann and Robin boundary conditions in terms of reflected diffusions and boundary local time (cf. [Dynkin, 1965](#); [Fleming and Soner, 2006](#)).

For the boundary value problem (4.1)–(4.2) considered in Section 3, the operator $\mathcal{L}$ and data (f, g) fit into the elliptic template (E.5) (with V encoding any zeroth-order term). In particular, in the Poisson, mean exit time, and committor benchmarks of the main text one typically has $V \equiv 0$, so that (E.6) reduces to

$$u(x) = \mathbb{E}^x \left[\int_0^{\tau_D} f(X_s) ds + g(X_{\tau_D}) \right],$$

which is of the form (4.7), with running and terminal functionals r and h determined (up to the sign convention for $\mathcal{L}$) by the source term f and boundary data g . In the algorithms of the main text, expectations such as (E.4) and (E.6) are approximated by simulating time-discretized trajectories of (E.1) (e.g. Euler–Maruyama) and averaging over samples. Weak and strong error estimates for such schemes can be found in ([Kloeden and Platen, 1992](#); [Glasserman, 2004](#)).

2 Linear equations with nonlocal generators

Let X be a Lévy process in $\mathbb{R}^d$ with Lévy triplet (b, a, ν) , where $b \in \mathbb{R}^d$, $a \in \mathbb{R}^{d \times d}$ is symmetric nonnegative definite, and ν is a Lévy measure. Its generator $\mathcal{L}$ acts on $C_c^2(\mathbb{R}^d)$ as

$$\begin{aligned} (\mathcal{L}\varphi)(x) &= b \cdot \nabla \varphi(x) + \frac{1}{2} \operatorname{Tr}(a D^2 \varphi(x)) \\ &\quad + \int_{\mathbb{R}^d \setminus \{0\}} \left[\varphi(x+y) - \varphi(x) - \mathbf{1}_{\{|y| \leq 1\}} y \cdot \nabla \varphi(x) \right] \nu(dy), \end{aligned} \quad (\text{E.7})$$

see, for instance, (Applebaum, 2009).

Linear parabolic and elliptic equations with the nonlocal operator $\mathcal{L}$ in (E.7) have the same abstract form as (E.3) and (E.5). Under suitable conditions on (b, a, ν) and the data, the solution u admits a Feynman–Kac representation identical to (E.4), with X now a Lévy (or more generally a Lévy-type) process (see, e.g., Dynkin, 1965; Applebaum, 2009). Monte Carlo approximation then proceeds by simulating jump–diffusion paths (using standard jump-adapted schemes) and averaging the corresponding functionals (Glasserman, 2004). In principle, FK-PINNs could be extended to this setting by supervising the network with such Lévy paths instead of pure diffusions.

3 Semilinear parabolic equations and BSDE representations

Consider the semilinear parabolic PDE

$$\begin{cases} \partial_t u(t, x) + (\mathcal{L}u(t, \cdot))(x) + F(t, x, u(t, x), \sigma(x)^\top \nabla u(t, x)) = 0, & (t, x) \in [0, T] \times \mathbb{R}^d, \\ u(T, x) = g(x), & x \in \mathbb{R}^d, \end{cases} \quad (\text{E.8})$$

where $\mathcal{L}$ is the diffusion generator (E.2), F is Lipschitz in (y, z) with linear growth, and g is bounded and Lipschitz. Let X solve (E.1) and consider the forward–backward system

$$dX_s = b(X_s) ds + \sigma(X_s) dB_s, \quad X_t = x, \quad (\text{E.9})$$

$$Y_s = g(X_T) + \int_s^T F(r, X_r, Y_r, Z_r) dr - \int_s^T Z_r dB_r, \quad s \in [t, T]. \quad (\text{E.10})$$

Under the above assumptions, there exists a unique adapted solution (Y, Z) to (E.10) (Pardoux and Peng, 1990; El Karoui et al., 1997), and the nonlinear

Feynman–Kac formula identifies

$$u(t, x) = Y_t^{t,x}, \quad (\text{E.11})$$

where $(X^{t,x}, Y^{t,x}, Z^{t,x})$ denotes the solution of (E.9)–(E.10) with $X_t^{t,x} = x$. Conversely, if u is a sufficiently regular solution of (E.8), then $(Y_s, Z_s) = (u(s, X_s), \sigma(X_s)^\top \nabla u(s, X_s))$ solves (E.10) (Pardoux and Peng, 1990; El Karoui et al., 1997).

Monte Carlo approximations for (E.8) discretize the forward SDE and the BSDE and estimate the conditional expectations in the backward recursion by regression-based methods (Bouchard and Touzi, 2004). In the present paper we only need the linear case where $F(t, x, y, z) = -V(t, x)y + f(t, x)$, which reduces to the classical FK form in Section 3.1, but the nonlinear Feynman–Kac formula suggests a natural path to extending FK-PINNs to general semilinear problems by coupling PINNs with Monte Carlo BSDE solvers.

4 Semilinear equations with polynomial nonlinearities and branching diffusions

Branching diffusion processes provide representations for semilinear equations of reaction–diffusion type. A prototype is

$$\partial_t u(t, x) = (\mathcal{L}u(t, \cdot))(x) + \lambda(u(t, x)^p - u(t, x)), \quad u(0, x) = u_0(x), \quad (\text{E.12})$$

with $\lambda > 0$, integer $p \geq 2$, and $\mathcal{L}$ the diffusion generator (E.2). Under appropriate integrability and non-explosion conditions, the solution of (E.12) can be written as

$$u(t, x) = \mathbb{E}^x \left[\prod_{i=1}^{N_t} u_0(X_t^{(i)}) \right], \quad (\text{E.13})$$

where $(X_t^{(i)})_{1 \leq i \leq N_t}$ are the positions of particles in a branching diffusion system with spatial motion governed by $\mathcal{L}$ and binary or p -ary branching at rate λ (see, e.g., Dynkin, 1991). More general semilinear PDEs with polynomial nonlinearities in u and ∇u admit representations in terms of marked branching diffusions (Henry-Labordère et al., 2019; Agarwal and Claisse, 2020).

Monte Carlo evaluation of (E.13) proceeds by simulating the associated random branching tree and averaging the weight functional, provided the variance is controlled (Henry-Labordère et al., 2019).

5 Hamilton–Jacobi–Bellman equations from stochastic control

Let A be a compact metric control set. For an A -valued progressively measurable control process α , consider the controlled diffusion

$$dX_s^{t,x,\alpha} = b(X_s^{t,x,\alpha}, \alpha_s) ds + \sigma(X_s^{t,x,\alpha}, \alpha_s) dB_s, \quad X_t^{t,x,\alpha} = x, \quad (\text{E.14})$$

with running cost $\ell : [0, T] \times \mathbb{R}^d \times A \rightarrow \mathbb{R}$ and terminal cost $g : \mathbb{R}^d \rightarrow \mathbb{R}$. The value function is

$$u(t, x) := \sup_{\alpha} \mathbb{E}^{t,x,\alpha} \left[\int_t^T \ell(s, X_s^{t,x,\alpha}, \alpha_s) ds + g(X_T^{t,x,\alpha}) \right], \quad (\text{E.15})$$

where the supremum ranges over all admissible controls. Writing $\mathcal{L}^a$ for the diffusion generator associated with the frozen control $a \in A$, the corresponding Hamilton–Jacobi–Bellman equation is

$$\begin{cases} \partial_t u(t, x) + \sup_{a \in A} \{(\mathcal{L}^a u(t, \cdot))(x) + \ell(t, x, a)\} = 0, & (t, x) \in [0, T] \times \mathbb{R}^d, \\ u(T, x) = g(x), & x \in \mathbb{R}^d. \end{cases} \quad (\text{E.16})$$

Under standard compactness, measurability, and regularity assumptions, u is the unique bounded viscosity solution of (E.16) and coincides with (E.15) (see [Fleming and Soner, 2006](#)).

Stochastic representations of the form (E.15) can be approximated numerically by simulation-based dynamic programming and related Monte Carlo methods for controlled diffusions; see ([Fleming and Soner, 2006](#)) for an overview. For high-dimensional parabolic PDEs and BSDEs, including HJB-type equations from stochastic control, deep learning-based methods can approximate the gradient of the value function (interpreted as a feedback control or policy) by a neural network and train it via Monte Carlo simulation of the underlying diffusion; see, for example, ([E et al., 2017](#)). While FK-PINNs in the present work are restricted to linear operators, the HJB representation suggests a possible extension in which the neural network is supervised by Monte Carlo estimates of value functions under sampled controls.

6 Fully nonlinear parabolic equations and second-order BSDEs

Consider a fully nonlinear parabolic PDE of the form

$$-\partial_t u(t, x) + F(t, x, u(t, x), Du(t, x), D^2u(t, x)) = 0, \quad u(T, x) = g(x), \quad (\text{E.17})$$

where F is nonlinear in the Hessian D^2u and satisfies suitable ellipticity, monotonicity, and regularity conditions ensuring a comparison principle and well-posedness in the viscosity sense. Second-order backward SDEs (2BSDEs) provide pathwise representations of solutions to (E.17) (Cheridito et al., 2007). Roughly speaking, given a forward diffusion X , one considers an adapted quadruple (Y, Z, Γ, A) solving a 2BSDE whose first component satisfies

$$Y_t^{t,x} = u(t, X_t^{t,x}), \quad (\text{E.18})$$

and conversely any such 2BSDE solution identifies the unique viscosity solution u of (E.17) (Cheridito et al., 2007). Numerical schemes discretize both the forward diffusion and the 2BSDE and approximate conditional expectations by Monte Carlo in an enlarged state space, providing a probabilistic route to numerical solutions of fully nonlinear equations (Cheridito et al., 2007). Extending FK-PINNs to this 2BSDE setting is an interesting direction for future research but lies beyond the scope of this paper.

7 Summary table

For convenience, Table E.1 summarizes the PDE classes discussed above, the corresponding operators, the probabilistic objects, and some standard references.

PDE class	Schematic form	Probabilistic object	Representation / references
Linear parabolic / elliptic (diffusion)	$\partial_t u + \mathcal{L}u - Vu + f = 0$, or $\mathcal{L}u - Vu + f = 0$ in D	Diffusion X with generator $\mathcal{L}$, exit time τ_D	Feynman-Kac expectations such as (E.4), (E.6); see (Dynkin, 1965; Karatzas and Shreve, 2014)
Linear nonlocal (Lévy / jump-diffusion)	$\partial_t u + \mathcal{L}u - Vu + f = 0$ with $\mathcal{L}$ as in (E.7)	Lévy or Lévy-type process X	Same Feynman-Kac structure as in the diffusion case with nonlocal generator (Dynkin, 1965; Applebaum, 2009)
Semilinear parabolic (BSDE)	$\partial_t u + \mathcal{L}u + F(t, x, u, \sigma^\top \nabla u) = 0$	Forward diffusion X + BSDE (Y, Z)	Nonlinear Feynman-Kac: $u(t, x) = Y_t^{t,x}$ as in (E.11) (Pardoux and Peng, 1990; El Karoui et al., 1997)
Semilinear (polynomial)	$\partial_t u + \mathcal{L}u = \Phi(t, x, u, \nabla u)$, polynomial	(Marked) branching diffusion system Φ	Branching representations such as (E.13) (Dynkin, 1991; Henry-Labordère et al., 2019; Agarwal and Claisse, 2020)
HJB (stochastic control)	$\partial_t u + \sup_{a \in A} \{\mathcal{L}^a u + \ell(t, x, a)\} = 0$	Controlled diffusion X^α	Value function representation (E.15) (Fleming and Soner, 2006)
Fully nonlinear parabolic	$-\partial_t u + F(t, x, u, Du, D^2 u) = 0$	Forward diffusion X + 2BSDE (Y, Z, Γ, A)	Fully nonlinear Feynman-Kac via 2BSDE, $u(t, x) = Y_t^{t,x}$ as in (E.18) (Cheridito et al., 2007)

Table E.1: PDE classes covered by standard probabilistic representations. Here $\mathcal{L}$ denotes a (possibly nonlocal) Markov generator, X the corresponding Markov or controlled process, and expectations $\mathbb{E}^{t,x}[\cdot]$ are taken with respect to the law of X started from (t, x) . The FK-PINNs studied in the main text use the first (linear diffusion) row, but the same Monte Carlo supervision principle could in principle be adapted to the other classes.

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